



EST4 Technical Reference Manual

P/N 3102301-EN • REV 001 • ISS 15JUL19



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Important information

Regulatory information

This product has been designed to meet the requirements of NFPA 72 National Fire Alarm Signaling Code, UL 864 Standard for Control Units and Accessories for Fire Alarm Systems, and CAN/ULC-S527 Standard for Control Units for Fire Alarm Systems.

Limitation of liability

To the maximum extent permitted by applicable law, in no event will United Technologies Corporation be liable for any lost profits or business opportunities, loss of use, business interruption, loss of data, or any other indirect, special, incidental, or consequential damages under any theory of liability, whether based in contract, tort, negligence, product liability, or otherwise. Because some jurisdictions do not allow the exclusion or limitation of liability for consequential or incidental damages the preceding limitation may not apply to you. In any event the total liability of United Technologies Corporation shall not exceed the purchase price of the product. The foregoing limitation will apply to the maximum extent permitted by applicable law, regardless of whether United Technologies Corporation has been advised of the possibility of such damages and regardless of whether any remedy fails of its essential purpose.

Installation in accordance with this manual, applicable codes, and the instructions of the authority having jurisdiction is mandatory.

While every precaution has been taken during the preparation of this manual to ensure the accuracy of its contents, United Technologies Corporation assumes no responsibility for errors or omissions.

Advisory messages

Advisory messages alert you to conditions or practices that can cause unwanted results. The advisory messages used in this document are shown and described below.

WARNING: Warning messages advise you of hazards that could result in injury or loss of life. They tell you which actions to take or to avoid in order to prevent the injury or loss of life.

Caution: Caution messages advise you of possible equipment damage. They tell you which actions to take or to avoid in order to prevent the damage.

Note: Note messages advise you of the possible loss of time or effort. They describe how to avoid the loss. Notes are also used to point out important information that you should read.

EST4 FCC compliance

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

3-MODCOM(P) FCC compliance

Cautions

- To ensure proper operation, this dialer must be installed according to the enclosed installation instructions. To verify that the dialer is operating properly and can successfully report an alarm, it must be tested immediately after installation, and periodically thereafter, according to the enclosed test instructions.
- In order for the dialer to be able to seize the phone line to report an alarm or other event when other customer equipment (telephone, answering system, computer modem, etc.) connected to the same line is in use, the dialer *must* be connected to a properly installed RJ-31X jack. The RJ-31X jack must be connected in series with, and ahead of, all other equipment attached to the same phone line. Series installation of an RJ-31X jack is depicted in the wiring diagram. If you have any questions concerning these instructions, you should consult your telephone company or a qualified installer.

Testing

When programming emergency numbers or making test calls to emergency numbers, remain on the line and briefly explain to the dispatcher the reason for the call. Perform programming and testing activities in the off-peak hours, such as early morning or late evenings.

Compliance

- **For equipment approved before July 23, 2001:** This dialer complies with Part 68 of the FCC rules. A label attached to the dialer contains, among other information, the FCC registration number and ringer equivalence number (REN) for this equipment. If requested, this information must be provided to the telephone company.
- **For equipment approved after July 23, 2001:** This dialer complies with Part 68 of the FCC rules and the requirements adopted by the Administrative Council for Terminal Attachments (ACTA). A label attached to the dialer contains, among other information, a product identifier in the format US:AAAEQ##TXXXX. If requested, this information must be provided to the telephone company.
- The plug and jack used to connect the dialer to the premises wiring and telephone network must comply with the applicable FCC Part 68 rules and requirements adopted by ACTA. The dialer must be connected to a compliant RJ-31X or RJ-38X jack using a compliant cord. If a modular telephone cord is supplied with the dialer, it is designed to meet these requirements. See installation instructions for details.
- A ringer equivalence number is used to determine how many devices you can connect to a telephone line. If the total REN value for all devices connected on a telephone line exceeds that allowed by the telephone company, the devices may not ring on an incoming call. In most (but not all) areas the total REN value should not exceed 5.0. To be certain of the total REN value allowed on a telephone line, contact the local telephone company.

For products approved after July 23, 2001, the REN is part of the product identifier in the format US:AAAEQ##TXXXX. The digits ## represent the REN without a decimal point. Example: 03 is an REN of 0.3. For earlier products the REN is listed separately.

- If the dialer is harming the telephone network, the telephone company will notify you in advance that temporary discontinuance of service may be required. If advance notice isn't practical, the telephone company will notify you as soon as possible. You will also be advised of your right to file a complaint with the FCC, if you believe it is necessary.
- The telephone company may make changes to its facilities, equipment, operations, or procedures that could affect the operation of the dialer. If this happens, the telephone company will provide advance notice in order for you to make necessary modifications to maintain uninterrupted service.
- If you are experiencing problems with the dialer, contact the manufacturer for repair or warranty information. If the dialer is harming the telephone network, the telephone company may request that you disconnect the dialer until the problem is resolved.

- The dialer contains no user serviceable parts. In case of defects, return the dialer for repair.
- You may *not* connect the dialer to a public coin phone or a party line service provided by the telephone company.

3-MODCOM(P) Industry Canada information

Note: The Industry Canada label identifies certified equipment. This certification means that the equipment meets certain telecommunications network protective, operational, and safety requirements. Industry Canada does not guarantee the equipment will operate to the user's satisfaction.

Before installing this equipment, users should ensure that it is permissible to be connected to the facilities of the local telecommunications company. The equipment must also be installed using an acceptable method of connection. The customer should be aware that compliance with the above conditions may not prevent degradation of service in some situations.

Repairs to certified equipment should be made by an authorized Canadian maintenance facility designated by the supplier. Any repairs or alterations made by the user to this equipment, or equipment malfunctions, may give the telecommunications company cause to request the user disconnect the equipment.

Caution: Users should not attempt to make connections themselves, but should contact the appropriate electric inspection authority, or electrician, as appropriate

Users should ensure for their own protection that the electrical ground connections of the power utility, telephone lines, and internal metallic water pipe system, if present, are connected together. This precaution may be particularly important in rural areas.

Note: The Load Number (LN) assigned to each terminal device denotes the percentage of the total load to be connected to a telephone loop which is used by the device, to prevent overloading. The termination on a loop may consist of any combination of devices subject only to the requirements that the sum of the Load Numbers of all the devices does not exceed 100.

Chapter 1

Introduction

Summary

This chapter provides information about this manual and other related documentation.

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About this manual

This manual provides information on how to install, program, and operate an EST4 control unit. It is organized as follows:

Chapter 1, Introduction: Provides information about this manual and other related documentation.

Chapter 2, Product description: Provides technical descriptions of the control unit and its operation. It also provides descriptions of the command menus.

Chapter 3, Operating instructions: Provides instructions for operating the life safety system from the control unit user interface. It is intended for those who might be expected to operate the control unit in a fire alarm emergency.

Chapter 4, Installation: Provides installation information for system components and applications. It is intended for those trained and authorized to maintain the life safety system.

Chapter 5, Supplementary applications: Provides technical descriptions of supplementary applications that can expand system capabilities.

Chapter 6, Preventive maintenance and testing: Provides maintenance schedules and testing procedures for life safety system. It is intended for those trained and authorized to maintain the life safety system.

Chapter 7, Service and troubleshooting: Provides instructions for servicing and troubleshooting the life safety system. It is intended for those trained and authorized to maintain the system.

Appendix A, System calculations: Provides worksheets for sizing standby batteries, and for calculating the maximum wire lengths for notification appliance circuits and intelligent addressable loops.

Appendix B, System addressing: Provides a list of logical addresses for system modules and devices.

Appendix C, Pseudo points: Provides pseudo point addresses, sources, and descriptions that can be used to troubleshoot pseudo point events.

Appendix D, UL 864 programming requirements: Summarizes the operation of the system.

Intended audience

The intent of this document is to provide trained and certified personnel with technical, operational, service, and maintenance information.

Lifesafety system limitations

The purpose of an automatic life safety system is to provide early detection and warning of a developing fire. There are a number of uncontrollable factors that can prevent or severely limit the ability of an automatic life safety system to provide adequate protection. As such, an automatic life safety system cannot guarantee against loss of life or loss of property.

Two main causes of system failures are improper installation and poor maintenance. The best way to minimize these types of system failures is to have only trained life safety system professionals design, install, test, and maintain your life safety system in accordance with national and local fire codes.

Life safety systems will not operate without electrical power. As fires frequently cause power interruption, we suggest that you discuss ways to safeguard the electrical system with your local fire protection specialist.

Chapter 2

Product description

Summary

This chapter provides descriptions of the fire alarm control unit (FACU) and its components, and the operator interface controls, indicators, and commands.

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System overview

The EST4 control unit can operate as a stand-alone control unit or in a Class B, Class A, Class X, Class N, or full mesh EST4 life safety network. Control unit cabinets are available in a variety of sizes and are designed to use modular hardware components that can meet most applications.

EST4 user interface modules mount onto the inner door UI frame assembly and hardware modules onto the backbox chassis rail for data processing, intrapanel communication of command and control data, response data, audio signal processing, and power distribution. Most field wiring is terminated using plug-in terminal strips for easy installation and servicing of modules. Refer to “Control unit components” on page 10 for a list of standard FACU components and optional equipment.

System size

Table 1 lists the maximum hardware capabilities for EST4 control units.

Table 1: EST4 hardware capabilities

Item	Maximum capacity
EST4 stand-alone	
Signaling line circuits (SLC)	10
Addressable detectors	1,250 (125 detectors each SLC)
Addressable modules	1,250 (125 modules each SLC)
Initiating device circuits	144 Class B
Notification appliance circuits	72 Class B
Reverse polarity outputs	3
EST4 network	
Remote annunciators and graphic annunciators	The networked system is capable of scaling to 40 emergency control centers with user interfaces (UI) for command and control, audio and firefighters telephone
Networked EST4 control units	40 nodes total Scale: 40 nodes; 100,000 addressable devices (40 nodes × 10 signaling line circuits × 250 devices) 23,040 annunciation indicators; 23,040 control switches 250 audio messages 40 command centers; 100 audio channels per notification control area, 200 audio channels per system

4-CU computer specifications

The 4-CU computer specifications provided in Table 2 below are the *minimum* recommended requirements for performing 4-CU functions. While the minimum requirements are acceptable, 4-CU performance can be improved by increasing computer memory and processor speed.

Two factors affect the amount of time it takes for a 4-CU database conversion to EST4 control units:

- Computer size (processor speed and memory)
- Project size (number of control units in a network and number of devices)

Example: For a computer with the minimum recommended requirements, a project with 12 control units and 4,000 devices may take only 10 minutes for a database conversion. However, a project with 30 control units and 25,000 devices may take 90 minutes for a database conversion.

Table 2: Minimum recommended 4-CU computer specifications

	Minimum requirement
Operating system	Windows 10 Professional (64 bit)
Processor [1]	7th Generation Core i5
Memory [1]	16 GB RAM
Hard drive	512 GB SSD (unencrypted)
Display/Monitor	17 inch, 1920 × 1080

[1] The project size affects the amount of time for a database conversion. Increasing the computer memory size and processor speed can reduce conversion time.

Cybersecurity

The EST4 life safety system provides levels of protection and the flexibility to deploy in a manner that can help with cybersecurity requests. The system's flexibility allows the installer to work with the building owner and local IT professionals to match the system to their IT and cybersecurity needs.

- **Airgap:** EST4 has multiple levels of physical security that can be used to meet simple airgap requirements in systems without firewalls.
- **Internal network:** EST4 has multiple levels of physical security that can be used, such as tamper switches and control unit keyed door locks.
- **Firewalls:** When external connections are needed, EST4 protects the life safety network by offering the fire listed 4-FWAL series firewall modules. The 4-FWAL separates the life safety network from outside networks.
- **Access and authentication:** Access to an EST4 system is based on industry-standard, best practices. First a physical barrier, the locked door, provides a barrier to the system. To gain access to higher level system operations, a seven-digit PIN is required. All control unit access is role-based; a user can only perform actions explicitly allowed by system configuration and the permissions policy set in the system configuration.

See *EST4 Network and Audio Application Guide* (P/N 3102306-EN) for detailed information on how EST4 addresses cybersecurity protection.

Network topologies

EST4 allows wiring topology to be determined by the needs of your installation. The various supported topologies can be combined on a single network as well as hybrid variations. For more information see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

Supported fire alarm system types

The EST4 fire alarm control unit is listed for use as the following types of fire alarm systems:

- Protected Premises (Local) Fire Alarm System
- Proprietary Supervising Station Alarm System
- Auxiliary Alarm System
- Remote Supervising Station Alarm System
- Smoke Control System
- Releasing Fire Alarm System
- Central Station Service Alarm System
- In-building Mass Notification System
- Emergency Communication

The minimum requirements to meet each service listing are shown below. For optional components, refer to *EST4 UL Listing Document* (P/N 3102629-EN).

Protected Premises (Local) Fire Alarm System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-COMREL common relay card (optional)
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display module(s)

Proprietary Supervising Station Alarm System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-COMREL common relay card (optional)
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display module(s)

Auxiliary system

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-COMREL common relay card (optional)
- 4-FWAL1 firewall module (at least one)
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display module(s)
- 3-MODCOM modem communicator module or 3-MODCOMP modem communicator plus pager module (at least one)
- 3-OPS off-premises signaling module

Remote Supervising Station Alarm System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-COMREL common relay option card
- 4-NET-TP(-HC) twisted pair controller or 4-NET-CAT Small Form-Factor Pluggable (SFP) network controller (one pair is needed for interunit network communication)
- 4-NET-SM -MM, -SMH, -SMU, -SMD SFP fiber-optic network controller (one pair is needed for interunit network communication; the 4-NET-SMU and 4-NET-SMD make one up/down pair)
- 4-FWAL-SM -MM -SMH -SMU -SMD -CAT SFP firewall adapters (one plus the appropriate 4-FWAL is required for non-interunit interfacing)
- 4-FWAL-1 -2 -3 -4 firewall module (one or more control-display modules required if control unit is used as a firefighter smoke control station (FSCS))
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display module(s)
- 3-MODCOM modem communicator module or 3-MODCOMP modem communicator plus pager module (at least one)
- 3-OPS off-premises signaling module (when used with outside wiring a UL listed secondary protector manufactured by DITEK, Model DTK36VLPSCP must be used)

Smoke Control System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display module(s) (one or more control-display modules required if control unit is used as an FSCS)

Releasing Fire Alarm System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display modules
- SIGA-REL releasing module

Central Station Service Alarm System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-NET-TP twisted pair 4-NET-CAT SFP network controller (one pair is needed for interunit network communication)
- 4-NET-SM -MM, -SMH, -SMU, -SMD SFP fiber-optic network controller (one pair is needed for interunit network communication; the 4-NET-SMU and 4-NET-SMD make one up/down pair)
- 4-FWAL-SM -MM -SMH -SMU -SMD -CAT SFP firewall adapters (one plus the appropriate 4-FWAL is required for non-interunit interfacing)
- 4-FWAL-1 -2 -3 -4 firewall module (one or more control-display modules required if control unit is used as an FSCS)
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display modules
- 3-MODCOM modem communicator module or 3-MODCOMP modem communicator plus pager module

In-building Mass Notification System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-NET-TP(-HC) twisted pair controller or 4-NET-CAT SFP network controller (one pair is needed for interunit network communication)
- 4-NET-SM -MM, -SMH, -SMU, -SMD SFP fiber-optic network controller (one pair is needed for interunit network communication; the 4-NET-SMU and 4-NET-SMD make one up/down pair)
- 4-FWAL-SM -MM -SMH -SMU -SMD -CAT SFP firewall adapters (one plus the appropriate 4-FWAL is required for non-interunit interfacing)
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display modules
- 4-MIC paging microphone
- 3-ZA20(A/B), 3-ZA40(A/B), or 3-ZA95 zone amplifier (at least one amplifier)

EVAC Emergency Communication System

- 3-CAB5B, 7B, 14B, or 21B control unit backbox
- 4-CAB8D, 16D, 24D, 24DRL or 3-CAB7D, 14D, 21D door assembly
- 3-PPS/M(-230) power supply
- 4-CPU central processor unit
- 4-ANNCPU annunciator central processor unit
- 4-ANNAUDTEL annunciator audio telephone interface module
- 4-LCD or 4-3LCD display module user interface (4-3LCD when not using 4-CABxD doors)
- 4-LCDAUDTEL audio and firefighter telephone control LCD display module user interface
- 4-24L, 4-24L12S, 4-24L18S, 4-24L24S, 3-24R, 3-24Y, 3-24G, 3-12RY, 3-12SG, 3-12SR, 3-12SY, 3-12/S1GY, 3-12/S1RY, 3-12/S2Y, 3-12/S2R, 3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3S2RY, 3-4/3SGYWR control-display modules
- 4-MIC paging microphone
- 3-ZA20(A/B), 3-ZA40(A/B), or 3-ZA95 zone amplifier (at least one amplifier)

Migrating an EST3 system to an EST4 system

EST3 life safety control units can be migrated to an EST4 life safety control unit by replacing certain modules and hardware. Equipment replacement requirements depends on your system design. Refer to the *EST3 to EST4 Migration Guide* (P/N 3102305-EN) for details on how to upgrade an EST3 life safety control unit to an EST4 life safety control unit.

Control unit components

This section describes the compatible equipment enclosures, modules, and accessories that can be used with the EST4 control unit.

Note: Option cards and modules are ordered separately and installed in the field.

Table 3: Equipment enclosures

Model	Description
Backboxes	
3-CAB series backboxes	Equipment backboxes: Semi-flush or surface mount backboxes that provide the housing for locally installed EST4 components and batteries. 3-CAB5B: Includes a built in rail assembly providing five local rail module spaces, one footprint for a power supply, and a 1/2 footprint. Space for two 10 Ah standby batteries. Use with 4-CAB8D series door assemblies. See installation sheet P/N 3102288 for specifications. 3-CAB7B: Space for one chassis assembly providing seven local rail module spaces. Space for two 18 Ah standby batteries. Use with 4-CAB16D series door assemblies. See installation sheet P/N 387557 for specifications. 3-CAB14B: Space for two chassis assemblies each providing seven local rail module spaces. Space for two 18 Ah standby batteries. Use with 4-CAB24D series door assemblies. See installation sheet P/N 387557 for specifications. 3-CAB21B: Space for three chassis assemblies each providing seven local rail module spaces. Space for two 18 Ah standby batteries. Use with 4-CAB24DL series door assemblies. See installation sheet P/N 387557 for specifications.
4-ANNMT series annunciator backboxes	Annunciator backboxes. Provides the housing for annunciator equipment. See installation sheet P/N 3102276-EN for specifications. 4-2ANNMT: 2 wide annunciator backbox includes a trim kit for surface mounting or semiflush mounting. Used with 4-2ANN annunciator assembly. 4-4ANNMT: 4 wide annunciator backbox includes a trim kit for surface mounting or semiflush mounting. Used with 4-4ANN annunciator assembly. 4-6ANNMT: 6 wide annunciator backbox includes a trim kit for surface mounting or semiflush mounting. Used with 4-6ANN annunciator assembly. 4-8ANNMT: 4 wide x two rows high annunciator backbox, surface mount. Used with 4-8ANN annunciator assembly. For semiflush mounting use 3-CAB5B. 4-16ANNMT: 8 wide x two rows high annunciator backbox, surface mount. Used with 4-16ANN annunciator assembly. For semiflush mounting use 3-CAB7B. 4-24ANNMT: 8 wide x three rows high annunciator backbox, surface mount. Used with 4-24ANN annunciator assembly. For semiflush mounting use 3-CAB14B.
Doors	
4-CAB8D series doors	4-CAB8D Series Door: Includes an inner and outer door. The outer door provides a viewing window and is secured with a key lock. The inner door incorporates one 4-4ANNFA UI frame assembly that provides mounting for the optional user interface modules (4-LCD, 4-LCDANN, 4-LCDAUDTEL, 24L series control-display modules, 4-FT, and 4-MIC). See installation sheet P/N 3102272-EN for specifications. Note: Use with the 3-CAB5B Equipment Backbox and the 4-8ANNMT. 4-CAB8D: Bronze door assembly 4-CAB8DR: Red door assembly

Model	Description
4-CAB16D series doors [1]	4-CAB16D Series Door: Includes an inner and outer door. The outer door provides a viewing window and is secured with a key lock. The inner door incorporates two 4-8ANNFA UI frame assemblies that provide mounting for the optional user interface modules (4-LCD, 4-LCDANN, 4-LCDAUDEL, 24L series control-display modules, 4-FT, and 4-MIC). See installation sheet P/N 3102273-EN for specifications. Note: Use with the 3-CAB7B Equipment Backbox and the 4-16ANNMT. • 4-CAB16D: Bronze door assembly • 4-CAB16DR: Red door assembly
4-CAB24D series doors [1]	4-CAB24D Series Door: Includes an inner and outer door. The outer door provides a viewing window and is secured with a key lock. The inner door incorporates three 4-8ANNFA UI frame assemblies that provide mounting for the optional user interface modules (4-LCD, 4-LCDANN, 4-LCDAUDEL, 24L series control-display modules, 4-FT, and 4-MIC). See installation sheet P/N 3102274-EN for specifications. Note: Use with the 3-CAB14B Equipment Backbox and the 4-24ANNMT. • 4-CAB24D: Bronze door assembly • 4-CAB24DR: Red door assembly
4-CAB24DL series doors [1]	4-CAB16D Series Door: Includes an inner and outer door. The outer door provides a viewing window and is secured with a key lock. The inner door incorporates three 4-8ANNFA UI frame assemblies that provide mounting for the optional user interface modules (4-LCD, 4-LCDANN, 4-LCDAUDEL, 24L series control-display modules, 4-FT, and 4-MIC). See installation sheet P/N 3102275-EN for specifications. Note: Use with the 3-CAB21B Equipment Backbox. • 4-CAB24DL: Bronze door assembly • 4-CAB24DRL: Red door assembly
4-2ANND	Annunciator door assembly: Includes an inner and outer door. The outer door has a viewing window and is secured with a key lock. The inner door incorporates a 4-2ANNFA UI Frame Assembly that provides mounting for user interface modules. The 4-2ANND mounts to the 4-2ANNMT backbox and is included with the 4-2ANN annunciator assembly. See installation sheet P/N 3102278-EN for specifications.
4-4ANND	Annunciator door assembly: Includes an inner and outer door. The outer door has a viewing window and is secured with a key lock. The inner door incorporates a 4-4ANNFA UI Frame Assembly that provides mounting for user interface modules. The 4-4ANND mounts to the 4-4ANNMT backbox and is included with the 4-4ANN annunciator assembly. See installation sheet P/N 3102289-EN for specifications.
4-6ANND	Annunciator door assembly: Includes an inner and outer door. The outer door has a viewing window and is secured with a key lock. The inner door incorporates a 4-6ANNFA UI Frame Assembly that provides mounting for user interface modules. The 4-6ANND mounts to the 4-6ANNMT backbox and is included with the 4-6ANN annunciator assembly. See installation sheet P/N 3102290-EN for specifications.

[1] The outer door can be modified to install a locally sourced New York City lock. For more information see *EST4 New York City Panel Lock Installation Sheet* (P/N 3102293-EN).

Table 4: Power supplies

Model	Description
3-BBC/M(-230)	Booster/Charger Supply: Provides additional power to the control unit, and shares the control unit's 24 VDC electrical load with the 3-PPS/M. The 3-BBC/M is comprised of a heat sink assembly and a monitor module. The booster supply monitor module provides the interface between the booster power supply and the control unit, making the required data and power connections to and from the rail chassis. The heat sink assembly mounts onto four studs on the back panel of the 3-CHAS7 rail chassis and the monitor module plugs into the chassis rail. See installation sheet P/N 270495-EN for specifications. Note: The module must be firmware version 3.63 or higher.

Model	Description
3-BPS/M(-230)	Booster Power Supply: Provides additional power to the control unit, and shares the control unit's 24 VDC electrical load with the 3-PPS/M. The 3-BPS/M is comprised of a heat sink assembly and a monitor module. The booster supply monitor module provides the interface between the booster power supply and the control unit, making the required data and power connections to and from the rail chassis. The heat sink assembly mounts onto four studs on the back panel of the 3-CHAS7 rail chassis and the monitor module plugs into the chassis rail. See installation sheet P/N 270495-EN for specifications. Note: The module must be firmware version 3.63 or higher.
3-PPS/M(-230)	Primary Power Supply: Provides the required power and related supervision functions for the control unit, as well as filtered and regulated power to the LRMs and 24 VDC for ancillary equipment. The 3-PPS/M is comprised of a heat sink assembly and a monitor module. The heat sink assembly mounts onto four studs on the back panel of the 3-CHAS7 rail chassis and the monitor module plugs into the chassis rail. See installation sheet P/N 270495-EN for specifications. Note: The module must be firmware version 3.63 or higher.

Table 5: Control unit sub-assemblies

Model	Description
Central processor units	
4-CPU	Central Processor Module: Processes all information from modules installed in the same cabinet and networked control units. Installs onto two spaces on the rail chassis assembly. See installation sheet P/N 3102258-EN for specifications. Note: The 4-COMREL Common Relay Module and 4-AUDTELS Audio IO and Telephone Riser Source Module can be used with the 4-CPU.
4-ANNCPU	Annunciator Central Processor Module: Processes all control information from switches on the displays, as well as processing the data received from the network for display. The module is used in FACU and remote annunciator enclosures. See installation sheet P/N 3102264-EN for specifications. Note: The 4-ANNAUDTEL Annunciator Audio Telephone Interface Module can be used with the 4-ANNCPU.
4-CPUGRPH	Graphic Annunciator Central Processor Module: Provides connection to 3-EVDVR(A) and 3-EVPWR(A) graphic annunciators. Installs in the graphic annunciator backbox. See installation sheet P/N 3102346-EN for specifications.
LCDs	
4-LCD	Main LCD Display Module: Provides the user interface for the fire alarm control unit. The module includes operator command controls, LED indicators, and a resistive touch screen (4.5 x 3.4 in., 18 bit color, 640 x 480 pixels) capable of displaying eight events (768 10-point characters). The module installs in the top, far left space on the inner door UI frame assembly and connects to the 4-CPU. See installation sheet P/N 3102261-EN for specifications.
4-3LCD	EST3 Upgrade Main LCD Display Module: Replacement for the 3-LCD when upgrading EST3 systems. Installs on the 4-CPU. See installation sheet P/N 3102259-EN for specifications.
4-LCDAUDTEL	Audio and Firefighter Telephone Control LCD Display Module: Provides the user interface for the firefighter telephone and microphone. Installs in the far left space on the inner door UI frame assembly and connects to the 4-ANNCPU. See installation sheet P/N 3102262-EN for specifications.

Model	Description
Chassis rail modules	
3-IDC8/4	Traditional zone I/O module: Provides eight Class B traditional direct connect Initiating Device Circuits (IDC) for compatible two-wire smoke detectors and dry contact initiating devices or four NAC circuits. Installs on the backbox chassis rail. See installation sheet 270492-EN for specifications. Note: The module must be firmware version 3.60 or higher.
3-MODCOM	Modem communicator module: Provides dialer communications between the control unit and remote locations over telephone lines. Installs on the backbox chassis rail. See installation sheet P/N 387476-EN for specifications. Note: The module must be firmware version 3.60 or higher. The firmware can be upgraded only by serial download.
3-MODCOMP	Modem communicator module with pager interface: Provides dialer communications between the control unit and remote locations over telephone lines. It also can send information to individual predefined pagers. Installs on the backbox chassis rail. See installation sheet P/N 387476-EN for specifications. Note: The module must be firmware version 3.60 or higher. The firmware can be upgraded only by serial download.
3-OPS	Off-premises signaling module: Provides three independent reverse polarity circuits for transmitting alarm, supervisory, and trouble signals to compatible receivers. Can also be configured to provide a single reverse polarity connection (reverses on alarm, opens on trouble) or for a local energy municipal box connection. Installs on the backbox chassis rail. See installation sheet P/N 270494 for specifications. Note: The module must be firmware version 3.00 or higher.
3-SDC1	Signature data circuit card: Provides a Class B or Class A signaling line circuit and resettable 24 VDC for powering conventional two-wire smoke detector circuits on Signature modules. Installs on the rail chassis. See installation sheet P/N 3101772-EN for specifications.
3-SDC1-HC	Signature loop controller card: Optional replacement card for the 3-SDC1 for installations where the signaling line circuit has more than 90 isolators and isolator bases (with SIGA1 sensors installed). Provides a Class B or Class A signaling line circuit and resettable 24 VDC for powering conventional two-wire smoke detector circuits on Signature modules. See installation sheet P/N 3102326-EN for specifications.
3-SDDC1	Dual Signature loop controller module: Provides two Class B or Class A Signature data loops for Signature detectors and modules. The module also provides a connection for powering conventional two-wire smoke detector circuits on Signature modules. Installs on the backbox chassis rail. See installation sheet P/N 3100584 for specifications. Note: The module must be firmware version 5.20 or higher. The firmware can be upgraded only by serial download.
3-SSDC1	Single Signature loop controller module: Provides one Class B or Class A Signature data loop for Signature detectors and modules. The module also provides a connection for powering conventional two-wire smoke detector circuits on Signature modules. Installs on the backbox chassis rail. See installation sheet P/N 3100584 for specifications. Note: The module must be firmware version 5.20 or higher. The firmware can be upgraded only by serial download.
3-ZA20(A/B)	20-watt zoned audio amplifier module: Provides a standard 25 or 70 VRMS audio circuit, a power-limited 24 VDC notification appliance circuit, and integrated 1000 Hz temporal tone generators. EST4 nodes store their prerecorded messages locally. Programmed messages become active when a locally generated alarm is initiated, even when a catastrophic network failure occurs. The zoned amplifier's temporal tone generators will produce a backup signal should the main CPU suffer a catastrophic failure. Installs on the backbox chassis rail. See installation sheet P/N 387463 for specifications. Note: The module must be firmware version 3.40 or higher.

Model	Description
3-ZA40(A/B)	40-watt zone audio amplifier module: Provides a standard 25 or 70 VRMS audio circuit, a power-limited 24 VDC notification appliance circuit, and integrated 1000 Hz temporal tone generators. EST4 nodes store their prerecorded messages locally. Programmed messages become active when a locally generated alarm is initiated, even when a catastrophic network failure occurs. The zoned amplifier's temporal tone generators will produce a backup signal should the main CPU suffer a catastrophic failure. Installs on the backbox chassis rail. See installation sheet P/N 387463 for specifications. Note: The module must be firmware version 3.40 or higher.
3-ZA95	95-watt zone audio amplifier module: Provides a standard 25 or 70 VRMS audio circuit and integrated 1000 Hz temporal tone generators. EST4 nodes store their prerecorded messages locally. Programmed messages become active when a locally generated alarm is initiated, even when a catastrophic network failure occurs. The zoned amplifier's temporal tone generators will produce a backup signal should the main CPU suffer a catastrophic failure. Installs on the backbox chassis rail. See installation sheet P/N 3100765 for specifications. Note: The module must be firmware version 3.40 or higher.
Riser and relay cards	
4-ANNAUDEL	Annunciator audio telephone interface option card: Used with the 4-ANNCPU to process information from the 4-FT Firefighter Telephone Master Handset and 4-MIC Paging Microphone. The option card installs on the 4-ANNCPU. See installation sheet P/N 3102325-EN for specifications.
4-AUDELS	Audio IO and Telephone Riser Source Module: Used with the 4-CPU to process information from the 4-FT Firefighter Telephone Master Handset and 4-MIC Paging Microphone. Provides a single telephone riser and/or up to four 1 VRMS Audio inputs/outputs to a 4-CPU. The option card and terminal board mount on the back side of the 4-CPU. See installation sheet P/N 3102279-EN for specifications.
4-COMREL	Common Relay Module: Provides common alarm, common trouble, and common supervisory relay outputs, and connections for end-of-line resistors. Installs on the 4-CPU. See installation sheet P/N 3102284-EN for specifications.
User interface modules	
4-24L series control-display modules	EST4 control-display modules: Provides additional user interface capability for the life safety system. Installs onto the inner door UI frame assembly. See installation sheet P/N 3102263-EN for specifications. • 4-24L: 24 LEDs (programmable as red, yellow, green, white, or blue) • 4-24L12S: 12 LED-switches with 2 LEDs per switch (programmable as red, yellow, green, white, or blue) • 4-24L18S: 6 groups of 3 LED-switches with 4 LEDs per switch (programmable as red, yellow, green, white, or blue) • 4-24L24S: 24 LED-switches with 24 LEDs (programmable as red, yellow, green, white, or blue)
4-FT	Firefighter Telephone Master Handset: Provides local telephone handset support to EST4 control units and remote annunciators. Installs onto the inner door UI frame assembly. Controls and indicators are provided via a 4-LCD or 4-LCDAUDEL display modules or 4-24L series control display modules. See installation sheet 3102267-EN for specifications.
4-MIC	Paging microphone: Provides local microphone paging support to an EST4 control unit. Installs onto the inner door UI frame assembly. Controls and indicators are provided via a 4-LCD or 4-LCDAUDEL display module. See installation sheet P/N 3102266-EN for specifications.

Table 6: Network sub-assemblies [1]

Model	Description
Firewall modules	
4-FWAL series firewall modules	Firewall modules: Blocks unauthorized access to the EST4 life safety network, while permitting outward communication. See installation sheet P/N 3102280-EN for specifications. • 4-FWAL1: Supports communication with third party devices and the CDR-3 bell coder. • 4-FWAL2: Supports communication with third party devices, CDR-3 bell coder, IP DACT dialer, and web services. • 4-FWAL3: Supports communication with third party devices, CDR-3 bell coder, web services, and email services. • 4-FWAL4: Supports communication with third party devices, CDR-3 bell coder, IP DACT dialer, web services, and email services.
Small Form-Factor Pluggable (SFP) modules	
4-FWAL-CAT	100 Mbps SFP network adapter. Provide 100Base-TX RJ-45 internetwork connections between the EST4 system and external or peripheral systems. The adapter mounts in one SFP slot in the 4-FWALx Series Firewall Module. See installation sheet P/N 3102322-EN for specifications.
4-FWAL series fiber-optics network adapters	Firewall SFP network adapters: Input/output devices for use on the 4-FWAL series firewall modules. The 4-FWAL series SFP controllers provide internetwork connections between the EST4 system and external or peripheral systems. The adapters mount in one SFP slot in the 4-FWALx Series Firewall Module. See installation sheet P/N 3102277-EN for specifications. 4-FWAL-SM: SFP network media interface, single-mode fiber-optic, with a 9/125 μ (G.652) fiber pair up to 6.2 miles (10 km). 4-FWAL-MM: SFP network media interface, multi-mode fiber-optic, with a 50/125 μ (OM3/OM4) fiber pair up to 1.24 mi. (2 km), 62.5/125 μ (OM1) fiber pair up to 0.62 mi. (1 km), or a 100/140 μ fiber pair up to 150 m. 4-FWAL-SMH: SFP network media interface, single-mode fiber-optic, high-power output, with a 9/125 μ (G.652) fiber pair up to 24.8 mi (40km). 4-FWAL-SMU: SFP network media interface, bi-directional, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-FWAL-SMU must be paired with a 4-FWAL-SMD. 4-FWAL-SMD: SFP network media interface, bi-directional, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-FWAL-SMD must be paired with a 4-FWAL-SMU.
4-NET-CAT	100 Mbps SFP Network Controller: Supports all command and control, live and recorded audio functions, and live firefighters telephone network functions. Plugs into one SFP slot in the 4-CPU, 4-ANNCPU, 4-CPUGRPH, or the 4-NET-AD modules. See installation sheet P/N 3102270-EN for specifications.
4-NET-TP	Twisted Pair SFP Network Controller: Supports all command and control, and live and recorded audio functions. The 4-NET-TP also supports live firefighters telephone network functions. Plugs into one SFP slot in the 4-CPU, 4-ANNCPU, 4-CPUGRPH, or the 4-NET-AD modules. See installation sheet 3102268-EN for specifications.
4-NET-TP-HC [2]	Twisted Pair SFP Network Controller High Capacity: Supports all command and control, and live and recorded audio functions. Plugs into one SFP slot in the 4-CPU, 4-ANNCPU, 4-CPUGRPH, or the 4-NET-AD modules. See installation sheet 3102268-EN for specifications.

Model	Description
4-NET series fiber-optics network controllers	Fiber-optic SFP network controllers: Provides intranetwork connections within the life safety system. The controllers mount in one SFP slot in the 4-CPU, 4-ANNCPU, 4-CPUGRPH, or the 4-NET-AD network controllers. See installation sheet P/N 3102271-EN for specifications. 4-NET-MM: Multi-mode fiber-optic, with a 50/125 μ (OM3/OM4) fiber pair up to 1.24 mi. (2 km), 62.5/125 μ (OM1) fiber pair up to 0.62 mi. (1 km), or a 100/140 μ fiber pair up to 150 m. 4-NET-SMH: Dual filament, single-mode fiber-optic, high-power output, with a 9/125 μ (G.652) fiber pair up to 24.8 mi (40km). 4-NET-SMU: Single filament, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-NET-SMU must be paired with a 4-NET-SMD. 4-NET-SMD: Single filament, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-NET-SMD must be paired with a 4-NET-SMU. 4-NET-SM: Dual filament, single-mode fiber-optic, with a 9/125 μ (G.652) fiber pair up to 6.2 miles (10 km).
Expander modules	
4-NET-AD	Network Adder Module: Provides additional 4-NET series SFP network connections, allowing additional network interconnections from a control unit or annunciator. See installation sheet 3102287-EN for specifications.
4-NET-XT [3]	Twisted Pair Full Duplex Network Extender Module: Provides long distance network data between nodes, typically using existing telephone single-pair wires. Supports all command and control, and live and recorded audio functions. See installation sheet 3102286-EN for specifications.

[1] EST4 supports network pass-through mode of operation on the 4-CPU, 4-ANNCPU, 4-NET-AD, and 4-CPUGRPH CPUs. Pass-through from the CPU module eth1 and eth0 network connections is supported from one SFP to another SFP of any type except a 4-NET-TP to a 4-NET-TP-HC connection. For more information see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

[2] The 4-NET-TP-HC supports a maximum of two simultaneous audio IP streams of live paging or 1 VRMS input in addition to prerecorded audio. A network with 4-NET-TP-HC connections supports a maximum of 15 nodes. Note that a 4-NET-XT is not for use with firefighter telephones.

[3] The 4-NET-XT supports a single audio IP stream of live paging or 1 VRMS input in addition to prerecorded audio. A network with 4-NET-XT connections supports a maximum of 15 nodes. The 4-NET-XT is not for use with firefighter telephones.

Table 7: Control unit accessories

Model	Description
4-USBHUB	Multi-port USB Hub Module: Provides isolated RS-232 for a CDR-3 connection, USB and terminal block connections, isolated host for system printer, and no isolated device USB connections. Installs in a half-footprint space in the enclosure. See installation sheet P/N 3102281-EN for specifications.
Tamper switches	Tamper switch: Used to detect an open cabinet door. See installation sheet P/N 387422-EN for specifications. 4-TAMP: Used with EST4 doors. Installs on the top or side of a 3-CAB7B, 3-CAB14B, 3-CAB21B. 3-TAMP: Used with EST3 doors. Installs on the top or side of a 3-CAB7B, 3-CAB14B, 3-CAB21B. 3-TAMP5: Used with EST3 or EST4 doors. Installs on the top or side of a 3-CAB5, 3-CAB5B. 3-TAMP5RCC: Used with 3-RCC enclosure doors. Installs on the top or side of a 3-RCCxR.
CDR-3	Bell Coder: Provides coded outputs in response to alarm conditions for systems requiring march time, temporal, or unique coded outputs in separate zones and decodes alarm codes embedded in printer messages received through RS-232 input. Installs on a half footprint space on the equipment backbox. See installation sheet P/N 3100023 for specifications.

Model	Description
PT-1S	Serial Printer: Connects to the fire alarm control unit to print system events such as status changes, active events, or reports. See installation sheet P/N 3100989 for specifications. Note: For an isolated connection, use the 4-USBHUB.

EST4 LCD display modules

An EST4 system uses LCD display modules to provide the user interface for the fire alarm control unit.

- The 4-LCD and 4-3LCD modules are main LCD display modules that include operator command controls, LED indicators, and an LCD screen.
- The 4-LCDAUDTEL Audio and Firefighter Telephone Control LCD Display Module includes an operator LCD screen.

Operator alert signal

In coordination with visual event notifications on the user interface, the control unit employs an audible signal to alert the operator of off-normal system conditions for active event conditions. Different operator alert signal patterns are assigned to the event queues. Table 8 below shows the signal patterns used in the US Market. Note that the market setting for your control unit may result in different patterns.

Note: The operator alert signal may automatically sound a reminder signal, if supported by the market.

Table 8: Operator alert patterns (US Market patterns shown)

Event queue	Operator alert pattern
Call In	3 pulses every 16 seconds
Control Request	2 pulses every 4 seconds
Alarm	3 pulses every 4 seconds
Emergency	3 pulses every 4 seconds
Supervisory	2 pulses every 4 seconds
Building	2 pulses every 4 seconds
Disable	1 pulse every 4 seconds
Trouble	1 pulse every 4 seconds
Ground	1 pulse every 4 seconds
Other	1 pulse every 16 seconds

User interface command controls and indicators

The 4-LCD and 4-3LCD user interface command controls and indicators are shown in the figure below. See Table 9 below for a description of each control and indicator.

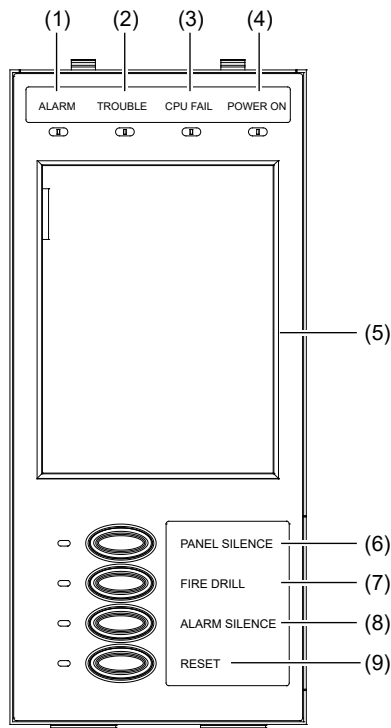


Table 9: User interface command controls and indicators descriptions

No.	Control/Indicator	Description
1	Alarm LED	The LED serves as a common alarm event indicator.
2	Trouble LED	The LED serves as a common trouble event indicator.
3	CPU Fail LED	The LED indicates that the CPU has detected a processor failure. Processor failures must be reset manually using the Reset button. A steady LED indicates that the CPU Fail pseudo point is active. An off LED indicates that CPU communications have been restored.
4	Power On LED	The LED indicates the power status for the control unit. The indicator is set by the selected market. Power Always On: A steady LED indicates that either primary power or standby power are present. An off LED indicates that both primary power and standby power are missing. Power Follows AC: A steady LED indicates that primary power is present and off when primary power is missing. An off LED indicates that standby power is present or is missing. Power Only If Available AC and Battery: A steady LED indicates that both primary power and standby power are present. An off LED indicates that either primary power or standby power are missing.
5	LCD screen	The LCD screen display provides information relevant to the current condition of the control unit, as well as operator controls. See “LCD screen indications” on page 21 for details on the LCD screen.

No.	Control/Indicator	Description
6	Panel Silence button and LED (button one default label and setting)	This configurable button is programmed by default as Panel Silence. The button can be configured as: Panel Silence: For local systems, pressing the Panel Silence button turns the operator alert signal off and acknowledges all events have been reviewed. For proprietary systems, the Panel Silence button is not operational. The operator alert signal only silences after each event has been individually acknowledged in the order displayed. The Panel Silence LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating the control unit is in an off-normal condition and the control unit has been placed in Panel Silence mode, and turns off when the system returns to normal. Drill: Pressing the Fire Drill button activates the system drill response. Pressing the button a second time cancels the drill response. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Alternate Alarm Sensitivity: Smoke detectors can operate using two levels of sensitivity: primary and alternate. Pressing the Alternate Sensitivity button switches Signature detectors to their alternate operating settings. Pressing the button a second time switches them back to primary sensitivity levels. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Acknowledge: Pressing the Acknowledge button acknowledges events. For proprietary systems, pressing the Acknowledge button acknowledges an event and turns the operator alert signal off when each event has been individually acknowledged in the order displayed. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the system returns to normal. Lamp Test: Pressing the Lamp Test button temporarily turns on all visual indicators on the local control unit and the LCD screen cycles through colors to reveal any dead pixels. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Note: If the local control unit is part of a command center, the lamp test is also executed on all nodes within the command center. System Function 1, System Function 2, System Function 3, or System Function 4: Pressing the System Function button activates the response programmed for function. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. None: No action assigned.

No.	Control/Indicator	Description
7	Fire Drill button and LED (button two default label and setting)	This configurable button is programmed by default as Drill (Fire Drill). The button can be configured in the 4-CU as: Drill: Pressing the Fire Drill button activates the system drill response. Pressing the button a second time cancels the drill response. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Panel Silence: For local systems, pressing the Panel Silence button turns the operator alert signal off and acknowledges all events have been reviewed. For proprietary systems, the Panel Silence button is not operational. The operator alert signal only silences after all events have been acknowledged. The Panel Silence LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating the control unit is in an off-normal condition and the control unit has been placed in Panel Silence mode, and turns off when the system returns to normal. Alternate Alarm Sensitivity: Smoke detectors can operate using two levels of sensitivity: primary and alternate. Pressing the Alternate Sensitivity button switches Signature detectors to their alternate operating settings. Pressing the button a second time switches them back to primary sensitivity levels. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Acknowledge: Pressing the Acknowledge button acknowledges events. For proprietary systems, pressing the Acknowledge button acknowledges an event and turns the operator alert signal off when all events have been acknowledged in the order displayed. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the system returns to normal. Lamp Test: Pressing the Lamp Test button temporarily turns on all visual indicators on the local control unit and the LCD screen cycles through colors to reveal any dead pixels. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. Note: If the local control unit is part of a command center, the lamp test is also executed on all nodes within the command center. System Function 1, System Function 2, System Function 3, or System Function 4: Pressing the System Function button activates the response programmed for function. The LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the command function is active, and turns off when the function ends. None: No action assigned.
8	Alarm Silence button and LED (button three)	This non-configurable button is programmed as Alarm Silence. Pressing the Alarm Silence button turns off the EVAC and ALERT channels, and all active audible and visible NACs if configured. Pushing the button a second time turns the notification appliance circuits back on. This button may be used to cancel the drill signal. The Alarm silence LED turns on as white while the button is pressed. When the button is released, the LED is on steady as yellow, indicating that the active notification appliance circuits have been silenced, and turns off when the system returns to normal. Note: Project configuration settings affect the operation of the Alarm Silence function.

No.	Control/Indicator	Description
9	Reset button and LED (button four)	This non-configurable button is programmed as Reset. Pressing the Reset button activates the system's reset sequence to restore the system to normal. The Reset LED turns on as white while the button is pressed. When the button is released, the LED flashes as green quickly during the smoke power-down phase, flashes slowly during the power-up phase, is on steady during the restoral phase, and turns off when the system has reset. Notes - The Reset button is disabled as long as the alarm silence inhibit timer is running. - The Reset button does not affect disabled points or manually overridden functions.

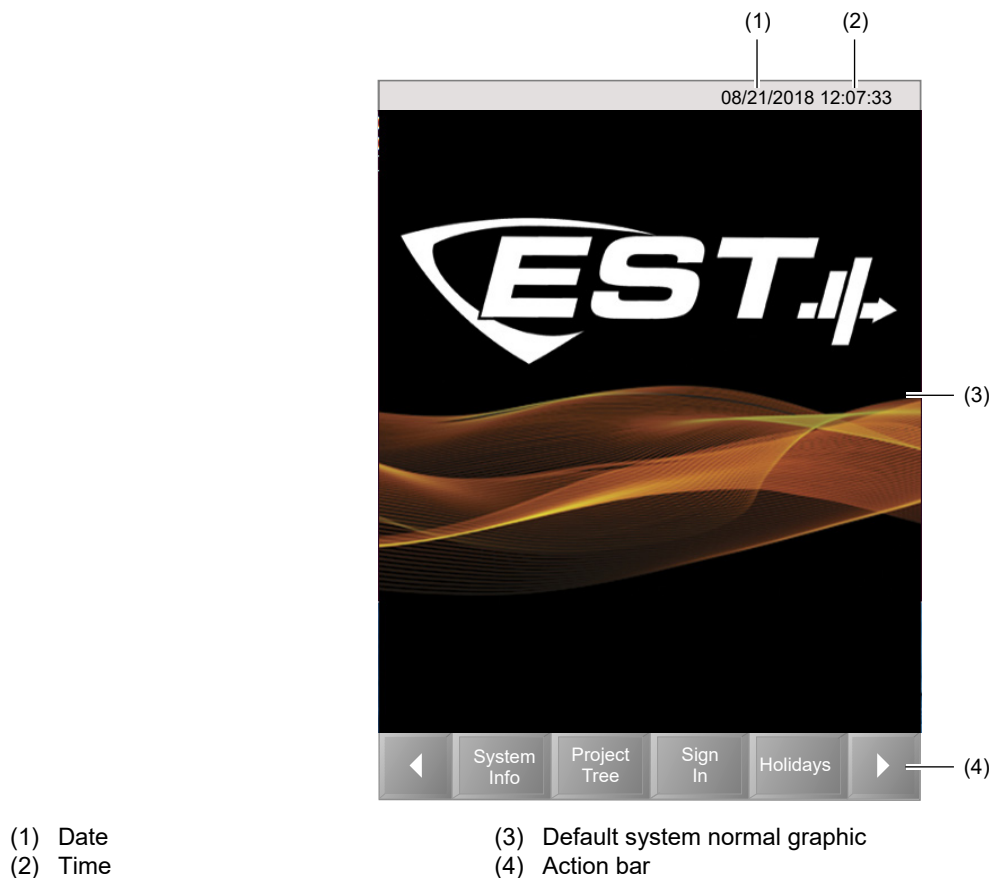
LCD screen indications

The 4-LCD, 4-3LCD, and 4-LCDAUDTEL display module LCD screen provides information relevant to the current functional condition of the control unit. There are two screen modes: system normal and system off-normal.

System normal screen

System normal means that the control unit is in the normal (standby) state. In this state, the LCD screen is clear of any event messages. Only the time, date, system normal graphic, and Action buttons are displayed.

Figure 1: System normal LCD screen

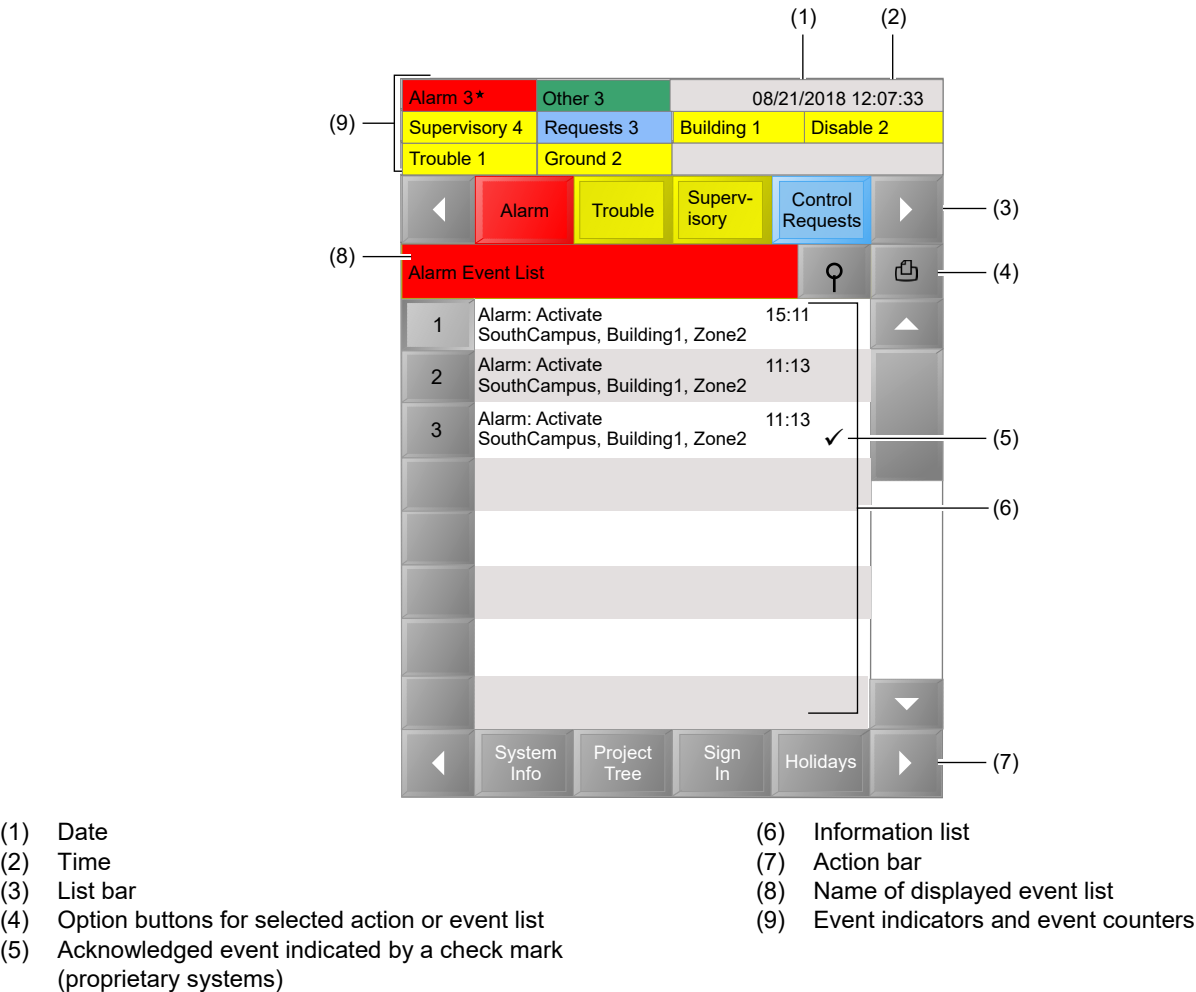


System off-normal

System off-normal means the control unit has changed to the alarm, trouble, supervisory, monitor, disabled, or test state. In this state, the screen displays event messages. The event messages provide details about what is happening in the system. The LCD screen displays the eight most recent highest priority events at one time.

Note: The off-normal LCD screen only displays when events are activated. It does not display when events are restored. In proprietary mode, restore events are displayed and all must be acknowledged in order to return to normal.

Figure 2: System off-normal LCD screen



Note: For a list of interface icon descriptions see Table 13 on page 26.

Event indicators and event counters

The event indicators (queues) that display on the LCD screen are determined by the 4-CPU properties configured in the 4-CU. Events in the queues can pass through to a configured printer, coder, or email service.

The event counters display the number of events in the event queue. A star designates that there are unacknowledged or new events since the control unit was last silenced.

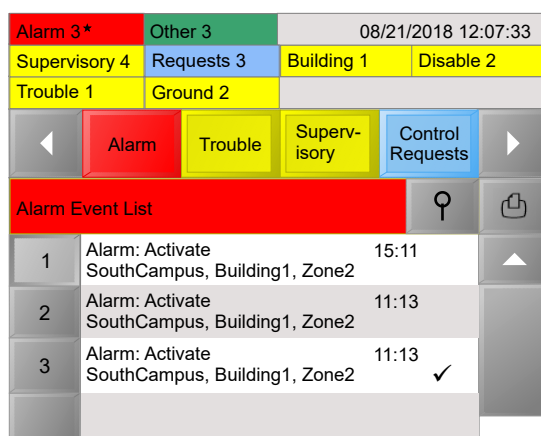
Note: The market setting determines which event types go into each event queue, and their priority (see Table 10 on page 23).

Event message lists

Event message lists allow you to view details of messages to help locate points that are in an abnormal state. When the system signals a status change, the control unit posts the event message for the point that activated the event in the appropriate event message list.

An event message consists of two lines of text, as shown in Figure 3 below. The first line displays the event type, current status, and time of the event. The second line displays the message text, which is either the address of the point that activated the event or, if programmed, a location description.

Figure 3: Event message



The event queues on the List bar display in order of priority based on programming and market settings. Event types are mapped to the queues based on the project's market setting. Table 10 below shows the event types that go into each event queue, and their priority, for US, Latin America, International, Middle East, and New Zealand markets.

Table 10: Event type-to-queue for US, Latin America, International, Middle East, and New Zealand markets

Event queue	Priority	Max Events	Event type
Call In	1	100	FirePhone
Control Request	2	100	Requested, Granted
Alarm	3	2,000	Alarm, Pull, Heat, Waterflow, Stagetwo, Zone, AND, Matrix, HeatAlarm
Emergency	4	2,000	Emergency, COAlarm, COAlarm3, ComboAlarm3
Supervisory	5	2,000	Supervisory, PreAlarm [1], Valve, Gatevalve, Tamper, COSupervisory, COSupervisory3, SmokeSupervisory
Building	6	1,000	Temperature, Power, Signal, Interlockfeedback, InterlockFBFailure, Interlock, Damper, Fan, Door
Disable	7	2,000	Disablement, SensorBypass, Testevent, ObjectRunning, Servicegroup
Trouble	8	2,000	TroubleOpen, TroubleShort, LocalTrouble, Dirtyhead, Devicecommunication, Internalfault, Badtype, Badpersonality, Unexpecteddevice, CommonTroubleinput, Devicecompatibility, Acknowledge, AccessTrouble, ACFail, EndOfLife, DeviceInitFault, BaseTypeFault, SensitivityFault, DeviceConfigurationFault, RiserFault, InvalidAddress, DuplicatesFault
Ground	9	500	Groundfault
Other	10	500	PreAlarm [1], AlarmVerify, Station, MaintenanceAlert, RelayConfirmation, Monitor, LocalMonitor, Switch, TimeControl, InstructionText, COMonitor, COMonitor3

[1] PreAlarm events appear in the Supervisory queue but are considered Monitor events, which do not trigger the supervisory relay. PreAlarm events appear in the "History" report as monitor events.

Table 11 below describes the behavior of event messages for Local and Proprietary modes.

Table 11: Event message behavior for Local and Proprietary modes

Scenario	Local mode	Proprietary mode
No new events have activated since the last time the control unit was silenced	The highest priority event queue displays.	If all events have been acknowledged, the highest priority event queue displays.
New events have activated since the last time the control unit was silenced	The first new, highest priority event displays.	The first unacknowledged event in the highest priority queue displays.
An active event is restored	The star that designated the new event no longer appears next to the event counter.	The star that designated the new event no longer appears next to the event counter and a check mark appears next to the event message.

Fire privilege levels

The number of users that can sign in to the control unit is configured in the 4-CU. The maximum number of users is 99. Certain user interface controls and functions are password protected and have a fire privilege level that is determined by the market setting or administrator programming.

The user fire privilege levels are summarized in Table 12 below. Each user is given a default password and fire privilege level that should be changed once the control unit is put into service. See “User administration” on page 38 for instructions.

Note: User sign in for initiating commands times out after a preconfigured period of time. When the user timeout period expires, the control unit reverts to the default fire privilege level. The time out period is configured in the 4-CU and can be set from 15 seconds to 8 hours. The typical user timeout setting is 5 minutes.

Table 12: Fire privileges by access level

Fire privilege level	Privileges
0 (default; no user sign in required)	• Reports • Write access function (1 hour or 8 hour access) • Print function • Reset command functions • Control request, grant, and deny • Panel Silence button [1][2] • Drill button [1][2] • Alternate Alarm Sensitivity button [1][2] • Acknowledge button [1][2] • Lamp Test button [1][2] • System Function button [1][2] • Alarm Silence button [1] • Reset button [1]
1	All level 0 privileges, plus: • Remote read function • Status holdup report (future feature)
2 (reserved for future features)	All level 0 and 1 privileges

Fire privilege level	Privileges
3	All level 0, 1, and 2 privileges, plus: • Freeze history function • Toggle alternate sensitivity function • System time and date • Device (enable/disable) • Logic groups (enable/disable) • Switch (enable/disable) • Relay (activate/restore) • LED (activate/restore) • Audio amp (activate/restore) • Audio message (play/stop)
4	All level 0, 1, 2, and 3 privileges, plus: • Clear history function • Holidays (add/delete) • Write access function (unlimited access) • Reboot (restart) function (local and networked control units) • Card (enable/disable) • Test (start/cancel) • Signature device test
5	All level 0, 1, 2, 3, and 4 privileges
9 (administrator)	All level 0, 1, 2, 3, 4, and 5 privileges, plus: • User administration (add, delete, assign fire privileges, modify PINs)

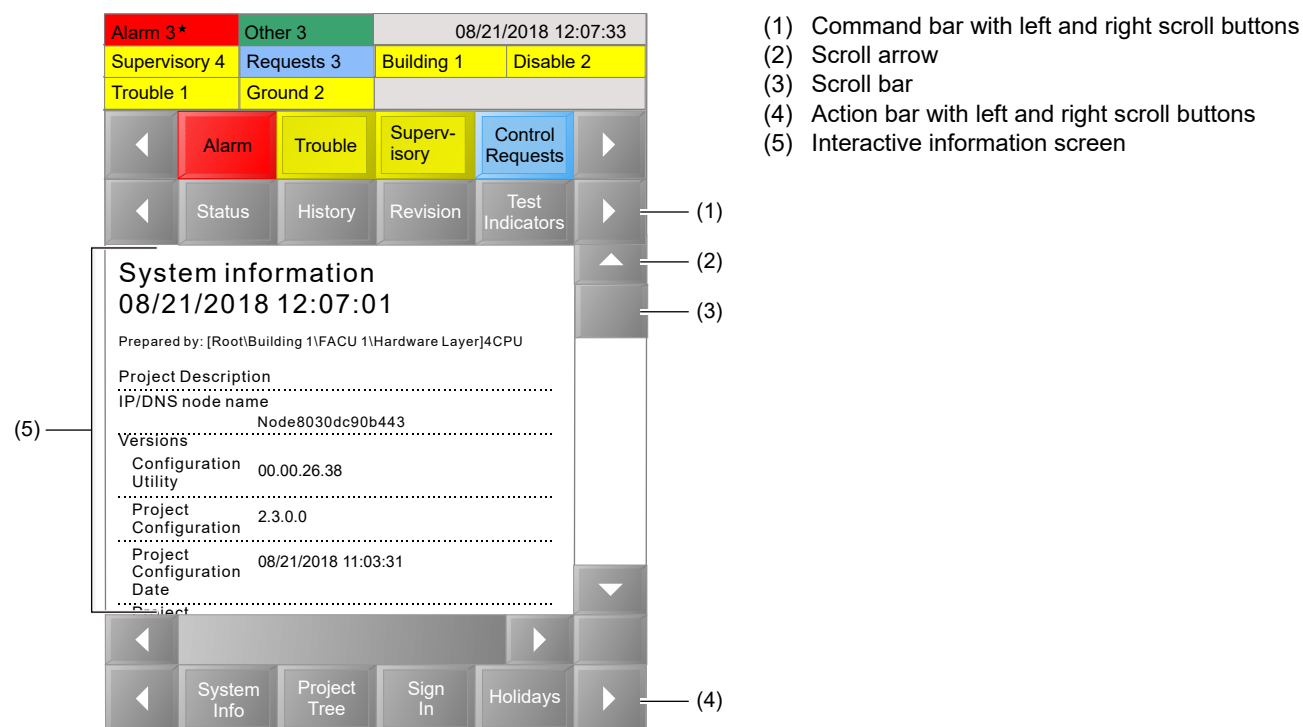
[1] May be programmed for a fire privilege level.

[2] System programming determines the function assigned to button one and button two on the 4-LCD user interface.

LCD screen operator controls

The LCD screen provides list buttons, action buttons, and command buttons that let you operate the control unit, configure various unit settings, and view system information and event activity.

Figure 4: LCD screen operator controls



LCD screen user interface icons and option buttons

The user interface includes several conditional icons, some of which act as operator controls (option buttons). Table 13 below provides descriptions of the icons that you may see.

Table 13: Operator interface icon and option button descriptions

Icon	Description	Icon	Description
★	Appears next to the event indicator counters. Designates there are unacknowledged or new events since the control unit was last silenced.	☺	Users administration button. Tap to show/hide user PIN.
⌨	Appears next to the signed in user ID.	📄	Users administration button. Tap to save user information.
🔄	Reports option button. Tap to refresh the report.	☑	Request control button. Tap to grant control.
←	Reports option button. Tap to go to previous page.	☒	Request control button. Tap to deny control.
→	- Reports option button. Tap to go to next page. - Keypad button. Tap to enter and go to next keypad screen.	📞	Fire phone connect button. Tap to connect a disconnected call.
🖨	Printer option button. Tap to send report to a configured printer.	📞	Fire phone disconnect button. Tap to disconnect a connected call.
♀	Details option button. Tap to access an event details report.	▶ ∞	Future feature

Icon	Description	Icon	Description
✓	Appears next to acknowledged events for proprietary mode systems.	► 1X	Future feature
✕	Keypad delete button. Tap to delete a character when using the screen keypads.	► 3X	Future feature
X	User administration button. Tap to delete a user.	■	Future feature
	Users administration button. Tap to modify user privileges.		

List bar

The List bar (Figure 4 on page 26) contains buttons for selecting which event messages you want to display on the screen. The market determines which event types are assigned to which list buttons. The List bar is only available when the system is off-normal screen.

The LCD screen displays up to four list buttons at a time. Tap the left or right scroll button to view the other list buttons. Up to ten lists may be available based on your control unit configuration and market setting.

Table 14: List bar button descriptions

List	Minimum fire privilege level	Description
Call In	0	Displays the Call In event list, which shows firefighter telephone call events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.
Control Requests	0	Displays the Control Requests event list, which shows a list of control requests and whether granted or denied, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.
Alarm	0	Displays the Alarm event list, which shows alarm events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.
Emerg (Emergency)	0	Displays the Emergency event list, which shows emergency events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.
Supervisory	0	Displays the Supervisory event list, which shows supervisory events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.
Building	0	Displays the Building event list, which shows building events stored in the list in the order, in which they were received. View an event's Details Report by tapping the event, and then the details button ⓘ. Print the Details Report by tapping the print button 🖨.

List	Minimum fire privilege level	Description
Disable	0	Displays the Disabled event list, which shows disabled events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button . Print the Details Report by tapping the print button .
Trouble	0	Displays the Trouble event list, which shows trouble events stored in the list in the order, in which they were received. View an event's Details Report by tapping the event, and then the details button . Print the Details Report by tapping the P print button .
Ground Fault	0	Displays the Ground Fault event list, which shows ground fault events stored in the list, in the order in which they were received. View an event's Details Report by tapping the event, and then the details button . Print the Details Report by tapping the print button .
Other	0	Displays the Other event list, which shows ancillary events stored in the list in the order in which they were received. View an event's Details Report by tapping the event, and then the details button . Print the Details Report by tapping the print button .

Action bar

The Action bar (Figure 4 on page 26) contains buttons for selecting system commands and changing administrative settings. Tapping action buttons opens system information and interactive screens, and command buttons that are used to apply system operations. The Action bar is available whether the system is normal or off normal.

The LCD screen displays up to four action buttons at a time. Tap the left or right scroll button to view the other action buttons.

Table 15: Action bar button descriptions

Action button	Minimum fire privilege level	Description
System Info	0	Displays the System information screen, and launches the Command bar that provides several system operation command buttons. When signed in as fire privilege level 1 to 5, or level 9 additional command system operation buttons appear.
Project Tree	0	Displays the project tree. Use the vertical scroll bar to view hidden sections of the expanded tree. - Recent Site Locations: Shows a maximum of 10 locations that the operator has viewed. - Top level: Expanding the top level of the project tree shows expandable nodes and branches that can hold hardware, operator, and communication devices that were configured in the 4-CU for the local control unit and networked control units. Tapping a device on the tree, displays its Device details report. For certain devices, the Command bar launches. The command buttons that appear are dependent on the fire privilege level of the user.
Sign In	0	Displays a keypad for entering your 7-digit PIN to access to user privileges. The first three digits of the PIN are your user ID followed by a 4-digit PIN.

Action button	Minimum fire privilege level	Description
Holidays	0	Displays the Holidays List screen. When signed in as fire privilege level 4 and higher, the Command bar launches to allow adding and deleting holidays.
Users	0	Displays the Users List screen. When signed in as fire privilege level 9 (administrator), the Command bar launches to allow applying user settings.
Request control	0	Activates a request to take control of the system and displays the Requests event indicator/counter, sounds the operator alert signal, and displays the Control Request List button. Tapping the Control Requests list button allows you to grant or deny a request. See “Request Grant Deny logics” on page 92.
Alternate language (Example: Española or English)	0	Switches the names on LCD screen buttons, lists, indicators, and the text in primary event message to the 4-CU configured alternate language. Notes - For the alternate language button to appear on the Action bar, the Alternate Language property must be configured in the 4-CU (Project > Edit Properties > Alternate Language). - Custom event message text does not switch to the alternate language. - The LCD screen defaults to the primary language at control unit startup.

Command bar

The Command bar (Figure 4 on page 26) appears when you tap certain action buttons or items on the LCD screen. For example, if you tap the System Info button, the Command bar appears containing several buttons that include allowing you to enable and disable devices and zones, activate and restore switch outputs, and view system reports.

The LCD screen displays up to four command buttons at a time. Tap the left or right scroll button to view the other command buttons.

Table 16: Command bar button descriptions

Action button	Command button	Minimum fire privilege level	Description
System Info	System Info	0	System information report. Lists the control unit firmware version, SDU and hardware configurations, and licensing information.
	Status	0	Status (all) report. Lists active events and disabled points from the selected node.
	Status Disabled	0	Status (disabled) report. Lists all addressable points that are in the disabled state.
	Status Test	0	Status (test) report. Lists off-normal devices under test.
	History	0	History – All Dates report. Lists the last 20,000 events and operator commands processed by the selected node.
	History Alarm	0	History – All Dates, Alarm report. Lists the event name, time, date, and location text for all alarm events.
	History Trouble	0	History – All Dates, Trouble report. Lists the event name, time, date, and location text for all trouble events.

Action button	Command button	Minimum fire privilege level	Description
	History Supervisory	0	History – All Dates, Supervisory report. Lists the event name, time, date, and location text for all supervisory events.
	Maintenance (all)	0	Maintenance report. Lists the %Dirty value for all of the detectors on a signaling line circuit for the selected node. For CO detectors the report includes CO Life Left Months or CO DAY Running. The report includes each detector's model type, primary and alternate alarm sensitivity values, and, if programmed, a location description.
	Maintenance Dirty	0	Maintenance (dirty) report. Lists all addressable smoke detectors that have a %Dirty value 20% and greater. A smoke detector that is more than 20% dirty should be noted for possible cleaning or replacing.
	Maintenance Not Clean	0	Maintenance (not clean) report. Lists all addressable smoke detectors that have a %Dirty value of 80% and greater. Smoke detectors that are more than 80% dirty should be cleaned or replaced as soon as possible.
	Maintenance (device)	0	Maintenance report. Lists the %Dirty value for a selected detector on a signaling line circuit. The report includes the detector's model type, primary and alternate alarm sensitivity value, and, if programmed, a location description.
	Revisions	0	Revisions report. Creates a report that lists the revision level of all the hardware and software components installed in a node selected in the project tree.
	1 Hour Access	0	Provides a validation code that unblocks the Remote Write command for one hour to permit database changes. The code must be entered in the 4-CU to allow database downloads from a programming computer to the 4-CPU in the EST4 control unit. When tapped, the operator alert sounds and the Other command button displays to show the "Remote write unlocked" local monitor event.
	8 Hour Access	0	Provides a validation code that unblocks the Remote Write command for eight hours to permit database changes. The code must be entered in the 4-CU to allow database downloads from a programming computer to the 4-CPU in the EST4 control unit. When tapped, the operator alert sounds and the Other command button displays to show the "Remote write unlocked" local monitor event.
	Cancel Access	0	Immediately restores the Remote Write command and prevents access to database changes.
	Reset	0	Resets the local control unit. When tapped, the Reset control button indicator flashes.
	Reset All	0	Resets all networked nodes. When tapped, the Reset control button indicator flashes.
	Test Indicators	0	Activates a test of all control unit LEDs and installed control-display module LEDs.
	Status Holdup	1	Future feature.
	Lock Incoming Network	1	Blocks web services for all 4-FWAL series nodes.
	Unlock Incoming Network	1	Unblocks web services for all 4-FWAL series nodes.

Action button	Command button	Minimum fire privilege level	Description
	Freeze History	3	Archives the most recent events (up to 10,000 events) so that they cannot be overwritten.
	Toggle Alternate Sensing	3	Switches detector alarm sensitivity levels from primary to alternate or from alternate to primary, whichever is currently active.
	Set Date Time	3	Opens a keypad that lets you set the system time and date.
	Clear History	4	Resets the event indicator counters on the LCD screen and erases the list of events that occurred on the control unit since it was placed into service or the last time the history file was cleared. This includes unfreezing and erasing archived history.
	Unlimited Access	4	Provides a validation code that unblocks the Remote Write command for an unlimited amount of time to permit database changes. The code must be entered in the 4-CU to allow database downloads from a programming computer to the 4-CPU in the EST4 control unit. When tapped, the operator alert sounds and the Other command button displays to show the "Remote write unlocked" local monitor event.
	Reboot node	4	Restarts the local control unit without removing power.
	Reboot All	4	Restarts all networked control units without removing power.
Project Tree	Enable	3	Enables a device selected in the project tree.
	Disable	3	Disables a device selected on the project tree.
	Activate	3	Activates AND groups and command lists selected in the project tree.
	Restore	3	Restores AND groups the command lists.
	On	3	Activates input and output devices selected in the project tree.
	Off	3	Restores the input and output devices.
	On	3	Turns on a control-display LED selected in the project tree to steady.
	Fast Blink	3	Turns on a control-display LED selected in the project tree to fast blink.
	Slow Blink	3	Turns on a control-display LED selected in the project tree to slow blink.
	Off	3	Turns off the control-display LED.
	► ∞	3	Future feature.
	■	3	Future feature.
	► 1X	3	Future feature.
	► 3X	3	Future feature.
	Activate SG	4	Starts a Service Group test for an alarm input device that is part of the service group selected in the project tree. Service groups allow alarm input devices to be activated without placing the system into alarm.
	Restore SG	4	Stops the Service Group test.
	CO Test On	4	Places a Signature CO detector or Signature optical smoke detector selected in the project tree into accelerated sensing rates for testing purposes.

Action button	Command button	Minimum fire privilege level	Description
	CO Test Off	4	Returns the selected Signature CO detector or Signature optical smoke detector to normal sensing rates.
	Test Activate 1	4	Places a Signature device selected in the project tree into alarm state for verification testing.
	Test Activate 2	4	Places a Signature device selected in the project tree into prealarm state for verification testing.
	Test trouble	4	Place a Signature device selected in the project tree into trouble state for verification testing.
	IPCID Test	4	Activates an IP CID-to-central monitoring station (CMS) selected in the project tree verification test.
	IPCIDMOD Test	4	Activates a MODCOM CID-to-CMS verification test.
	IPSIA Test	4	Activates a MODCOM SIA-to-CMS verification test.
Holiday	Add Holiday	4	Opens a keypad that lets you add holiday schedules.
	Delete Holiday	4	Displays a list of scheduled holidays that can be deleted.
Users	Add User	9	Allows the administrator to add a new user ID number (011 to 099). When adding a user the administrator assigns their personal identification number (PIN) and fire privilege level.
	Delete	9	Allows the administrator to delete a user. Note: User IDs 001 to 005 and 009 (administrator) cannot be deleted.
	Change Privileges	9	Allows the administrator to modify a user PIN and fire privilege level.

Chapter 3

Operating instructions

Summary

This chapter provides instructions for operating the life safety system from the EST4 control unit's user interface.

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Operating states

The EST4 control unit operates in the normal, alarm, disable, supervisory, trouble, monitor, test, and drill states.

Note: This section describes the operation of a stand-alone control unit configured using default settings. Actual operation may vary depending on your project's configuration.

Normal state

The system operates in the normal (standby) state in the absence of any events. In the normal state, the LCD screen shows the time, date, and LCD background graphic (see Figure 1 on page 21).

Alarm state

The control unit enters the alarm state when a point signals an alarm condition. For example, when a smoke detector, pull station, or waterflow switch is activated.

Output of the alarm state

Upon entering the alarm state, the control unit:

- Activates all supervised and unsupervised common alarm outputs
- Changes over the alarm relay contacts on a 4-COMREL, if installed
- Activates the first alarm pseudo point
- Changes the active state for the point that signaled the alarm event

Indication of the alarm state

To indicate it is in the alarm state, the control unit:

- Sounds the operator alert signal
- Turns on the Alarm LED on the LCD user interface
- Posts the event activated by the point into the appropriate event list queue (refer to Table 10 on page 23 for a list of event types-to-queue)

Disable state

The control unit enters the trouble state when a point signals a disabled condition.

When a point is disabled, the control unit does not process any of the point's status changes and the point remains in its current state. For example, if an audible device type in the normal state was disabled and subsequently activated, the audible device type would not turn on until it was enabled. Conversely, if an active audible device type were disabled and subsequently restored, the audible device type would not turn off until it was enabled.

If a point is disabled and the cause of the trouble changes while the point is disabled, the point's original trouble event message may not update when the point is enabled. This is dependent on the point state or the 4-CU configuration setting for the Restore Event on Disable property.

Note: View the Status Disabled report to identify a disabled point.

Output of the disable state

Upon entering the disable state, the control unit:

- Activates the first trouble pseudo point
- Activates the first disable pseudo point
- Changes the active state for the point that signaled the disable event

Indication of the disable state

To indicate it is in the disable state, the control unit:

- Sounds the operator alert signal
- Turns on the Trouble LED on the LCD user interface
- Posts the event activated by the point into the appropriate event list queue (refer to Table 10 on page 23 for a list of event types-to-queue)

Supervisory state

The control unit enters the supervisory state when a point signals a supervisory condition.

Output of the supervisory state

Upon entering the supervisory state, the control unit:

- Activates all supervised and unsupervised common supervisory outputs
- Activates the supervisory signal output on a 3-OPS, if installed
- Changes over the supervisory relay contacts on a 4-COMREL, if installed
- Activates the first supervisory pseudo point
- Changes the active state for the point that signaled the supervisory event

Indication of the supervisory state

To indicate it is in the supervisory state, the control unit:

- Sounds the operator alert signal
- Posts the event activated by the point into the appropriate event list queue (refer to Table 10 on page 23 for a list of event types-to-queue)

Trouble state

The control unit enters the trouble state when a point signals a trouble condition.

Output of the trouble state

Upon entering the trouble state, the control unit:

- Changes over the trouble relay contacts on a 4-COMREL, if installed
- Activates the first trouble pseudo point
- Changes the active state for the point that signaled the trouble event

Indication of the trouble state

To indicate it is in the trouble state, the control unit:

- Sounds the operator alert signal
- Turns on the Trouble LED on the LCD user interface
- Posts the event activated by the point into the appropriate event list queue (refer to Table 10 on page 23 for a list of event types-to-queue)

Monitor state

The control unit enters the monitor state when a point signals a monitor condition.

Output of the monitor state

Upon entering the monitor state, the control unit:

- Activates the first monitor pseudo point
- Changes the active state for the point that signaled the monitor event

Indication of the monitor state

To indicate it is in the monitor state, the control unit:

- Sounds the operator alert signal
- Posts the event activated by the point into the appropriate event list queue (refer to Table 10 on page 23 for a list of event types-to-queue)

Note: Monitor indications are restored automatically when the monitor input is restored.

Test state

The control unit enters the test state when a service group is activated.

Output of the test state

Upon entering the test state, the control unit:

- Activates the first monitor pseudo point
- Activates the first trouble pseudo point
- Activates the first test pseudo point
- Changes the active state for the service group that was activated

While in the test state:

- When a member of an active service group signals an active event, the control unit executes the service group's active test response
- When a member of an active service group signals a trouble event, the control unit executes the service group's trouble test response

Note: If you do not program a trouble test response, the control unit executes the active test response instead.

Indication of the test state

To indicate it is in the test state, the control unit:

- Sounds the operator alert signal
- Turns on the Trouble LED on the LCD user interface
- Activates the Trouble event indicator and displays an event message in the Trouble list queue for the first test pseudo point, provided there are no higher priority events
- Activates the Other event indicator and displays an event message in the Other list queue for the service group that was activated, provided there are no higher priority events

Drill state

The drill function activates the system notification appliances generally for conducting a fire drill. In this state, an alarm is not transmitted to the central monitoring station.

Output of the drill state

Upon entering the drill state, the control unit:

- Changes the active state for the device that activated

Indication of the drill state

To indicate it is in the drill state, the control unit:

- Activates all audible and common alarm output devices
- Activates all configured visual devices

Control unit power up

Initial power up

When you power up the EST4 control unit for the first time, the LCD screen may begin to show event messages on the screen as the 4-CPU microprocessor begins communicating with devices. You can use the Panel Silence button to silence the operator alert signal. For proprietary systems, use the Acknowledge button, if programmed, to acknowledge any events.

Notes

- Before applying power to the control unit, make sure the standby batteries are not connected to the 3-PPS/M Power Supply.
- The 3-PPS/M Power Supply should already be installed and mains AC (primary power) wired to the input terminals (TB1).

To power up the control unit for the first time:

1. Apply power to the control unit.
2. Connect the batteries to the battery wiring terminal on the 3-PPS/M Power Supply.
3. Press the Panel Silence button, and the Acknowledge button, if necessary.
4. Download the database as instructed in “Downloading a project database” on page 43.

- For a network system, clear any faults between control units.
- Verify proper operation. Refer to “Routine maintenance and tests” on page 127 for Initial and Reacceptance testing.

User administration

Several EST4 user interface features are accessible without requiring a user sign in. However, password protection is used to regulate access to certain user interface controls and commands.

The EST4 control unit automatically includes six default user IDs, each having default fire privileges and PINs. The control unit can have up to 99 users.

The user access time out can be configured in the 4-CU. The default setting is 5 minutes. When user access times out, the control unit returns to the default fire privilege level (0).

Table 17 below provides a brief summary of the default user IDs and their fire privileges levels, and default PIN.

Table 17: Default user IDs, fire privileges, and PINs

User IDs	Default fire privilege level [1]	Default PIN [2]
None (no user sign in required) [3]	0	–
001	1	1234
002	2	1234
003	3	1234
004	4	1234
005	5	1234
009 (administrator)	9	1234
011 to 099	As set by the User 009 administrator	As set by the User 009 administrator Note: Users 011 to 099 can change their own PIN.

[1] Refer to Table 12 on page 24 for a list of fire privileges by level.

[2] It is recommended that you change the default passwords once the control unit is put into service.

[3] Permission allowing access to system functions and features without logging on must be configured in the 4-CU.

Adding users

Note: Only the administrator (user 009) can add new users.

To add a user:

- Sign in with fire privilege 9 (user 009), and then tap Users on the Action bar.
- Tap Add User on the Command bar, and then use the interface keypad to enter the 3-digit user ID (011 to 099) for the new user. The screen automatically changes to the PIN screen.
- Enter the 4-digit PIN for the user. You can show the PIN by tapping the show/hide button .
- Tap the enter button  on the interface keypad to display the User: *nnn* Fire Privilege list.
- Tap the desired level, and then tap the save button . The screen changes to the Users List and shows the new user.

Deleting users

Note: Only the administrator (user 009) can delete users.

To delete a user:

1. Sign in with fire privilege 9 (user 009), and then tap Users on the Action bar.
2. Tap Delete on the Command bar.
3. Tap the user number to be deleted, and then tap the delete button . The screen changes to the Users List and no longer includes the deleted user number.

Changing user PINs

Note: The administrator (user 009) can change all users PIN. Users 011 to 099 can change their own PIN.

To change user passwords:

1. Sign in, and then tap Users on the Action bar.
 User 009: Can change all users (001 to 099) PIN
 Users 011 to 099: Can change their own PIN
2. Tap Privileges on the Command bar.
3. On the Modify list, tap the user ID number, and then tap the edit button .
4. Enter the new 4-digit PIN for the user, and then tap the save button . You can show the PIN by tapping the show/hide button .
5. On the User: *nnn* Fire Privilege list, tap the desired level, and then tap the save button .

Changing fire privilege levels

Note: Only the administrator (user 009) can change fire privilege levels.

1. Sign in with fire privilege 9 (user 009), and then tap Users on the Action bar.
2. Tap Privileges on the Command bar.
3. On the Modify list, tap the user ID number, and then tap the edit button .
4. Tap the enter button  button on the keypad to display the User: *nnn* Fire Privilege list.
5. Tap the desired fire privilege level, and then tap the save button .

Changing the LCD screen to an alternate language

For control units configured in the 4-CU for an alternate language, a language button displays on the Action bar. The button name shows the alternate language configured for the control unit. For example, Española or English.

When the alternate language button is tapped, the names on LCD screen buttons, lists, and indicators and the text in primary event messages switches to the configured alternate language.

Notes

- For the alternate language button to appear on the Action bar, the Alternate Language property must be configured in the 4-CU (Project > Edit Properties > Alternate Language).

- Custom event message text does not switch to the alternate language.
- The LCD screen defaults to the primary language at control unit startup.

To change the LCD screen to the alternate language:

1. Tap the alternate language button on the Actions bar (example: Española or English).

Obtaining and recording node MAC addresses

Each node CPU module in the EST4 system has a unique MAC address that is used to identify the node in the EST4 network. The node module's MAC address must be entered in the 4-CU for the CU to be able to communicate with the control unit.

The following are node-modules: 4-CPU, 4-ANNCPU, 4-CPUGRPH, 4-NET-AD, and 4-FWALx. See *4-CU Help* for information on entering MAC addresses in the 4-CU.

The MAC address can be obtained from node's product label or by querying the network.

To manually record the MAC address from the node-module's product label:

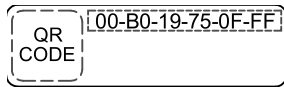
1. From the 4-CU, print a node list to use for recording the MAC addresses.

In the Navigation pane, select the system branch under Project Tree.

On the Node List tab, format the list grid, and then select any row in the grid.

On the Home tab, in the Operations group, click Print.

2. From the control unit, before you install the node-module locate the product label on the front, bottom left corner of the module.



Manually record the MAC address shown on the label onto the node list sheet.

— or —

Scan the QR code and save it to record later.

3. From the 4-CU, enter the MAC address for each node. See *4-CU Help* for information on entering MAC addresses in the 4-CU.

To query the network:

1. Connect the Type B end of a USB cable to the J14 USB port on the 4-CPU. See Figure 5.
2. Connect the Type A end of the cable to a standard USB port on the computer with the project database.
3. From the control unit LCD screen:

On the Action bar, tap System Info.

On the Command bar, tap 1 Hour Access or 8 Hour Access to obtain a 4-CU required access code that will allow read access.

4. From the 4-CU:

On the Home tab, in the Communication group, click Enter Access Code.

Enter the Access Code, and then click Validate.

On the Home tab, in the Communication group, click Query Network, and then Refresh. The query returns network information that includes node MAC addresses. The window remains open until you close it.

In the Navigation pane, select the system branch under Project Tree.

On the Node List tab, format the list grid, and then select the node for which you want to enter the MAC address.

On the Edit Properties tab, enter or copy and paste the MAC Address property from the Query Network screen.

5. From the control unit LCD screen, when finished entering all MAC addresses:

On the Action bar, tap System Info.

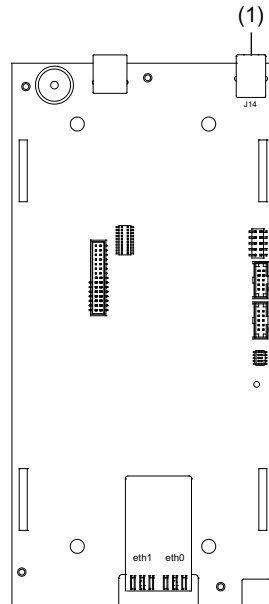
On the Command bar, tap Cancel Access, to block read access.

Downloading the 4-CU firmware and project database to the control unit

You can download firmware and project databases from the project computer to the control unit through a USB connection. The control unit CPU is blocked by default to disallow unauthorized database changes. An authorization code must be obtained from the control unit and entered into the 4-CU that activates the Remote Write Unlock command and allows the download.

Note: This function should only be used by the installer or service provider. Changes to the life safety system must be tested and may require local authority approval.

Figure 5: Download port on the CPU



(1) USB Type B port

Notes

- A USB 2.0, Type A to B cable is required.
- The Windows-based Remote Network Driver Interface Specification (RNDIS) driver is required on the project computer to provide the programming interface between the computer and CPU.

- To avoid possible failure of the download process due to a login/logout error, any Wi-Fi or other network adapters should be disabled on the project computer. Only the USB RNDIS connection should be enabled during the download process.
- A Remote Write Unlocked local monitor event appears in the Other queue and the operator alert signal sounds when the control unit is unlocked. Both are restored to normal after the control unit is restored to its default setting when the access is cancelled by the programmer or the access period expires.

Downloading firmware to the control unit

The control unit allows you to download the 4-CU firmware into the CPU directly from a computer through the USB Type B port on the 4-CPU.

There are three steps in downloading the firmware to the control unit:

- Step 1: Download the firmware file.
- Step 2: Import the firmware into the 4-CU.
- Step 3: Download the firmware to the control unit.

Step 1: Download the firmware file:

1. Download the firmware file to any location on the computer with the 4-CU:
Enter www.edwardsfiresafety.com in your web browser, click Login to log on to the My-Eddie website.
Locate the 4-CU firmware file in Media Type > Software, and then download the file.

Step 2: Import the firmware into the 4-CU and update each node CPU:

1. From the 4-CU:
Click the Firmware Management tab.
On the Home tab, in the Project group, click Import Firmware.
Browse to the location of the firmware file (.enc file type).
Select the file, and then click Open.
Click Import Firmware.
Click the Project Management tab, and then open the project.
In the Navigation pane, for *each* node update its CPU firmware version:
Select the node, and then click the Hardware Layer tab.
Select the CPU module, and then click the Edit Properties tab.
In the LRM Configuration group, Firmware Version property select the new firmware.

Step 3: Download the firmware to the control unit

Note: An RNDIS driver is required for the PC port to communicate with the control unit. If necessary, install it before proceeding.

1. Disable or disconnect any Wi-Fi or other network adapters on the project computer. Only the USB RNDIS connection should be enabled during the download process.
2. Connect the Type B end of a USB cable to the J14 USB port on the 4-CPU. See Figure 5.
3. Connect the Type A end of the cable to a standard USB port on the computer with the project database.
4. From the control unit LCD screen:
On the Action bar, tap System Info.

On the Command bar, tap 1 Hour Access or 8 Hour Access to obtain a 4-CU required access code that will allow you to download to the control unit.

5. From the 4-CU:

On the Project Management tab, open the project.

On the Home tab, in the Communication group, click Enter Access Code.

Enter the access code, and then click Validate.

On the Home tab, in the Communication group, click Query Network, and then Refresh to confirm communication with the node(s).

On the Home tab, in the Communication group, click Transmit To Panel. A progress bar displays in the lower right side of the CU window. The control unit reboots when the download is complete.

6. From the control unit LCD screen, when finished downloading:

On the Action bar, tap System Info.

On the Command bar, tap Cancel Access.

Downloading a project database

The control unit allows you to download the project database into the CPU directly from a computer through the USB Type B port on the 4-CPU.

Note: An RNDIS driver is required on the project computer to provide the programming interface between the computer and CPU. If necessary, install it before proceeding and disconnect

To download the project database:

1. Disable or disconnect any Wi-Fi or other network adapters on the project computer. Only the USB RNDIS connection should be enabled during the download process.

2. Connect the Type B end of a USB cable to the J14 USB port on the 4-CPU. See Figure 5.

3. Connect the Type A end of the cable to a standard USB port on the computer with the project database.

4. From the control unit LCD screen:

On the Action bar, tap System Info.

On the Command bar, tap 1 Hour Access or 8 Hour Access to obtain a 4-CU required access code that will allow you to download to the control unit.

5. From the 4-CU:

On the Project Management tab, open the project.

On the Home tab, in the Communication group, click Enter Access Code.

Enter the access code, and then click Validate.

On the Home tab, in the Communication group, click Query Network, and then Refresh to confirm communication with the node(s).

On the Home tab, in the Communication group, click Transmit To Panel. A progress bar displays in the lower right side of the CU window. The control unit reboots when the download is complete.

6. From the control unit LCD screen, when finished downloading:

On the Action bar, tap System Info.

On the Command bar, tap Cancel Access.

Setting the system date and time

The control unit incorporates a system clock to time stamp events and to activate time controls. The time is presented in 24-hour format. The date format is determined by the 4-CU project property setting. The default format is presented in month-day-year.

Note: Only users with fire privilege 3 and higher can access the system clock function.

To set the system date and time:

1. Sign in with fire privilege 3 or higher, and then tap System Info on the Action bar.
2. On the Command bar, tap Set time.
3. Use the interface keypad to enter date and time.

The default format is YYYYMMDDhhmmss. Where,

YYYY = 4-digit year

MM = 2-digit month

DD = 2-digit day

hh = 2-digit hour

mm = 2-digit minutes

ss = 2-digit seconds

Note: The time is entered in 24-hour format, for example:

000000 = midnight

010000 = 1:00 a.m.

120000 = noon

130000 = 1:00 p.m.

235900 = 11:59 p.m.

The date and time, displayed at the top of the user interface screen, immediately changes.

Silencing the operator alert signal

The control unit sounds the operator alert signal when an event message is posted into one of the event message queues. Pressing the Panel Silence button or acknowledging the event message silences the operator alert signal. The operator alert signal automatically re-sounds when a new event message is posted or when the system panel silence cancel timer expires (typically 24 hours).

For proprietary systems, the Panel Silence button is not operational. The operator alert signal only silences after each event has been individually acknowledged in the order displayed.

Notes

- For nonlatching events, the operator alert signal automatically silences when the event is restored. For example, when a trouble clears.
- Pressing the Panel Silence button may also silence the operator alert signal on remote annunciators, if the system is configured to do so.
- The control unit panel silence function can be programmed for a fire privilege level. The default setting is no PIN required.

To silence the operator alert signal:

1. Press the Panel Silence operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Silencing alarm signals

WARNING: Death or serious injury. The protected premises may be occupied. Do not silence alarm signals or reset the control unit unless you are authorized to do so and only after all occupants have been evacuated.

Pressing the Alarm Silence operator control button silences all audible alarm signals and, if configured, all visual alarm signals. Pressing the button a second time turns back on the signals.

Pressing the Alarm Silence operator control button *does not* silence alarm signals under the following conditions:

- When a waterflow alarm switch is active and the system is configured to prevent silencing alarm signals when a waterflow alarm switch is active
- When the system is configured to delay the silencing of alarm signals, in which case the Alarm Silence button may not be operational for up to five minutes following the first alarm event

Silenced outputs automatically re-sound when:

- The Alarm Silence button is pressed a second time
- Another alarm input activates
- Another alarm input in the same zone activates, unless the system is configured to prevent alarm signals from re-sounding

Note: The alarm silence function can be programmed for a fire privilege level password. The default setting is no PIN required.

To silence alarm signals:

1. Press the Alarm Silence operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Acknowledging events

When an event occurs, it is important that you acknowledge it and review the issue. The control unit handles the event by showing it on the LCD screen as an event message, sounding the operator alert signal, placing the event in the appropriate queue, and, when configured, flashing the Acknowledge LED.

If an operator control button has been programmed in the 4-CU as Acknowledge, pressing the Acknowledge button acknowledges events.

For proprietary systems, pressing the Acknowledge button acknowledges an event and turns the operator alert signal off after all events have been individually acknowledged in the order displayed.

Note: The acknowledge function can be programmed for a fire privilege level password. The default setting is no PIN required.

To acknowledge an event:

1. Press the Acknowledge operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Resetting the life safety system

WARNING: Death or serious injury. The protected premises may be occupied. Do not reset the life safety system until the proper authorities have determined that the threat of fire is no longer present.

Resetting the life safety system restores the system to its normal state, provided all latched inputs have been restored before the end of the reset cycle.

If alarm signal initiating devices have not been restored before the end of the reset cycle:

- Active alarm signals will remain active
- Silenced alarm signals will remain silenced

Notes

- The Reset button may be inoperable for up to three minutes following the first alarm event.
- The Reset button does not affect disabled points or manually overridden functions.
- The reset function can be programmed for a fire privilege level password. The default setting is no PIN required.

To reset the life safety system:

1. Press the Reset operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Rebooting the life safety system

Rebooting the system reinitializes it without removing power.

Note: Only users with fire privilege 4 and higher can access the reboot function.

To reboot the system:

1. Sign in with fire privilege 4 or higher, and then tap System Info on the Action bar.
2. Tap the desired reboot button on the Command bar.
 - Reboot node: Reboots the local control unit
 - Reboot all: Reboots all networked control units

Clearing the alarm history

Caution: Clearing the control unit history file permanently deletes all history data for the control unit.

The alarm history counter keeps track of how many times the control unit has entered the alarm condition. Clearing the history resets the event indicator counters on the LCD screen (Figure 1 on page 21) and erases the list of events that occurred on the control unit since it was placed into service or the last time the history file was cleared.

Notes

- Only users with fire privilege 4 and higher can access the clear history function.
- Clearing the alarm history also unfreezes and erases archived history.

To clear the alarm history:

1. Sign in with fire privilege 4 or higher, and then tap System Info on the Action bar.
2. On the Command bar, tap Clear History.

Freezing the events history

When the Freeze History command is activated, the system archives the most recent events (up to 10,000) so that they cannot be overwritten.

Note: Only users with fire privilege 3 and higher can access the freeze history function.

To freeze event history:

1. Sign in with fire privilege 3 or higher, and then tap System Info on the Action bar.
2. On the Command bar, tap Freeze History.

To view or print events in the freeze history archive:

1. Connect to the control unit to the 4-CU, as described in “System reports uploaded to the 4-CU” on page 52.
2. After confirming 4-CU communication with the node(s), from the control unit LCD screen:
On the Action bar, tap System Info.
On the System Info screen, find History Frozen and note the date and time that the Freeze history command was activated.
3. From the 4-CU:
On the Reports tab, in the Navigation pane, select the Project Tree branch.
On the Project Tree Configuration tab, click the Node list tab, and then select the node from which you want the freeze history events.
On the Reports tab, in the System Reports group, click Events, and then select Alarm History.
In the Alarm History dialog box, configure the report properties.
 - History Types: Select All, Alarm, Supervisory, Trouble, or Test.
 - Start Time: Enter the desired begin date and time.
 - End Time: Enter the History Frozen information noted in step 2 above.

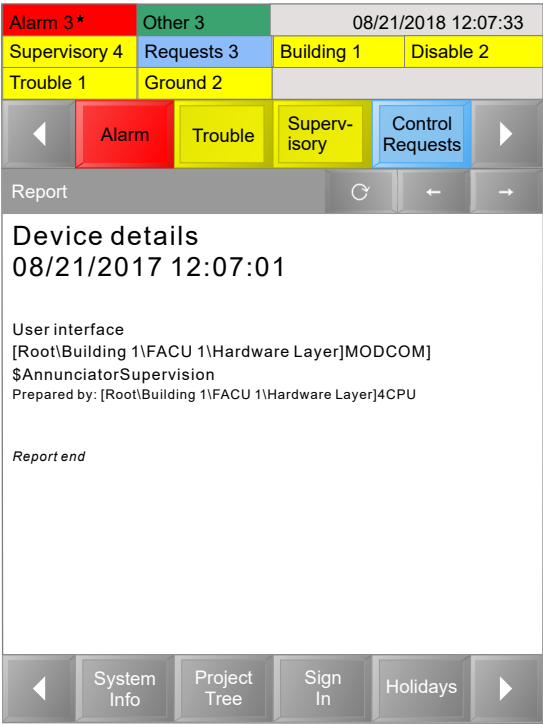
Click Run. The history report displays listing the selected alarm types that occurred within the specified timeframe.

- 4. From the control unit LCD screen, when finished:
 - On the Action bar, tap System Info.
 - On the Command bar, tap Cancel Access, to block read access.

Viewing event details

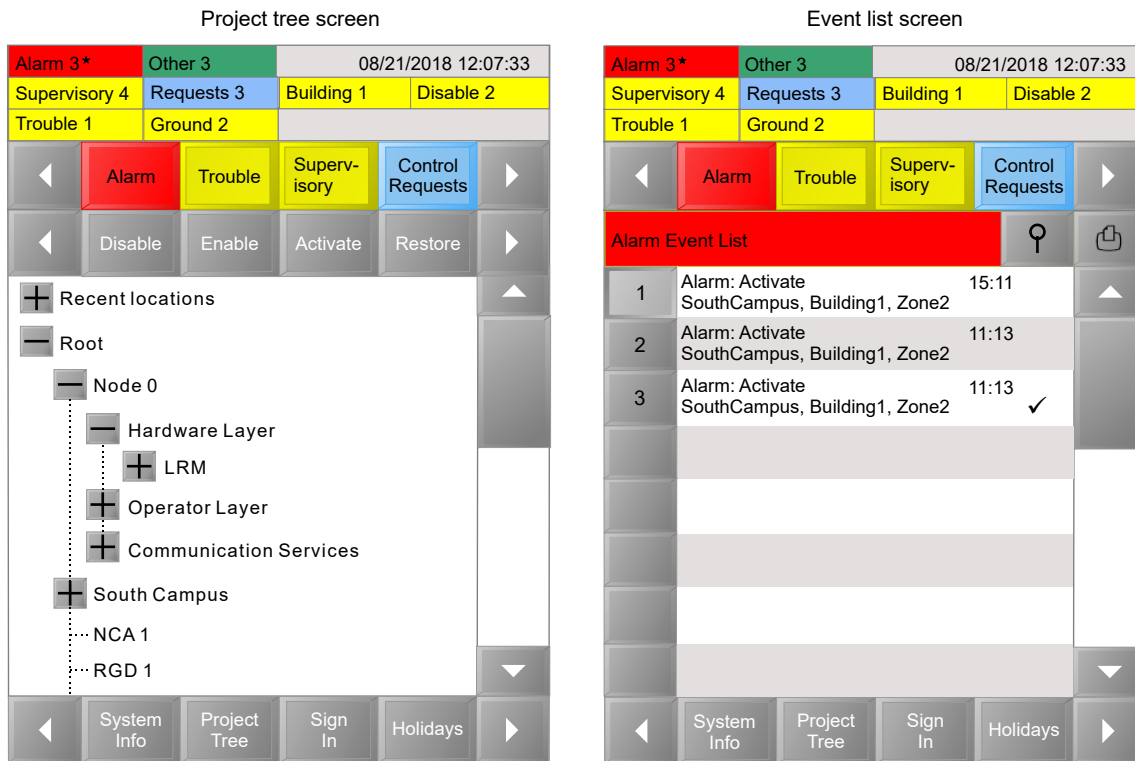
An event or device details report provides the type of event, address of the device that generated the event and, if programmed, the location.

Figure 6: Event details report



Event details can be accessed from the project tree or event list.

Figure 7: Viewing event details

**To view device details from the project tree:**

1. On the Action bar, tap Project Tree.
2. Expand the project tree by tapping the expand button , and then locate the target device or pseudo point.
3. Tap on the device/pseudo point. The Device Details Report displays.

To view device details from an event list:

1. Tap an event List button. For example, Alarm.
2. In the list, tap the event number, and then tap the details button . The Device Details Report displays.

Device details

If a device activation causes an event, the Details screen shows the active device's logical address and the off-normal state.

Logic group details

If a group activation causes an event, the Details screen shows the state of the device, device address, and device message, which is usually the device location.

Instruction text details

You can program the system to include detailed instructions for certain events. When specific devices go into alarm, the system generates a related monitor event. If you select the monitor event, then press Details, the instruction text is displayed.

Viewing and printing system reports

Reports are used to check the current condition or history of the EST4 life safety network. EST4 system reports can be viewed and printed from either the control unit LCD screen or from the computer running the 4-CU programming utility.

System reports on the control unit

Table 18 below lists the reports that you can access from the control unit. The report displays on the LCD screen and can also be printed to a local printer.

Table 18: Reports accessed from the control unit

Report type	Description
Status	The status reports listed below are available to help you determine the current state of the system. • Status: Lists all active events and disabled points from the selected node. • Status Disabled: Lists of addressable points that are disabled. • Status Test: Lists all off-normal devices under test. • Status Holdup: Future feature.
Maintenance	A set of device maintenance reports are available to help you determine if any addressable smoke detector devices require maintenance. Each report gives you the option to display it on the LCD screen or print it to a local printer. The device maintenance reports are listed below. • Maintenance Not Clean: Lists all addressable smoke detectors that have a %Dirty value of 80% and greater. Smoke detectors that are more than 80% dirty should be cleaned or replaced as soon as possible. Note: The %Dirty value is an indication of a smoke detector's ability to compensate for environmental conditions. Smoke detectors with higher percentages are less able to compensate. • Maintenance Dirty: Lists all addressable smoke detectors that have a %Dirty value of 20% and greater. A smoke detector that is more than 20% dirty should be noted for possible cleaning or replacing. • Maintenance: Lists the %Dirty value for all of the smoke detectors on a signaling line circuit. The report also includes each smoke detector's model type, primary and alternate alarm sensitivity values, and, if programmed, a location description. • Maintenance Device: Lists the %Dirty value for a single smoke detector. For CO detectors the report includes CO Life Left Months or CO DAY Running. The report also includes the smoke detector's model type, primary and alternate alarm sensitivity values, and, if programmed, a location description. To view this report you must select the device from the Project Tree.

Report type	Description
History	History reports list the most recent events or operator instructions processed by the control unit for the previous month to current or since its history was cleared. To view all months, use the 4-CU reports feature or a web browser. The event or system command name, address, time and date of occurrence, and the source that initiated the event or command are included. When the Freeze History command is activated, the system archives the most recent events (up to 10,000) so that they cannot be overwritten. They can, however, be erased when the Clear history command is activated. To view or print a Freeze History archive, see “Freezing the events history” on page 47. When the Clear History command is activated, event indicator counters reset on the LCD screen and the list of events that occurred on the control unit since it was placed into service, or the last time the history file was cleared, is erased. This includes unfreezing and erasing archived history. The report is structured with the most recent event or instruction listed first. The report can be displayed for viewing on the LCD screen or printed to a local printer. The history reports are listed below. • History: All Dates report: Lists the last 20,000 events and operator commands processed by selected node • History Alarm: All Dates, Alarm report: Lists the event name, time, date, and location text for all alarm events • History Trouble: All Dates, Trouble report: Lists the event name, time, date, and location text for all trouble events. • History Supervisory: All Dates, Supervisory report: Lists the event name, time, date, and location text for all supervisory events.
Revisions	The Revisions report provides system database information and installed hardware information. The hardware that shows on the list is dependent on the devices installed in the control unit cabinet. The initial Revisions report is described below. • Alarm history count: Shows the total number of times that the control unit has gone into alarm since the alarm history was cleared. • Market: Shows the 4-CU market setting. • Configuration version: Shows the 4-CU project version number. • Configuration time stamp: Shows the date and time that the 4-CU project was created or modified. • Configuration schema: Shows the 4-CU firmware-to-software interface version. • Configuration messaging: Shows the 4-CU internal messaging protocol version. • Firmware: Shows the 4-CU firmware version. • LRM x shows: – LRM type – Firmware version and date – Bootloader version and date – 4-CU database version and date • CDM xx Annunciator shows: – Control-display module (CDM) type – Firmware version and date • CPU shows: – CPU type – Firmware version and date – Bootloader version and date.

Report type	Description
System	The System Info report provides a project description that includes the control unit MAC address required for communication between the 4-CU and control unit, and project versioning numbers. Component licensing and copyright information is also included in the report. The System Info report content is described below. Project Description • IP/DNS node name: Shows the MAC address of the installed CPU. • Versions: – Configuration Utility: Shows the 4-CU software version. – Project Configuration: Shows the 4-CU project version number. – Project Configuration Date: Shows the date and time that the database was downloaded. – Project Configuration ID: Shows the database serial number created when the database was converted. – Configuration schema: Shows the 4-CU software-to-firmware interface version. – Message Schema: Shows the 4-CU internal messaging protocol version. – Firmware: Shows the 4-CU firmware version. – Archive Enabled: Shows whether the project is configured to automatically archive project files. Licensing: Shows licensing information and copyright information for various components used in the 4-CU software and firmware.

To view or print reports from the control unit LCD:

1. On the Action bar, tap System Info.
2. On the Command bar, tap the desired report button. The report displays on the screen.

Tap the  (print) button to send the report to a local printer.

To view device reports:

See “Viewing event details” on page 48.

System reports uploaded to the 4-CU

Table 19 below lists the system reports that can be uploaded to the 4-CU, and then viewed and printed.

Table 19: System reports accessed from the 4-CU

Report type	Description
Details	The Details report lists the type of event, address of the device that generated the event and, if programmed, the location of a system event.
Status	The status reports listed below are available to help you determine the current state of the system. • All Off normal: Lists all active events and disabled points from the selected node. • Disabled: Lists of addressable points that are disabled. • Test: Lists points in an active logic group that are in the active or trouble state. • Status Holdup: Future feature

Report type	Description
Maintenance	A set of device maintenance reports are available to help you determine if any addressable smoke detector devices require maintenance. - Not Clean: Lists all addressable smoke detectors that have a %Dirty value of 80% and greater. Smoke detectors that are more than 80% dirty should be cleaned or replaced as soon as possible. Note: The %Dirty value is an indication of a smoke detector's ability to compensate for environmental conditions. Smoke detectors with higher percentages are less able to compensate. - Dirty: Lists all addressable smoke detectors that have a %Dirty value of 20% and greater. A smoke detector that is more than 20% dirty should be noted for possible cleaning or replacing. - All: Lists the %Dirty value for all of the smoke detectors on a signaling line circuit. The report also includes each smoke detector's model type, primary and alternate alarm sensitivity values, and, if programmed, a location description. - Device: Lists the %Dirty value for a single smoke detector. For CO detectors the report includes CO Life Left Months or CO DAY Running. The report also includes the smoke detector's model type, primary and alternate alarm sensitivity values, and, if programmed, a location description. To view this report you must select the device from the Project Tree.
Events	Alarm History: Alarm History reports list the events or operator instructions processed by the control unit for a user-specified timeframe. - All: Lists the last 20,000 events and operator commands processed by selected node - Alarm: Lists the event name, time, date, and location text for all alarm events - Trouble: Lists the event name, time, date, and location text for all trouble events. - Supervisory: Lists the event name, time, date, and location text for all supervisory events. - Test: Lists the event name, time, date, and location text for all test events. Event Queue: Lists the event name, time, date, and location text for all events that were mapped to the user-specified queue.
Informational	Information reports provide reports Revisions: Lists system database and installed hardware information. See "Revisions Report" in Table 18 on page 50 for the details on the information that displays. System: Lists the control unit MAC address, project versioning numbers, component licensing and copyright information. See "System Info" in Table 18 on page 50 for the details on the information that displays.

To view and print systems reports uploaded to the 4-CU:

1. Connect the Type B end of a USB cable to the J14 USB port on the 4-CPU. See Figure 5 on page 41.
2. Connect the Type A end of the cable to a standard USB port on the computer with the project database.
3. From the control unit LCD screen:

On the Action bar, tap System Info.

On the Command bar, tap 1 Hour Access or 8 Hour Access to obtain a 4-CU required access code that will allow read access.
4. From the 4-CU:

On the Home tab, in the Communication group, click Enter Access Code.

Enter the Access Code, and then click Validate.

On the Home tab, in the Communication group, click Query Network, and then Refresh to confirm communication with the node(s). The window remains open until you close it.

On the Reports tab, in the Navigation pane, select the Project Tree branch.

On the Project Tree Configuration tab, click the Node list tab, and then select the node for which you want the upload system information.

On the Reports tab, in the System Reports group, click desired report. See Table 19 on page 52 for a description of the available reports.

If required, enter information when prompted. The report displays on the screen. You can also print or save the report. Refer to *4-CU Help* for instructions.

5. From the control unit LCD screen, when finished:

On the Action bar, tap System Info.

On the Command bar, tap Cancel Access, to block read access.

Disabling and enabling devices

Devices include input and output circuits, detectors, and modules. Disabling a device isolates it from the system. While the device is disabled, the EST4 control unit logs the status change signals, but is prevented from processing the signals until the device is enabled. For example, the control unit does not activate an alarm event when you activate a disabled detector, but it will after the detector is enabled.

The control unit keeps track of how many times you disable a device without enabling it. You must enable a device the same number of times you disable it in order to return the device to its original condition. The Status Disabled report provides a counter that shows the number of times the device was manually disabled.

Notes

- You cannot disable a device configured as a common alarm output.
- Disabling all of the devices in a Zone group automatically disables the Zone group. Enabling each device in a Zone group automatically enables the Zone group.
- Disabling the device address for the dialer or a dialer account deletes all event messages sent to that account before they are transmitted. The dialer still transmits the account's test-abnormal message and any message that was in the dialer queue before the account was disabled.
- When you enable a device, all indicators and outputs activated by the device will reactivate.

To disable a device:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target device.
3. Tap the device, and then tap Disable on the Command bar.

To enable a device:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target device.
3. Tap the device, and then tap Enable on the Command bar.

Note: Enable the device the same number of times that you disabled it. If necessary, view the Status Disabled report to see the manual disable counter (Action Bar > System Info > Status Disabled).

Disabling and enabling local rail modules (LRMs)

Local rail modules can include chassis rail cards and modules, and UI frame assembly control-display modules. Disabling a module isolates it from the system. While the module is disabled, the EST4 control unit logs the status change signals, but is prevented from processing the signals until the module is enabled.

Note: The control unit keeps track of how many times you disable a module without enabling it. You must enable a module the same number of times you disable it in order to return the module to its original condition. The Status Disabled report provides a counter that shows the number of times the device was manually disabled.

To disable an LRM:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target LRM.
3. Tap the module, and then tap Disable on the Command bar.

To enable an LRM:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target LRM.
3. Tap the module, and then tap Enable on the Command bar.

Note: Enable the device the same number of times that you disabled it. If necessary, view the Status Disabled report to see the manual disable counter (Action Bar > System Info > Status Disabled).

Disabling and enabling logic groups

A group is a *logic* device that is created in the 4-CU. Groups are required in order to execute certain system functions, but groups bear no physical relationship to the system. For example, smoke detectors can be assigned to the same Zone group even though they are not attached to the same wire run.

Disabling a group isolates it from the system. While the group is disabled, the EST4 control unit is prevented from processing status change signals from every device in the group until the device is enabled. For example in a Zone group, the control unit does not activate an alarm event when you activate a disabled detector, but it will after the detector is enabled.

The control unit keeps track of how many times you disable a logic group without enabling it. You must enable a logic group the same number of times you disable it in order to return the group to its original condition. The Status Disabled report provides a counter that shows the number of times the device was manually disabled.

Note: If you disabled a Zone logic group by disabling all of the devices in the zone, enabling the zone enables all of the devices in the zone.

The logic groups are listed below.

- **AND group:** A collection of devices that are grouped in the database to provide a group response that is separate from that of its member devices. An AND group activates when a specified number of devices change to a specified state. The specified state can be alarm, supervisory, trouble, monitor, or not active (NA). AND groups can be configured to signal an alarm, supervisory, trouble, or monitor condition upon activation.
- **Matrix group:** A collection of devices that are grouped in the database to provide a unique system response when a specified number of its members signal an alarm event or when more than one device in a defined search radius signals an alarm event. A member's alarm, supervisory, trouble, or monitor state's active status serves as an input for activating the AND group.

- **Service group:** A collection of devices that are grouped together in the database to provide a unique response for testing purposes. When enabled, the Service group automatically disables the member device's normal alarm response, and provides a common alternate test response.
- **Zone group:** A collection of input devices that are grouped in the database to provide a unique response separate from their individual device responses. Zone groups can be configured to go into alarm when any member of the group goes active or when any device in the group goes into trouble.
- **Instruction Text group:** A collection of devices that are grouped in the database to provide additional detailed instructions or warnings when any device in the group changes to a qualified active state.

To disable a group:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target group.
3. Tap the group, and then tap Disable on the Command bar.

To enable a group:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target group,
3. Tap the group, and then tap Enable on the Command bar.

Note: Enable the device the same number of times that you disabled it. If necessary, view the Status Disabled report to see the manual disable counter (Action Bar > System Info > Status Disabled).

Disabling and enabling time controls

Time Controls are *logic* devices that are created in the 4-CU. They are configured to set up automatic starting and stopping of system events based on time and date. The controls run in the background and do not require any operator action. In the event you need to disable a control, you can do so from the control unit.

To disable a time control:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target time control.
3. Tap the time control, and then tap Disable on the Command bar.

To enable a time control:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target time control.
3. Tap the group, and then tap Enable on the Command bar.

Activating alarm signals manually

The EST4 drill feature lets you activate alarm signals manually without putting the control unit into alarm. When you activate a drill, all audible alarm signals turn on and, if configured, all visual alarm signals, but other automatic fire alarm responses are not activated. The alarm signals remain active until the drill is canceled.

Notes

- To activate a drill using a 4-LCD operator control button, either button one or two must be programmed in the 4-CU as Drill.
- The drill function can be programmed for a fire privilege level password. The default setting is no PIN required.

To activate a drill:

1. Press the Drill operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

To cancel a drill:

1. Press the Drill operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Testing the system and devices

Test commands are used to perform periodic inspection tests on the life safety system.

Performing a lamp test

Use the Lamp Test operator button on the 4-LCD display module user interface to verify the operation of the LCD screen and LED indicators. The lamp test command temporarily turns on the operator alert signal, all LED indicators, and every pixel on the LCD screen. If the local control unit is part of a command center, the lamp test is also executed on all nodes within the command center.

Notes

- To activate a lamp test using an operator control button, either LCD button 1 or 2 must be programmed in the 4-CU as Lamp Test.
- A control-display module switch can also be programmed to activate the LampTestOn command.
- The lamp test function can be programmed for a fire privilege level password. The default setting is no PIN required.

To activate a lamp test:

1. Press the Lamp Test operator control button.
2. If prompted, enter the fire privilege PIN using the interactive keypad.

Testing Signature devices

Use commands accessed from the control unit LCD screen to remotely place a Signature device into the alarm, prealarm, monitor, supervisory, or trouble condition for testing purposes. Signature devices include all detectors and modules. To test a Signature device, the device must be connected to a signaling line circuit.

Note: For latching devices, you must reset the control unit to restore the tested device to its normal state. Nonlatching devices restore automatically without resetting the control unit.

To place a Signature CO detector into test state:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Signature CO detector.
3. Tap the detector, and then tap CO Test On on the Command bar.

To cancel the test, tap CO Test Off on the Command bar.

To place a Signature optical smoke detector into test state:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Signature optical smoke detector.
3. Tap the detector, and then tap CO Test On on the Command bar.

To cancel the test, tap CO Test Off on the Command bar.

To place a Signature device into test state:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Signature device.
3. Tap the device, and then tap Test Trouble on the Command bar.

To place a Signature device into alarm, monitor, or supervisory state:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Signature device.
3. Tap the device, and then tap Test Activate 1 on the Command bar.

To place a Signature device into prealarm state:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Signature device.
3. Tap the device, and then tap Test Activate 2 on the Command bar.

Testing alarm input devices

In order to test an alarm input device, the device must be part of a Service group that was created in the 4-CU. Service groups allow alarm input devices to be activated without placing the system into alarm. The protected premises may be divided into more than one Service group to make testing possible without leaving the entire premises unprotected.

Note: The alarm input test automatically times out after approximately 1-hour of inactivity.

To put a Service Group into test:

1. Sign in with fire privilege 4 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target Service group.
3. Tap the Service group, and then tap Activate SG on the Command bar. To cancel the test, tap Restore SG on the Command bar.

Changing output states

Use the On and Off commands to change the output state of relays, NAC circuit outputs, and audio amplifiers.

Note: Only users with fire privilege 3 can access this function.

To activate a output state:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target device.
3. Tap the device, and then tap On on the Command bar.

To restore a relay output state:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target device.
3. Tap the device, and then tap Off on the Command bar.

Switching the smoke detector alarm sensitivity threshold

Intelligent addressable smoke detectors are configured with two alarm sensitivity thresholds: primary and alternate. The alarm sensitivity setting, determines how much smoke is needed for the automatic fire detectors to sense a fire alarm condition. This allows you to increase or reduce an individual detector's sensitivity at various times of the day, dependent upon, environmental conditions, occupancy, manufacturing processes, etc.

A time control is commonly used to automatically switch alarm sensitivity thresholds. However, you can manually switch alarm sensitivity thresholds by using a command button.

Primary alarm sensitivity threshold

Typically, the primary alarm sensitivity threshold is set to a lower threshold. This threshold is commonly used for a daytime operation to reduce the occurrence of nuisance alarms when a facility is occupied, or when environmental conditions may create prealarm conditions.

Alternate alarm sensitivity threshold

The alternate alarm sensitivity threshold sets the *secondary threshold* at which the smoke detector activates an alarm event. Typically, the alternate threshold is set to a higher sensitivity threshold. This threshold is commonly used for a nighttime or weekend operation, when the facility is unoccupied.

Alarm sensitivity settings

Alarm sensitivity settings are expressed in percent of smoke obscuration per foot. The setting defines the threshold at which the detector will change to the alarm state when the smoke in its sensing chamber exceeds the obscuration per foot threshold. The alarm sensitivity levels for Signature devices are described below.

Note: When smoke detectors having both ionization and photoelectric elements are used, the sensitivity setting applies to both elements.

- **Most:** Activates an alarm event when the smoke level reaches approximately 1.0 %/ft obscuration (0.7 %/ft for ionization detectors)
- **More:** Activates an alarm event when the smoke level reaches approximately 2.0 %/ft obscuration (1.0 %/ft for ionization detectors)
- **Normal:** Activates an alarm event when the smoke level reaches approximately 2.5 %/ft obscuration (1.2 %/ft for ionization detectors)
- **Less:** Activates an alarm event when the smoke level reaches approximately 3.0 %/ft obscuration (1.4 %/ft for ionization detectors)
- **Least:** Activates an alarm event when the smoke level reaches approximately 3.5 %/ft obscuration (1.6 %/ft for ionization detectors, 2.46 %/ft for a SIGA-SD Duct Smoke Detector)

Switching the sensitivity threshold

To activate the alternate alarm sensitivity settings:

1. Sign in with fire privilege 3 or higher, and then tap System Info on the Action bar.
2. On the Command bar, tap Toggle Alternate Sensing.

To restore the primary alarm sensitivity settings:

1. Sign in with fire privilege 3 or higher, and then tap System Info on the Action bar.
2. On the Command bar, tap Toggle Alternate Sensing.

Adding and deleting holidays

The Holiday time control is a special time control used to program up to 99 holidays on a per control unit basis. Holiday time controls supersede the normal CU-programmed system time controls on dates that are designated as holidays.

To add a holiday:

1. Sign in with fire privilege 4 or higher, and then tap Holidays on the Action bar.
 2. Tap Add on the Command bar
 3. Use the interface keypad to enter the 2-digit month and 2-digit day (MMDD). For example, 0101
- The Holiday list appears, showing the added holiday.

To delete a holiday:

1. Sign in with fire privilege 4 or higher, and then tap Holidays on the Action bar.
 2. Tap Delete on the Command bar
 3. Tap the number of the holiday to be deleted, and then tap the delete button X.
- The Holiday list appears, no longer showing the deleted holiday.

Control-Display modules

Control-display modules provide additional operator interface capability. The 4-24L series modules provide pockets for inserting custom labels. The labels can be printed using the 4-CU configuration utility.

Model [1]	Description
4-24L	24 indicators
4-24L12S	24 indicators and 12 switches
4-24L18S	24 indicators and 18 switches
4-24L24S	24 indicators and 24 switches

The buttons on a control-display module use one of three available operating modes that are database configured.

- **Toggle:** The state of the button changes each time the button is pushed (i.e. off to on or on to off). Toggle buttons are commonly used to control two-state operations such as on/off, open/close, speaker select, telephone select, etc.
- **Interlocked:** Adjacent toggle buttons that operate as a group. Pushing any button in the group turns the output of the other two buttons off and turns its own output on. The interlocked mode is commonly used for hand-off-auto control of HVAC systems. An interlocked button in the on state can be turned off without activating a second button by pressing the On button a second time. The output of the on button remains on during control unit reset. It must be manually returned to auto when no longer required.
- **Momentary:** The button is on only while pressed by the operator. Momentary buttons are typically used to activate functions that are self-restoring. Examples include lamp test, function reset, and test sequence.

You may find multiple button modes on a single control-display module. Consult your site-specific documentation for additional information.

Disabling and enabling control-display modules

Disabling a control-display module isolates it from the system. While disabled, changes to the module's state are not processed. When the module is disabled, a Disabled Active event shows in the Trouble queue.

Enabling a control-display module re-establishes it as part of the system. When enabled, any changes in state that occurred while the module was disabled are not processed.

To disable a control-display module:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target control-display module.
3. Tap the control-display, and then tap Disable on the Command bar.

To enable a control-display module:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target control-display module.
3. Tap the control-display module, and then tap Enable on the Command bar.

Disabling and enabling control-display module elements

Disabling and enabling control-display module switches

To disable a control-display module switch:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target switch.
3. Tap the switch, and then tap Disable on the Command bar.

To enable a control-display module switch:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target switch.
3. Tap the switch, and then tap Enable on the Command bar.

Disabling and enabling control-display module LEDs

To disable a control-display module LED:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target LED indicator.
3. Tap the LED indicator, and then tap Disable on the Command bar.

To enable a control-display module LED:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target LED indicator.
3. Tap the LED indicator, and then tap Enable on the Command bar.

Setting a control-display module LED mode

To set a control-display module LED mode:

1. Sign in with fire privilege 3 or higher, and then tap Project Tree on the Action bar.
2. Expand the project tree by tapping the expand button , and then locate the target LED indicator.
3. Tap the LED indicator, and then tap one of the following: Steady, Fast Blink, Slow Blink, Off.

Chapter 4

Installation

Summary

This chapter provides installation information for system components and applications that supplement the instructions provided on individual component installation sheets.

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System installation sequence

Follow these basic instructions when installing a control unit. Refer to the installation sheet that came with a component for specific instructions.

Notes

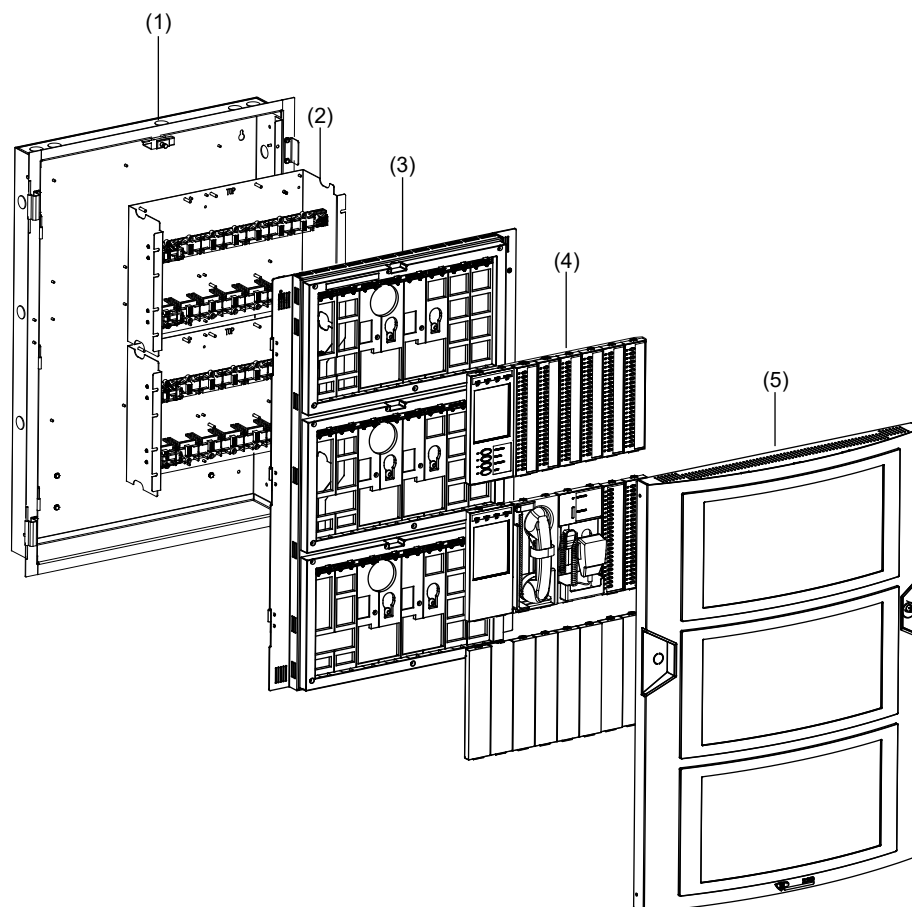
- Make sure the installation location is free from construction dust and debris, and is not subject to extreme temperature ranges and excess humidity.
- Ensure sufficient floor and wall space, to avoid obstructions during installation and servicing.
- When installing the cabinet, use fasteners that can support the full weight of the cabinet, including standby batteries.
- Be sure to tighten the fasteners firmly to prevent the cabinet from vibrating.
- Remove a control unit component from its protective antistatic packaging only for inspection or installation.
- Do not connect standby batteries until initial control unit power up (see “Initial power up” on page 37).

Sequence list

Notes

- Install all FACU components as instructed on the installation sheet that came with the component.
 - Record MAC addresses and locations required during 4-CU programming (see “Obtaining and recording node MAC addresses” on page 40).
1. Unpack the equipment and make sure it is not damaged. Be sure to remove the installation sheets from component accessory bags.
 2. Install the backbox at the required location and pull all the required wiring through the conduit into the backbox.
 3. Install the local rail module (LRM) chassis assembly.
 4. Verify the field wiring. See “Preliminary field wiring testing.”
 5. Install the 3-PPS/M Power Supply and any project-required half-footprint modules.
 6. Install any project required option cards on the 4-CPU, and then install the 4-CPU.
 7. Install all project required option cards and local rail modules on the chassis rail.
 8. Install the inner door, and then install all user interface modules on the UI frame assembly.
 9. Connect field wiring.
 10. Power up the control unit and download an optional initial startup database. See “Initial power up” on page 37.
 11. Define the cabinet configuration, device loops, option cards, etc. for a final 4-CU database, and then download it to the control unit. See “Downloading a project database” on page 43.
 12. Install the outer door assembly onto the backbox.

Figure 8: Exploded view of EST4 cabinet (3-CAB14B backbox and 4-CAB24D door shown)



- | | |
|----------------------|----------------------------|
| (1) Backbox | (4) User interface modules |
| (2) Chassis assembly | (5) Outer door |
| (3) Inner door | |

Note: See Table 6 on page 15 for a list of user interfaces and chassis rail modules.

Component installation

An EST4 enclosure houses all of the control unit components. Enclosure components can include power supplies, batteries, electronics and communication rails, user interfaces, option modules, and audio subsystem components needed for an EST4 system to operate as a standalone unit or part of a network.

Enclosure backboxes

The enclosure backbox holds the primary power supply module, footprint option cards, chassis rail for hardware LRMs and optional audio subsystem components, and standby batteries for the control unit.

See Table 3 on page 10 for a list of backboxes used in an EST4 system.

Local rail module chassis

3-CHAS7 LRM chassis

The 3-CHAS7 chassis is used in an EST4 enclosure to provide the mounting, internal power, and data distribution for up to seven local rail modules, including power supply modules, CPU node modules, and one half-footprint module.

A 4-BRKT-CB mounting bracket can be mounted on the back plane of the 3-CHAS7. The bracket provides space for one 4-ANNCPU, 4-FWAL1, 4-FWAL2, 4-FWAL3, 4-FWAL4, or a 4-NET-AD module. For more information about the bracket, see *4-BRKT-CB Mounting Bracket Installation Sheet* (P/N 3102323-EN).

The first two slots of the *first* 3-CHAS7 in the control unit are used for a 4-CPU central processor unit. See Table 5 on page 12 for a list of CPU modules and Table 6 on page 15 for firewall modules. The last five slots on the first 3-CHAS7 are used for optional LRMs. See Table 5 on page 12 for a list of chassis rail modules that can be installed.

The seven slots on a second or third 3-CHAS7 chassis in a control unit are used for additional optional rail modules.

Notes

- You *cannot* install local rail modules in slots 3 and 4 on the 3-CHAS7 rail when a 4-FT firefighter telephone is installed on the inner door UI frame assembly.
- For more information see *3-CHAS7 Local Rail Module Chassis Installation Sheet* (P/N 270484-EN).

3-CAB5B backbox with LRM chassis

A 3-CAB5B backbox can be used in an EST4 enclosure to provide the mounting, internal power, and data distribution for up to five local rail modules, including power supply modules and CPU node modules.

The first two slots of the chassis rail are used for a 4-CPU central processor unit. The last three slots are used for optional LRMs. See Table 5 on page 12 for a list of chassis rail modules that can be installed.

Notes

- You cannot install a 4-ANNCPU, 4-FWAL1, 4-FWAL2, 4-FWAL3, 4-FWAL4, or 4-NET-AD module in a 3-CAB5B backbox when a 4-CPU is installed.
- You can install a 4-ANNCPU if removing the 3-CAB5B chassis rail to use the backbox with a 4-CAB8D door as a semiflush mounted annunciator cabinet.
- You cannot install local rail modules in slots 4 and 5 on the 3-CAB5B LRM rail when a 4-FT firefighter telephone is installed on the inner door UI frame assembly.
- For more information see *3-CAB5B Equipment Backbox* (P/N 3102288-EN).

Enclosure doors

The EST4 enclosure door consists of an inner and outer door. The outer door may have one or more viewing windows and is secured with a key lock. The inner door isolates the operator from the panel electronics and wiring. The inner door also incorporates UI frame assemblies for mounting user interface modules.

See Table 3 on page 10 for a list of EST4 equipment enclosure doors.

Inner door UI frame assembly user interface modules

User interface modules installed on the UI frame assembly rail include LCD user interface modules, control-display modules, 4-MIC paging microphone, and 4-FT firefighter telephone. The EST4 control unit uses the Controller Area Network (CAN) serial communication protocol for user interface module communication through the UI frame assembly rail to the CPU. See Table 5 on page 12 for a list of LCD display modules and user modules.

When a user module is not installed on the UI frame assembly you can install a 4-FIL blank plate to cover the slot. For more information about the filler plate, see *4-FIL Blank Plate Installation Sheet* (P/N 3102285-EN).

Notes

- For a 3-CHAS7 LRM chassis, when a 4-FT firefighter telephone is installed on the UI frame assembly, you *cannot* install local rail modules in slots 3 and 4 on the LRM rail behind the 4-FT.
- For a 3-CAB5B backbox LRM rail, when a 4-FT firefighter telephone is installed on the inner door UI frame assembly, you *cannot* install local rail modules in slots 4 and 5 on the LRM rail behind the 4-FT.
- When a 4-LCDx user interface module is not installed on the UI frame assembly, a 4-CABL0542 flex cable is used to connect the control unit's CPU to the UI frame assembly rail. For example, a command center can be comprised of multiple control units and only one of the control unit houses the LCD user interface module. The flex cable is for use in 3-CAB5B, 3-CAB7B, 3-CAB14B, and 3-CAB21B backboxes. For more information about the 4-CABL0542, see *4-CABL Series Flex Cable Installation Sheet* (P/N 3102265-EN).

Outer door tamper switch

For added enclosure security, a tamper switch can be installed in the backbox that will detect an open enclosure door. For more information see *Tamper Switches Installation Sheet* (P/N 387422-EN).

Outer door New York city lock

For 4-CAB16D, 4-CAB24D, and 4-CAB24DL doors, the outer door lock can be modified to accommodate installation of a New York City Lock.

For more information see *EST4 New York City Lock Installation Sheet* (P/N 3102293-EN).

4-CPU

Before installing the 4-CPU, refer to the *4-CPU Central Processor Module Installation Sheet* (P/N 3102258-EN) for additional installation information and wiring instructions.

WARNINGS

- **Electrocution hazard.** To avoid personal injury or death from electrocution, make sure the distribution circuit providing mains AC is rendered inoperative prior to connecting mains input wiring to the 3-PPS/M Power Supply.
 - **Electrocution hazard.** To avoid personal injury or death from electrocution, remove all sources of power and wait 11 minutes to allow stored energy to discharge before installing or removing equipment.
-

Caution: Circuit boards are sensitive to electrostatic discharge (ESD). To avoid damage, follow ESD handling procedures.

Notes

- Each 4-CPU has a label with a unique MAC address printed on it in text and in the form of a QR code. This unique MAC address is used by the 4-CU configuration utility to identify the 4-CPU (node). Prior to installing the 4-CPU, record the MAC address in project documentation or enter it in the 4-CU. See "Obtaining and recording node MAC addresses" on page 40.
- The 4-CPU can be mounted to the 3-CHAS7 or the 3-CAB5B chassis.
- All wiring is power-limited and supervised.
- Maintain 0.25 in. (6 mm) separation between power-limited and nonpower-limited wiring at all times. Refer to the backbox documentation for details.

The accessory modules and flex cables that can be attached to the 4-CPU are listed in Table 20 below. Refer to the documentation that is provided with the module for installation details.

Table 20: Accessories

Model	Description
4-AUDTELS	The 4-AUDTELS Audio I/O and Telephone Riser Source Module provides a single telephone riser, paging microphone support, and up to four 1 VRMS audio channels that can be programmed for either input or output operation.
4-COMREL	The 4-COMREL Common Relay Module provides three common relays (alarm, supervisory, and trouble) with duplicated terminals for installing EOLRs.
4-CABL0542	The 4-CABL0542 connects the 4-CPU to the UI rail when no 4-LCD display module is installed on a 4-CAB8D(R), 4-CAB16D(R), 4-CAB24D(R), or 4-CAB16D(R)L inner door.

3-PPS/M Primary Power Supply

One 3-PPS/M primary power supply is required for each control unit. The power supply provides the required power and related supervisory functions for the control unit as well as filtered, regulated power, and 24 VDC output for operating notification appliances and ancillary equipment. The 3-PPS/M(-230) comprises a heat sink assembly and a monitor module. The monitor module plugs into the rail, and the heat sink and the heat sink assembly mount onto the rail chassis. For more information see *Power Supply Installation Sheet* P/N 270495-EN) for installation instructions and specifications.

WARNINGS

- Electrocutation hazard. To avoid personal injury or death from electrocution, make sure the distribution circuit providing mains AC is rendered inoperative prior to connecting mains input wiring to the 3-PPS/M Power Supply.
- Electrocutation hazard. To avoid personal injury or death from electrocution, remove all sources of power and wait 11 minutes to allow stored energy to discharge before installing or removing equipment.

Caution: Circuit boards are sensitive to electrostatic discharge (ESD). To avoid damage, follow ESD handling procedures.

Notes

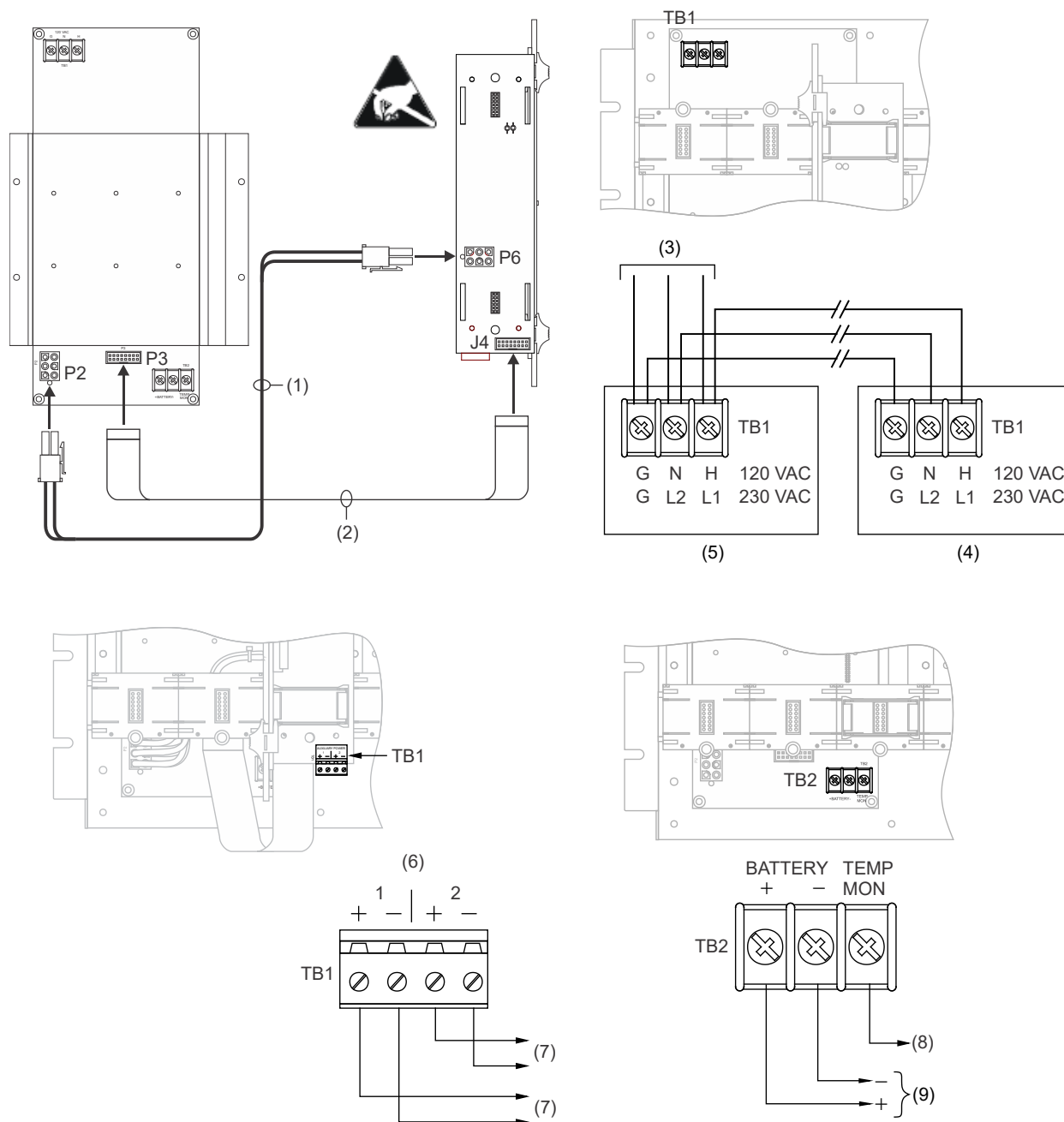
- Mains input and battery wiring are supervised and nonpower-limited.
- Connect the mains power wiring after all power supplies have been installed. No more than one primary power supply and three booster supplies may be connected to a single mains AC circuit.
- Mains power wiring must be double insulated and connected only to a dedicated 120 V or 230 V mains power distribution circuit with its own disconnect device.
- Each set (pair) of batteries must be connected to a separate battery charging circuit, either on a 3-PPS/M(-230) or on a 3-BBC/M(-230).
- Each power supply must have its own separate pair of wires connecting it to a set of standby batteries. Daisy chaining battery connections from power supply to power supply is not allowed.
- All battery wiring must be the same length and wire gauge.
- All batteries connected to the same control unit must have the same ampere-hours rating, be from the same manufacturer, and have the same manufacturing date code.

- Batteries greater than 17 Ah and any additional standby batteries must be installed in an external battery cabinet. If an external battery cabinet is used to house standby batteries, the cabinet must be installed within three feet and in the same room as the control unit.

Each power supply must be connected to only one set of standby batteries. Refer to Table 21 below for power supply and standby battery requirements. A 3-PPS/M(-230) and a 3-BPS/M(-230) can be wired to the same set of batteries as well as a 3-BBC/M(-230) and a 3-BPS/M(-230) but never a 3-PPS/M(-230) and a 3-BBC/M(-230).

Table 21: Power supply and standby battery requirements

Total current	Power supplies required	Battery sets required
7 A	1 3-PPS/M(-230)	1 set, 65 Ah max.
14 A	1 3-PPS/M(-230) 1 3-BPS/M(-230)	1 set, 65 Ah max.
	1 3-PPS/M(-230) 1 3-BBC/M(-230)	2 sets, 65 Ah max.
21 A	1 3-PPS/M(-230) 2 3-BPS/M(-230)	1 set, 65 Ah max.
	1 3-PPS/M(-230) 2 3-BBC/M(-230)	3 sets, 65 Ah max.
28 A	1 3-PPS/M(-230) 3 3-BPS/M(-230)	1 set, 65 Ah max.
	1 3-PPS/M(-230) 3 3-BBC/M(-230)	4 sets, 65 Ah max.

Figure 9: 3-PPS/M(-230) Power Supply

- | | |
|---|---|
| (1) Cable harness P/N 250187 | (6) Auxiliary power |
| (2) Ribbon cable P/N 250189 (primary monitor module), P/N 250188 (booster monitor module) | (7) 24 VDC |
| (3) Dedicated 120 V or 230 V mains distribution circuit | (8) Not used |
| (4) Last booster supply in the same cabinet | (9) To plus and minus terminals on cabinet battery or to plus and minus terminals on power distribution bus in remote cabinet |
| (5) Primary power supply | |

On-board terminals and connectors facilitate mains input wiring, NAC/AUX wiring, and battery wiring. The 24 VDC rechargeable battery circuit on the power supply board has the capacity to charge up to two 65 Ah sealed lead acid batteries.

Auxiliary power riser

The 3-PPS/M provides a 24 VDC auxiliary power riser on the power supply monitor module. The auxiliary DC output rating is listed below.

- Special applications: Two 24 VDC connections at 3.5 A maximum. Wiring is power-limited and supervised for ground faults and shorts. For a list of devices you can connect to special application circuits, refer to the control unit compatibility list.
- Regulated applications: One 24 VDC connection at 3.5 A maximum for all circuits except the regulated NAC and one 24 VDC connection at 1.0 A maximum for regulated NACs to 3-IDC8/4 modules only. Wiring is power-limited and supervised for ground faults and shorts.

Low Battery Cutoff (LBC)

To protect the power supply from damage caused by extreme discharge you can specify in the 4-CU how long the system runs on low battery power before automatically shutting down. In LBC mode, when the power supply indicates the battery has reached its lower limit (approximately 19.5 VDC) the system automatically shuts down, ensuring that an extreme load is not placed on individual power supplies.

Standby batteries

Standby batteries are used as secondary power for a control unit, as defined by NFPA 72. See Table 22 below for a list of standby batteries and backbox capacity.

Notes

- Each set (pair) of batteries must be connected to a separate battery charging circuit, either on a 3-PPS/M(-230) or on a 3-BBC/M(-230).
- All battery wiring must be the same length and wire gauge.
- All batteries connected to the same control unit must have the same ampere-hours rating, be from the same manufacturer, and have the same manufacturing date code.
- Standby batteries greater than 17 Ah and any additional batteries must be installed in an external battery cabinet. If an external battery cabinet is used to house standby batteries, the cabinet must be installed within 3-foot of the control unit and in the same room.

Table 22: Standby batteries and backbox capacities

Backbox model	Battery capacity	Battery voltage (VDC) Ah capacity @ 20 hr rate	Battery dimensions TH × CH × CW × CD [1]	Catalog number EST model
3-CAB5B	2	12 VDC 10 Ah	6.48 × 6.32 × 5.28 × 3.15	12V10A
3-CAB7B	2	12 VDC 18 Ah	6.56 × 6.50 × 7.13 × 3.0	12V17A
3-CAB14B	2	12 VDC 18 Ah	6.56 × 6.50 × 7.13 × 3.0	12V17A
3-CAB21B	2	12 VDC 18 Ah	6.56 × 6.50 × 7.13 × 3.0	12V17A
3-RCC7R	2	12 VDC 18 Ah or	6.56 × 6.50 × 7.13 × 3.0	12V17A
	2	12 VDC 26 Ah or	6.94 × 6.94 × 6.44 × 4.88	12V24A
	2	12 VDC 51 Ah or	7.88 × 7.88 × 10.13 × 5.16	12V50A
	2	12 VDC 65 Ah [2]	7.44 × 7.44 × 14.25 × 5.0	12V65A
3-RCC14R	2	12 VDC 18 Ah or	6.56 × 6.50 × 7.13 × 3.0	12V17A
	2	12 VDC 26 AH or	6.94 × 6.94 × 6.44 × 4.88	12V24A
	2	12 VDC 51 Ah or	7.88 × 7.88 × 10.13 × 5.16	12V50A
	2	12 VDC 65 Ah [3]	7.44 × 7.44 × 14.25 × 5.0	12V65A

Backbox model	Battery capacity	Battery voltage (VDC) Ah capacity @ 20 hr rate	Battery dimensions TH × CH × CW × CD [1]	Catalog number EST model
3-RCC21R	2	12 VDC 18 Ah or	6.56 × 6.50 × 7.13 × 3.0	12V17A
	2	12 VDC 26 Ah or	6.94 × 6.94 × 6.44 × 4.88	12V24A
	2	12 VDC 51 Ah or	7.88 × 7.88 × 10.13 × 5.16	12V50A
	2 or 4	12 VDC 65 Ah [4]	7.44 × 7.44 × 14.25 × 5.0	12V65A
BC-1(R)	2	12 VDC 18 Ah or	6.56 × 6.50 × 7.13 × 3.0	12V17A
	2	12 VDC 26 Ah or	6.94 × 6.94 × 6.44 × 4.88	12V24A
	2	12 VDC 40 Ah	6.69 × 6.69 × 7.75 × 6.50	12V40A
BC-2(R)	2	12 VDC 18 Ah	6.56 × 6.50 × 7.13 × 3.0	12V17A

[1] TH = terminal height, CH = case height, CW = case width, CD = case depth

[2] Requires one 3-BATS shelf in addition to dedicated battery space to support two 12V 65 AH standby batteries - no chassis space available in this configuration.

[3] Requires one 3-BATS shelf in addition to dedicated battery space to support two 12V 65 AH standby batteries - one chassis space available in this configuration.

[4] Requires one 3-BATS shelf in addition to dedicated battery space to support two 12V 65 AH standby batteries - two chassis spaces available in this configuration.

— or —

Requires three 3-BATS shelves in addition to dedicated battery space to support four 12V 65 AH standby batteries - no chassis space available in this configuration

3-MODCOM(P) Modem Communicator module

The 3-MODCOM(P) Modem Communicator is a chassis rail module that connects the EST4 system to a public switched telephone network or cellular network when used with a compatible cellular capture module. The MODCOM(P) has two 8-position modular jacks for connecting two loop-start lines.

Note: Refer to the control unit compatibility list for a list of compatible cellular capture modules. Refer to the *Cellular Capture Module Configuration Application Guide* (P/N 3102371-EN) for model-specific programming and wiring requirements.

The 3-MODCOM(P) can support 255 accounts and communicate with 80 receivers in any of the following protocols:

- Contact ID
- SIA DCS
- TAP (3-MODCOMP only)

The 3-MODCOM(P) configuration settings made in the 4-CU define the line properties, receiver attributes, account parameters, and transmission details.

Line types

The 3-MODCOM(P) can be configured as a one- or two-line dialer, a modem, or a modem and one- or two-line dialer.

Note: For UL listed or FM approved installations, you must configure the 3-MODCOM(P) as a two-line dialer. Both lines must have supervision (line-cut detection) selected.

Central Monitoring Station dialing

The 3-MODCOM(P) electronically dials receivers in the CMS using either pulse or tone dialing, as specified in the 4-CU. The module dials the stored CMS telephone number using the same digits that would be used if a person were dialing from the premises with an ordinary telephone.

Call timers

The 3-MODCOM(P) sends messages in order of their priority. Messages may include device and user ID information regarding events, such as openings, closings, alarms, and tamper or trouble events. The module waits for acknowledgement that each message sent has been received. Where necessary, the 3-MODCOM(P) can be configured to begin dialing without waiting for a dial tone. This option is used in areas where the telephone line has an absent or erratic dial tone.

Transmission process

The 3-MODCOM(P) includes features that provide an appreciable level of transmission integrity. Multiple telephone lines and multiple telephone numbers help to ensure that a call to the receiver gets through.

The 3-MODCOM(P) module sequences through the following basic steps to contact the CMS receiver.

1. The 3-MODCOM(P) seizes one of the telephone lines and puts the line on-hook for a minimum of 3 seconds.

This cuts off any ongoing call and disconnects the line from any telephone or dialing devices that are connected downstream.

Note: The module tries to select an unused line for its first two attempts.

2. The 3-MODCOM(P) takes the line off-hook and waits for a dial tone.

The Line 1 or Line 2 LED indicates.

If a dial tone is not received by the configured time, the module goes on-hook, increments the attempt counter, and continues to alternate lines and numbers until a dial tone is acquired.

If the 3-MODCOM(P) is configured with two telephone numbers and only one telephone line, it will make four attempts using the first telephone number, then four attempts using the second telephone number. This alternation of telephone numbers continues as needed until a connection is made or the configured number of dial attempts has been made.

Note: In areas where the telephone system has no dial tone, or where the dial tone is erratic, you can configure the 3-MODCOM(P) to dial without waiting for a dial tone.

3. The 3-MODCOM(P) dials the CMS using the programmed dialing mode and telephone number.
4. The 3-MODCOM(P) waits for a handshake message from the CMS, indicating that a connection has been established.

If a handshake is not received within 40 seconds the module puts the telephone line on-hook and waits for the configured period.

After the wait, processes 2 through 4 are repeated. If the module is still unable to contact the receiver, it seizes the other telephone line and repeats two attempts. If still unable to contact the receiver, it switches back to the first telephone line and attempts to contact the receiver using the secondary telephone number. If still unable to contact the receiver, the module continues to alternate lines and numbers until the configured maximum number of attempts have been reached. If the maximum number of attempts is reached, the module sends a trouble message to the CPU. The module retries the full number of attempts if another event is activated or makes one attempt if a configured period (Wait Time Between Attempts) expires.

5. When the call is completed, ringing is detected by the CMS dialer-receiver (DACR). The DACR goes off-hook and transmits a handshake.
6. If the handshake matches the desired transmit format, the 3-MODCOM(P) transmits, in the specified format, all premises event data.

The Line 1 or Line 2 LED flashes to indicate data is being transmitted.

7. The 3-MODCOM(P) waits for an acknowledgement and a shutdown signal from the CMS receiver, then puts the line on-hook, ending the call.

The Line 1 or Line 2 LED stops indicating.

Installing the 3-MODCOM(P)

When installing the 3-MODCOM(P), follow these general steps:

1. Identify suitable telephone company lines and services.
2. Install the 3-MODCOM(P) module on the electronics chassis.
3. Connect the 3-MODCOM(P) to telephone company lines.
4. Download configuration data from the 4-CU.
5. Make test transmissions to verify proper operation.

Refer to the *3-MODCOM and 3-MODCOMP Modem Communicator Installation Sheet* (P/N 387476-EN) for installation instructions, wiring, and specifications.

POTS line requirements

The 3-MODCOM(P) is supplied with two 7-foot cables that are 8-conductor, flat telephone cables, with 8-position modular plugs on both ends. One end of the cable plugs into the 3-MODCOM(P). The other end plugs into an RJ-31X jack. You must obtain the RJ-31X jack locally.

Note: For cellular capture module line requirements, refer to the *3-MODCOM and 3-MODCOMP Modem Communicator Installation Sheet* (P/N 387476-EN). For cellular capture module model-specific programming and wiring requirements, refer to the *Cellular Capture Module Configuration Application Guide* (P/N 3102371-EN).

WARNING: Failure to use an RJ-31X jack violates FCC and NFPA regulations. A telephone connected directly to the incoming telephone line without the proper use of the RJ-31X jack will cause a telephone company trouble when used and possibly prevent the dialer from getting through to the CMS receiver in an emergency.

- The RJ-31X jack must be used to connect each line of the 3-MODCOM(P) to the switched telephone network. One jack is required for each telephone line.

The 8-position jack has a special jumper between terminals 1 and 4, and 5 and 8. This jumper is in place when the plug is removed from the jack.

Removing the plug re-establishes connection to the premises telephones. Inserting the plug opens the jumper and connects the 3-MODCOM(P), which provides a series connection to the telephones.

Refer to the *3-MODCOM and 3-MODCOMP Modem Communicator Installation Sheet* (P/N 387476-EN) for a diagram of the jack wiring.

- 3-MODCOM(P) dialers can be used for most applications that use telephone lines. The exceptions are:
 - The central station telephone number cannot be dialed directly (using access numbers and area code where necessary) without operator interception of the call
 - Multiparty service (a party line) exists
 - Operator assistance is required to complete a telephone call and a foreign exchange cannot be introduced
 - Connection is not established within 38 seconds following completion of dialing
- The 3-MODCOM(P) dialer circuit is compatible with any switched telephone network that employs direct dialing (local) and Direct Distance Dialing (DDD), without operator interception of the call.
- Operator interception occurs in some areas where message billing is not completely automatic. Where operator interception is involved, you must obtain a foreign exchange (FX) connection from the central station exchange to the exchange serving the customer. The FX provides a local number for calling the central station without toll billing. A WATS or ground-start line connection must not be used for this purpose because the line cannot be supervised.
- The 3-MODCOM(P) includes a feature that prevents jamming by an incoming telephone call. The feature is based on a telephone service option referred to as called party disconnect. This option lets the receiver of a call disconnect by hanging up the telephone for a period of time, even if the caller stays on the line. The time

required for disconnect varies in different areas, but is usually between 18 and 90 seconds. Called party disconnect is available in most areas. To determine whether the called party disconnect control is available in the area to be served, consult the local telephone company.

In areas not having called party disconnect, the 3-MODCOM(P) module is vulnerable to jamming by an incoming call. To minimize the possibility of jamming, we recommend that the customer order a separate, unlisted number for exclusive use of the 3-MODCOM(P) module. The customer should keep this number confidential. In the case of the two-line dialer, two premises telephone numbers would have to be busied by incoming calls to jam the system.

Progressive anti-jamming measures would entail the use of one unlisted telephone number, or two unlisted numbers for maximum dialer integrity.

- The 3-MODCOM(P) must be connected to the incoming line ahead of all connected equipment on that line, but just behind the demarcation block. This puts the control unit telephone connection in series, assuring that all telephones, answering machines, and FAX machines are disconnected during dial-out to the CMS. This requirement is necessary so the 3-MODCOM(P) dialer circuit can seize the line for its exclusive use in the event of an alarm.
- Using a telephone line that is considered essential for conducting business at the site is not preferred. The dialer must be the first connection in line because it seizes the line and disconnects all other equipment when making a call. If connection will be made to a telephone company line that is also used for normal business purposes, advise the customer that the telephone service will be disrupted for a few minutes during the connection period.
- If the incoming lines to the protected premises involve a rotary telephone line arrangement, make the connection to the line having the highest number. This will create the least interference with business lines.
- In areas where the telephone company requires their own connector block be installed, it should be wired as per the USOC RJ-31X or RJ-38X configuration. (The RJ-38X configuration is identical to RJ-31X except for a jumper between 2 and 7 that is used in some residential applications but is not used by the 3-MODCOM(P).)
- When the 3-MODCOM(P) is configured as a two-line dialer module, two incoming lines must be used and connections must be made to each line.

Connecting the 3-MODCOM(P) to telephone company lines

Notes

- For the installation of a system in compliance with NFPA 72, the 3-MODCOM(P) must be connected to loop-start telephone lines. If the site has ground-start lines, two separate loop-start lines must be installed for the dialer.
- If the installation is for a certified life safety system or a burglar alarm system in compliance with NFPA 72, the telephone company line must be of the called party disconnect type (also called timed-release disconnect). This feature permits the communication module to seize the line and dial out, even when the telephone company line is in use.
- For connecting the 3-MODCOM(P) to a cellular capture module for communication over a digital cellular network, refer to the *3-MODCOM and 3-MODCOMP Modem Communicator Installation Sheet* (P/N 387476-EN). For cellular capture module model-specific programming and wiring requirements, refer to the *Cellular Capture Module Configuration Application Guide* (P/N 3102371-EN).

To determine the type of telephone company line:

1. Disconnect the line pair and connect the lines to a test meter.

If the line is equipped for loop-start, the meter should read 48 to 52 VDC between the lines.

If the line is equipped for ground-start, the meter will read 0 VDC between the lines, 48 to 52 VDC between one line and ground, and 0 VDC between the other line and ground.

To determine whether the telephone line supports called party disconnect:

1. Have someone telephone the premises from the outside.
2. Hang up the telephone that received the call, but have the individual who placed the call remain on the line.
3. After 40 seconds, pick up the called telephone again to determine whether the caller has been disconnected.

Downloading the configuration database

After installing the 3-MODCOM(P) and configuring its database in the 4-CU, download the database. Refer to "Downloading a project database" on page 43 for download instructions.

Note: For UL listed or FM approved installations, you must configure the 3-MODCOM(P) as a two-line dialer and both lines must have supervision (line-cut detection) selected.

Testing the transmission

The 4-CU provides a report that lists all CMS codes that can be transmitted from the 3-MODCOM(P). Give this report to the appropriate CMS.

After the CMS has programmed the central monitoring database, perform transmission tests as required by the local authority having jurisdiction (AHJ) and CMS.

Note: Transmission failures are latched at the control unit. You must reset the control unit in order to clear them.

Failover operation

To address the possibility of a communication failure or device trouble, you can create a failover operation in the 4-CU for the 3-MODCOM(P). Failover operates by enabling and disabling various accounts defined for the project. On detection of a fault or trouble, project rules disable accounts on the failed 3-MODCOM(P) and enable matching accounts on the backup 3-MODCOM(P).

Failover operation results in a system that is resistant to trouble arising from telephone lines, 3-MODCOM(P)s, or the 4-CPU module. The operation can be limited to a single control unit or can span two or more control units anywhere in a network.

In systems with a single 3-MODCOM(P) you can include a second 3-MODCOM(P) that acts as a redundant unit. In systems with two or more 3-MODCOM(P)s, you can program the system so that the units back up each other, while still handling their normal traffic.

Preliminary field wiring testing

It is recommended that you test all circuits before they are connected to the control unit components. Table 23 below indicates the recommended tests and acceptable test results.

Note: Individual devices are not checked as part of these tests. All equipment installed on field circuits must be individually tested to ensure proper operation when the system is running.

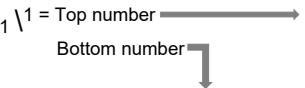
Table 23: Field wiring tests

Circuit type	Test
DC notification appliance circuit	1. Measure the resistance between conductors. The circuit resistance should be infinite if no devices are installed on the circuit. The circuit resistance should be approximately 15 kΩ when the polarized notification appliances and the end-of-line resistor are correctly installed. 2. Reverse the meter leads. The circuit resistance between conductors should read approximately 10 Ω to 20 Ω. If the resistance reading is still approximately the same value when the meter leads are reversed, one or more polarized devices are installed incorrectly. 3. Measure the resistance between each conductor and earth ground. The resistance should be infinite.
Audio notification appliance circuit	1. Measure the resistance between conductors. The circuit resistance between conductors should be infinite if no devices are installed on the circuit. The circuit resistance should be approximately 15 kΩ when the polarized notification appliances and the end-of-line resistor are correctly installed. 2. Reverse the meter leads. The circuit resistance between conductors should still read approximately 15 kΩ. 3. Measure the resistance between each conductor and earth ground. The circuit resistance between a conductors and earth ground should be infinite.
Signature SLCs	1. With field wiring disconnected, verify the continuity of each conductor. Each conductor should measure less than 38 Ω. 2. Measure the resistance between conductors. The circuit resistance between conductors should be infinite if no devices are connected to the circuit. The circuit resistance between conductors should be between approximately 18 kΩ (250 devices) and 4.5 MΩ (1 device) when devices are installed. 3. Measure the resistance between each conductor and earth ground. The circuit resistance between a conductors and earth ground should be infinite.
Traditional initiating device circuits	1. Verify the continuity of each conductor. 2. Measure the resistance between conductors. The circuit resistance between conductors should be infinite if no devices are connected to the circuit. The circuit resistance between conductors should be approximately 4.7 kΩ when devices are installed. 3. Measure the resistance between each conductor and earth ground. The circuit resistance between a conductors and earth ground should be infinite.
Earth Ground	Measure the resistance between the earth ground terminal and a convenient water pipe or electrical conduit. The circuit resistance should be less than 0.1 Ω .

Circuit compatibility

The circuit compatibility matrix in Table 25 on page 79 shows which circuit types may occupy the same conduit or be combined together, where permitted by code.

Table 24: Circuit compatibility matrix

Legend 1 = No restrictions 2 = Twisted-pair cable 3 = Twisted-pair, shielded cable 4 = CAT4 or better  = Do not combine Gray box = Power-limited circuit White box = Allowed in same conduit by manufacturer * 													Circuit wiring specifications	
	24 VDC power-limited	4-NET-TP twisted-pair circuit	4-NET-TP-HC twisted-pair circuit	4-NET-CAT Cat5x circuit	25 VRMS audio power-limited	25 VRMS audio nonpower-limited	70 VRMS audio power-limited	70 VRMS audio nonpower-limited	Signature signaling line circuit	Traditional two-wire IDC	4-AUDTELS firephone riser	AC mains, nonpower-limited	4-NET series fiber-optic circuit	
24 VDC power-limited														Size conductors per acceptable voltage drop and load
4-NET-TP twisted-pair circuit														Circuit resistance: 90 Ω max Circuit capacitance: 0.09 μF max.
4-NET-TP-HC twisted-pair circuit														Circuit resistance: 90 Ω max. Circuit capacitance: 0.3 μF max.
4-NET-CAT CAT5x circuit														CAT5e or greater
25 VRMS audio power-limited														Size conductors per acceptable voltage drop
25 VRMS audio nonpower-limited														Size conductors per acceptable voltage drop
70 VRMS audio power-limited														Size conductors per acceptable voltage drop
70 VRMS audio nonpower-limited														Size conductors per acceptable voltage drop
Signature signaling line circuit														Circuit resistance: 76 Ω max. Circuit capacitance: 0.5 μF max.
Traditional two-wire IDC														Circuit resistance: 50 Ω max.
4-AUDTELS firephone riser														Wire type: Twisted-pair, shielded cable
AC mains, nonpower-limited														
4-NET series fiber-optics circuit														Single mode and multimode available

* Install and wire devices in accordance with applicable national and local codes, ordinances, and regulations.

UL 864 notification appliance circuit signal synchronization

Table 25 below lists the installation requirements for systems that must meet UL 864 signal synchronization requirements.

Table 25: Installation requirements for UL 864 signal synchronization

Circuit	Installation requirements
3-SSDC1 and 3-SDDC1	Emergency evacuation signals are synchronized on a “per loop” basis when used with SIGA-CC1S or SIGA-MCC1S modules (personality code 25) or SIGA-CC2A or SIGA-MCC2A modules (personality code 19 or 25) and Genesis or Enhanced Integrity notification appliances. Emergency evacuation signals are synchronized on a “per module” basis when used with SIGA-CC1, SIGA-CC2A, SIGA-CC2A or SIGA-MCC2A modules (personality code 5), G1M or G1M-RM Genesis Signal Master modules, and Genesis notification appliances. Synchronization is supported between two daughter cards on the same 3-SDDC1 module. NACs on the individual daughter cards are synchronized as described above.
3-IDC8/4	Emergency evacuation signals are synchronized on a “per circuit” basis when you use G1M or G1M-RM Genesis Signal Master modules, and Genesis notification appliances. To silence audible appliances separately, use two NAC channels from the 3-IDC8/4 to provide separate audible and visible power to the NAC. In this configuration, the signal silence function operates as defined in your project. See Figure 10 on page 80 for typical wiring. Configure the audible notification appliances for temporal or steady output as desired.
SIGA-CC1, SIGA-MCC1, SIGA-CC1S, and SIGA-MCC1S	Signature CC1 modules do not generate temporal signals, they simply turn the NAC circuit on or off. You must configure the notification appliances for temporal or steady output as desired.
3-ZA20, 3-ZA40, and ZA95 amplifier modules	Emergency evacuation signals are synchronized on a “per circuit” basis when you use G1M or G1M-RM Genesis Signal Master modules, and Genesis notification appliances.
G1M and G1M-RM	The G1M and G1M-RM Genesis Signal Master modules can be used to synchronize NACs consisting of Genesis appliances. They can also be used to synchronize mixed NACs consisting of Genesis and Enhanced Integrity appliances, but the first appliance must be a Genesis device and the Genesis Signal Master module must be mounted on this device. G1M and G1M-RM Genesis Signal Master modules cannot be used to synchronize NACs consisting of Enhanced Integrity appliances.

Typical circuits

The circuit diagrams provided in this section use the term *zone* to indicate notification zones as defined in UL 864 (an area that is covered by notification appliances that are activated simultaneously).

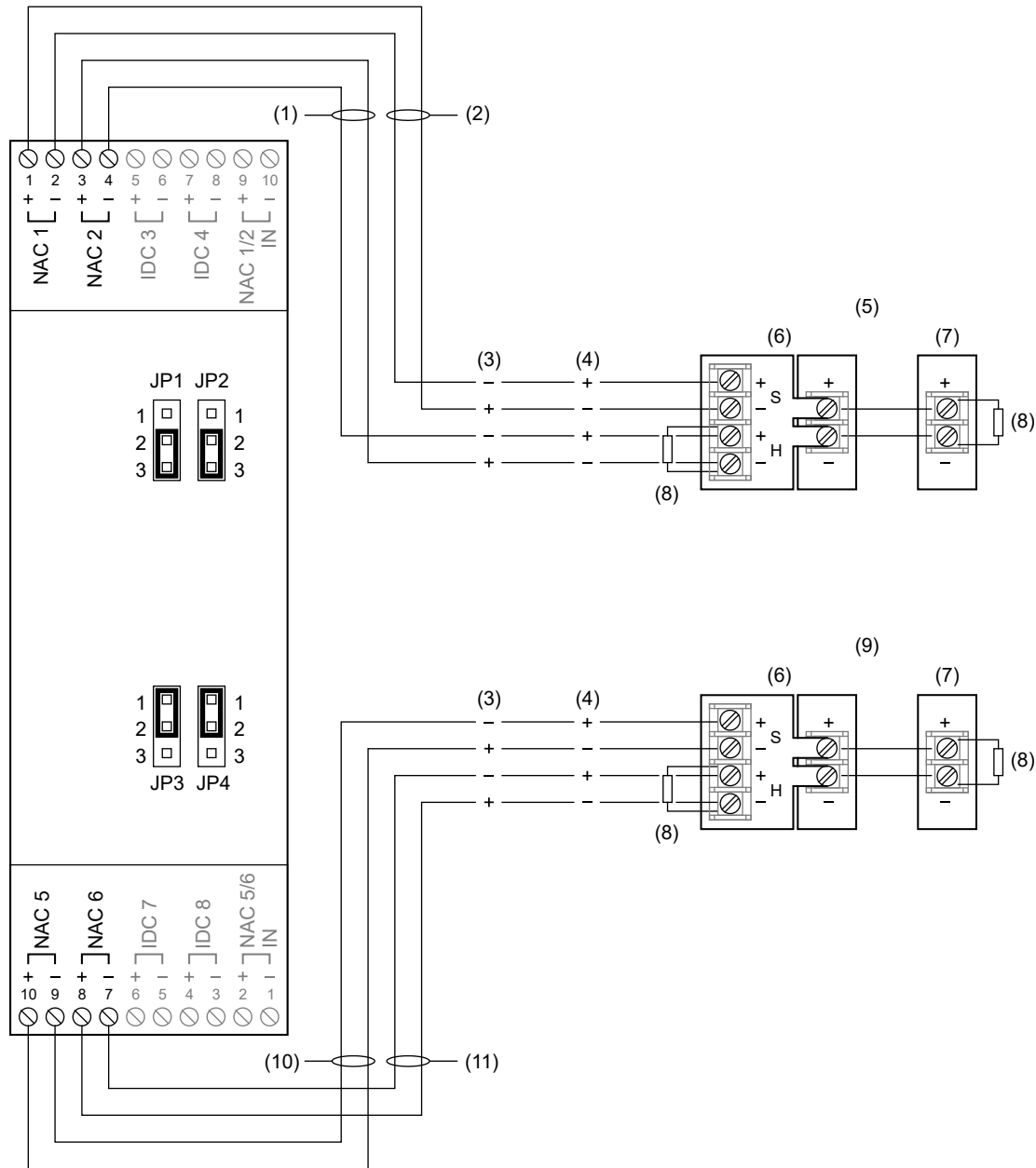
Figure 10 on page 80 shows a typical application of the 3-IDC8/4 module for supporting two notification zones. In this example, power is being supplied from the chassis rail and the jumpers (JP1 through JP4) on the module are set accordingly. Refer to the *3-IDC8/4 Traditional Zone I/O Module Installation Sheet* (P/N 270492-EN) for additional information on the 3-IDC8/4.

It is also possible to create a similar application that uses external power, supplied to NAC 1/2 IN and NAC 5/6 IN on the 3-IDC8/4. Refer to the *3-IDC8/4 Installation Sheet* (P/N 270492-EN) for wiring details and jumper settings.

In Figure 10, both zones are configured with separate NAC circuits for audible and visible appliances. NAC 1 and NAC 5 are programmed as visible device types, and NAC 2 and NAC 6 as audible device types. This means that the signal silence function can be configured to silence only the horns.

Separating the visible and audible devices is optional and may not be required for your project. Refer to the Genesis Signal Master installation sheets for additional configurations and wiring details.

Figure 10: Typical 3-IDC8/4 card NAC wiring

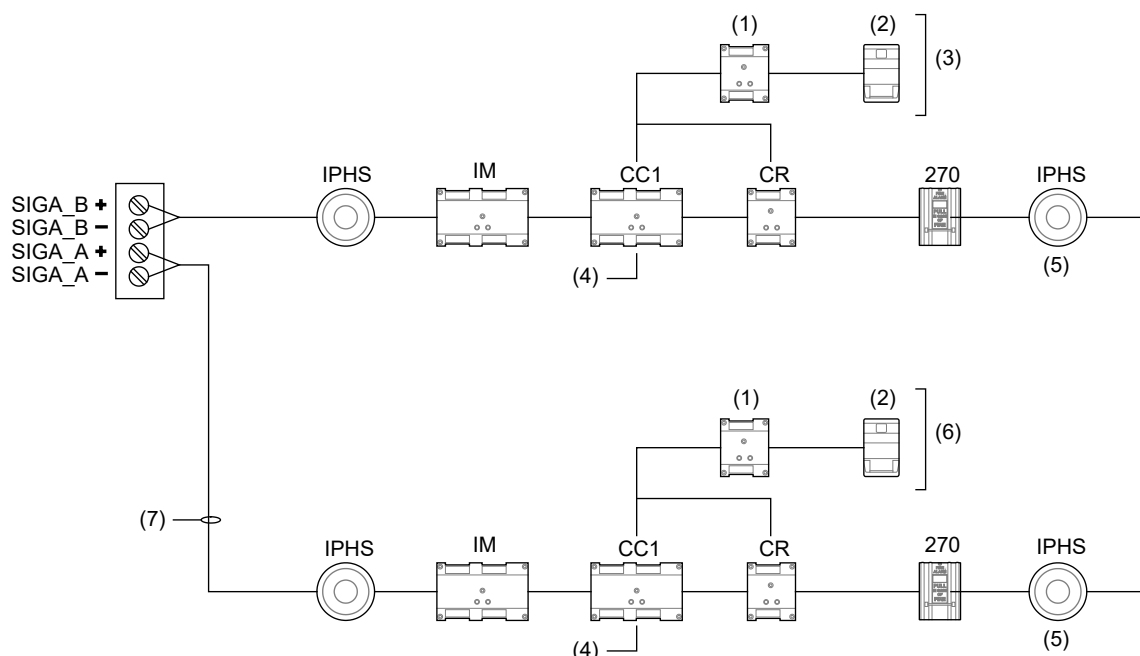


- (1) NAC 2, Device type: Audible, Label: Zone_1_Horns
- (2) NAC 1, Device type: Visible, Label: Zone_1_Strobes
- (3) Normal
- (4) Active
- (5) Zone 1
- (6) Temporal horn/strobe and G1M module

- (7) Temporal horn/strobe
- (8) 15 kΩ EOLR
- (9) Zone 2
- (10) NAC 5, Device type: Visible, Label: Zone_2_Strobes
- (11) NAC 6, Device type: Audible, Label: Zone_2_Horns

Figure 11 below shows a Signature SLC wired as Class A or X for notification circuit signal synchronization.

Figure 11: Class A Signature wiring for signal synchronization



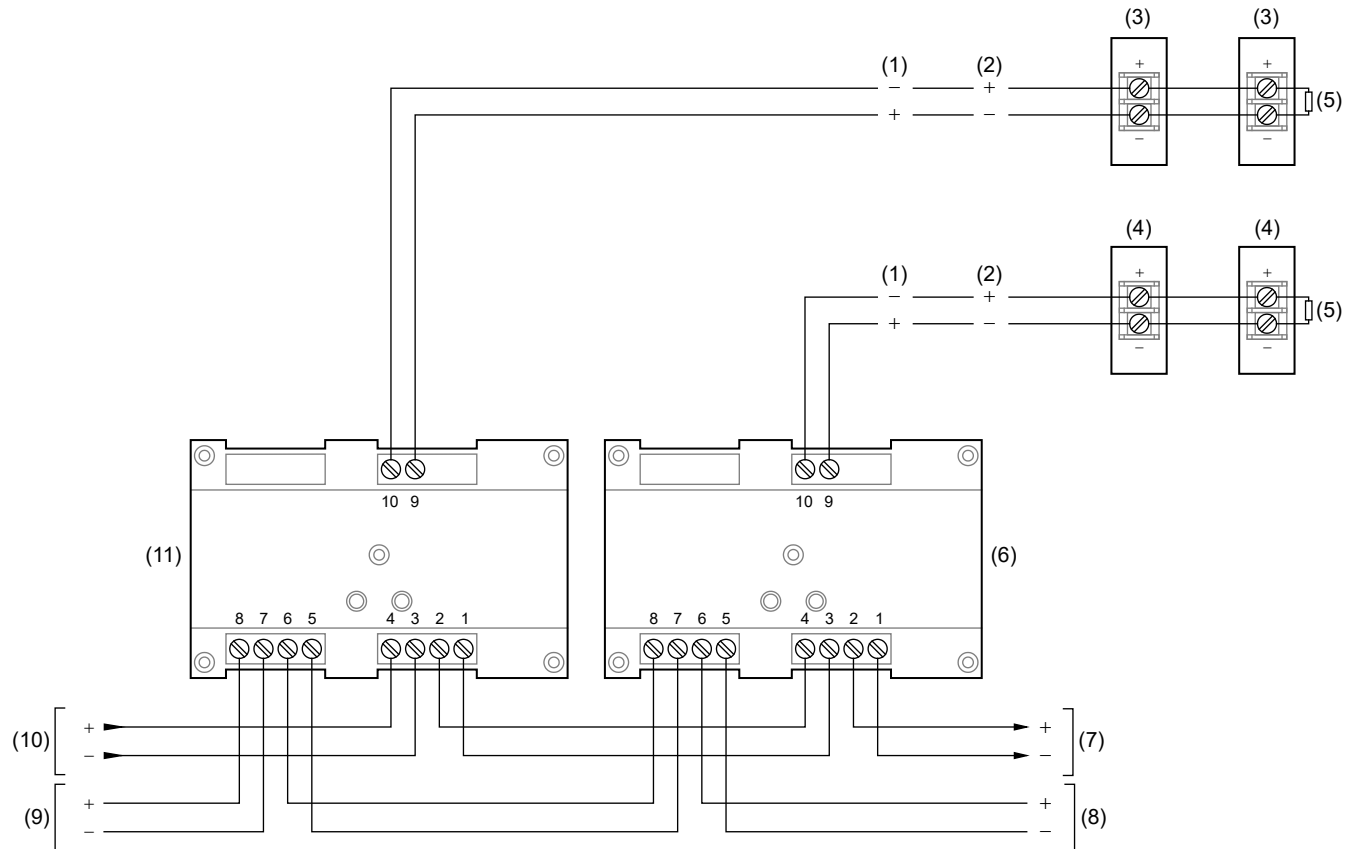
- | | |
|--------------------------|----------------------------|
| (1) Sync module | (5) Isolator base |
| (2) Temporal horn/strobe | (6) Notification zone 2 |
| (3) Notification zone 1 | (7) Class A / X (required) |
| (4) AUX riser | |

Notes

- SLC pathways that provide signaling outputs to more than one fire notification zone must prevent a single break, single ground, or wire-to-wire fault from adversely affecting more than one zone.
Exception: SLC pathways installed for survivability from attack by fire in accordance with NFPA 72.
- For Class A wiring, isolator modules and isolator detector bases are required to prevent wire-to-wire shorts on the signaling line circuit wiring from adversely affecting other segments of the loop. Do not install more than 50 addressable devices between isolators, per NFPA 72.
- For Class X wiring, un-isolated devices must be mounted in a cabinet with isolators on the incoming and outgoing wiring.
- For Class X wiring, isolator modules and isolator detector bases are required to prevent wire-to-wire shorts on the signaling line circuit wiring from adversely affecting any devices of the loop.

Figure 12 on page 82 shows two NACs on a Signature data loop. Each NAC is controlled by a SIGA-CC1S module, one for audible appliances and one for visible appliances. The SIGA-CC1S modules provide signal synchronization for both NACs.

As in Figure 10 on page 80, this configuration allows the audible appliances to be silenced independently of the visible appliances. This operation is optional and may or may not be required for your project.

Figure 12: Typical SIGA-CC1S NAC wiring

- | | |
|---------------------------------|---|
| (1) Normal | (7) Data out to next device |
| (2) Active | (8) AUX riser to next device or riser monitor |
| (3) Temporal horn | (9) AUX riser from previous device |
| (4) Strobe | (10) Data in from previous device |
| (5) 47 kΩ EOLR | (11) CC1S: |
| (6) CC1S: | Device type: Audible |
| Device type: Visible | Personality: (5) Riser Selector |
| Personality: (5) Riser Selector | Label: ZONE_1_HORNS |
| Label: ZONE_1_STROBES | |

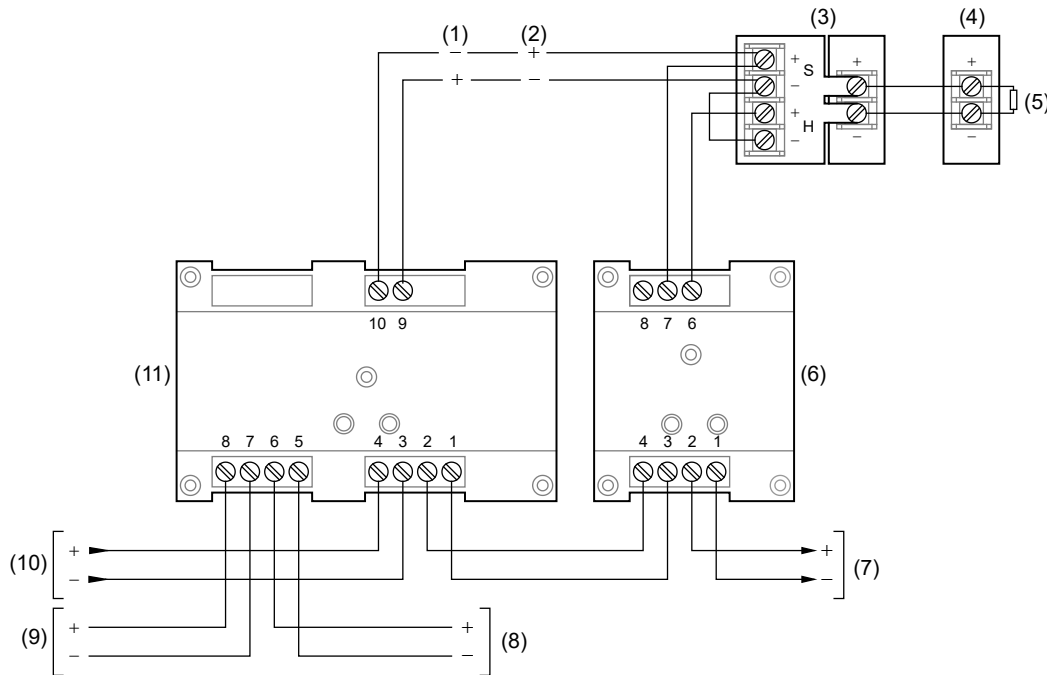
Figure 13 below shows a single SIGA-CC1 switching a NAC on or off. The G1M module provides signal synchronization for the temporal horn/strobe appliances.

As in earlier examples, this circuit allows for independent silencing of the audible appliances. This operation is provided by the SIGA-CR module, which opens or closes the circuit between S+ and H+ on the G1M module. In this case, you must program the operation of the SIGA-CR in the 4-CU. The CU settings for signal silence operation will not determine the operation of the audible appliances in this NAC.

Note also that this application could be implemented with a SIGA-CC1S module. The SIGA-CC1S provides signal synchronization compatible with the operation of the G1M module.

The advantage to using a SIGA-CC1S module is that the NAC would then be synchronized with other NACs on the Signature SLC.

Figure 13: Typical SIGA-CC1 NAC wiring



- | | |
|---|---|
| (1) Normal | (7) Data out to next device |
| (2) Active | (8) AUX riser to next device or riser monitor |
| (3) Temporal horn/strobe and G1M module | (9) AUX riser from previous device |
| (4) Temporal horn/strobe | (10) Data in from previous device |
| (5) 47 kΩ EOLR | (11) CC1: |
| (6) CR: | Device type: Visible |
| Device type: Dry Contact | Personality: (5) Riser Selector |
| Personality: (8) Dry Contact | Label: ZONE_1_NAC |
| Label: ZONE_1_NAC_SILENCE | |

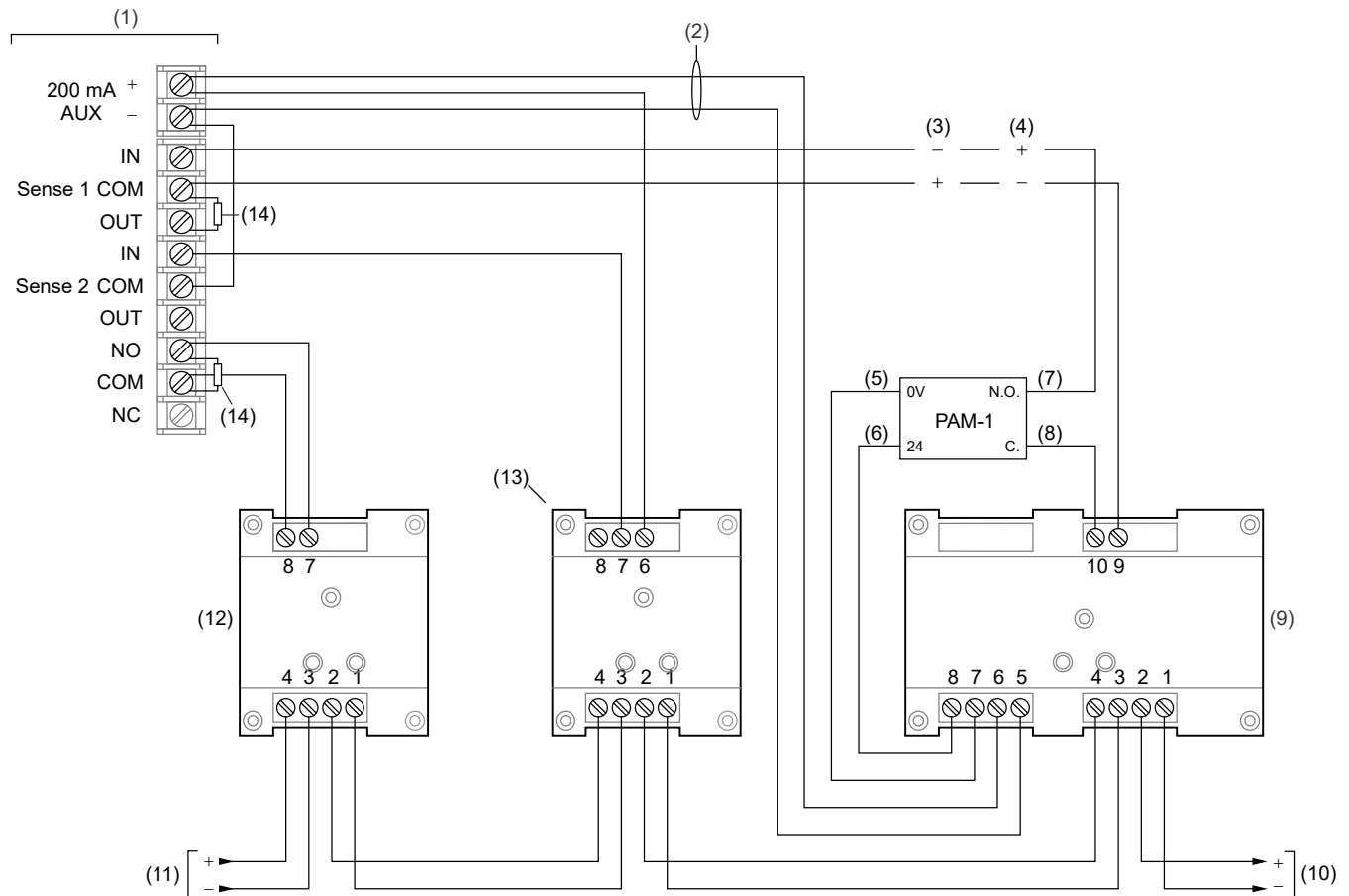
Figure 14 on page 84 shows an auxiliary/booster power supply being used to power the NAC, to provide synchronization and horn silence capability. Because the auxiliary/booster supply has the ability to silence the horn circuit, this application can be created using only the Signature wiring.

The SIGA-CT1 module monitors the power supply for AC failure. The SIGA-CR module signals the power supply to turn the horns on or off. The SIGA-CC1 module signals the power supply when the system goes into alarm, turning the NAC on.

Notes

- The power supply can only synchronize the notification appliances to which it is connected. If you need to synchronize several similar NACs on the same Signature loop, you can use a SIGA-CC1S module in place of the SIGA-CC1.
- All wiring is supervised and power-limited unless otherwise noted.
- Install a PAM1 or equivalent listed relay only when you are required to supervise the 200 mA AUX circuit wiring.
- Configure Sense 1 and Sense 2 operation for Genesis Master mode and NAC operation for “continuous.” See the documentation received with the auxiliary/booster supply for details.
- Use a CC1S if you want to maintain signal synchronization across multiple auxiliary/booster supplies on the same Signature loop.

Figure 14: Using an auxiliary/booster supply for horn silence capability with two wires



- (1) Auxiliary/booster supply
 (2) Not supervised
 (3) Normal
 (4) Active
 (5) White
 (6) Red
 (7) Orange
 (8) Blue
 (9) CC1 or CC1S:
 Device type: Visible
 Personality: (5) Riser Selector
 Text 1: REMOTE_SUPPLY
 Text 2: HRNS_&_STROBES

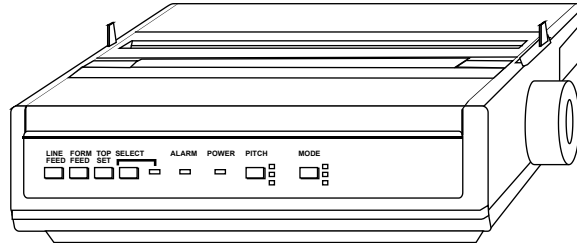
- (10) Data out to next device
 (11) Data in from previous device
 (12) CT1:
 Device type: AC Power
 Personality: (3) Active B
 Text 1: REMOTE_SUPPLY
 Text 2: HRNS_&_STROBES
 (13) CR:
 Device type: NSCommonAlarmOutput
 Personality: (8) Dry Contact
 Text 1: REMOTE_SUPPLY
 Text 2: HRNS_&_STROBES
 (14) 47 kΩ EOLR

Connecting a PT-1S printer

The PT-1S serial printer can be connected to an EST4 control unit by way of the USB port on the CPU or the 4-USBHUB to provide a printout of system events such as status changes, active events, and reports. The printer is an 80-character line width, freestanding printer that uses standard form feed paper.

The EST4 Printer Service communications service is used to allow the transmission of system events through the EST4 control unit to the printer. For more about the Printer Service and other communication services see “Communication services” on page 105.

Figure 15: PT-1S serial printer



Printer manufacturer’s documentation is provided with the printer that contains detailed instructions for assembling the printer, installing the print drivers, configuring the serial interface, and testing the printer.

Printer configuration settings

Configuring the printer’s serial interface

Configure the printer’s serial interface settings using printer controls. Instructions for configuring the settings are provided in the printer manufacturer’s documentation and can also be found in *PT-1S Fire Alarm Printer Installation Sheet* (P/N 3100989-EN).

The printer’s serial interface default and permissible settings are shown in Table 26 below.

Table 26: PT-1S serial interface settings

Item	Default settings	Permissible settings
Parity	None	None, Even
Serial Data 7/8 bits	8 bits	8 bits
Protocol	Ready/Busy	Ready/Busy [1], X-ON/X-OFF [2]
Diagnostic Test	No	No
Busy Line	SSD–	DTR
Baud rate	9600 bps	4800
DSR Signal	Valid	Invalid
DTR Signal	Ready on Power Up	Ready on Power Up
Busy Time	200 ms	200 ms

[1] For supervised printers

[2] For unsupervised printers

Configuring the Printer Service port setting

Allowing an EST4 control unit to send events to a connected PT-1S printer is managed by the EST4 Printer Service communications service that is configured in the 4-CU. The USB Port property in the Printer Service specifies the 4-USBHUB port to which the printer is physically connected.

To configure the Printer Service port setting:

1. From the 4-CU:

In the Navigation pane, on the Project Tree select the node that contains the Printer Service.

On the Node Configuration tab, in the Communication Services list select the Printer Service.

On the Edit Properties tab, for the USB Port property enter one of the following:

2-1.1 for a printer connected directly to the CPU

— or —

2-1.5 or 2-1.6 (dependent on port used) for a printer connected through a 4-USBHUB

Wiring the PT-1S

Wire the printer in accordance with applicable national and local codes, ordinances, and regulations. For more information about the PT-1S, see *PT-1S Fire Alarm Printer Installation Sheet* (P/N 3100989-EN).

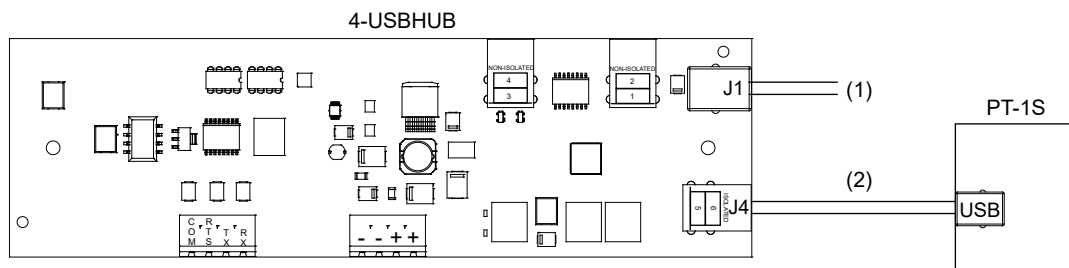
WARNING: Electrocution hazard. To avoid personal injury or death from electrocution, remove all sources of power and allow stored energy to discharge before installing or removing equipment.

Notes

- USB printer connections are power-limited and may or not be supervised, depending on the control unit.
- Locate supervised USB printers in the same room as the equipment to which they connect.
- Locate unsupervised USB printers in the same room and within 20 ft. (6.1 m) of the equipment to which they connect. Enclose wiring in conduit or equivalent protection against mechanical injury.

For printers located further than 20 ft of the equipment, use

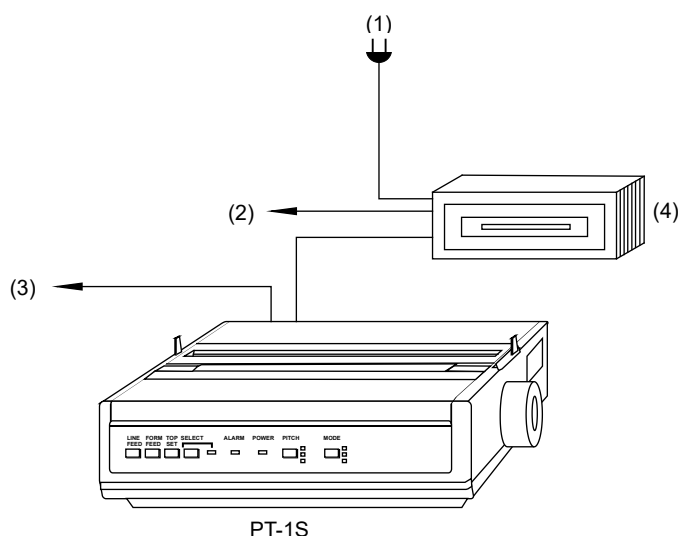
- Printers that incorporate isolation can be connected to any EST4 CPU, Type A port.
- Printers that do not incorporate isolation and cause a ground fault when connected to the CPU, require a customer supplied isolation dongle. This configuration is not UL/ULC approved.
- For more information about the 4-USBHUB, see *4-USBHUB Multi-port USB Hub Module Installation Sheet* (P/N 3102281-EN).
- If the printer is required to operate during brownout conditions and AC power failures, install a UL Listed (UTRZ) uninterruptible power supply that can maintain printer operating voltage for at least 24 hours. See “System printer standby power supply” on page 87.
- For ULC printer applications in which a UPS is used as the emergency power supply, install this product and the UPS in a lockable enclosure in accordance with CAN/ULC-S527.

Figure 16: Typical printer wiring using the 4-USBHUB

- (1) From a 4-CPU, 4-ANNCPU, 4-NET-AD, or a 4-FWAL-1,-2, -3, -4 module that supplies primary 24 VDC and communication to the 4-USBHUB at this connection. A USB 3.0 cable (supplied with the 4-USBHUB) is required
- (2) USB 2.0 A to B cable (customer supplied)

System printer standby power supply

For a printer using an uninterruptible power supply the UPS should be UL Listed for fire protection (UTRZ) and provide 120 VAC at 50/60 Hz for at least 24 hours.

Figure 17: Printer to uninterruptible power supply wiring

- (1) 120 VAC, 14 A circuit
- (2) UPS trouble contact monitor circuit
- (3) To the 4-USBHUB J4 connection
- (4) Uninterruptible power supply

Connecting a CDR-3 for coded tone output

The CDR-3 Bell Coder module can be connected to the 4-USBHUB module to provide a coded or march time tone to the audio system. For more information see *CDR-3 Bell Coder Installation Sheet* (P/N 310023-EN).

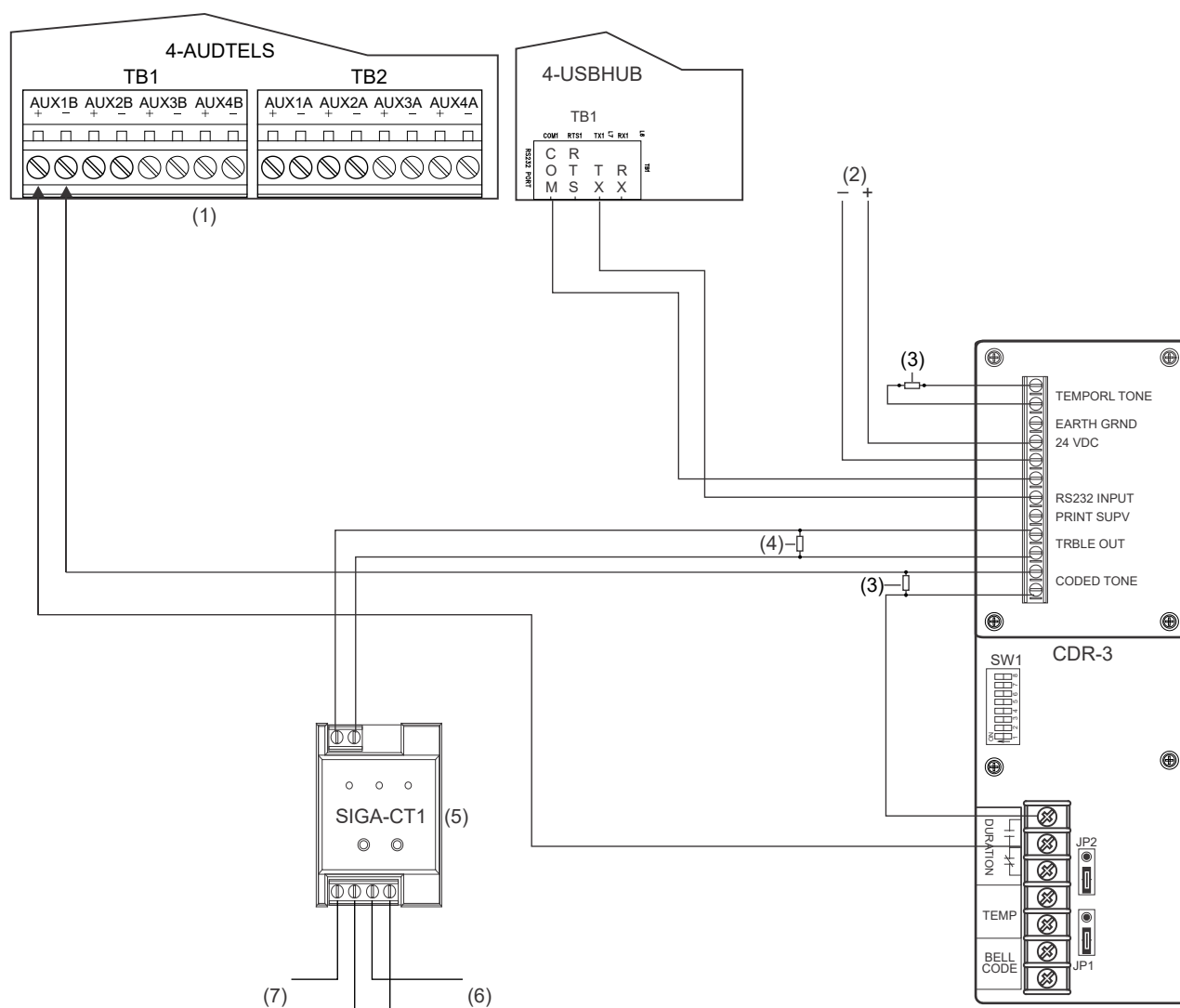
The EST4 Coder Service communications service is used to allow communication through the EST4 control unit to the bell coder. The Coder Service properties are configured in the 4-CU. For more about the Coder Service and other communication services see “Communication services” on page 105.

Wiring the CDR-3

Wire the CDR-3 in accordance with applicable national and local codes, ordinances, and regulations. For more information about the

WARNING: Electrocution hazard. To avoid personal injury or death from electrocution, remove all sources of power and allow stored energy to discharge before installing or removing equipment.

Figure 18: CDR-3 wiring



Legend

- (1) 4-AUDTELS mounted on a 4-CPU
- (2) From a regulated 24 VDC power supply that is UL/ULC listed for fire protective signaling systems
- (3) 10 kΩ EOLR (P/N EOL-10)
- (4) 47 kΩ EOLR (P/N EOL-47)
- (5) Configure the SIGA-CT1 as a non-latching input circuit with the Monitor device type
- (6) SLC OUT
- (7) SLC IN

Notes (see additional figure notes on next page)

- Wiring is supervised and power-limited.
- Maintain 0.25 in. (6 mm) separation between power-limited and nonpower-limited wiring at all times.

- For more information about the 4-USBHUB, see *4-USBHUB Multi-port USB Hub Module Installation Sheet* (P/N 3102281-EN).
- For more information see *4-AUDTELS Audio IO and Telephone Riser Source Module Installation Sheet* (P/N 3102279-EN).
- For more information see *SIGA-CT1 Single Input Module Installation Sheet* (P/N 387021P-EN).

Adjusting amplifier output levels

The audio system works best if the prerecorded tones and messages have roughly the same volume, or amplitude. The process of establishing a common maximum amplitude is sometimes called *normalizing*.

We suggest that you normalize your tones and messages to a maximum amplitude of 1 V peak-to-peak, or an average of 0.7 VRMS using a sound editor, and then import them into the 4-CU.

Before starting the process of adjusting the amplifier output levels, you need the following:

- A project database that contains a 1 kHz tone WAV file
- An installed control-display module with a switch programmed to turn on the 1 kHz tone
- An RMS voltmeter (Fluke 83 or equivalent)

To adjust the output level:

1. Disconnect the field wiring to all the zoned audio amplifier modules in the control unit.
2. Place an RMS meter across the amplifier's TB2 NAC/B+ and NAC/B– terminals.
3. Press the control-display module switch programmed to turn on the 1 kHz tone.
4. Adjust the amplifier's gain pot until the RMS meter displays the configured output level (25 or 70 VRMS).
5. Reconnect the amplifier's field wiring.
6. Press the control-display module switch again and verify that the output level remains the same.
7. If the output level needs readjusted, repeat steps 1 through 6.

Chapter 5

Supplementary applications

Summary

This chapter describes supplementary applications that can be controlled or monitored locally to expand your life safety system capabilities.

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Notification Control Areas

NCA programming allows the programmer to group nodes that are installed throughout the premises. Typically, there is one NCA per building with one primary control location and multiple remote control locations. NCAs provide a structure for the distributed audio system and are used to define the audio subsystem's coverage area.

A basic NCA has one or more command centers for controlling the operations of the members of the NCA. In some installations multiple NCAs can be managed by one or more CCSs.

Notification Control Area, Command Center, and Request Grant Deny logic devices are configured in the 4-CU that can allow implementation of single or multiple command centers for a protected premises. The logic group functionalities are independent operations that can have an audio subsystem. At any given time, one command center in an NCA can be granted control so that a user can initiate fire responses.

For more information about audio applications, components, and examples see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

Notification Control Area logics

NCA logic devices are used to define the audio subsystem's coverage area. NCA programming defines the device nodes within a premises, and allows the programmer to group nodes that are installed throughout the premises. Typically, there is one NCA per building with one primary control location and multiple remote control locations in each.

A basic NCA has one or more autonomous control units and local operator consoles for managing the NCA. Multiple NCAs can be managed from one of more command centers, also referred to as central control stations (CCSs).

Refer to *4-CU Help* for information on configuring an NCA logic device.

Command Center logics

Command center logic devices are used to define control locations within a notification control area. Command center programming works in association with NCAs and RGD logic devices to act as a Central Control Station. The control center resides in a notification control area and is the control unit used to request, grant, or deny control of an NCA, whether the one in which it resides or another NCA.

Refer to *4-CU Help* for information on configuring a Command Center logic device.

Request Grant Deny logics

Request Grant Deny logics is a programming method working in association with NCAs and command centers to arbitrate demands for control of audio communications in the life safety system. Request to control a system and permission to grant or deny the request can be made from the 4-LCD display module user interface, or from programmed control-display module switches.

Refer to *4-CU Help* for information on configuring a Request Grant Deny logic device.

Port blocking of audio traffic

Port blocking allows users to specify whether audio messages from other NCAs on a network connection should be blocked. This is helpful where the user wants to page to a remote NCA from a CCS that is different from the NCA which the CCS belongs to, preventing the audio from traversing to the CCS's own NCA.

Refer to *4-CU Help* for information on configuring Ethernet (eth0, eth1) and USB (usb0, usb1) block mode.

Mass notification system (MNS) emergency communication

Emergency communication/mass notification systems typically perform notification operations in the following order of priority:

1. Special suppression activation
2. MNS signaling
3. Fire life safety signaling
4. Other property notification

Signals from the MNS system must have the capability of overriding all other notification signals, except suppression, but whether MNS signals have priority over fire alarm signals must be determined by risk analysis.

Once MNS operation is activated, a trained operator must maintain manual control over the system at all times.

MNS paging

In MNS signaling mode, the 4-AUDTELS redirects the page signal source to the Emergency channel. Any amplifier connected to either the Page or Emergency audio channels will receive the page. The Emergency (MNS) audio channel can be configured in the 4-CU for MNS signaling to be higher priority than EVAC and Alert.

For information on making live MNS announcements see “4-MIC live mass notification signaling” on page 100.

MNS Email

When configured for mass notification, the 4-FWAL series firewall module is provided with connections via a primary port to a UL/ULC Listed DRMNS email server.

Transmit MNS events and fire events to separate accounts so that the appropriate authorities are dispatched.

When transmitting MNS events, turn off the auto generate CID feature on accounts intended for MNS transmission.

The overall MNS system with a 4-FWAL series firewall module (with email capability) shall have a maximum of five email addresses that can connect to a UL/ULC Listed DRMNS email server.

MNS IP Dialer, DACT, and CCM

Transmit MNS events and fire events to separate accounts so that the appropriate authorities are dispatched.

When employing MNS service, you may use nonstandard CID codes. However, you must notify the central station of the nonstandard MNS CID codes so that the appropriate authorities can be informed.

Do not use a cellular connection to transmit MNS events.

Security and data protection levels

Table 27: UL 2572 second edition and ULC-S576 security and data protection levels

UL 2572 requirement	Security levels	Level employed
Approved Security Function for FIPS PUB 140-2	N/A	No encryption employed
Communication Security	0 to 3	0 [1][4]
Stored Data Security	0 to 2	0 [1]
Access Control Security	0 to 3	0 or 1 [2]
Physical Security Access	0 to 3	1 [3]
Audit Control	1 to 3	Not provided

[1] Meets communication security level 1 only if 4-FWAL series firewall module employs a dedicated MNS connection, where only MNS events are permitted to be transmitted. All other non-MNS events must use another connection.

[2] Security is employed. Access by way of the front panel is controlled by a four-digit PIN (1000 PIN combinations). Access automatically times out after reaching the configured user time out period. The default setting is five minutes. Security level 1.

[3] Security is employed. Access is restricted by a locked cabinet door.

[4] In order to transmit MNS email, an EST4 email capable 4-FWAL series firewall module must be installed in the same secured room as the UL/ULC listed DRMNS email server.

Audio subsystem

The audio subsystem consists of a variety of signal sources and amplifiers, that can add emergency voice and alarm communications to your EST4 life safety system. Live paging, playing prerecorded audio, support for 1 VRMS audio input and output, and telephone conversations between master and remote firefighter telephone handsets are supported in the EST4 system. The audio system is programmed using the 4-CU. For more programming information, refer to the *4-CU Help*.

Note: For more information about audio applications, components, and examples see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

Audio subsystem components

Note: For more information about audio subsystem components in an EST4 network with NCAs, see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

4-AUDTELS Audio IO and Telephone Riser Source Module: The 4-AUDTELS provides provides up to four line-level inputs or outputs and one firephone riser, and supports one 4-MIC connection and one 4-FT connection

For more information about the 4-AUDTELS, see *4-AUDTELS Audio IO and Telephone Riser Source Module Installation Sheet* (P/N 3102284-EN).

4-ANNAUDTEL Annunciator Audio Telephone Interface Module: The 4-ANNAUDTEL mounts on the 4-ANNCPU card and supports one 4-MIC connection and one 4-FT connection.

For more information about the 4-ANNAUDTEL, see *4-ANNAUDTEL Annunciator Audio Telephone Interface Module Installation Sheet* (P/N 3102325-EN).

4-MIC Paging Microphone: The 4-MIC provides paging capability local to the node, within an NCA, or over an entire network. The 4-MIC assembly incorporates push-to-talk (PTT) LED that illuminates when the PTT button is pressed, and a microphone audio level LED that illuminates red if the voice level is too high.

For more information about the 4-MIC, For more information see *4-MIC Paging Microphone Installation Sheet* (P/N 3102266-EN) for technical specifications.

4-FT Firefighter Telephone: The 4-FT firefighter telephone handset provides the capability of making phone connections between a remote phone handset on a riser (typically uses SIGA-CC1(S) modules) and firefighter's telephones 4-FT. An EST4 node and a NCA can be connected to a *single* telephone riser. The maximum simultaneous phone connections is six, five remote plus one master.

- Five phone off-hook limit: The 4-AUDTELS telephone riser supports up to five telephones off-hook in addition to the master handset at the control unit at any one time. The flexibility of the EST4 system permits any number of phones to be wired on a single phone circuit, as long as they are not all used simultaneously.
- One phone per circuit: The advantages of installing a single firefighter phone station or jack on a SIGA-CC1 module (personality code 6) are numerous. The system provides complete control and annunciation phone/circuit. Installing a single phone on a circuit permits the operator to immediately identify the exact location of the calling party. Because the 4-AUDTELS will only permit five circuits to be connected simultaneously, the maximum number of off-hook handsets can never be exceeded. Should a branch telephone circuit be damaged during a fire, the fault will not affect other phone circuits. When there is only one phone per circuit, troubleshooting of faults is simplified.

For more information about the 4-FT, see *4-FT FireFighter Telephone Master Handset Installation Sheet* (P/N 3102267-EN) for technical specifications.

Central processor modules

4-CPU Central Processor Module: The 4-CPU provides connections for a firefighter telephone, a paging microphone, and up to four 1 VRMS audio channels that can be programmed for either input or output operation (requires optional 4-AUDTELS).

For more information about the 4-CPU, For more information see *4-CPU Central Processor Module Installation Sheet* (P/N 3102258-EN).

4-ANNCPU Annunciator Central Processor Module: The 4-ANNCPU is a control element for the LCD display module user interface and control-display modules connected to it.

For more information about the 4-ANNCPU, see *4-ANNCPU Annunciator Central Processor Module Installation Sheet* (P/N 3102264-EN).

Table 28: 4-ANNCPU accessories

Model	Description
4-ANNAUDTEL	The 4-ANNAUDTEL Annunciator Audio Telephone Interface Module is used with the 4-ANNCPU to process information from the 4-FT Firefighter Telephone Master Handset and 4-MIC Paging Microphone.
4-CABL0504	Used to connect the 4-ANNCPU to a 4-LCDAUDTEL or a 4-LCD when the 4-ANNCPU is installed in a 4-2ANNMT, 4-4ANNMT, or 4-6ANNMT backbox.
4-CABL0509	Used to connect the 4-ANNCPU to a 4-LCDAUDTEL when mounted horizontally on a 4-MPLT plate
4-CABL0542	Used to Connect the 4-ANNCPU to the UI rail when no 4-LCDAUDTEL or a 4-LCD is installed on a 4-CAB8D(R), 4-CAB16D(R), 4-CAB24D(R), or 4-CAB16DL(R) inner door.

For systems using audio, a 4-ANNAUDTEL accessory module is mounted on the top, front side of the 4-ANNCPU to control audio functions for a firefighter telephone and a paging microphone.

Refer to Table 29 on page 96 to determine when a 4-ANNAUDTEL or 4-AUDTELS is required for audio applications.

Table 29: Audio interface card matrix

Installed CPU model	4-MIC / 4-FT installed next to:	Required audio module	
		4-ANNAUDTEL Annunciator Audio Telephone Interface	4-AUDTELS Audio I/O and Telephone Riser
4-CPU	4-CPU		X
4-ANNCPU	4-ANNCPU	X	
4-CPU and 4-ANNCPU	4-CPU		X
4-CPU and 4-ANNCPU	4-ANNCPU	X	
4-CPU and 4-ANNCPU	4-CPU and requires a telephone riser		X
4-CPU and 4-ANNCPU	4-ANNCPU and requires a telephone riser	X	X

Amplifiers

Amplifiers are designed to feed a single audio zone and provide a 24 VDC visible notification appliance circuit. Amplifier modules are available in 20, 40, and 95 watt versions, with supervised Class B or Class A audio output circuits. The amplifier is configurable for either 25 or 70 VRMS output. An independent supervised Class B or Class A, 24 VDC, 3.5 A notification appliance circuit (NAC) is also provided on the 20 and 40 watt amplifiers to drive notification appliances. In addition, automatic backup amplifiers can be added on a switched common backup configuration.

Each audio power amplifier has an integral demultiplexer, making the eight audio channels available to the amplifier's input, as directed by the system programming. Each amplifier also contains circuitry that handles routine signal processing functions such as channel priority.

The amplifier's output is a dedicated, supervised, 25 or 70 VRMS speaker circuit, which covers one audio zone in the protected facility. For more information about audio applications, see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

Table 30: Amplifier fault conditions

Possible fault condition	Amplifier operation
Amplifier loses communication with CPU	If the control unit is configured for stand-alone operation, the amplifier automatically switches to the EVAC channel and outputs its 1 kHz temporal tone when the control unit detects an alarm. If the control unit is not configured for stand-alone operation, the amplifier will not output any signal.
Control unit loses communication with network data riser	Amplifier switches to the EVAC channel only in response to the local control unit's programming uses the default EVAC message.
Amplifier plays a backup tone when an audio message plays	A short on the 4-CPU voltage controlled oscillator (VCO) causes the amplifier to play a backup tone when the audio message plays.
Control unit loses communication with network	Amplifier switches to the EVAC channel in response to the system programming. For EVAC, the amplifier uses its 1 kHz temporal tone. For Alert, the amplifier uses its 1 kHz 20 bps tone.

Audio types

EST4 supports live paging using the 4-MIC, 4-FT, and 1 VRMS input. Prerecorded audio can be played over multiple logical channels. The prerecorded audio will not be distributed over the network, instead each node will have a copy of the audio messages and the playing is synchronized across nodes configured to play the same messages. The system also provides two-way, wired telephone communications capability between a 4-FT handset and multiple portable telephone riser handsets.

Refer to *EST4 Network and Audio Application Guide* (P/N 3102306-EN) for details.

Audio signal priority

Each audio channel type has a priority level associated with it. Table 31 below shows the priority of each channel. The Page channel will only go active when the microphone PTT button is pressed or when a page from a 4-FT is selected.

For more information see *EST4 Network and Audio Application Guide* (P/N 3102306-EN).

Table 31: Audio signal priority levels

Channel	Priority
Page	1
Emergency [1]	2
EVAC	3
Alert	4
Auxiliary	5
General	6
Other	6

[1] The priority in Table 31 above applies if the user chooses Mass Notification to have higher priority than Fire in the project configuration. If fire is chosen, Evacuate and Alert will have higher priority than Emergency.

Paging functions

When an alarm is detected, the EVAC and Alert message channels are selected for the amplifiers in the appropriate areas in the facility, as directed by the system rules. If a specific evacuation message has been programmed to play in response to the alarm, the proper channel and message are selected and sent to the amplifiers. In the case of paging, since the manual controls should always override the automatic controls, if the user chooses to page to Evacuate or Alert, the prerecorded message will stop playing until the paging is done. The prerecorded message will then continue in a synchronized manner.

In addition to standard paging functions (Page to Evacuation, Page to Alert, All Call, and All Call Minus), EST4 also includes Page to Other and Page to Emergency. Table 32 on page 98 shows the different paging functions and to which channel type the system will page when selecting any of the paging options.

Table 32: Paging functions

Paging type	Channels
Page to All call [1]	Paging + Evacuate + Alert + Auxiliary + General
Page to All call minus [1]	Paging + Auxiliary + General
Page to Evacuate	Paging + Evacuate
Page to Alert	Paging + Alert
Page to Emergency (Mass Notification)	Paging + Emergency
Page to Other	Paging + Other

[1] This will activate idle audio amplifier and pre-amp circuits to the Paging channel.

The *All Call* function is used to broadcast live-voice instructions to the entire facility at the same time. When the All Call paging function is activated, the page signal is connected to every audio channel to ensure that the amplifier receives the signal regardless of which channel is selected. Note that all channels receive the same signal. Any amplifier on the system, regardless of the audio channel selected, will receive the page. All amplifiers that were previously idle will power up and receive the page.

The *All Call Minus* function is used to send a page to all areas *not* automatically receiving the EVAC or alert signals. In high rise applications, All Call Minus is an effective way to quickly select stairwells. Activating the All Call Minus Switch causes the 4-AUDTELS to enter the All Call Minus function. In this function, the 4-AUDTELS redirects the page signal source to the auxiliary and general channels. The Page, Auxiliary and General audio channels all receive the page signal. Any amplifier connected to the Page, Auxiliary or General audio channels will receive the page. Any amplifiers that were previously idle will power up and receive the page. The EVAC and Alert channels are connected to their respective signal sources, as in the normal mode.

The *Page to EVAC* function is used to send a page to the areas automatically receiving the evacuation signal. Activating the EVAC switch causes the 4-AUDTELS/ANNAUDTEL to enter the Page to EVAC function. In this function, the 4-AUDTELS redirects the page signal source to the EVAC channel. The page and EVAC audio channels both receive the page signal. Any amplifier connected to either the Page or the EVAC audio channels will receive the page, as controlled by the 4-CPU. The Alert, Auxiliary, and General channels are connected to their respective signal sources, as in the normal mode.

The *Page to Alert* function is used to send a page to the areas automatically receiving the alert signal. Activating the Alert switch causes the 4-AUDTELS to enter the Page to Alert function. In this function, the 4-AUDTELS redirects the page signal source to the Alert channel. The page and alert audio channels both receive the page signal. Any amplifier connected to either the page or alert audio channels will receive the page. The EVAC, Auxiliary, and General channels are connected to their respective signal sources, as in the normal mode.

The *Page to Emergency* function is used for Mass Notification systems. Activating the Page to Emergency switch causes the 4-AUDTELS to enter the Page to Emergency function. In this function, the 4-AUDTELS redirects the page signal source to the Emergency channel. Any amplifier connected to either the page or Emergency audio channels will receive the page.

The *Page to Other* function is a quick way to reach people in stairwells and elevators. Activating the Page to Other switch causes the 4-AUDTELS to enter the Page to Other function. In this function, the 4-AUDTELS redirects the page signal source to the Other channel. Any amplifier connected to either the page or Other audio channels will receive the page.

Automatic messaging

One of the features of the 4-AUDTELS is the method used to monitor the integrity of the digital audio system. When an audio messaging system is configured, default audio messages are recorded for the EVAC and Alert channels. The text of default messages should be generic in nature, and should not include location specific instructions. When the system is in the normal condition, the 4-AUDTELS continuously transmits default

messages over the network. The zone amplifiers use the default messages to verify their operational integrity, as well as the integrity of the riser wiring.

When an alarm is detected, the EVAC and Alert message channels are selected by the amplifiers in the appropriate areas in the facility, as directed by the system rules. If a specific evacuation message has been programmed to play in response to the alarm, it is sent out over the EVAC channel. Location specific evacuation messages contain information and instructions that should only be used for a specific alarm location. Should a second alarm from another location be received, the evacuation message playing as a result of the first alarm may not be appropriate for the second alarm.

Note: In the event of conflicting messaging instructions caused by multiple alarm events, the system will play the default evacuation message whenever two or more different messages are requested at the same time on the evacuation channel. By reverting to the generic default evacuation message in multiple alarm location scenarios, no one can be misdirected by the wrong message. Default messages also play during alarms when no location specific message has been requested.

4-MIC microphone live voice announcements

The basic tasks in responding to an emergency event are:

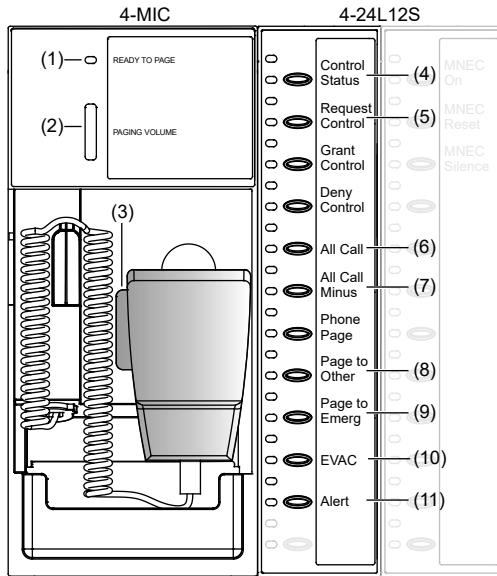
- Use All Call mode to announce the arrival of the fire department, making any necessary announcements.
- Use Page to Evac mode to reinforce the evacuation of the occupants in areas receiving the evacuation signal.
- Use Page to Alert mode to notify the areas not in immediate danger to prepare to evacuate, or that evacuating people may enter their safe area.
- Use All Call Minus mode to make announcements to areas of the facility not receiving the EVAC or Alert signals, as required. Stairwells are typical areas to use the All Call Minus page mode.

Notes

- All paging functions require a 4-24L series control-display module.
- Refer to *4-CU Help* for information on writing rules for the required control-display module indicators and switches.
- Your system may be configured differently than described below.

To make a live voice announcement:

1. For a control unit programmed as a command center, request for control of the EST4 system may be required.
 - Using the operator touch screen:
On the Action bar tap Request Control, and then tap the Control Request list button. The information screen will show the control status. When ready, proceed to step 2.
 - Using a 4-24L series control-display module (see Figure 19 on page 100):
If the top Control Status LED (4) is steady, proceed to step 2. If not, press Request Control (5); its LED flashes while requesting control. When the top Control Status LED is steady, proceed to step 2.

Figure 19: 4-MIC voice announcements using control-display module switches and indicators

Note: 4-24L12S indicators and switches functions require 4-CU rules programming.

2. Remove the microphone from its hook, and then press the areas to receive the page.
 - All Call (6): Broadcasts to all areas whether or not they are sounding a signal.
 - All Call Minus (7): Broadcasts to areas not currently sounding the evacuation signal or the alert signal.
 - Page to Other (8): Broadcasts live voice messages to areas such as stairwells and elevators.
 - Page to Emerg (9): Broadcasts live voice messages to areas defined for mass notification announcements.
 - EVAC (10): Broadcasts to areas currently sounding the evacuation signal.
 - Alert (11): Broadcasts to areas configured to areas currently sounding the alert signal.
3. Press and hold the microphone push-to-talk switch (PTT) (3). The Ready to Page LED (1) flashes during a preannouncement tone. When the LED is on steady, begin the announcement. Adjust your voice level so that the red LED on the Paging Volume meter (2) *does not* illuminate.
4. Release the PTT switch when the announcement is finished and return the microphone to its hook.

4-MIC live mass notification signaling

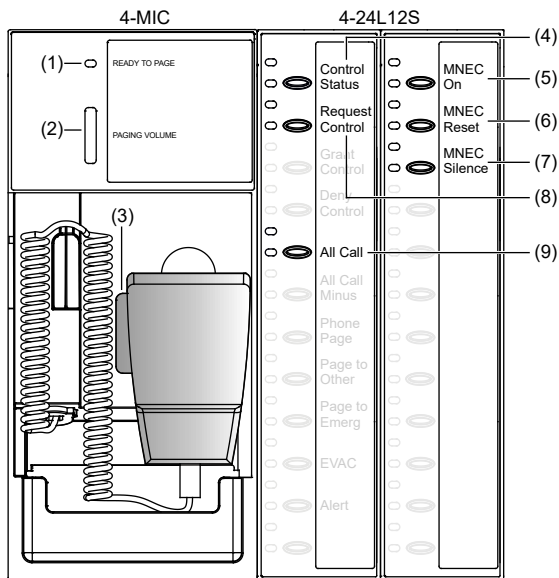
In MNS signaling mode, the 4-AUDTELS redirects the page signal source to the Emergency channel. Any amplifier connected to either the Page or Emergency audio channels will receive the page.

Notes

- All paging functions require a 4-24L series control-display module.
- Refer to *4-CU Help* for information on writing rules for the required control-display module indicators and switches.
- Your system may be configured differently than described below.

To make a live mass notification announcement:

- For a control unit programmed as a command center, request for control of the EST4 system may be required.
 - Using the operator touch screen:
On the Action bar tap Request Control, and then tap the Control Request list button. The information screen will show the control status. When ready, proceed to step 2.
 - Using a 4-24L series control-display module (see Figure 20 below):
If the top Control Status LED (4) is steady, proceed to step 2. If not, press Request Control (8); its LED flashes while requesting control. When the top Control Status LED is steady, proceed to step 2.

Figure 20: 4-MIC live MNS signaling using control-display module switches and indicators

Note: 4-24L12S indicators and switches functions require 4-CU rules programming.

- Press the MNEC On button (5) on the control-display module, to activate the function.
- Remove the microphone from its hook, and then press the All Call button (9) on the control-display module.
- Press and hold the microphone push-to-talk switch (PTT) (3). The Ready to Page LED (1) flashes during a preannouncement tone. When the LED is on steady, begin the announcement. Adjust your voice level so that the red LED on the Paging Volume meter (2) *does not* illuminate.
- Release the PTT switch when the announcement is finished and return the microphone to its hook.

To silence mass notification signals:

- Press MNEC Silence (figure item 7 above). The bottom LED indicates silence is active.

Note: This action only silences mass notification signals. To silence fire alarm signals, see “Silencing alarm signals” on page 45.

To reset the system when mass notification operation is active:

- Press MNEC Reset (figure item 6 above) on the control-display module. The bottom LED indicates the reset is active.

Note: This action only resets the system when mass notification operation is active. To reset the system for fire events, see “Resetting the life safety system” on page 46.

4-FT firefighter telephone live voice announcements

The EST4 system includes a phone page feature that permits individuals with access to a remote firefighter telephone to make announcements over the emergency voice/alarm communications system.

Notes

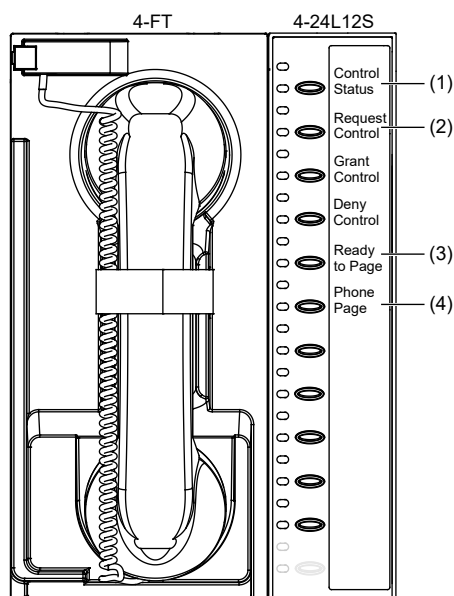
- All 4-FT live voice announcement functions require a 4-24L series control-display module.
- Refer to *4-CU Help* for information on writing rules for the required control-display module indicators and switches.
- Your system may be configured differently than described below.

To make a live voice announcement:

1. For a control unit programmed as a command center, request for control of the EST4 system may be required. Use a 4-24L series control-display module to request control. See Figure 21 below.

If the top Control Status LED (1) is steady, proceed to step 2. If not, press Request Control (2); its LED flashes while requesting control. When the top Control Status LED is steady, proceed to step 2.

Figure 21: 4-FT live voice announcements using control-display module switches and indicators



Note: 4-24L12S indicators and switches functions require 4-CU rules programming.

2. Remove the telephone handset from its hook.
3. The Ready to Page LED (item 3) on the control-display module flashes during a preannouncement tone. When the LED is on steady, press the Phone Page button (figure item 4 above) and begin your announcement.
4. When finished, place the telephone handset on its hook.

4-FT firefighter telephone two-way communication

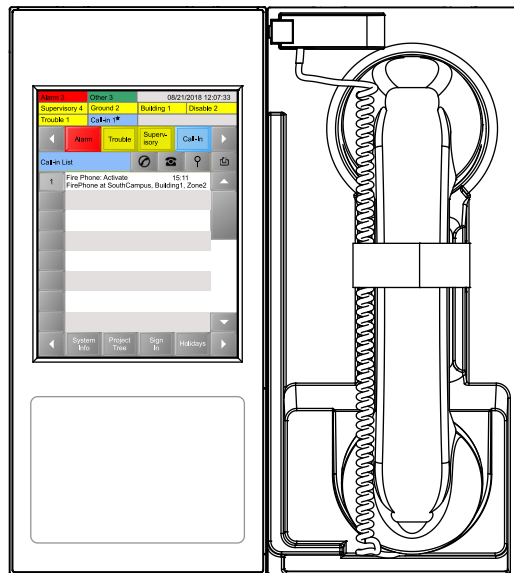
The 4-FT firefighter telephone provides an analog telephone riser for totally independent two-way communications between the fire command station and firefighter telephone stations / jack telephones installed at strategic locations throughout the protected facility.

Answering incoming calls

An incoming call is initiated when a 4-FT firefighter telephone is taken off-hook or plugged into a telephone jack, generating a visual and audible incoming call signal at the fire command station. The caller hears a tone until the handset is connected to the system. The fire command station operator manually connects the incoming phone call to the phone riser to complete the call. The firefighter telephone can also be used as a page source.

The incoming call sounds the control unit operator alert signal and displays the Call-in List button on the LCD screen that, when tapped, displays a Call-in list of all incoming calls in the order in which they were received. Call-in icons also appear that serve as operator controls for connecting and disconnecting calls.

Figure 22: 4-FT installed with a 4-LCDAUDEL Audio and Firefighter Telephone Control Display Module



To answer a call:

1. For a control unit programmed as a command center, request for control of the EST4 system may be required. Use a 4-24L series control-display module to request control. See Figure 21 on page 102.
If the top Control Status LED (1) is steady, proceed to step 2. If not, press Request Control (2); its LED flashes while requesting control. When the top Control Status LED is steady, proceed to step 2.
2. From the operator touch screen, on the List bar, tap Call-in. See Figure 22 above.
3. In the Call-in List, locate the desired call, and then tap the call-in event number.
4. Tap the connect button  to connect the call to the master (local) telephone handset. The event's status changes to Connected.
5. Remove the telephone from its hook and communicate with the caller.
6. To add another call-in location to the conversation, in the Call-in list tap the location's event number, and then tap Connect again.

Up to five phone circuits can be connected, in addition to the master telephone.

Disconnecting calls

When the caller is ready to hang up or remove a phone from its jack, disconnect the call.

To disconnect a call:

1. On the Call in List, locate the call to be disconnected, and then tap the call-in event's number.
2. Tap the disconnect button  to disconnect the call. The event status changes to Disconnected.

When the *remote telephone handset* is hung up or removed from the phone jack, the call will be removed from the Call-in list.

3. Place the *master telephone handset* on its hook. The Call-in event indicator and Call-in list no longer appear on the LCD screen.

Testing the audio system integrity

Activate prerecorded audio messages to test the integrity of the digital audio system.

To test the audio system integrity:

1. On the Action bar, tap Project Tree.
2. Expand the project tree by tapping the expand button , and then locate an audio device.
3. Tap the device, and then on the Command bar tap Activate.
4. When finished, on the Command bar tap Restore.

Remote annunciation

4-ANN series remote annunciator control units provide remote annunciation options for your life safety network, and other applications. Depending on your remote annunciator control unit configuration the system can provide emergency annunciation, system monitoring and control, local microphone paging, local telephone communications, LED annunciation, and switch controls.

Standard components

4-ANNCPU annunciator control unit central processor unit

The 4-ANNCPU is the control element for the LCD display module user interface and control-display modules installed in a remote annunciator. The module processes all control information from switches on the displays installed within the cabinet as well as processing the data received from the network for display. The module ships with the inner and outer door assembly, and installs on the inner door frame assembly. For more information see *4-ANNCPU Annunciator Central Processor Module Installation Sheet* (P/N 3102264-EN).

4-LCDANN remote annunciator LCD display module user interface

The 4-LCDANN Remote Annunciator LCD Display Module provides a 4.5 × 3.4 in. VGA color touch screen display capable of displaying 24 lines of 40 characters for system monitoring and includes operator controls and indicators. The LCD display module ships with the inner and outer door assembly, and installs on the inner door frame assembly. For more information see *4-LCDANN Remote Annunciator LCD Display Module Installation Sheet* (P/N 3102260-EN).

Graphic annunciation

A 4-CPUGRPH Graphic Annunciator Central Processor Module is the control element for graphic driver interface modules installed in graphic annunciators. The 4-CPUGRPH consists of a CPU module, a Graphic Interface Card, and a mounting plate.

The 4- CPUGRPH module provides:

- One USB host port
- One USB device port
- Connections for two SFP (small form-factor pluggable) network adapters
- Connection to 3-EVDVR drivers

For more information about see *4-CPUGRPH Graphic Annunciator Central Processor Module Installation Sheet* (P/N 3102346-EN)

Communication services

EST4 communications services may be used to allow transmission of system events through an EST4 control unit to external devices such as central monitoring stations, computers, printers, and smartphones. Communication occurs over Ethernet or fiber SFP input/output network adapters using communication services described in this section.

Table 33 below shows the maximum number of services that can be configured for a node type. The services can be any combination of different services but cannot exceed the maximum number.

Table 33: Communication services and node types

Service	Chassis rail modules						
	4-CPU (maximum services: 2)	4-ANNCPU (maximum services: 2)	4-CPUGRPH (maximum services: 2)	4-FWAL1 [1] (maximum services: 8)	4-FWAL2 [1] (maximum services: 8)	4-FWAL3 [1] (maximum services: 8)	4-FWAL4 [1] (maximum services: 8)
Coder	1	1	1	1	1	1	1
Printer [2]	2	2	2	2	2	2	2
ECP	0	0	0	8	8	8	8
Email	0	0	0	0	0	8	8
IP Receiver	0	0	0	0	8	0	8
Web Browser	0	0	0	0	8	8	8

[1] Uses the External Communications Protocol XML (ECPxml) that requires customer licensing to allow communication with the control unit.

[2] Two printers are supported when a 4-USBHUB is used. One printer is supported when a direct connection is used. One printer is supported when a Coder Service is also used.

Coder Service

The Coder Service allows an EST4 control unit to send events to a connected CDR-3 bell coder by way of RS-232 on a 4-USBHUB. For information about wiring the CDR-3 see “Connecting a CDR-3 for coded tone output” on page 87.

The CDR-3 connection is supported on EST4 CPUs and firewall modules.

One coder service can be configured for the following nodes:

- 4-CPU. For more information about the 4-CPU, see *4-CPU Central Processor Module Installation Sheet* (P/N 3102258-EN).
- 4-ANNCPU. For more information about the 4-ANNCPU, see *4-ANNCPU Annunciator Central Processor Module Installation Sheet* (P/N 3102264-EN).
- 4-NET-AD. For more information about the 4-NET-AD, see *4-NET-AD Network Adder Module Installation Sheet* (P/N 3102258-EN).
- 4-CPUGRPH. For more information about the 4-CPUGRPH, see *4-CPUGRPH Graphic Annunciator Central Processor Module Installation Sheet* (P/N 3102346-EN).
- 4-FWAL1, 4-FWAL2, 4-FWAL3, 4-FWAL4. For more information about the 4-FWAL, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

Printer Service

The Printer Service allows an EST4 control unit to send events to a connected PT-1S printer by way of a direct connection to the USB port on the CPU or the USB port on a 4-USBHUB. For more information about wiring the PT-1S printer, see “Connecting a PT-1S printer” on page 85.

The printer connection is supported on EST4 CPUs and firewall modules.

Up to two printer services can be configured for the following nodes when a 4-USBHUB is used (only one printer is supported for a direct connection):

- 4-CPU. For more information about the 4-CPU, see *4-CPU Central Processor Module Installation Sheet* (P/N 3102258-EN).
- 4-ANNCPU. For more information about the 4-ANNCPU, see *4-ANNCPU Annunciator Central Processor Module Installation Sheet* (P/N 3102264-EN).
- 4-NET-AD. For more information about the 4-NET-AD, see *4-NET-AD Network Adder Module Installation Sheet* (P/N 3102258-EN).
- 4-CPUGRPH. For more information about the 4-CPUGRPH, see *4-CPUGRPH Graphic Annunciator Central Processor Module Installation Sheet* (P/N 3102346-EN).
- 4-FWAL1, 4-FWAL2, 4-FWAL3, 4-FWAL4. For more information about the 4-FWAL, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

ECP Service

The ECP Service allows external third-party applications to connect to an EST4 control unit by way of a TCP port. A network of control units can support up to 10 ECP connections for third-party communication.

ECP connections are supported on 4-FWAL1, 4-FWAL2, 4-FWAL3, 4-FWAL4 firewall modules. For more information about 4-FWAL modules, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

Up to eight ECP services can be configured for the firewall module. Refer to *4-CU Help* for instructions on configuring the Web Browser services.

Before starting the configuration process in the 4-CU, you need to gather configuration settings from the remote interface administrator. To guide you through what information to gather, download the *EST4 Network Services Configuration Worksheet* (P/N 3102627-EN) from the My-Eddie website. To access the website, enter www.edwardsfiresafety.com in your web browser, click Login to log on to the My-Eddie website, and then locate the worksheet in Media Type > Installation Sheets.

ECP over IP communication with FireWorks

The control unit can be configured to communicate system events to a FireWorks UL5W7 or later workstation that supports IP connections using the Ethernet connection (Gateway Type III). The user-configured External Command Protocol service allows transmission of system events over IP.

Email Service

The Email Service allows an EST4 control unit to communicate system events to email or text addresses through SMTP servers. Email messages allow communication of multiple or single system events; text messages allow single system events.

For text messaging, a text message is routed to the recipient's wireless service provider phone number email address through a configured SMTP server. The wireless service provider then transmits the message to the recipient's text account. Check with the wireless provider for their carrier gateway address and any additional carrier requirements.

The email connection is supported on 4-FWAL3 and 4-FWAL4 firewall modules. The firewall module supports eight email services, one account per email service (up to 20 email addresses per email account). For more information about 4-FWAL modules, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN). For instructions on configuring Email services, refer to *4-CU Help*.

Before starting the configuration process in the 4-CU, you need to gather configuration settings from the local IT administrator. If no local IT administrator or local SMTP server is available and a public email service will be used, you need to contact the Internet service provider for configuration settings. To guide you through what information to gather, download the *EST4 Network Services Configuration Worksheet* (P/N 3102627-EN) from the My-Eddie website. To access the website, enter www.edwardsfiresafety.com in your web browser, click Login to log on to the My-Eddie website, and then locate the worksheet in Media Type > Installation Sheets.

EST4 Trust Level property

The SMTP server Trust Level property is configured on the Email Service properties tab in the 4-CU. You can choose to authenticate the server using a digital certificate or choose no authentication. Because some commercial services may not be included in the "Root Certification" option, it is recommended that you choose "No Authentication".

Note: The "EST4 Certificate" option in the 4-CU Trust Level property is a future feature.

To configure the Trust Level:

1. From the 4-CU:

On the Home tab, in the Navigation pane, on the Project Tree select the node that has the email service.

On the Node Configuration tab, click the Communication Services tab, select the email service, and then click the Edit Properties tab.

In the Communication Service Configuration group, Trust Level property, select No authentication.

EST4 email supervision property

EST4 supervises the email server periodically by default. You can disable the feature on the Email Service properties tab in the 4-CU. Even when disabled the system will still report a fault on the LCD if an email fails to send.

To turn off email supervision:

1. From the 4-CU:

On the Home tab, in the Navigation pane, on the Project Tree select the node that has the email service.

On the Node Configuration tab, click the Communication Services tab, select the email service, and then click the Edit Properties tab.

In the Communication Service Configuration group, Supervision Interval property, set the interval to time to 00:00:00:00 (dd:hh:mm:ss).

Note: The system will still indicate a fault when supervision has been disabled and an email fails to send.

Third-party email provider settings

When using email providers such as AOL, Gmail, iCloud, or Yahoo that offer two-factor authentication, it may be necessary to configure your email service to accept messages from the EST4. For example when using Gmail, the email account should be set to “Allow less secure apps or devices” to send email. This property may be found under your Email Account, Security Settings.

Some email providers, support multiple TCP ports. Check that you are using the proper TCP port for your email server. Use the TCP port that supports TLS v1.2 connections. Example, Gmail provides TCP ports 465 and 587 for secure connections, but only port 587 supports the TLS v1.2 protocol used by an EST4 system.

IP Receiver service

The IP Receiver service allows an EST4 control unit to communicate system events to a central monitoring station (CMS) by way of Ethernet as an alternative to, or in conjunction with, an IP-DACT dialer. A network of control units can support up to 100 IP Receiver accounts (75 maximum if all are configured to automatically generate events).

The IP-DACT connection is supported on 4-FWAL2 and 4-FWAL4 firewall modules. For more information about 4-FWAL modules, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

Up to eight IP Receiver services can be configured for the firewall module. Refer to *4-CU Help* for instructions on configuring the Web Browser services.

Before starting the configuration process in the 4-CU, you need to gather configuration settings. If the control unit will be configured for CMS and FireWorks communication you will need to gather settings from both of those system administrators. To guide you through what information to gather, download the *EST4 Network Services Configuration Worksheet* (P/N 3102627-EN) from the My-Eddie website. To access the website, enter www.edwardsfiresafety.com in your web browser, click Login to log on to the My-Eddie website, and then locate the worksheet in Media Type > Installation Sheets.

Web Browser service

The Web Browser service allows end user browsers to access system reports from an EST4 control unit by way of a TCP port.

The browser connection is supported on control units with a 4-FWAL2, 4-FWAL3, or 4-FWAL4 firewall module installed. For more information about 4-FWAL modules, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

Up to eight Web Browser services can be configured for the firewall module. Refer to *4-CU Help* for instructions on configuring the Web Browser services.

Before starting the configuration process in the 4-CU, you need to gather configuration settings from the local IT administrator. To guide you through what information to gather, download the *EST4 Network Services Configuration Worksheet* (P/N 3102627-EN) from the My-Eddie website. To access the website, enter www.edwardsfiresafety.com in your web browser, click Login to log on to the My-Eddie website, and then locate the worksheet in Media Type > Installation Sheets.

Centralized audio applications

Centralized audio applications are supported on EST4 systems. The 4-AUDTELS Audio IO and Telephone Riser Source Module provides the 1 VRMS audio required for connection to banked audio amplifiers.

One audio amplifier (3-ZAxx, IB3-250, or IB3-125) is required for each audio channel to be operated simultaneously. The output of each amplifier is reduced from 25 VRMS to the 1 VRMS input level using the 3-ATPINT interface card, and then fed into the input of the banked amplifiers.

The audio riser output of the banked amplifiers is directed to the appropriate areas using Signature Series modules. The SIGA-CC1 module, Figure 37 on page 123, is used for single channel systems and the SIGA-CC2 module, Figure 38 on page 124, for two channel systems. EST4 audio system programming requires that the Signature modules controlling the audio signals be programmed in addition to the programming required for the audio amplifier(s) supplying the audio signal.

Table 34 below lists publications that provide additional information for devices referenced in this section. Refer to the publication for specific device settings, terminal connections, specifications, and installation procedures.

Table 34: Related publications for centralized audio applications

Device	Document P/N	Document name
3-ATPINT	387284-EN	3-ATPINT Interface Card Installation Sheet
3-ZA20A, 3-ZA20B, 3-ZA40A, 3-ZA40B	387463-EN	3-ZA20A, 3-ZA20B, 3-ZA40A, 3-ZA40B Zoned Audio Amplifiers Installation Sheet
4-AUDTELS	3102279-EN	4-AUDTELS Audio IO and Telephone Riser Source Module Installation Sheet
ATP, ATP-220	3101001-EN	ATP and ATP-220 Amplifier Terminal Panel Installation Sheet
ATPC	270219	Amplifier Terminal Panel Cabinet (ATPC) Installation Manual
IB3-125	403-352	1B3125 125-watt Power Amplifier Installation Sheet
IB3-250	427-06-00018	1B3250 250-watt Power Amplifier Installation Sheet
SIGA-CC1	387022P-EN	SIGA-CC1 Single Input Signal Module Installation Sheet
SIGA-CC2	387040P-EN	SIGA-CC2 Dual Input Signal Module Installation Sheet
SIGA-CC2A	3101168	SIGA-CC2A Dual Input Signal Module Installation Sheet
SIGA-CR	387023P-EN	SIGA-CR Control Relay Module Installation Sheet
SIGA-MAB	387356P-EN	SIGA-MAB Class A/B Input/Output Module Installation Sheet
SIGA-MCC1	387352P	SIGA-MCC1 Single Input Signal Module Installation Sheet
SIGA-MCC2	387353P	SIGA-MCC2 Dual Input Signal Module Installation Sheet
SIGA-MCC2A	3101340	SIGA-MCC2A Dual Input Signal Module Installation Sheet
SIGA-RM1	387347	SIGA-RM1 Signature Series Riser Monitor Module Installation Sheet
SIGA-UMI	3102191-EN	SIGA-UMI Universal Class A-B Module Installation Sheet

Amplifier Terminal Panel Cabinet (ATPC)

For a centralized audio application, the ATPC is used to house an RKU series equipment rack, Amplifier Terminal Panel (ATP), 3-ATPINT Interface card, Dukane 125 W (model IB3-125) or 250 W (model IB3-250) audio amplifiers, standby batteries and, in some situations, an external battery charger.

The ATPC can be located up to 3,000 ft. (914 m) from 3-ZAxx amplifiers supplying audio signals.

RKU series equipment racks

UL Listed RKU series equipment racks are used to house the banked Dukane amplifiers and associated equipment. The equipment racks are designed to support standard 19-inch (48.26 cm) wide rack-mount components. They are constructed of 16 gauge steel and finished in either white or black enamel. Refer to Table 35 below for a list of RKU equipment rack models and measurement specifications.

Interior-facing louvers on the RKU side panels and the back door provide ventilation for installed equipment, while maintaining a flush outside surface for side-by-side stacking of multiple racks. The louvered back door attaches to the cabinet with spring hinges, allowing easy field access and door removal. A key lock is provided on the door for added security.

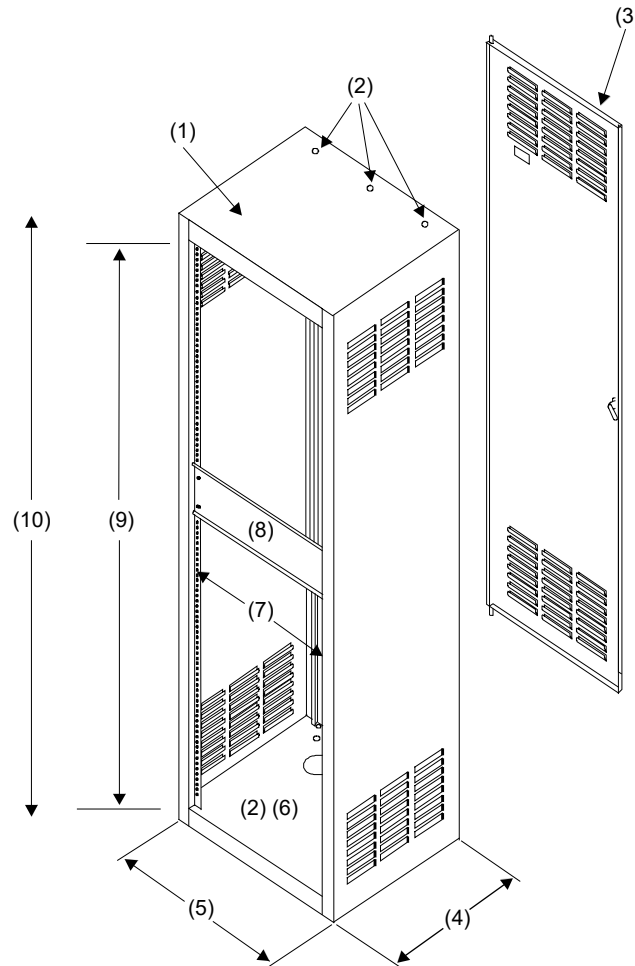
Six conduit knockouts are available on the top end panel (three on top, three on the flange), and six on the bottom end panel (three on the bottom, three on the flange). Three cable access holes are located on the bottom end panel for routing wiring to cabinet components. See Figure 23 on page 111.

Multiple racks can be installed side-by-side where additional cabinet capacity is required.

Table 35: RKU series equipment rack measurement specifications

Model	Width	Depth	Height	Rack Space
RKU-36(B)	22.31 in. (56.7 cm)	18.50 in. (47.0 cm)	41.06 in. (104.3 cm)	36.75 in. (93.3 cm)
RKU-42(B)	22.31 in. (56.7 cm)	18.50 in. (47.0 cm)	46.31 in. (117.6 cm)	42.00 in. (106.7 cm)
RKU-61(B)	22.31 in. (56.7 cm)	18.50 in. (47.0 cm)	65.56 in. (166.5 cm)	61.25 in. (155.6 cm)
RKU-70(B)	22.31 in. (56.7 cm)	18.50 in. (47.0 cm)	74.31 in. (188.7 cm)	70.00 in. (177.8 cm)
RKU-77(B)	22.31 in. (56.7 cm)	18.50 in. (47.0 cm)	81.31 in. (206.5 cm)	77.00 in. (195.6 cm)

Figure 23: RKU equipment rack



- | | |
|--|---|
| (1) Top | (6) Three 2.88-inch (7.3 cm) cable access holes |
| (2) Six 0.5 in (1.27 cm) and 0.75 in (1.91 cm) conduit knockouts | (7) 19.0 in rack mount |
| (3) Rear door with key lock | (8) Support bar |
| (4) Depth (all RKU models): 18.5 in (47.0 cm) | (9) Rack space measurement (see Table 35 on page 110) |
| (5) Width (all RKU models): 22.31 in (56.7 cm) | (10) Height measurement (see Table 35 on page 110) |

Note: The equipment mounting rails on the front of the rack are recessed 0.63-inch (1.59 cm).

ATP Amplifier Terminal Panel and ATPINT Interface card

ATP

The ATP supervises AC power to audio amplifiers and ensures that they can operate as needed during AC power failures or brownouts. If a power failure or brownout occurs, and the amplifiers are active, a power relay transfers battery power to the amplifiers. The amplifiers only draw battery power when in use. Two ATP models are available:

- ATP terminal panel, 120 VAC
- ATP-220 terminal panel, 220 VAC

The ATP enters a trouble state if any of the following events occur:

- ATP brownout or loss of AC power
- Low battery charge or missing battery (with J2 enabled)

- Ground fault (if ground fault detection J1 is enabled)
- Fuse failure

Notes

- The ATP must have a 3-ATPINT installed to work with the EST4 system.
- The ATP supports two 1B3-250 Dukane amplifiers maximum.

Figure 24: ATP, front view

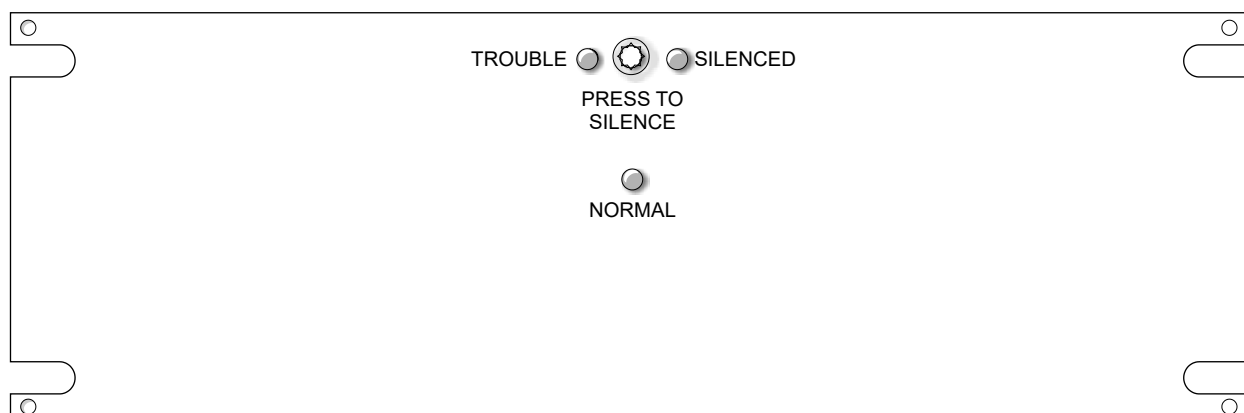
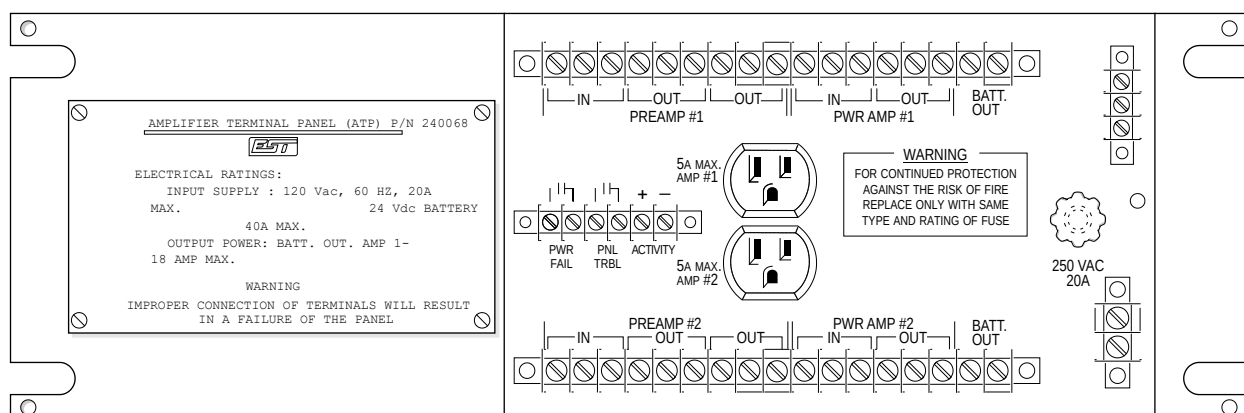


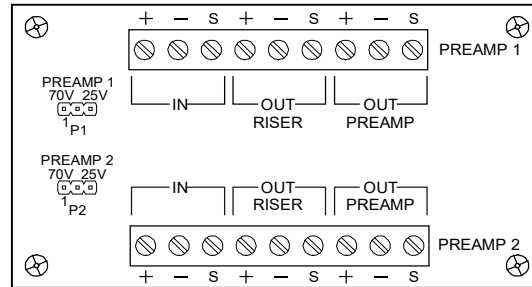
Figure 25: ATP, rear view



3-ATPINT

The 3-ATPINT Interface Card is a 25 V and 70 V adapter card for the ATP. The 3-ATPINT is required when using distributed (high voltage) output for an audio amplifier as the audio source for the ATP.

The 3-ATPINT is designed for use with audio source amplifiers that use 24 VDC output circuit supervision with an EOL resistor. Multiple 3-ATPINT cards can be connected to a common source amplifier using Class B or Class A supervision, as provided by the sourcing amplifier.

Figure 26: 3-ATPINT Interface card

ATP with 3-ATPINT operation

An ATP with installed 3-ATPINT is mounted in an RKU rack, and provides termination for the power amplifier's audio power and control signals. The ATP has an integral battery charger capable of charging 40 Ah (maximum) of standby batteries. The charger is fully supervised and provides a silenceable trouble buzzer and trouble contacts. One ATP is required for every two amplifiers.

When a brownout condition is sensed at the ATP, the trouble contacts and AC fail contacts are closed and a programmed supervisory zone reports the condition to the EST4 system. The EST4 system is designed to provide +24 VDC to the ATP's audio activity input by-way-of control relay, enabling standby power only when primary power to the amplifiers has failed and audio is active during an alarm condition.

Battery backup

When multiple ATPs share a common battery or when a backup amplifier is supplied from a single battery source, an external battery charger must be used. Battery power is applied to the amplifiers by the ATP. The internal battery charger supports two standby batteries of up to 40 Ah capacity.

Caution: Do not connect the battery chargers of multiple ATPs in parallel to increase the charger current.

When calculating the battery size required to support the amplifiers, the alarm current must be known.

- Each 250 W amplifier connected to the system draws 20 amperes at 24 VDC at full load
- Each 125 W amplifier connected to the system draws 10 amperes at 24 VDC at full load

The amplifiers draw no current in standby mode. NFPA 72 specifies that designing the system to provide 15 minutes of evacuation alarm at full load is the equivalent of 2 hours of emergency operation; however, the local AHJ or local codes can modify the amount of time for which standby power must be provided.

Dukane audio amplifiers

Two Dukane amplifiers models are available:

- 1B3-125, 125 watt output
- 1B3-250, 250 watt output

Both models operate from 120 VAC, 50/60 Hz, as well as 24 VDC battery standby. The amplifiers are mounted in an ATPC.

Notes

- The 1B3-250 amplifier should be loaded to no more than 72% of rated capacity. The amp is derated by 28% to allow for continuous operation and line loss averages.
- The ATP supports two 1B3-250 amplifiers maximum.

Figure 27: Dukane 1B3-250, front view

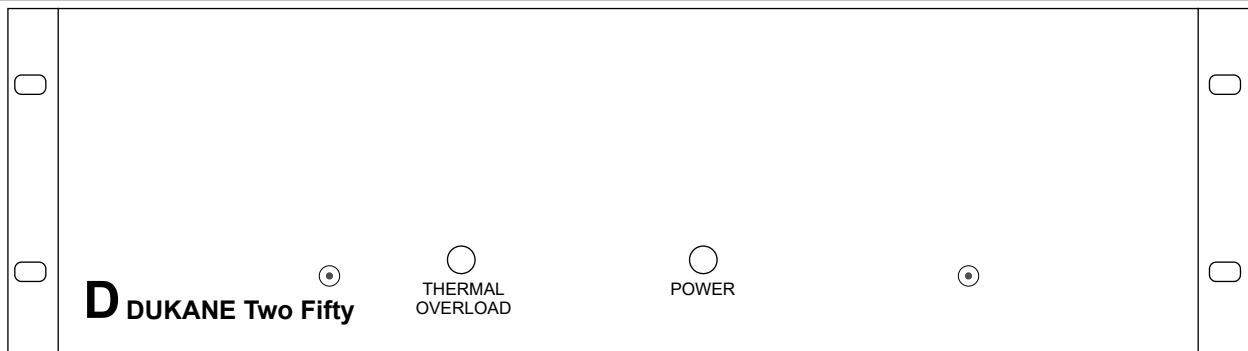
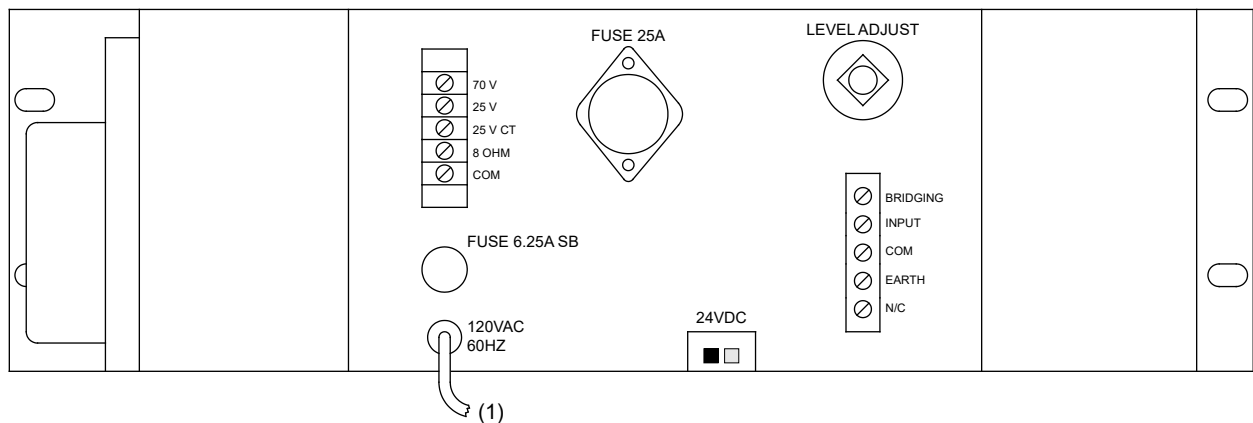


Figure 28: Dukane 1B3-250, rear view



(1) To ATP power receptacle

Figure 29: Dukane 1B3-125, rear view

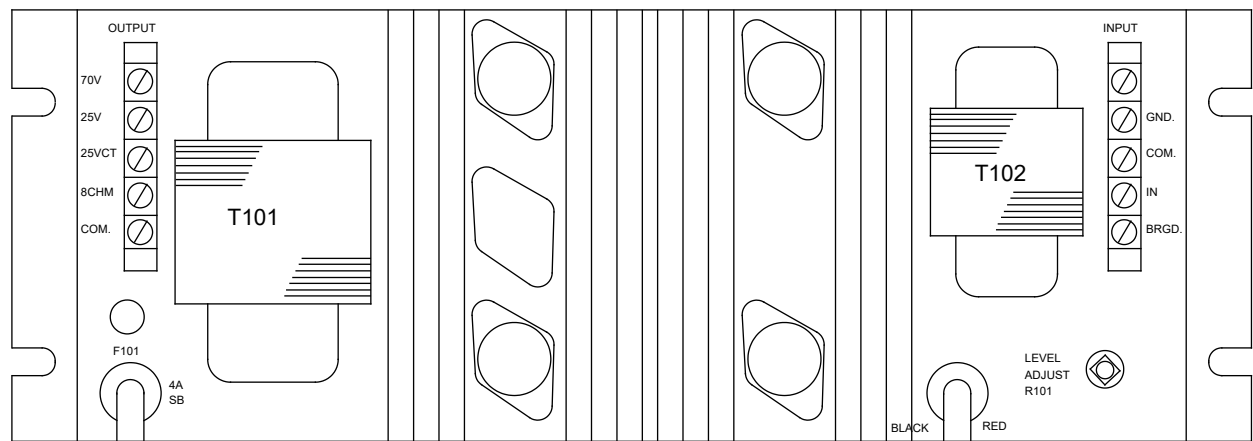
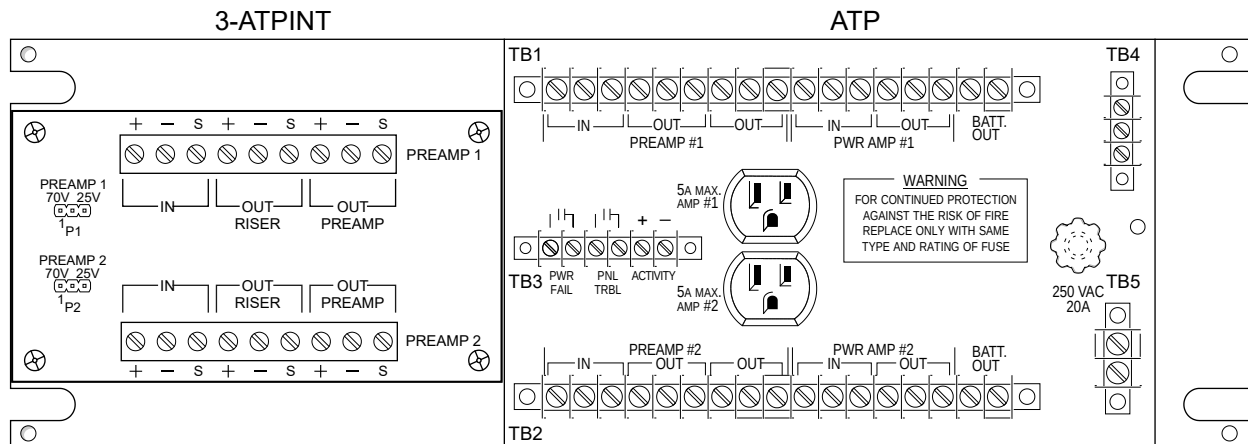


Table 36: Dukane 1B3-125 and 1B3-250 amplifiers specifications

	1B3125	1B3-250
Rated output power	125 W	250 W (180 W max. loaded)
Max. signal input	1 VRMS	1 VRMS
Input impedance	75 k Ω	75 k Ω
Output voltage	25 or 70 VRMS	25 or 70 VRMS
Primary power	120 VAC, 60 Hz	120 VAC, 60 Hz
Battery power	24 VDC	24 VDC
AC power consumption		
Standby	27 W	48 W
Full load	360 W	700 W
DC power consumption		
Standby	0 W (when using the ATP)	0 W (when using the ATP)
Full load	11.5 A	20 A
Dimensions (H × W × D)	5.25 × 19.0 × 6.625 in (13.3 × 48.3 × 16.8 cm)	8.5 × 19 × 15 in (21.6 × 48.3 × 38.1 cm)
Weight	22.5 lb (10.1 kg)	55 lb (24.9 kg)

The output of the amplifier must be set for the proper value by adjusting Level Adjust on the back of the amplifier. For 1,000 Hz generated tones, the amplifier must adjust for 70 VRMS output by using an RMS voltmeter.

ATP and 3-ATPINT terminal connections and jumper settings

Figure 30: ATP with 3-ATPINT installed, rear view

ATP terminal connections

Table 37: ATP TB1 and TB2 terminals for connecting audio signals to the ATP and the two amplifiers

Terminal	Description
Preamp IN	<i>Do not use if 3-ATPINT is installed.</i> Provides input from the fire control unit audio subsystem. If multiple ATP modules are used, connect to the <i>Preamp Out</i> terminal of the previous ATP.
Preamp OUT	<i>Do not use if 3-ATPINT is installed.</i> Connects to the <i>Preamp In</i> terminals of the next ATP. If this is the last ATP, terminate as indicated in Table 39.
Preamp OUT	<i>Do not use if 3-ATPINT is installed.</i> Connects to the High Z input terminal of the Dukane amplifier.
Power Amp IN	Connects to the 70 V or the 25 V output of the Dukane amplifier.
Power Amp OUT	Connects to a SIGA-RM1 Signature Series Riser Monitor Module configured as a riser monitor. Each riser can supply 180 W, max.

Table 38: ATP TB3 terminals

Terminal	Description
ATP power fail	Normally open relay contacts that close when primary power to the amplifiers is lost or in brownout condition. Connect to a SIGA-CT1 Signature Series Single Input Module configured as an ACfail device type. Make sure that jumper JP1 is in position 1 as shown in Figure 32 on page 118 so that the ATP does not create an additional delay. If ACfail device type is not used to monitor the ACfail of the ATP, then JP1 must be installed in position 2 where the built in ACfail delay on the ATP is used and this contact is to be supervised by a supervisory input either on the control unit or via a Signature Series input module configured as a supervisory input.
Panel trouble	Normally open relay contacts close when any of the following problems occur: • Failure of the battery charger circuit (if enabled) • Blown fuse or circuit breaker • Ground fault on battery terminals • Brownout or loss of AC power
Activity	When this input is active, power relay contacts transfer battery power to the BATT OUT terminals. Connect these terminals to a SIGA-CR Signature Series Control Relay Module to provide 24 VDC during any system alarm state or user activation of the paging system.

Table 39: ATP preamplifier termination

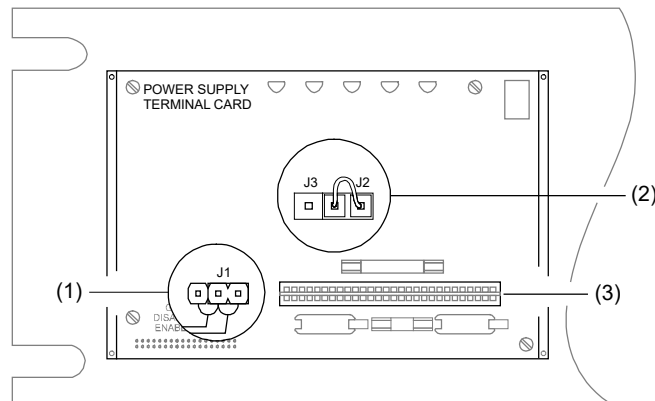
Amplifiers per channel	EOL resistor value	Number
0	10 k Ω , 0.5 W	260002
1–5	1.8 k Ω , 0.5 W	260044
6–10	2.2 k Ω , 0.5 W	260045
11–15	3.0 k Ω , 0.5 W	260046
16–20	3.9 k Ω , 0.5 W	260001
21–25	6.8 k Ω , 0.5 W	260047

ATP jumper settings

Two jumpers for configuring the ATP are found on the power supply terminal card. Jumper J1 is used to enable or disable the ground fault detector. Jumper J3-J2 is used to enable or disable the internal battery charger. For more information see Figure 31 below and Table 40 below.

One jumper (JP1) is found on the power supply card. Jumper JP1 is used to impose a 150-minute delay on trouble indication when an AC power loss or brownout occurs. This delay is used to prevent false trouble reports in areas where brief power failures occur frequently. See Figure 32 on page 118 and Table 41 on page 118 for details.

Figure 31: ATP power supply terminal card placement of J1 and J3-J2 jumpers



- (1) Jumper J1, enlarged
- (2) Jumper J3-J2 with the jumper installed in the J2 position
- (3) Connector for the power supply card

Table 40: ATP power supply terminal card jumper positions

Jumper	Function
J1	Ground fault detector jumper: • ENABLE position enables the ground fault detector • DISABLE position disables the ground fault detector
J3-J2	Internal battery charger jumper: • JP2 position enables the internal battery charger • JP3 position disables the internal battery charger

Figure 32: ATP power supply card placement of JP1 jumper

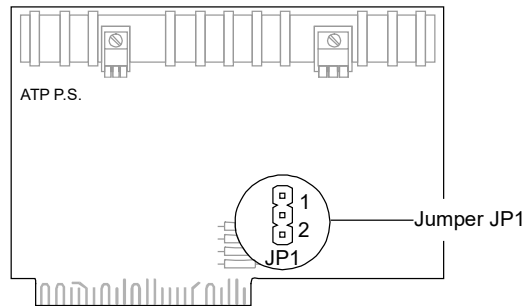


Table 41: ATP power supply card jumper positions

Jumper	Position	Response to AC power failure or brownout
JP1	Position 1	The ATP goes into trouble immediately
JP1	Position 2	The ATP waits for 150 minutes. If power has not been restored to normal levels before 150 minutes, the ATP goes into trouble.

3-ATPINT terminal connections

Table 42: 3-ATPINT terminals

Terminal	Description
IN riser	To audio source amplifier 25 or 70 VRMS output, or previous 3-ATPINT riser output
OUT RISER	25 or 70 VRMS output to next 3-ATPINT IN riser or EOL resistor
OUT PRE AMP	Low level audio to audio power amplifier input

3-ATPINT jumper settings

Two jumpers are used for configuring the 3-ATPINT input voltage. See Figure 33 below and Table 43 below for details.

Figure 33: 3-ATPINT P1 and P2 jumpers positions

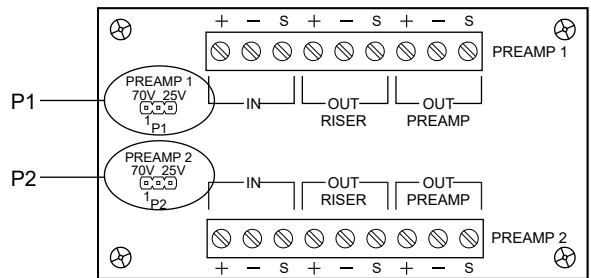


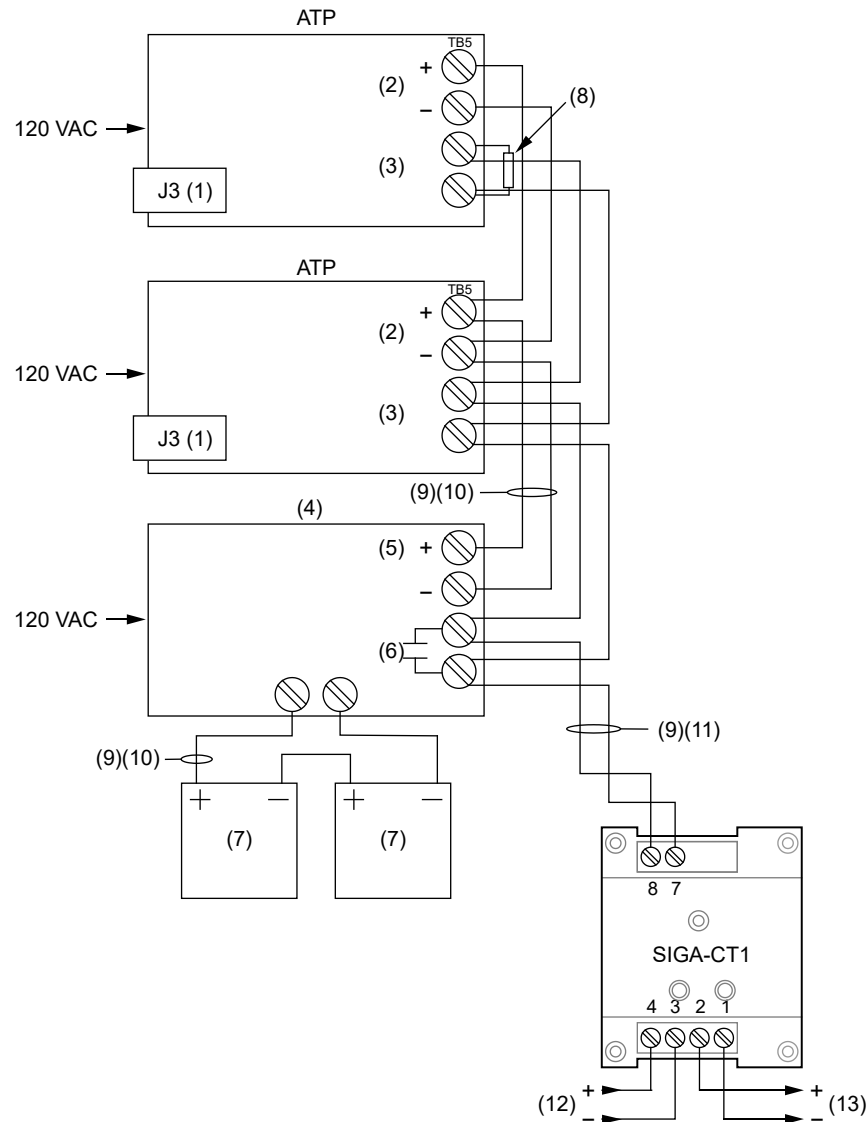
Table 43: 3-ATPINT jumper positions

Jumper	Position	Function
P1	Position 1/2	Sets PREAMP 1 input voltage to 70 VRMS
	Position 2/3	Sets PREAMP 1 input voltage to 25 VRMS
P2	Position 1/2	Sets PREAMP 2 input voltage to 70 VRMS
	Position 2/3	Sets PREAMP 2 input voltage to 25 VRMS

ATP external battery charger

When multiple ATPs are connected to a common battery set, disable the ATP internal battery charger by installing J3 and removing J2 on the power supply terminal board (Figure 31 on page 117). Use an external battery charger that can charge up to 160 Ah standby batteries, as shown in Figure 34 below.

Figure 34: ATP external battery charger wiring



Legend

- | | |
|---|--|
| (1) Disable charger jumper J3 on the ATP power supply terminal card (see Figure 31 on page 117) | (7) Standby battery (24 VDC 160 Ah max.) |
| (2) Battery IN | (8) UL/ULC Listed 47 kΩ end-of-line resistor |
| (3) Panel trouble | (9) Supervised |
| (4) External battery charger | (10) Nonpower-limited |
| (5) Load | (11) Power-limited |
| (6) Battery trouble | (12) Data IN |
| | (13) Data OUT |

Notes

- Maximum wire size: 10 AWG (5.26 sq mm)
- External battery charger must be installed in same enclosure as the ATP

Wiring diagrams

This section includes examples of wiring diagrams used in centralized audio applications.

See specific device installation sheets for more information about a device. You can find publication part numbers listed in Table 34 on page 109.

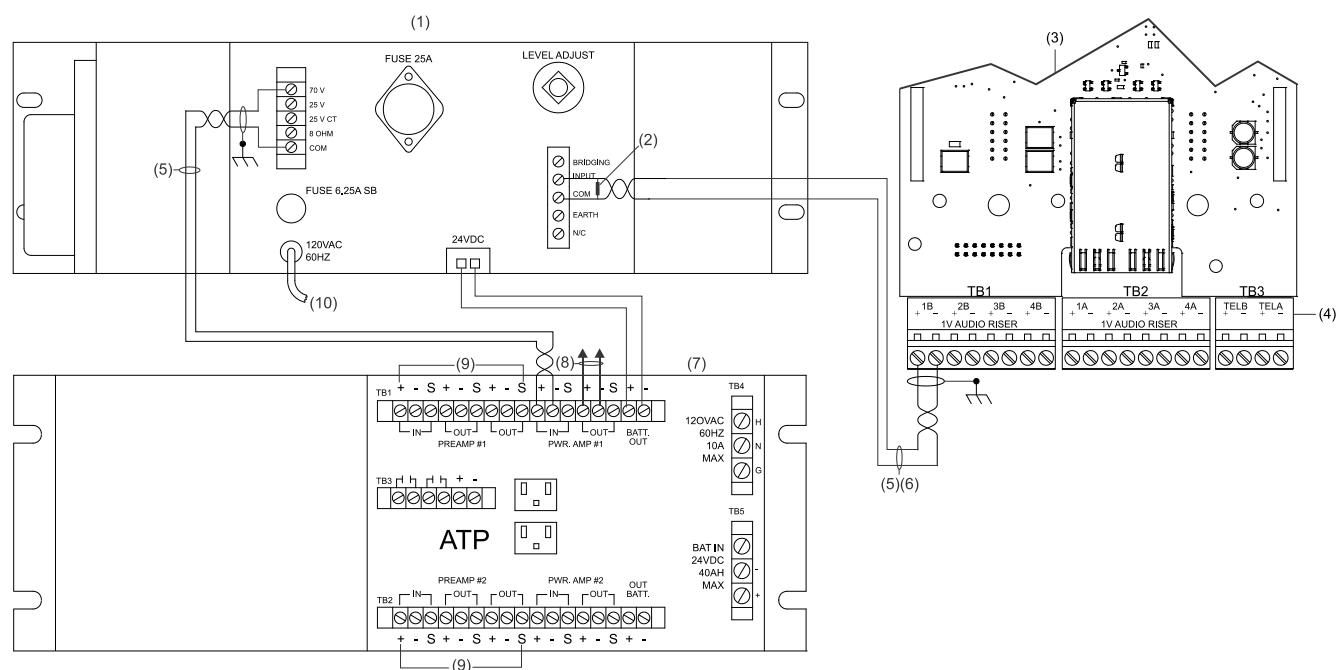
Notes

- Follow power-limited or nonpower-limited wiring practices as determined by the amplifier providing the audio signal.
- Wiring between the output of each 3-ZA20 and its associated banked amplifier input should be twisted-pair, shielded cable and can be configured for Class A or Class B integrity monitoring.

4-AUDTELS wiring for 1 VRMS to Dukane amplifiers

Note: See *ATP and ATP-220 Amplifier Terminal Panel Installation Sheet (P/N 3101001-EN)* for more information on to Dukane amplifiers.

Figure 35: 4-AUDTELS – ATP and 1 VRMS to Dukane ampler (1B3-250 shown)



Legend

- | | |
|---|----------------------------------|
| (1) 1B3-250 Dukane amplifier | (6) Twisted-pair, shielded cable |
| (2) 15 kΩ EOLR (P/N EOL-15) | (7) ATP |
| (3) 4-AUDTELS | (8) Audio riser out |
| (4) 4-AUDTELS terminal board | (9) Do not use |
| (5) 1 VRMS out to Dukane amplifier wiring (Class B) | (10) To ATP power receptacle |

Notes

- Maintain 0.25 in. (6 mm) separation between power-limited and nonpower-limited wiring at all times.
- For Canadian mass notification installations, a minimum of 4 hours of standby battery power is required in conjunction with an automatic starting, engine driven generator.
- The 1B3-250 Dukane amplifier should be loaded to no more than 72% of rated capacity. The amp is derated by 28% to allow for continuous operation and line loss averages.

Backup audio amplifier

Various methods are available to provide a backup amplifier in the event that a primary amplifier fails. Depending upon the local AHJ, a single backup amplifier can be required for each primary amplifier or a single backup per bank of amplifiers.

Figure 36 on page 122 provides an example of centralized banked audio with a backup amplifier. In the example:

- Five Dukane 1B3-250 audio amplifiers are used, four primary amplifiers and one backup amplifier, and one SIGA-RM1 riser module for each amplifier.

- For rules programming:

There is an AND logic device group for *each* SIGA-RM1 riser module that has the First Alarm pseudo point and SIGA-RM1 in the group. Because there is a SIGA-RM1 for each of the four primary Dukane audio amplifiers, there are four AND groups.

The SIGA-RM1 for the backup amplifier is for information only to verify that the backup amplifier is operational (no rules programming is needed).

The AND group activates a SIGA-RM1 Command List that first disables itself, then disables all SIGA-RM1 modules, then activates the SIGA-CR transfer relay of the failed amplifier, and then enables itself.

A System Reset rule enables all the RM1 modules.

Rules for Figure 36 example:

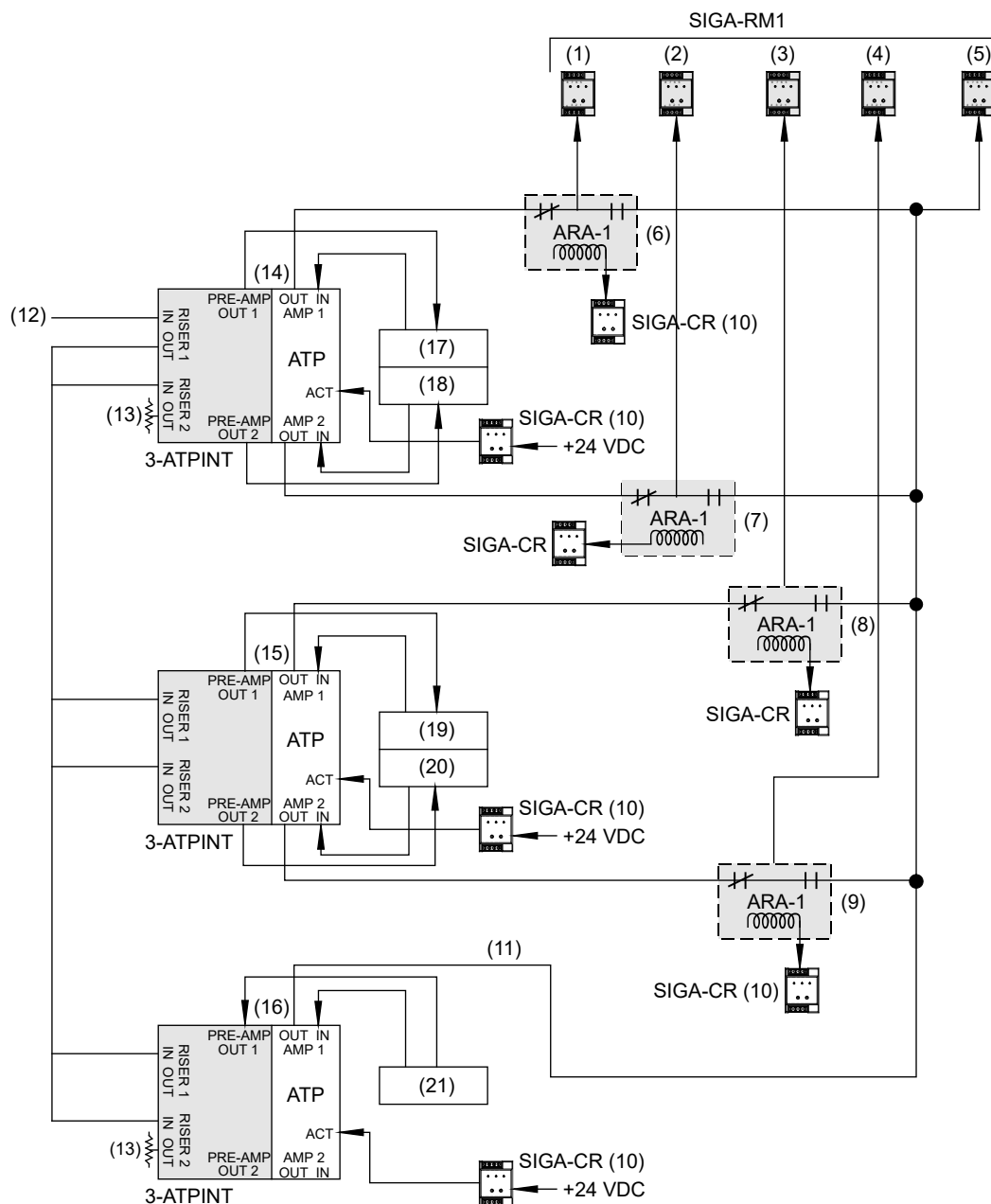
```
[STARTUP]
STARTUP:  AMPON 'EST3_ZA20_AMPLIFIER'  to 'GENERAL CHANNEL',
          MSGON 'SUPERVISORY_TONE_or_MESSAGE'  to 'GENERAL CHANNEL';

[AND GROUP]
ACTIVATION 'RM1-1'  :  ACTIVATE 'RM1-1_COMMAND_LIST';

[COMMAND LIST]
ACTIVATION 'RM1-1_COMMAND_LIST'  :      DISABLE 'RM1-1_COMMAND_LIST',
                                         DLYA 0,
                                         DISABLE 'RM1-*',
                                         DLYA 0,
                                         ON 'SIGA-CR_BACKUP_RELAY',
                                         DLYA 0,
                                         ENABLE 'RM1-1_COMMAND_LIST';

[SYSTEM RESET]
RESET  :  ENABLE 'RM1-*' ;
```

Figure 36: Example wiring for centralized banked audio with one backup amplifier



Legend (see figure notes on next page)

- | | |
|--|---|
| (1) SIGA-RM1 riser reports audio amplifier 1 failure | (13) UL/ULC Listed 15 KΩ end-of-line resistor |
| (2) SIGA-RM1 riser reports audio amplifier 2 failure | (14) ATP1: AMP 1 and AMP 2 |
| (3) SIGA-RM1 riser reports audio amplifier 3 failure | 3-ATPINT: PRE-AMP OUT 1 and PRE-AMP OUT 2 |
| (4) SIGA-RM1 riser reports audio amplifier 4 failure | (15) ATP2: AMP 3 and AMP 4 |
| (5) SIGA-RM1 riser provides information only to verify backup audio amplifier 5 is operation | 3-ATPINT: PRE-AMP OUT 3 and PRE-AMP OUT 4 |
| (6) ARA-1 relay activates on amplifier 1 failure | (16) ATP3: AMP 5 and AMP 6 |
| (7) ARA-1 relay activates on amplifier 2 failure | 3-ATPINT: PRE-AMP OUT 5 and PRE-AMP OUT 6 |
| (8) ARA-1 relay activates on amplifier 3 failure | (17) Dukane 1B3-250 audio amplifier 1 |
| (9) ARA-1 relay activates on amplifier 4 failure | (18) Dukane 1B3-250 audio amplifier 2 |
| (10) SIGA-CR defined as nonsupervised output | (19) Dukane 1B3-250 audio amplifier 3 |
| (11) Backup riser | (20) Dukane 1B3-250 audio amplifier 4 |
| (12) From 3-ZA20/B | (21) Dukane 1B3-250 backup audio amplifier |

Figure 36 notes

- SIGA-RM1 modules report riser troubles to the network controller.
- The ARA-1 relay installs on an ARM-8 module. For more information about the ARA-1, see *ARM-8 Auxiliary Relay Module Installation Sheet* (P/N 270753).

Branch speaker wiring

Signature modules are used to connect individual floor branch speaker circuits to the main riser. Table 44 below lists the single channel and dual channel modules that can be used for Class B and Class A branch speaker wiring.

Table 44: Branch speaker wiring single and dual channel Signature modules

Channels	Class B Signature modules	Class A Signature modules
Single channel	SIGA-CC1, SIGA-MCC1, SIGA-UM, SIGA-MAB, SIGA-CC2, SIGA-MCC2	SIGA-UM, SIGA-MAB
Dual channel	SIGA-CC2, SIGA-MCC2	SIGA-CC2A, SIGA-MCC2A

Figure 37: Typical Class B, single channel wiring using a SIGA-CC1

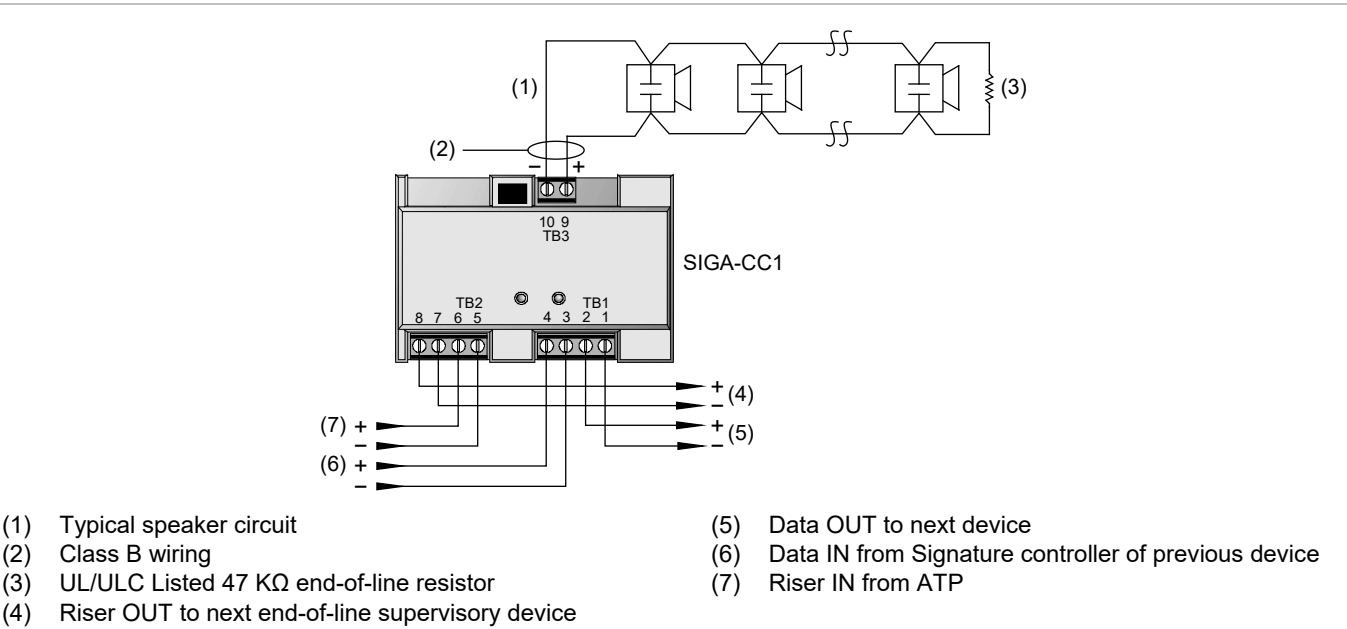
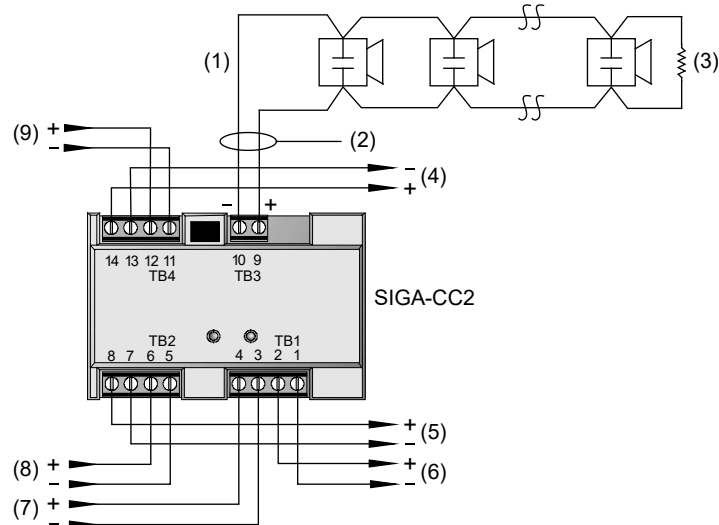
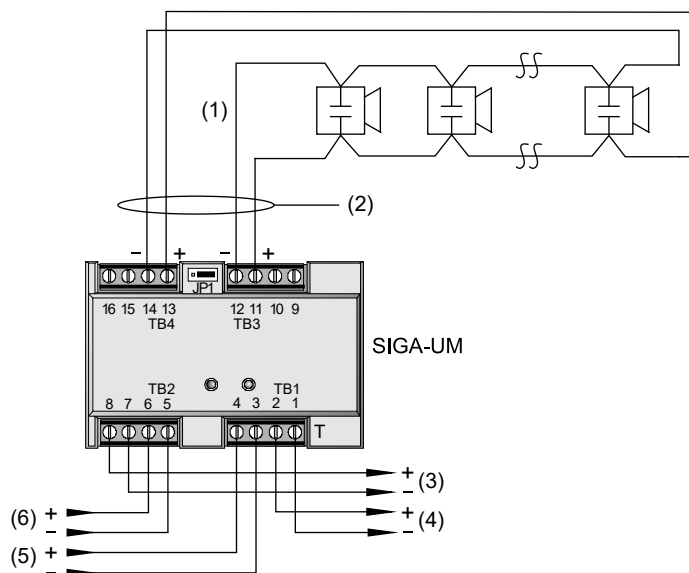


Figure 38: Typical Class B, dual channel wiring using a SIGA-CC2

- | | |
|---|--|
| (1) Typical speaker circuit | (6) Data OUT to next device |
| (2) Class B wiring | (7) Data IN from Signature controller of previous device |
| (3) UL/ULC Listed 47 KΩ end-of-line resistor | (8) Channel 1 (input 1) riser IN from ATP |
| (4) Channel 2 (input 2) riser OUT to next device or next end-of-line supervisory device | (9) Channel 2 (input 2) riser IN from ATP |
| (5) Channel 1 (input 1) riser OUT to next device or next end-of-line supervisory device | |

Figure 39: Typical Class A single channel wiring using a SIGA-UM Module

- | | |
|---|--|
| (1) Typical speaker circuit | (4) Data OUT to next device |
| (2) Class A wiring | (5) Data IN from Signature controller of previous device |
| (3) Channel 1 (input 1) riser OUT to next device or next end-of-line supervisory device | (6) Channel 1 (input 1) riser IN from ATP |

Chapter 6

Preventive maintenance and testing

Summary

This chapter provides instruction for maintaining and testing the EST4 life safety system.

Content

Visual inspections 126

Routine maintenance and tests 127

 Maintenance schedule 127

System trouble and maintenance log 133

Record of completion 134

Visual inspections

Perform visual inspections in accordance with Table 45, or more often if required by the local AHJ. See Table 47 on page 128 for test methods.

Table 45: Visual inspection schedule

Component	Frequency	Recommended procedure
Radiant energy fire detectors	Monthly	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance. Clean if necessary.
Supervisory signal devices	Monthly	Verify that the module's green LED flashes. Ensure there are no changes that may adversely affect equipment performance.
Waterflow devices	Monthly	Verify that the module's green LED flashes. Ensure there are no changes that may adversely affect equipment performance.
Batteries	Semiannually	Inspect batteries for corrosion or leakage. Verify that the battery connections are tight and secure. Clean the connections, if required. Replace batteries every 5 years, or sooner if conditions warrant.
Control unit trouble signals	Semiannually	Ensure there are no changes that may adversely affect equipment performance.
Emergency voice/alarm communication equipment	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Remote annunciators	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Duct detectors	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Electromechanical releasing devices	Semiannually	Ensure there are no changes that may adversely affect equipment performance.
Fire extinguishing systems or suppression systems	Semiannually	Ensure there are no changes that may adversely affect equipment performance.
Fire alarm boxes	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Heat detectors	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance. Clean if necessary.
Smoke detectors	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance. Clean if necessary.
Interface equipment	Semiannually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Alarm notification appliances	Semiannually	Verify that the module's green LED flashes. Ensure there are no changes that may adversely affect equipment performance.
Supervising station life safety system transmitters	Semiannually	Ensure there are no changes that may adversely affect equipment performance.
Control unit	Annually	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
Fiber optic cable connections	Annually	Inspect the cables for any visible signs of damage, loose connections, or other changes that may adversely affect performance.

Routine maintenance and tests

Perform routine maintenance and tests in accordance with Table 46 below or more often if required by the local AHJ. See Table 47 on page 128 for test methods.

Notes

- Before starting testing, notify all areas where the alarm sounds and off premises locations that receive alarm and trouble transmissions that testing is in progress.
- Keep records of all testing and maintenance on the protected premises for a period of at least five (5) years.
- A complete check of installed field wiring and devices should be made at regular intervals, in accordance with NFPA 72 and ULC 524 requirements. This includes testing all alarm and supervisory alarm initiating devices and circuits, and any off premise connections.
- Control unit operation should be verified in the alarm, supervisory, and trouble modes.
- For SIGA and SIGA2 Signature Series detector testing and maintenance procedures, refer to *Signature Series Detector Application Bulletin* (P/N 270145).

Maintenance schedule

Table 46: Routine maintenance schedule

Component	Initial and Reacceptance	Other
Control equipment [1]	Y	Quarterly/Annually
Batteries [2]	Y	Annually
Control unit trouble signals	Y	Annually
Fiber optic cable connections	Y	Annually
Emergency voice/alarm communication equipment	Y	Annually
Remote annunciators	Y	Annually
Smoke detectors	Y	Annually
Heat detectors	Y	Annually
Fire alarm boxes	Y	Annually
Supervisory signal devices (except valve tamper switches)	Y	Quarterly
Waterflow devices	Y	Semiannually
Valve tamper switches	Y	Semiannually
Fire extinguishing systems or suppression systems	Y	Annually
Interface equipment	Y	Annually
Audible notification appliances	Y	Annually
Textual audible notification appliances (speakers)	Y	Annually
Visible notification appliances	Y	Annually
Off-premises transmission equipment	Y	Quarterly
Supervising station fire alarm system transmitters	Y	Annually

[1] Test control equipment quarterly when it is not connected to a supervising station.

[2] Replace batteries every five years or sooner if conditions warrant.

Table 47: Routine maintenance and tests

Component	Test	Test methods
Control unit	Visual inspection	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
	Initial and Reacceptance	1. Verify that the control unit indicates open, short, and ground faults for all notification appliance circuits, initiating device circuits, and signaling line circuits. 2. Verify that the control unit activates all evacuation signals and auxiliary functions according to the site specific software. 3. Verify that all controls and indicators function as intended. 4. Disconnect the primary (mains) power. Verify that the control unit indicates an AC power failure.
	Quarterly/Annual	Test one-fourth of the entire system every three months such that the entire system is tested in a one year period, or test the entire system once each year.
Standby batteries	Visual inspection	Inspect batteries for corrosion or leakage. Verify that the battery connections are tight and secure. Clean the connections, if required. Replace batteries every 5 years or sooner if conditions warrant.
	Initial and Reacceptance	1. With the control unit powered up, and with the batteries connected and fully charged, verify that the voltage across the battery terminals is the correct voltage in accordance with the battery manufacturer's specifications. 2. With the control unit under full load, disconnect the primary (mains) power. Wait until the standby operation time requirement passes then activate all alarm signals. Verify that the alarm signals remain active for at least 5 minutes for horns and strobes or 15 minutes for audio in the US, whichever is greater.
	Annual	1. With the control unit powered up, and with the batteries connected and fully charged, verify that the voltage across the battery terminals is the correct voltage in accordance with the battery manufacturer's specifications. 2. Test the capacity of the batteries using a battery tester suitable for the amp-hour rating of the batteries.
Smoke detectors	Visual inspection	1. Verify that the detector's green LED flashes if programmed to do so. 2. Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance. Clean if necessary.
	Initial and Reacceptance	1. Remove the detector from its base. Verify that the control unit displays a trouble message that correctly identifies the detector. 2. Activate the detector. Verify that the detector's red LED flashes and the control unit displays an alarm message that correctly identifies the detector. 3. If the detector is installed in a relay base, verify the correct operation of the relay. 4. Run a Device Maintenance Report on all the smoke detectors in the system. Verify that all sensitivity levels fall within acceptable limits. Keep a printed copy for your records.

Component	Test	Test methods
Heat detectors	Annual	1. Activate the detector. Verify that the detector's red LED flashes and the control unit displays an alarm message that correctly identifies the detector. 2. If the detector is installed in a relay base, verify the correct operation of the relay. 3. Run a Device Maintenance Report on all the smoke detectors in the system. Verify that all sensitivity levels fall within acceptable limits. Keep a printed copy for your records.
	Visual inspection	1. Verify that the detector's green LED flashes. 2. Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance. Clean if necessary.
	Initial and Reacceptance	Caution: Directing heated air at a single point may permanently damage the heat detector. Wave the hair blower slowly back and forth approximately 1 in. from the heat entry slots. 1. Remove the detector from its base. Verify that the control unit displays a trouble message that correctly identifies the detector. 2. Activate the detector using a commercial grade (1200 to 1500 W) hair blower. Verify that the detector's red LED flashes and the control unit displays an alarm message that correctly identifies the detector. 3. If the detector is installed in a relay base, verify the correct operation of the relay.
Duct detectors	Annual	Caution: Directing heated air at a single point may permanently damage the heat detector. Wave the hair blower slowly back and forth approximately 1 in. from the heat entry slots. 1. Activate the detector using a commercial grade (1200 to 1500 W) hair blower. Verify that the detector's red LED flashes and the control unit displays an alarm message that correctly identifies the detector. 2. If the detector is installed in a relay base, verify the correct operation of the relay.
	Visual inspection	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
	Initial and Reacceptance	Activate the equipment. Make sure the control unit correctly identifies the device.
Fire alarm boxes	Semiannual	Activate the equipment. Make sure the control unit correctly identifies the device.
	Visual inspection	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
	Initial and Reacceptance	Activate the equipment. Make sure the control unit correctly identifies the device.
Fire alarm boxes	Semiannual	Activate the equipment. Make sure the control unit correctly identifies the device.

Component	Test	Test methods
Alarm input modules (except waterflow switch inputs)	Visual inspection	Verify that the module's green LED flashes. Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	1. Open the circuit. Verify that the control unit displays a trouble message that correctly identifies the module. 2. Short each side of the circuit to ground one at a time. Verify that for each short the control unit displays a ground fault message that correctly identifies the module. 3. Activate the module. Verify that the module's red LED flashes and the control unit displays an alarm message that correctly identifies the module.
	Annual	Activate the module. Verify that the red LED flashes and the control unit displays an alarm message that correctly identifies the module.
Waterflow switch input modules	Visual inspection	Verify that the module's green LED flashes. Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	1. Open the circuit. Verify that the control unit displays a trouble message that correctly identifies the module. 2. Short each side of the circuit to ground one at a time. Verify that for each short the control unit displays a ground fault message that correctly identifies the module. 3. Activate the module. Verify that the module's red LED flashes and the control unit displays an alarm message that correctly identifies the module.
	Semiannual	Activate the module. Verify that the module's red LED flashes and the control unit displays an alarm message that correctly identifies the module.
Supervisory input modules (except valve tamper inputs)	Visual inspection	Verify that the module's green LED flashes. Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	1. Open the circuit. Verify that the control unit displays a trouble message that correctly identifies the module. 2. Short each side of the circuit to ground one at a time. Verify that for each short the control unit displays a ground fault message that correctly identifies the module. 3. Activate the module. Verify that the module's red LED flashes and the control unit displays a supervisory message that correctly identifies the module.
	Quarterly	Activate the module. Verify that the module's red LED flashes and the control unit displays a supervisory message that correctly identifies the module.
Valve tamper input modules	Visual inspection	Verify that the module's green LED flashes. Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	1. Open the circuit. Verify that the control unit displays a trouble message that correctly identifies the module. 2. Short each side of the circuit to ground one at a time. Verify that for each short the control unit displays a ground fault message that correctly identifies the module. 3. Activate the module. Verify that the module's red LED flashes and the control unit displays a supervisory message that correctly identifies the module.
	Semi-annual	Activate the module. Verify that the module's red LED flashes and the control unit displays a supervisory message that correctly identifies the module.
Releasing modules	Visual inspection	Verify that DS2 flashes and DS4 is on. Ensure that there are no changes that may adversely affect equipment performance.

Component	Test	Test methods
	Initial and Reacceptance	WARNING: Disconnect all wiring on TB4 (RELEASE 1 and RELEASE 2) when servicing or testing the system. Disabling points does not prevent activation of the release circuits. Failure to follow these instructions may result in loss of life, serious injury, or property damage. 1. Verify that the control unit indicates open, shorts, and ground faults for each of the circuits. 2. Verify that the release initiation circuit activates the release circuits as intended, and that all required signals are indicated on the control unit. 3. Verify that the manual release switch, if used, activates the release circuits as intended, and that all required signals are indicated on the control unit. 4. Verify that the abort switch, if used, prevents the release circuits from activating as intended.
	Semiannual	WARNING: Disconnect all wiring on TB4 (RELEASE 1 and RELEASE 2) when servicing or testing the system. Disabling points does not prevent activation of the release circuits. Failure to follow these instructions may result in loss of life, serious injury, or property damage. 1. Verify that the release initiation circuit activates the release circuits as intended, and that all required signals are indicated on the control unit. 2. Verify that the manual release switch, if used, activates the release circuits as intended, and that all required signals are indicated on the control unit. 3. Verify that the abort switch, if used, prevents the release circuits from activating as intended.
Audible notification appliances	Visual inspection	Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	Measure and record sound pressure levels throughout the protected area to ensure the minimum audibility requirements of NFPA 72 are met. Confirm synchronization of outputs within each notification zone per NFPA 72.
	Annual	Measure and record sound pressure levels throughout the protected area to ensure the minimum audibility requirements of NFPA 72 are met. Confirm synchronization of outputs within each notification zone per NFPA 72
Textual audible notification appliances (speakers)	Visual inspection	Ensure there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	Measure and record sound pressure levels of the required tone(s) preceding the textual audible (voice) message throughout the protected area to ensure the minimum audibility requirements of NFPA 72 are met. Verify that audible information is distinguishable and understandable.
	Annual	Measure and record sound pressure levels of the required tone(s) preceding the textual audible (voice) message throughout the protected area to ensure the minimum audibility requirements of NFPA 72 are met. Verify that audible information is distinguishable and understandable.
Visible notification appliances	Visual inspection	Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	Verify that the appliance locations are in accordance with the approved layout and are set for the correct candela rating. Verify that each appliance flashes. Verify synchronization of the light output between devices as required by NFPA 72.
	Annual	Verify that each appliance flashes. Verify synchronization of the light output between devices as required by NFPA 72.

Component	Test	Test methods
Off-premises transmission equipment	Visual inspection	Ensure that there are no changes that may adversely affect equipment performance.
	Initial and Reacceptance	1. Activate an alarm initiating device. Verify that the off-premises location receives an alarm signal. 2. Create a trouble condition. Verify that the off-premises location receives a trouble signal. 3. Activate a supervisory device. Verify that the off-premises location receives a supervisory signal. 4. If the module is configured to transmit alarm signals and trouble signals over the same dedicated pair of wires, create a trouble condition, and then activate an alarm initiating device. Verify that the off-premises location receives an alarm signal and a trouble signal.
	Semiannual	1. Activate an alarm initiating device. Verify that the off-premises location receives an alarm signal. 2. Create a trouble condition. Verify that the off-premises location receives a trouble signal. 3. Activate a supervisory device. Verify that the off-premises location receives a supervisory signal. 4. If the module is configured to transmit alarm signals and trouble signals over the same dedicated pair of wires, create a trouble condition, and then activate an alarm initiating device. Verify that the off-premises location receives an alarm signal and a trouble signal.
Digital alarm communicator transmitter (DACT)	Visual inspection	Inspect the equipment for any visible signs of damage or other changes that may adversely affect performance.
	Initial and Reacceptance	1. Activate an alarm input while using the primary telephone line for a telephone call. • Verify that the supervising station receives the correct signal. • Verify completion of the transmission attempt occurs within 90 seconds. 2. Disconnect the primary telephone line and connect the secondary telephone line. • Verify that the control unit indicates a DACT trouble message. • Verify that the DACT transmits the trouble signal to the supervising station within 4 minutes of detecting the fault. 3. Disconnect the secondary telephone line and connect the primary telephone line. • Verify that the control unit indicates a DACT trouble message. • Verify that the DACT transmits the trouble signal to the supervising station within 4 minutes of detecting the fault.
	Semiannual	Same as initial and reacceptance testing.

System trouble and maintenance log

[illegible]

Record of completion

NFPA 72 requires a Record of Completion be filled out at the time of system acceptance and approval, and revised when changes to the system are made. You can download a copy of the form from the NFPA website (www.nfpa.org).

After completing the Record of Completion form, mount it near the fire alarm control unit or give it to the building representative.

Chapter 7

Service and troubleshooting

Summary

This chapter provides instructions for servicing and troubleshooting the life safety system. It is intended for those trained and authorized to maintain the system.

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System repairs

The EST4 life safety system is made up of modular assemblies that are easily installed and maintained. Because of the modular design, component level field repairs mainly consist of isolating a fault to the circuit card in an assembly and replacing the defective card.

The following table provides a general guideline of recommended spare components to have on hand.

Table 48: Recommended spares list

Minimum of 1 each or 10% of the quantity installed	Minimum of 3 each or 10% of the quantity installed
• Power supply • Option cards • Amplifiers (if no backup installed in system) • Printer ribbon	• Monitor modules • Control modules • Heat detectors • Photoelectric smoke detectors • Detector base • Breakglass replacement for pull stations • Breakglass replacement for warden stations • Horn, bell, strobe, and speaker

Service and repair of system components centers around the following assumptions:

- Qualified technicians possessing a complete understanding of the system hardware and functions will perform maintenance.
- Only certified maintenance technicians will service the equipment.
- Maintenance technicians will have a readily available supply of replacement parts.

Precautions

Removing or replacing circuit board modules

When removing or replacing circuit modules, always remember to:

- First disconnect the battery then remove AC power from the control unit. Removing or replacing circuit modules when power is applied will damage the equipment.
- Avoid applying excessive force to the snap-rivet fasteners that lock the plug-in modules in place.

Handling static sensitive circuit board modules

Many of the control unit components are circuit boards that are sensitive to electrostatic discharge. To avoid damage to the board, take the following precautions:

- Use only approved grounding straps that are equipped with a 1 MΩ resistive path to earth ground.
- Remove a circuit board module from its protective antistatic packaging only for inspection or installation.
- Always hold circuit modules by the sides. Avoid touching component leads and connector pins.

Hardware problems

Hardware problems are typically identified by an intermittent or total failure of a device. The problem may occur within an equipment cabinet or with field wiring and devices.

The quickest way to locate a hardware problem is by selectively isolating portions of the system and observing the results of the isolation. By isolating smaller and smaller portions of the system, hardware faults can usually be located.

Substituting hardware

WARNING: Electrocution hazard. To avoid personal injury or death from electrocution, remove all sources of power and allow stored energy to discharge before installing or removing equipment.

Caution: Circuit boards are sensitive to electrostatic discharge (ESD). To avoid damage, follow ESD handling procedures.

The control unit's CPU automatically configures the firmware, bootstrap, and configuration database of replaced 3-SxDC1, 3-MODCOM(P), 4-ANNAUDTEL, 4-AUDTELS, and control display modules. No download from the 4-CU is required.

The following is a guideline for substitutions and replacements.

- A chassis rail module must be replaced with the same model number module.
- A UI frame assembly control-display module must be replaced with the same module number control-display module.
- A substitute module should be installed in the same card location as the module it is replacing.

Adding hardware

When hardware is added to a cabinet, a portion of the network configuration database must also be changed. The extent of the changes depends on the rule relationships between the added component and the rest of the network. Once the new hardware has been configured in the 4-CU, revised copies of the database must then be downloaded to the control unit.

Hardware troubleshooting

3-PPS/M Power Supply troubleshooting

Under most conditions, a defective power supply will be identified by the system, and annunciated as a trouble. The system may continue to operate nearly normally, as the battery connected to the faulty supply will automatically be switched into the circuit, as the load demands.

You can connect the control unit to the PC running the 4-CU and perform a power supply diagnostics check. Refer to the *4-CU Help* for details on using diagnostics tools.

Table 49: Voltage specifications

Test Point	Voltage
Output ratings	
Special applications	
Total	24 VDC at 7.0 A (internal and auxiliary outputs)
Internal DC	24 VDC at 7.0 A max.
Auxiliary DC	Two 24 VDC at 3.5 A max.
	Power-limited and supervised for ground faults and shorts
	For special applications, see the panel compatibility list
Regulated applications	
Total	24 VDC at 4.5 A (internal and auxiliary outputs)
Auxiliary DC	Two: one 24 VDC at 3.5 A maximum for all circuits except the regulated NAC; one 24 VDC at 1.0 A max. for regulated NACs to 3-IDC8/4 modules only
	Power-limited and supervised for ground faults and shorts
Input AC voltage	120 VAC, +10%, -15%, 3.0 A, 50 to 60 Hz 230 VAC, +10%, -15%, 1.5 A, 50 to 60 Hz (-230 only)
Battery charging	
Current	10 to 65 Ah
Type	1.5 or 3 A, temperature compensated dual rate

Table 50: Power Supply troubleshooting

Problem	Possible cause
Auxiliary voltage low	There is an excessive load
Batteries will not charge	- The system is in alarm mode - The NAC/AUX outputs on the power supply are configured for AUX and are loaded with more than 7.0 A - The wrong battery type is configured in the 4-CU - The battery is shorted - The battery is not wired correctly to the 3-PPS/M
System will not operate on batteries	- The battery voltage is low - The batteries were connected before AC power was energized - The batteries are defective

EST4 node troubleshooting

EST4 provides the following node types:

- 4-CPU Central Processor Module
- 4-ANNCPU Annunciator Central Processor Module
- 4-NET-AD Network Adder Module
- 4-FWALx Firewall Module
- 4-CPUGRPH Graphic Annunciator Central Processor Module

4-CPU Central Processor Module troubleshooting

The 4-CPU is a node module that processes all information from modules installed in the same cabinet and from other nodes in the life safety network.

The 4-CPU module incorporates:

- One USB 3.0, Type A connector to connect supported USB devices (5 VDC)
- One USB 3.0, Type B connector device port (usb0) to connect to a PC
- Two SFP (small form-factor pluggable) network adapter connectors (eth0, eth1)
- A network backup power connection for a 24 VDC battery source, to maintain network and audio communication in the event that the control unit needs to be powered down for service.
- Several LEDs that indicate activity on a communication path. See Figure 40 below for the LED locations on the module and Table 51 below for LED descriptions.

EST4 network node count: 1

For more information, see *4-CPU Central Processor Module Installation Sheet* (P/N 3102258-EN).

Figure 40: 4-CPU LEDs

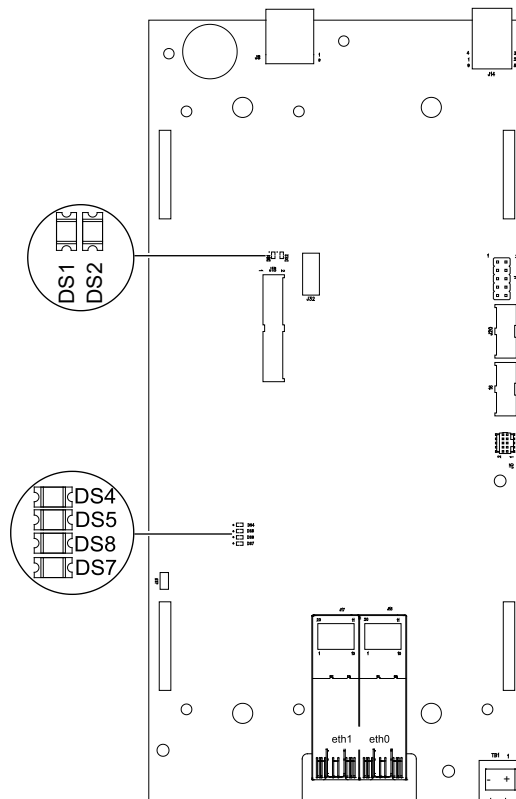


Table 51: 4-CPU LED descriptions

LED	State	Descriptions
DS1	ON - blinking	Rail communications transmit activity
DS2	ON - blinking	Rail communications receive activity
DS4	ON – blinking [1]	eth0 (J15, Figure 40 above) Network communications activity
DS5	ON – solid [2]	eth0 (J15, Figure 40 above) Network link established
DS7	ON – blinking [1]	eth1 (J17, Figure 40 above) Network communications activity
DS8	ON – solid [2]	eth1 (J17, Figure 40 above) Network link established

[1] LED indicates activity between the 4-CPU and the SFP, not the field wiring even if the remote connection has been removed or has failed.

[2] LED indicates the link between the 4-CPU and the SFP, not field wiring even if remote connection has been removed or has failed.

Table 52: 4-CPU troubleshooting

Problem	Possible cause
LEDs DS1 or DS2 is off	• CPU is not firmly seated on the chassis rail • CPU failure
LEDs DS4 and DS5 or DS7 and DS8, individually or pairs, are off	• SFP controller connection to J15 or J17 is loose • Loop was not properly terminated • Improper wire used • Ground fault has been detected
4-USBHUB connected peripheral RS-232 device is inoperative	• 4-USBHUB connection to J8 is loose • Peripheral device is off-line or improperly configured
Data or audio communication is inoperative	• SFP controller connection to J15 or J17 is loose • Improper wire used
LCD display module user interface is inoperative and the power LED is off	• Flex cable between the LCD display module and CPU is loose or defective • CPU is defective • LCD display module is defective • CPU is not configured in the 4-CU for the LCD display module • No power to the control unit

4-ANNCPU Annunciator Central Processor Module troubleshooting

The 4-ANNCPU is an annunciator node module that provides user interface and audio capability.

The 4-ANNCPU module incorporates:

- One USB 3.0, Type A connector to connect supported USB devices (5 VDC)
- One USB 3.0, Type B connector device port (usb0) to connect to a PC
- Two SFP (small form-factor pluggable) network adapter connectors (eth0, eth1)

EST4 network node count: 1

For more information about the 4-ANNCPU, see *4-ANNCPU Annunciator Central Processor Module Installation Sheet* (P/N 3102264-EN).

Figure 41: 4-ANNCPU LEDs

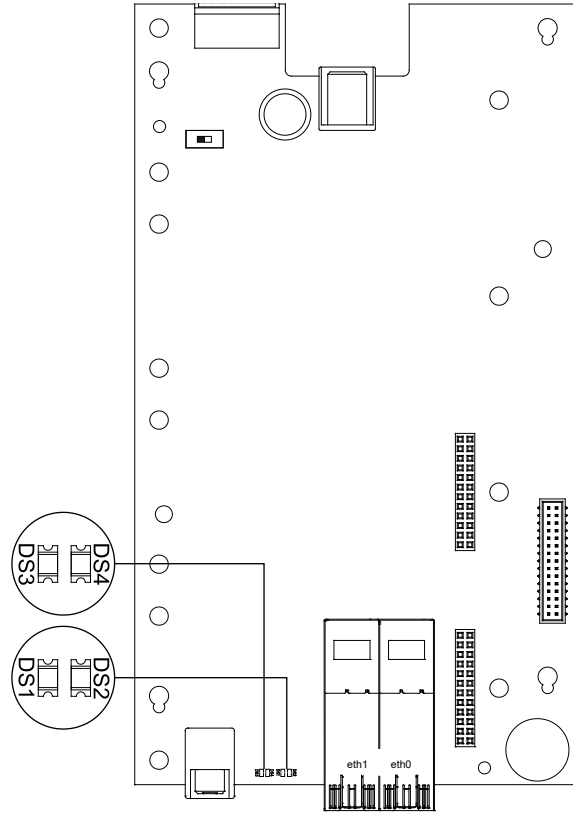


Table 53: 4-ANNCPU LED descriptions

LED	State	Descriptions
DS1	ON – blinking	eth0 Network communications activity
DS2	ON – solid	eth0 Network link established
DS3	ON – blinking	eth1 Network communications activity
DS4	ON – solid	eth1 Network link established

Table 54: 4-ANNCPU troubleshooting

Problem	Possible cause
LEDs DS1 and DS2 or DS3 and DS4, individually or pairs, are off	- SFP controller connection is loose - Loop was not properly terminated - Improper wire used - Ground fault has been detected
4-USBHUB connected peripheral RS-232 device is inoperative	4-USBHUB connection to J8 is loose - Peripheral device is off-line or improperly configured - Lack of continuity on the network wiring
Data or audio communication is inoperative	- SFP controller connection is loose - Improper wire used
Redundant 24 VDC power is inoperative	When an 4-ANNCPU is installed in an annunciator cabinet, switch S2 is not positioned to OFF

Problem	Possible cause
LCD display module user interface is inoperative and the power LED is off	• Flex cable between the LCD display module and 4-ANNCPU is loose or defective • 4-ANNCPU is defective • LCD display module is defective • 4-ANNCPU is not configured in the 4-CU for the LCD display module • No power to the control unit

4-NET-AD Network Adder Module troubleshooting

The 4-NET-AD Network Adder Module provides additional 4-NET series SFP network connections. Using the 4-NET-AD in an enclosure allows two additional network interconnections. A single enclosure can support a maximum of six network interconnections.

The 4-NET-AD module incorporates:

- One USB 3.0, Type A connector
- One USB 3.0, Type B connector
- Two SFP (small form-factor pluggable) network adapter connectors

EST4 network node count: 1

For more information about the 4-NET-AD, see *4-NET-AD Network Adder Module Installation Sheet* (P/N 3102287-EN).

Figure 42: 4-NET-AD LEDs

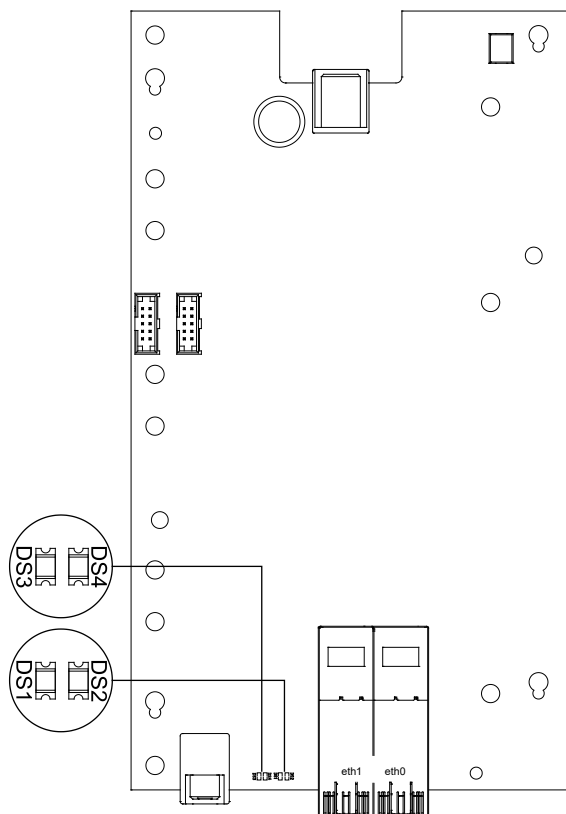


Table 55: 4-NET-AD LED descriptions

LED	State	Descriptions
DS1	ON – blinking	eth0 Network communications activity
DS2	ON – solid	eth0 Network link established
DS3	ON – blinking	eth1 Network communications activity
DS4	ON – solid	eth1 Network link established

Table 56: 4-NET-AD troubleshooting

Problem	Possible cause
LEDs DS1 and DS2 or DS3 and DS4, individually or pairs, are off	• SFP controller connection is loose • Improper wire used • Ground fault has been detected
Data or audio communication is inoperative	• SFP controller connection is loose • Improper wire used

4-FWAL1/2/3/4 Firewall Module troubleshooting

The 4-FWAL series firewall modules block unauthorized access to the EST4 life safety network, while permitting outward communication. Refer to for supported connections.

The 4-FWAL series modules provide:

- One USB host port
- One USB device port
- Connections for two SFP (small form-factor pluggable) network adapters

EST4 network node count: 1

For more information about the 4-FWAL1/2/3/4, see *4-FWAL Series Firewall Module Installation Sheet* (P/N 3102280-EN).

Figure 43: 4-FWAL1/2/3/4 LEDs

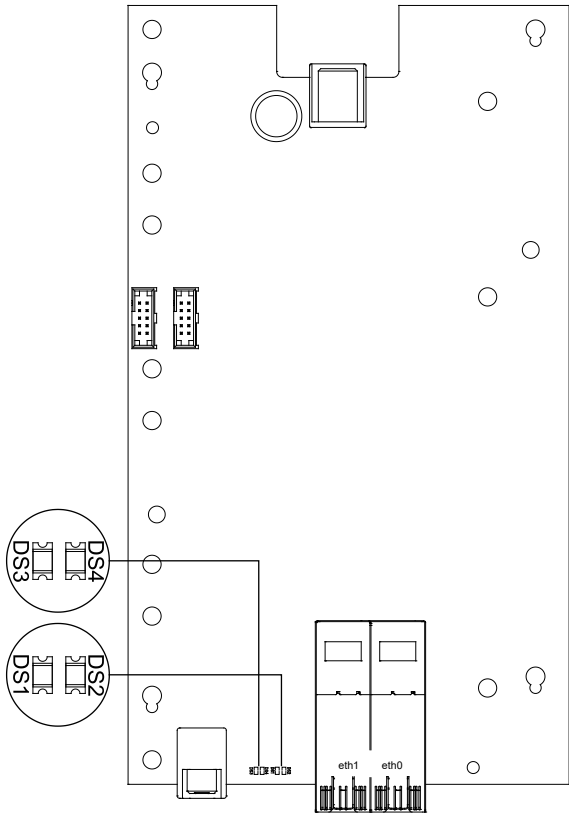


Table 57: 4-FWAL1/2/3/4 LED descriptions

LED	State	Descriptions
DS1	ON – blinking	eth0 Network communications activity
DS2	ON – solid	eth0 Network link established
DS3	ON – blinking	eth1 Network communications activity
DS4	ON – solid	eth1Network link established

Table 58: 4-FWAL1/2/3/4 troubleshooting

Problem	Possible cause
LEDs DS1 and DS2 or DS3 and DS4, individually or pairs, are off	• SFP controller connection is loose • Improper wire used • Ground fault has been detected
Data or audio communication is inoperative	• SFP controller connection is loose • Improper wire used
The control unit reported a communication service trouble	The configured DHCP address changed (i.e., the network configuration updated) before the lease expired. The node must be rebooted in order to acquire the new IP address.

4-CPUGRPH Graphic Annunciator Central Processor Module troubleshooting

The 4-CPUGRPH is a node that provides an interface to a graphical annunciator.

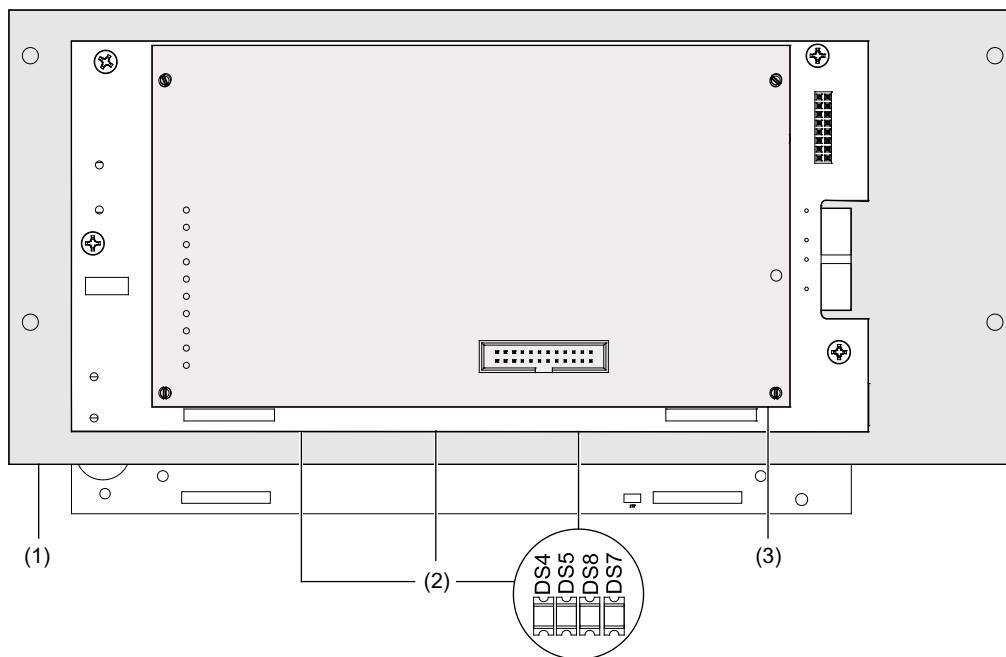
The 4-CPUGRPH module incorporates:

- One USB 3.0, Type A connector
- One USB 3.0, Type B connector
- Two SFP (small form-factor pluggable) network adapter connectors
- Network backup power connection: In the event that the panel needs to be powered down for service, a 24 VDC battery source can be connected to TB1 to maintain network and audio communications during the servicing.

EST4 network node count: 1

For more information about the *4-CPUGRPH*, see *4-CPUGRPH Graphic Annunciator Central Processor Module* (P/N 3102346-EN).

Figure 44: 4-CPUGRPH LEDs



- (1) Mounting plate
 (2) CPU (DS4, DS5, DS7, and DS8 LEDs are located on the back of the CPU card)
 (3) Graphic interface card

Table 59: 4-CPUGRPH LED descriptions

LED	State	Descriptions
DS4	ON – blinking	eth0 Network communications activity
DS5	ON – solid	eth0 Network link established
DS7	ON – blinking	eth1 Network communications activity
DS8	ON – solid	eth1Network link established

Table 60: 4-CPUGRPH troubleshooting

Problem	Possible cause
LEDs DS4 and DS5 or DS7 and DS8, individually or pairs, are off	- SFP controller connection is loose - Improper wire was used - Ground fault has been detected
Data or audio communication is inoperative	- SFP controller connection is loose - Improper wire used
Not communicating with graphic annunciator	Ribbon cable connection is loose at connector from the graphic interface card to the 3-EVPWR(A)

Network data riser

Network wiring originates from eth0 and eth1 network SFP connections on the EST4 nodes. Network wiring connections to other nodes can be made between eth0 and eth0 or between eth 0 and eth1. To function properly, the SFP eth port must be enabled using the 4-CU.

The data network riser is isolated at each node. This prevents a shorted data loop from interrupting communication on the entire loop.

When trying to isolate trouble on a data network riser, remember that both shorted and open network segments will interrupt communication between two nodes. In this case, the EST4 life safety network will reconfigure and operate as two independent sub-networks.

Fiber optics network troubleshooting

The 4-NET series fiber-optic SFP network controllers and 4-FWAL series fiber-optics SFP network adapters provide IPv6 fiber, Class A, Class B, or Class X and mesh networking topologies. The controllers mount in one of two available SFP slots in a EST4 CPU node.

LEDs on the 4-CPU module indicate activity on the fiber communication path. See Figure 40 on page 139 for the LED locations on the module and Table 51 on page 139 for LED descriptions.

Note: If a control unit must be powered down for service, connect a backup power source to the 24 VDC terminals (TB1) on the electronics card to maintain network communication.

For more information, see *4-NET Series Fiber Optics SFP Network Controllers Installation Sheet* (3102271-EN).

Table 61: EST4 fiber-optics SFP network controllers

Model	Description
4-NET-SM	SFP network media interface, dual filament, single-mode fiber-optic, with a 9/125 μ (G.652) fiber pair up to 8.7 miles (14 km).
4-NET-MM	SFP network media interface, multi-mode fiber-optic, with a 50/125 μ (OM3/OM4) fiber pair up to 1.24 mi. (2 km), 62.5/125 μ (OM1) fiber pair up to 0.62 mi. (1 km), or a 100/140 μ fiber pair up to 150 m.
4-NET-SMH	SFP network media interface, dual filament, single-mode fiber-optic, high-power output, with a 9/125 μ (G.652) fiber pair up to 24.8 mi (40 km).
4-NET-SMU	SFP network media interface, single filament, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-NET-SMU must be paired with a 4-NET-SMD.
4-NET-SMD	SFP network media interface, single filament, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-NET-SMD must be paired with a 4-NET-SMU.
4-FWAL-SM	SFP network media interface, single-mode fiber-optic, with a 9/125 μ (G.652) fiber pair up to 8.7 miles (14 km).

Model	Description
4-FWAL-MM	SFP network media interface, multi-mode fiber-optic, with a 50/125 μ (OM3/OM4) fiber pair up to 1.24 mi. (2 km), 62.5/125 μ (OM1) fiber pair up to 0.62 mi. (1 km), or a 100/140 μ fiber pair up to 150 m.
4-FWAL-SMH	SFP network media interface, single-mode fiber-optic, high-power output, with a 9/125 μ (G.652) fiber pair up to 24.8 mi (40 km).
4-FWAL-SMU	SFP network media interface, bi-directional, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-FWAL-SMU must be paired with a 4-FWAL-SMD.
4-FWAL-SMD	SFP network media interface, bi-directional, single-mode fiber-optic with a 9/125 μ (G.652) fiber up to 6.2 miles (10 km). The 4-FWAL-SMD must be paired with a 4-FWAL-SMU.

CAT5 and twisted-pair network troubleshooting

The 4-NET-CAT and 4-NET-TP(-HC) twisted pair SFP network controllers provide IPv6 copper, Class A, Class B, or Class X and mesh networking topologies. The controllers mount in one SFP slot in a EST4 CPU.

LEDs on the 4-CPU module indicate activity on the communication path. See Figure 40 on page 139 for the LED locations on the module and Table 51 on page 139 for LED descriptions.

Note: If a control unit must be powered down for service, connect a backup power source to the 24 VDC terminals (TB1) on the 4-CPU electronics card to maintain network communication.

For more information about the 4-NET-CAT, see *4-NET-CAT 100 Mbps SFP Network Controller Installation Sheet* (3102270-EN).

For more information about the 4-NET-TP or 4-NET-TP-HC, see *4-NET-TP(-HC) Twisted Pair SFP Network Controller Installation Sheet* (3102270-EN).

4-24L series control-display modules troubleshooting

The 4-24L series control-display modules mount on the UI frame assemblies attached to the inner door.

Performing a lamp test will quickly isolate hardware problems from programming problems with any control-display module. See “Performing a lamp test” on page 57.

Table 62: Control-display module troubleshooting

Problem	Possible cause
The indicators and buttons are inoperative	- There is no power to the control unit - The control-display module is not properly seated on the UI frame assembly - The control-display module is defective - The control-display module is not defined in the 4-CU
The indicators respond incorrectly	- The control-display or indicator is not defined in the 4-CU - A rule governing indicator operation is not correctly written
A button does not perform the expected function	- The control-display or button is not defined in the 4-CU - A rule governing button operation is not correctly written

3-ZAxx zoned amplifier modules troubleshooting

Table 63: 3-ZAxx Zoned Audio Amplifier module troubleshooting

Problem	Possible cause
The audio output level is too low	- A jumper is set for 25 VRMS but connection is to a 70 VRMS circuit - The gain is set too low
There is no audio output or is extremely low	- An open fuse needs replaced (not field replaceable) - The gain is set too low - The prerecorded message/tone is too low - There is an incorrect channel activation
The audio output level is too high	- A jumper is set for 70 VRMS but connection is to a 25 VRMS circuit - The gain is set too high
The amplifier current is limiting	- The audio circuit overloaded - The input level to the 4-MIC is too high
An incorrect amplifier version is reported by the CPU	- The jumpers are installed incorrectly - An incorrect model amplifier installed

3-OPS Off-Premises Signal module troubleshooting

Table 64: 3-OPS Off-Premises Signal module troubleshooting

Problem	Possible cause
The module is in trouble state	- The master box circuit is open or not reset - The reverse polarity circuit is open - A 3.6 kΩ EOL resistor is not installed on unused circuits
The remote receiver indicates a loop trouble and does not receive an alarm	- The circuit polarity is reversed - A circuit is open - There is excessive circuit resistance - A receiver is incompatible - The module is defective
The remote receiver does not indicate a loop trouble and does not receive an alarm	- The module is not defined in the 4-CU - A receiver is incompatible - The module is defective

3-IDC8/4 Initiating Device Circuit module troubleshooting

Table 65: 3-IDC8/4 Initiating Device Circuit module troubleshooting

Problem	Possible cause
The module is in trouble state	- A 4.7 kΩ EOL resistor is not installed on unused IDC circuits - A 15 kΩ EOL resistor is not installed on unused NAC circuits - There is no communication with the CPU module - The module is not defined in the 4-CU - The field wiring connector is not plugged into the module

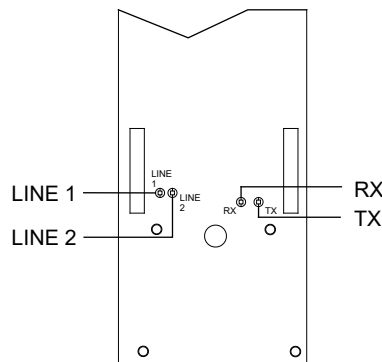
Problem	Possible cause
NAC output is not working	- The jumpers are installed incorrectly - An external source is configured but not connected - The circuit is overloaded - The circuit is silenced - The circuit is shorted - A polarized device is defective or reversed on the circuit - Output was not programmed
The IDC circuit is not working	- There is an incompatible two-wire smoke detectors - There is excessive wiring resistance or capacitance - There is an open circuit on the line

3-MODCOM(P) Modem Communicator module troubleshooting

LED indicator diagnostics

LINE 1 and LINE 2 LEDs on the 3-MODCOM(P) provide diagnostic information. See the tables below for a description of the LEDs and their dialing and data transmission states.

Figure 45: 3-MODCOM(P) LED indicators



Label	Description	Label	Description
LINE 1	Indicates line 1 telephone activity	RX	Indicates receive activity
LINE 2	Indicates line 2 telephone activity	TX	Indicates transmit activity

Table 66: 3-MODCOM(P) LINE 1 and LINE 2 LED states

LED state	LINE 1 description	LINE 2 description
Off	There is no activity	There is no activity
On	LINE 1 has been seized	LINE 2 has been seized
Slow flash	Dialer or modem data is being passed on LINE 1	Dialer data is being passed on LINE 2 (modem data is passed only on LINE 1)
Slow flash (both LEDs)	The application code or configuration code is downloading from the CPU or 4-CU	
Fast flash	There is ringing on LINE 1 (flashing pattern detected)	N/A (LINE 2 does not have ring detection)

Audible diagnostics

Obtain an audio amplifier device locally for listening to the distinctive sounds associated with dialing, receiving handshakes, transmitting data, and receiving acknowledgements. Place a 0.1 μ F, 200 V or greater capacitor in series with one of the leads. Alternately, you can use a lineman's handset in monitor mode.

During downloading from a remote computer, you will hear the distinct sound of modems establishing a connection, and then a series of rapid chirps as data is transmitted.

Note: Remove the audio amplifier when you finish troubleshooting.

Common problems

Evaluation of visual and audible indications will usually serve to isolate the source of trouble. Before replacing a 3-MODCOM(P), investigate the following common causes of a module problem.

- The module is not properly seated on the electronics chassis, or one or more connector pins are bent away from the associated connectors
- A modular telephone plug is not connected to the appropriate line 1 or line 2 jack, is not fully seated, or is not connected at the telephone block
- The module is configured with incorrect CMS telephone numbers
- The telephone line is faulty

If the module and telephone line are okay, check the CMS telephone number by dialing it using a standard telephone plugged directly into the RJ-31X jack. (The jack will accommodate a standard modular phone plug.) You should:

1. Hear a dial tone when going off-hook
2. Lose the dial tone after dialing the first digit
3. Hear the receiver ringing
4. Hear the CMS receiver go off-hook and send a handshake tone

Typical problems dialing the CMS involve missing or incorrect area codes, the need to dial 1 for long distance, or missing line access codes (for example, dialing 9 for an outside line).

If the receiver answers, check that it is sending out the correct handshake. For Contact ID, the handshake signal consists of two short tones of different frequency. For TAP, there should be a modem-type exchange of handshake messages.

If the receiver sends the correct handshake and the 3-MODCOM(P) transmits data but the receiver does not send an acknowledgement, check that the receiver is compatible with the desired protocol. (SIA DCS standard is available from the Security Industry Association). Typical problems involve an incompatible format or data message.

If the handshake and acknowledge signals are audible, check that the correct account number was configured in the 3-MODCOM(P) and the code being sent was correctly programmed in the CMS computer.

Where a 3-MODCOM(P) module is suspected of being faulty, try substituting a known good one that has been properly programmed.

4-MIC paging microphone troubleshooting

Figure 46:4-MIC

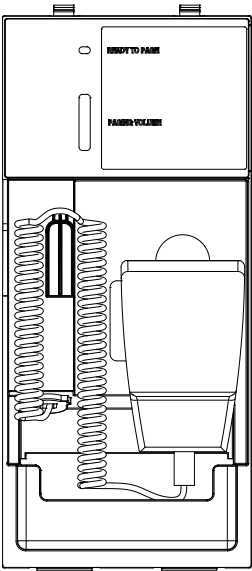


Table 67: 4-MIC troubleshooting

Problem	Possible cause
Microphone functionality is not working	• The 4-CPU is missing the 4-AUDTELS or the 4-ANNCPU is missing the 4-ANNAUDTEL cards • The 4-MIC is not properly seated in the UI frame assembly • The 4-MIC is not programmed • The 4-MIC is defective
Page audio is distorted	• The operator is speaking too loud into the microphone • The amplifier gain is set too high • A speaker is located too close to the 4-MIC

4-FT fire fighter telephone troubleshooting

Figure 47:4-FT

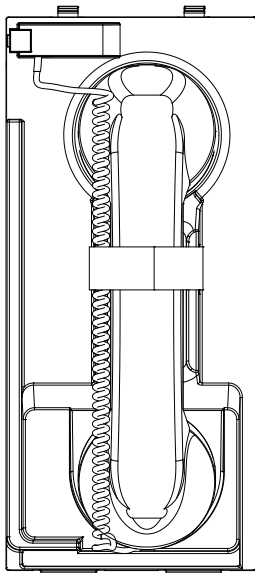


Table 68: 4-FT troubleshooting

Problem	Possible cause
Telephone functionality is not working	• The 4-CPU is missing the 4-AUDTELS or the 4-ANNCPU is missing the 4-ANNAUDTEL cards • The 4-FT is not properly seated in the UI frame assembly • The 4-FT is not programmed • The 4-FT is defective
Page audio is distorted	• The operator is speaking too loud into the telephone handset • The amplifier gain is set too high

Pseudo points

A pseudo point is an input or output point that is not a physical device. For example, ground fault and communication fault notifications. When a pseudo point event occurs an event message displays on the LCD screen. Viewing the event message details provides the pseudo point address.

Refer to the pseudo point tables in Appendix C “Pseudo points table” on page 197 where you can cross-reference the pseudo point address to its source and read a brief description.

Note: For instructions on how to view event details, see “Viewing event details” on page 48.

Understanding Signature data loops

Operation

The advanced features of the Signature controller module perform a number of advanced operations. These operations are not always apparent from the control unit. Table 69 lists a number of Signature data loop conditions and describes the loop's operational responses.

Table 69: Signature data loop operation

When you:	Response
Remove a detector, and then reinstall the detector in the same base	- A system trouble displays on the LCD screen when the detector is removed. The message shows the detector's label or address. - The system restores when the detector is reinstalled.
Remove a module or pull station, and then reinstall the module/pull station in the same location	- A system trouble displays on the LCD screen when the module/pull station is removed. The message shows the device's label or address. - The system restores when the module/pull station is reinstalled.
Remove a detector, and then install a different detector of the same type in the same base	- A system trouble displays on the LCD screen when the detector is removed. The message shows the detector's label or address. - The Signature loop controller remaps the loop, replacing the serial number of the old detector with the serial number of the new detector, when mapping is enabled. All of the old detector's sensitivity and verification settings are transferred to the new detector. The system returns to normal when mapping is finished. - The communication fault for the old detector remains, when mapping is disabled.
Remove a module or pull station, and then reinstall a different module/pull station of the same type in the same location. Note: A SIGA-UM replacement module must have jumper JP1 set in the same position as the original module.	- A system trouble displays on the LCD screen when the module/pull station is removed. The message shows the device's station label or address. - The Signature loop controller remaps the loop, replacing the serial number of the old device with the serial number of the new device, when mapping is enabled. If the devices are modules (not pull stations), the old module's personality codes are transferred to the new module. The system returns to normal when mapping is finished. - The communication fault for the old detector remains, when mapping is disabled.
Remove a detector, and then reinstall a different detector type in the same base.	- A system trouble displays on the LCD screen when the detector is removed. The message shows the detector's label or address. - The Signature loop controller remaps the loop, replacing the serial number of the old detector with the serial number of the new detector, when mapping is enabled. All of the old detector's sensitivity and verification settings are transferred to the new detector. The new detector will be operational but a system trouble displays, indicating a device type mismatch. To clear the trouble, the new detector type must be assigned to the base using the 4-CU and the database downloaded to the control unit. - The communication fault for the old detector remains, when mapping is disabled.

When you:	Response
Remove a module or pull station, and then reinstall a different module/pull station type in the same location.	• A system trouble displays on the LCD screen when the module/pull station is removed. The message shows the device's station label or address. • The Signature loop controller remaps the loop, replacing the serial number of the old device with the serial number of the new device, when mapping is enabled. The new device is <i>not</i> operational. A system trouble displays, indicating a device type mismatch. To clear the trouble, the new device type must be defined using the 4-CU and the database downloaded to the control unit. • The communication fault for the old detector remains, when mapping is disabled. • If a single address module is replaced with a dual address module or vice versa, a map fault is generated by the address count mismatch.

Signature data loop troubleshooting basics

The 3-SDC1 loop controller card provides one Class B or Class A signaling line circuit that supports up to 125 detectors and 125 module addresses. The card also provides resettable 24 VDC for powering conventional two-wire smoke detector circuits on Signature Series modules. When a device is removed from the loop, the loop controller recognizes the change and the control unit processes the information.

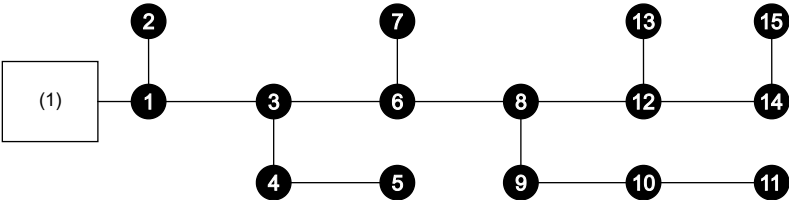
Note: The 3-SDC1-HC is an optional replacement card for the 3-SDC1 and should be used for installations where the signaling line circuit has more than 90 isolator modules and isolator bases (with SIGA1 sensors installed).

Isolating loop and device problems

The process of isolating a problem on a Signature data loop is similar to that used on a conventional fire alarm Initiating Device Circuit (IDC). An accurate and complete wiring diagram of the loop installation is the best troubleshooting aid available. When the diagram is used in conjunction with the information provided by the control unit, you can easily isolate open conditions or defective devices. The loop shown in Figure 48 below will be used to illustrate basic troubleshooting techniques.

Note: When troubleshooting Class A loops, disconnect the loop from the return (loop A) terminals and temporarily jumper both loop A terminals to the respective loop B terminals. You can then troubleshoot the loop as a Class B loop.

Figure 48: Normal Signature data loop topology

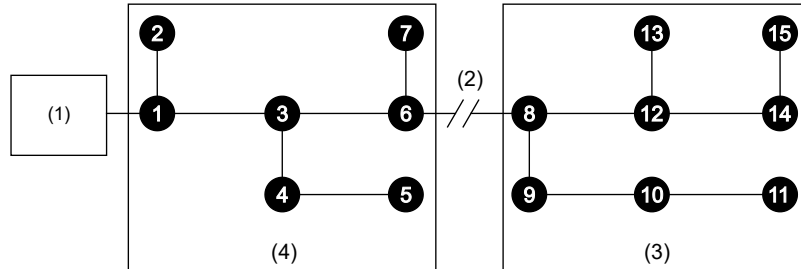


(1) Signature loop controller

Open circuit conditions

On a loop with an open fault, the Signature modules communicate with devices up to the break. The control unit LCD screen displays a trouble condition for all devices beyond the break. Figure 49 shows devices 1 through 7 continuing to operate and devices 8 through 15 reporting device troubles.

Figure 49: Open fault on the data loop



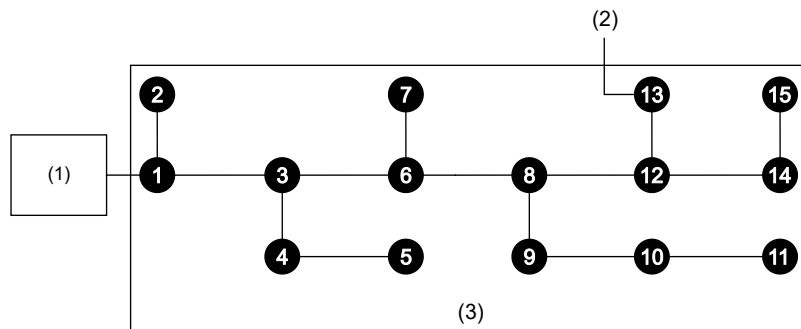
- (1) Signature loop controller
- (2) Break in the loop
- (3) Devices in trouble
- (4) Devices operating normally

In Figure 49, a wire break or intermittent connection between devices 6 and 8 is the most probable cause of the failure. Other possible causes include a device failure in devices 9 through 15, failure to define them in the loop controller's database, or failure to define them correctly in the 4-CU.

Short circuit conditions

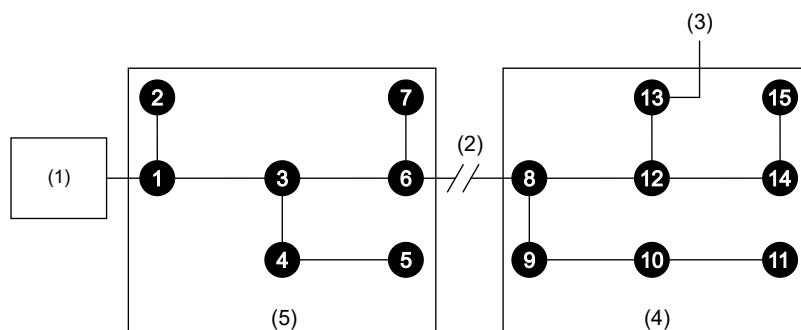
Short circuit conditions require selective isolation of portions of the loop to systematically narrow down the fault's location. A shorted circuit typically reports to the control unit a trouble condition on all devices, as shown in Figure 50.

Figure 50: Wiring short on the data loop



- (1) Signature loop controller
- (2) Wiring short location
- (3) Devices in trouble

To isolate the short, open the data loop at a point that will disconnect approximately 50% of the installed devices as shown in Figure 51.

Figure 51: Isolating a short on the data loop

- | | |
|---------------------------------------|---------------------------------|
| (1) Signature loop controller | (4) Devices remain in trouble |
| (2) Open in loop to isolate the short | (5) Devices return to operation |
| (3) Wiring short location | |

After opening the loop, if some of the devices restore, the short is located on the portion of the loop that has been disconnected. If no devices restore, the short has been isolated to the first 50% of the loop.

Reconnect the previously isolated portion of the loop. If during the first isolation process some devices restored, open the loop at a location *electrically farther* from the loop controller, and then repeat the analysis. If during the first process no devices restored, open the loop at a location *electrically closer* to the loop controller, and then repeat the analysis.

Continue increasing or decreasing the number of devices on the opened loop leg until you isolate the device or wire segment causing the problem.

Distinguishing short circuits from off-hook conditions in telephone risers

If local regulations require the ability to distinguish between a short circuit and an off-hook condition in a telephone riser, you must configure the circuit so that it functions as a four-state telephone. Compatible riser selector modules and telephone sets are listed below.

- SIGA-CC1 riser selector
- SIGA-CC1S riser selector
- SIGA-CC2 riser selector
- SIGA-MCCA riser selector
- SIGA-MCC1 riser selector
- SIGA-MCC1S riser selector
- Portable handset and receptacle (P/N 6830-3 and 6833-4) telephone module
- Remote telephone and wall box, Break Glass (P/N 6830-4 and 6831-1, or 6831-3) telephone module
- Remote telephone and wall box, Nonbreak Glass (P/N 6830-4 and 6831-2, or 6831-4) telephone module

Note: For instructions on configuring a four-state telephone, refer to the installation sheet supplied with the SIGA input or output module.

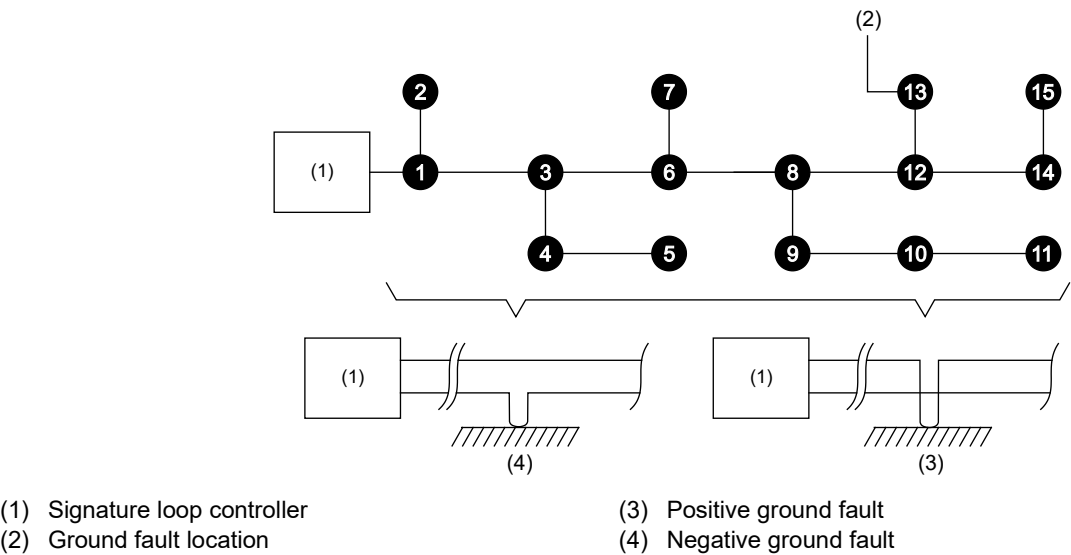
Ground fault conditions

Ground fault conditions require selective isolation of portions of the data loop to systematically narrow down the fault's location. A loop with a ground fault (approximately 10 kΩ or less to ground) activates the Ground event queue indicator on the LCD screen. The conditions can occur on the Signature loop, the 24 VDC smoke power circuit, or the input circuits to 3-SDC1(-HC), 3-SSDC1, and 3-SDDC1 loop controllers. The general location of a ground fault can be determined by viewing indications and messages on the control unit user interface (see Table 70).

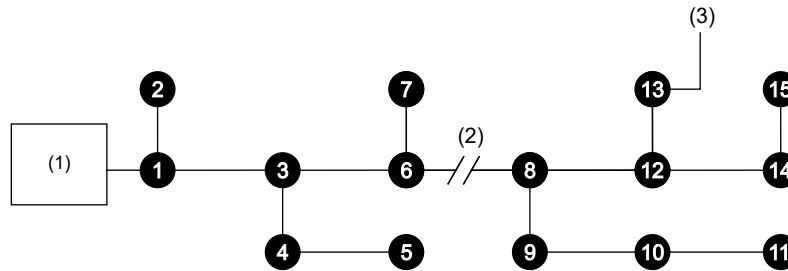
Table 70: Ground fault indications

Control unit user interface indications	Ground fault location
The Ground event queue indicator activates but no event message shows in the Ground Fault event list	• Loop controller circuit • 24 VDC smoke power circuit
The Ground event queue indicator activates and the event message appears in the Ground Fault event list	Positive leg of the input circuit for the device

Figure 52: Data loop ground faults



To isolate the ground fault, open the suspect loop (both conductors) at a point that will disconnect approximately 50% of the installed devices as shown in Figure 53. A similar technique is used on smoke power or module input circuits.

Figure 53: Data loop ground fault isolation

- (1) Signature loop controller (ground fault LED Off)
 (2) Both conductors open insulates the ground fault
 (3) Ground fault

After opening the loop, if the GND Fault event clears, the ground fault is located on the portion of the loop that has been disconnected. If the LED remains on and no devices restore, the short has been isolated to the first 50% of the loop.

Reconnect the previously isolated portion of the loop, and then open the loop at a new point. If during the first isolation process the GND Fault event cleared, open the loop at a location *electrically farther* from the loop controller, and then repeat the analysis. If during the first process the GND Fault event remained, open the loop at a location *electrically closer* to the loop controller, and then repeat the analysis.

Continue increasing or decreasing the number of devices on the opened loop leg until you will isolate the device or wire segment causing the problem.

Notes

- The ground fault detection circuitry requires approximately 30 to 40 seconds response time when the fault is removed.
- The EST4 control unit performs a ground fault test for 2 seconds at 18-second intervals. If the system is working properly, the voltage between earth ground and logic negative should be between 12.3 VDC and 16.8 VDC during the 2-second test. The system reports a ground fault when the voltages are less than 12.3 VDC and more than 16.8 VDC. In a non-faulted system, the voltage outside the 2-second test period may float randomly. If the system is faulted, then the voltage is likely to be a fixed value such as 3 or 19 VDC.

Substituting Signature Series devices

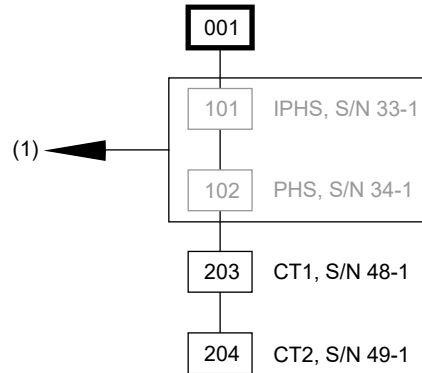
When substituting a *known good* detector or module in place of a suspect device, the following scenarios can take place.

- With mapping enabled, if the substituted device is the same model as the suspect device, the system accepts it with no further operator action. When the substituted device is installed, the system goes into trouble. When the quantity of devices defined on the loop is reached, the system automatically remaps the loop, stores the revised information, and returns to normal. This process may take a few minutes.
- With mapping enabled, if the substituted device is a different model than the suspect device, when the device count is correct, the Signature controller module automatically remaps the loop. A trouble occurs at the address of the suspect device as the result of a map fault because the known good device's parameters differ from those of the removed suspect device. You must accept the parameters of the known good device to remove the fault. These can be changed later.
- If the substituted device is a DH or DS detector, the system automatically disables the mapping feature. As a result, a Map Fault and Mapping Disabled monitor event is generated. You must remove the detector and restart the control unit to restore mapping operations.

Detectors

When one or more devices are removed from a Signature loop for servicing, as shown in Figure 54, the control unit LCD screen displays a trouble condition for each device. If the detector is removed from an isolator base, the isolator will transfer.

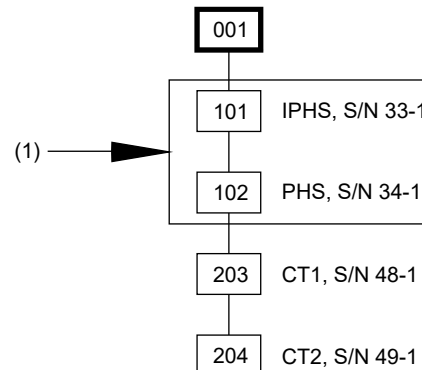
Figure 54: Detectors removed for service



(1) Removed detectors

If the devices are returned to their original locations, as shown in Figure 55, the map supervision function recognizes the detectors have been returned as originally installed and mapped, and no additional action is taken.

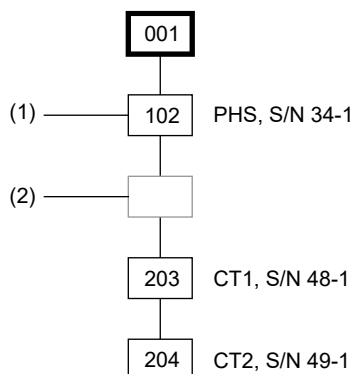
Figure 55: Detectors returned to original location



(1) Detectors returned

If the devices are returned to different locations, the map supervision function recognizes that previously mapped serial numbers occupy new map locations. When the mapping supervision function recognizes the need to remap the loop, the control unit is put in the map pending state. In the map pending state, the control unit automatically remaps the loop when the quantity of reinstalled devices is equal to or greater than the quantity of devices defined in the original map.

In Figure 56, the PHS, S/N 34-1 detector originally installed at address 102 has been installed in the location originally occupied by the IPHS, S/N 33-1 detector.

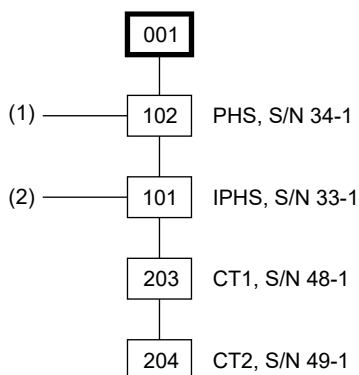
Figure 56: One detector returned to a new location on the loop

- (1) PHS, S/N 34-1 detector installed in a new location
 (2) IPHS, S/N 33-1 detector not installed

Until all devices are returned on the loop and the loop automatically remaps, the original S/N-to-control unit address correlation is still valid. Figure 56, shows that the device address moves with the detector until the loop is remapped. In this example, relocating the PHS detector temporarily relocated address 102. Until all devices are returned and the loop remapped, testing a relocated detector will cause the control unit to respond as though the detector was still installed in its original location.

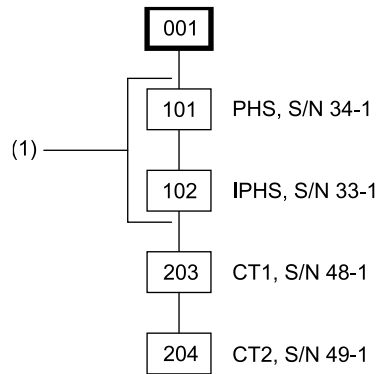
During mapping, all devices remain operational and are capable of initiating an alarm. Figure 57 below shows that both the IPHS and the PHS detectors retain their old S/N-to-control unit address correlations while the loop is mapping. Mapping activity is indicated on the control unit LCD screen.

Once mapped, the mapping supervision function automatically correlates a control unit address to a specific map location until manually changed using the CU configuration utility.

Figure 57: Both detectors returned to new locations

- (1) PHS, S/N 34-1 detector returned to service in new location before remapping
 (2) IPHS, S/N 33-1 detector returned to service in new location before remapping

Figure 58 shows the map after remapping. Note that the new S/N-to-control unit address correlations have been made. The IPHS detector is now correlated with address 102 and the PHS detector is correlated with address 101. The relocated devices will now respond as programmed for the original address location.

Figure 58: Both detectors remapped

(1) Remapped detectors PHS, S/N 34-1 and IPHS, S/N 33-1

When a factory-new detector replaces an in-service detector, the new detector is operational with a default address of 00 until it is mapped. When the loop is remapped, the new detector is given the address assigned to its map location. If a factory-new detector is added over and above the expected number of devices on the loop, it is operational with a default address of 00. However, the control unit reports a trouble because the *actual map* contains one more device than the *expected map*.

Modules

When a module is replaced with another module of the same type, automatic remapping assigns the replacement module the personality code of the module originally installed at that map location.

If a module is replaced with a module of a different type one of three things can happen.

1. If a single address module, such as the SIGA-CT1 or SIGA-CC1, is replaced with a different type of single input module, the loop remaps all devices. However, the new module type will not operate because of incompatible personality codes. A map fault generates because the actual device differs from the expected device. The new device type must be defined in the 4-CU and the database downloaded into the control unit. The map fault will then clear.
2. If a single address module is replaced with a dual address module, the control unit unsuccessfully attempts to remap all devices. A map fault generates because the actual device differs from the expected device, and the dual address module does not operate. To clear the map fault, define the new module using the 4-CU, and then download the database into the control unit.
3. If a dual address module is replaced with a single address module, the control unit never automatically attempts to remap all devices because the control unit does not see enough devices (one address less) on the loop. A manual remap will also be unsuccessful. A map trouble shows on the control unit LCD screen as the control unit remains in map pending mode. To clear the trouble, define the new module using the 4-CU, and then download the database into the control unit.

Notes

- Do not replace factory-programmed devices such as pull stations and MM1 modules with a SIGA-CT1.
- For mapping purposes, give all manual pull stations the device type *Pull*, regardless of their model numbers.

Device type replacement

If a different Signature device model is substituted for a suspect device, when the device count is correct the Signature controller module automatically remaps the loop. A trouble occurs at the address of the suspect device as the result of a map fault, because the known good device's parameters differ from those of the suspect device that was removed from the loop. To clear the map fault, accept the parameters of the known good device. You can change the parameters later.

Note: Signature Series devices require a solid connection at their terminals. If a wire can wiggle, it will be subject to contact resistance variations due to temperature changes. A loose wire can result in an intermittent connection, which will affect communication between the Signature devices and the control module. Using a proper size screwdriver, securely tighten all wiring connections.

Substituting Signature loop controller modules

When substituting a known good Signature Series loop controller module in place of a suspect module, it must be defined in the 4-CU and the database downloaded into the control unit.

The Signature loop controller module has two separate memories.

1. The first memory stores the firmware code that makes the module operate. If there is a problem with the firmware, or if an upgrade has been issued, download the new firmware into the module from the 4-CU. When upgrading the firmware, do not download the bootstrap data unless specifically instructed to do so.
2. The second memory stores the Signature loop controller configuration settings that were compiled in the 4-CU. If you suspect that the module itself is bad, download the loop configuration settings for the loop to which the substitute module is connected.

Substituting Signature Series devices on a nonmapping loop

When nonmapping DH and DS detectors are configured on a Signature loop, mapping is disabled for the loop. When substituting a known good detector or module in place of a suspect device, one of the following scenarios can take place.

1. If the substituted device is the same model as the suspect device and preprogrammed with the same device address, the system accepts it with no further action. When the substituted device is installed, the system goes into trouble. When the quantity of devices defined on the loop is reached, the system stores the revised information, updates the device's personality code and other parameters, and returns to normal. This process may take a few minutes.
2. If the substituted device is a different model than the suspect device but is preprogrammed with the same device address, the system accepts it but generates a device fault indicating the difference in models. The details of the trouble indicate a "bad type fault". To remove the fault, you must accept the model of the known good device using the 4-CU or replace the substituted device with a model that matches the suspect device.
3. If the substituted device is preprogrammed with a different device address, a trouble occurs at the address of the suspect device because it was removed. The system may also generate a device trouble if the substituted device address matches another device already configured on the loop or it may generate an "unprogrammed device" trouble. You must remove the substituted device, and then use the SIGA-PRO tool or the 4-CU to reprogram its address to remove the fault.

Signature loop controller troubleshooting

Table 71 provides a list of possible problems that may be detected by the loop controller module. For information on identifying and locating Signature SLC problems, see "Signature mapping diagnostics" on page 164.

Notes

- Do not replace factory-programmed devices such as pull stations and MM1 modules with a SIGA-CT1.
- For mapping purposes, give all manual pull stations the device type *Pull*, regardless of their model numbers.

Table 71: Signature Series loop controller troubleshooting

Problem	Possible cause
An open is detected on the loop	• The loop is incorrectly wired or a connector is loose • A detector or isolator base is defective • A conductor is broken • A device is not installed on the loop • A device is not defined in the 4-CU
A short is detected on the loop	• The loop is incorrectly wired (often crossed wires on a device base) • A detector, detector base, or module is defective • The insulation between conductors is nicked
A ground fault is detected on the loop	• There is a pinched wire between the device and electrical box • The wire insulation is nicked
An internal trouble is detected on the loop	• A detector is defective or dirty • A CO module is defective or not seated correctly • A photo or heat sensor is reading too high or too low • A light source is affecting a detector • There is a short between a detector and detector base • An isolator or relay base is defective • Devices are drawing too much or too little current during mapping
A maintenance alert is detected on the loop	A detector is nearing end of life
The system reports an “End of life ACT” event	A CO module needs replacement

Signature device troubleshooting

Module LED indicators

Each Signature Series module has a red and green status LED. These LEDs are useful when trying to determine the communication and alarm or active status of Signature devices.

Table 72: Signature Series module status LEDs

LED status	Device status
Green flashing	Normal communication
Red flashing	Alarm or active (either input of dual input modules)
Red and green steady	Stand-alone alarm or active (either input of dual input modules)

Common causes of Signature Series module and device problems

Common causes for Signature Series modules and devices not responding correctly or reporting trouble events are listed below.

- The module/device is installed in the wrong location or is not addressed correctly
- The module/device is not defined in the 4-CU database
- An incorrect personality code was programmed into the module
- The jumper is in the wrong position
- 24 VDC for smoke power is low or missing
- Inputs 1 and 2 were switched
- There is a ground fault
- There is incorrect wiring
- There are mapping errors
- A short or open is detected on the output circuit
- The polarized device is installed in reverse
- There is an incorrect or missing EOL resistor
- The module/device is defective

Signature mapping diagnostics

The 4-CU mapping feature allows you to perform diagnostics on all devices connected to a Signature loop controller module. When mapping is turned on for the signaling line circuit, you can upload data from the SLC to the 4-CU to view device information and placement on the circuit. By using the mapping tool you can compare actual data with expected data to remove any conflicts between the configured map and the current map. See “Uploading Signature diagnostics from the control unit” on page 166 for instructions on uploading data.

Refer to Table 73 for suggested corrective actions. Refer to *4-CU Help* for details on how to identify and locate Signature module and device problems.

Table 73: Signature mapping errors diagnostics

Problem	Suggested corrective action
The mapping command failed either because the sensor did not draw current or it was not possible to obtain stable mapping data from the signaling line circuit	• Verify that wiring is correct • Verify that devices are operational • Review 4-CU Signature Diagnostics to identify failed devices • Check for loose wiring connections at the devices or T-taps • Check for faulty device(s)
While mapping a chain from a device back to the Signature loop controller, the chain was built with holes in it	• Review 4-CU Signature Diagnostics to identify failed devices • Review 4-CU Signature Diagnostics to compare the serial numbers or addresses with the actual wiring to identify the conflict
Map tables are inconsistent	• Upload the current map • Compare the current map with the expected map • Write the map back to the loop controller • Ensure loop wiring is correct

Problem	Suggested corrective action
The actual Signature data loop map does not match the expected map	• Compare the current map with the expected map • Write the map back to the loop controller • Ensure loop wiring is correct
The system continues to remap the data loop	• An intermittent connection is causing one or more devices to lose then reestablish communication with the loop controller • There is a defective device or detector base
Device address assignment failed	Review the serial number or address; if missing, replace the device
The map in use has invalid data (map supervision failure)	Wait for automatic map reconstruction to complete before continuing
Mapping supervision detected a change on the loop (a map rebuild was scheduled)	Wait for automatic map reconstruction to complete before continuing
The mapping command failed because the sensor did not draw current, or it was not possible to obtain stable mapping data from the loop	• Check for loose wiring • Check for a defective device
Mapping supervision detected that the supervised device's type changed (a map fault was flagged)	• Replace the device • Correct the loop controller programming
Mapping aborted because a short or open was detected on the loop wiring	• Check for an open or short on a Class A loop • Check for a short across the entire Class B loop • A reset may restart mapping
Control unit startup is not able to recreate the current map	Wait for the automatic map reconstruction to complete before continuing
Mapping is disabled	Enable mapping
While mapping a chain from a device back to the loop controller, the chain appears to have two devices at the same location	• Check for faulty wiring or a faulty device on the loop • Review 4-CU Signature Diagnostics to identify the conflict
More than 125 end-of-line devices are detected on the loop	• Correct the wiring • Remap the loop • Reduce the number of T-taps
Mapping detected a difference between the device at the end-of-line and the devices in its chain	• Review 4-CU Signature Diagnostics to identify the conflict • Review 4-CU Signature Diagnostics to compare the serial numbers or addresses with the actual wiring to identify the conflict

Uploading Signature diagnostics from the control unit

The control unit allows you upload data from the SLC to the 4-CU for troubleshooting devices on the circuit.

Note: An RNDIS driver is required for the PC port to communicate with the control unit. If necessary, install it before proceeding.

To download the project database:

1. Connect the Type B end of a USB cable to the J14 USB port on the 4-CPU. See Figure 5 on page 41.
2. Connect the Type A end of the cable to a standard USB port on the computer with the 4-CU.
3. From the control unit LCD screen:
On the Action bar, tap System Info.
On the Command bar, tap 1 Hour Access Code or 8 Hour Access Code to obtain a 4-CU required access code that will allow you to download to the control unit.
4. From the 4-CU:
On the Project Management tab, open the project.
On the Home tab, in the Communication group, click Enter Access Code.
Enter the Access Code, and then click Validate.
On the project tree, select the Signature loop controller, and then the SLC to be mapped.
On the Home tab, in the Communication group, click Mapping Options, and then choose Upload Signature Data from > Panel. A progress bar displays in the lower right side of the CU window during the upload.
Note: Refer to *4-CU Help* for details on other options on the Mapping Options list.
5. From the control unit LCD screen, when finished uploading:
On the Action bar, tap System Info.
On the Command bar, tap Cancel Access Code, to activate the Remote Write Lock command.
6. Refer to *4-CU Help* for details on using the Signature diagnostics data and reports.

Appendix A

System calculations

Summary

This appendix provides instructions and worksheets for calculating wire lengths and sizing standby batteries.

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Network data riser limits

Cumulative data network capacitance refers to the total capacitance of all copper wire used for the data riser. The cumulative capacitance of data networks must be within certain limits to permit stable network communications.

The data network retransmits data bit-by-bit. This method of retransmitting data restores the amplitude of a bit at each node, but any distortions in bit timing are passed through to the next node. Data network communication faults begin to occur at about 23% distortion of bit timing.

A fiber link in a data network electrically isolates two nodes, but distortions in bit timing are not restored by the fiber segment. Distortions in bit timing are passed through the fiber to the next node. The bit transition time of fiber modules is fast enough to be neglected in determining the maximum wire length that can be used in the data network.

For EST4 network media risers and cable specifications, see *EST4 Audio and Network Application Guide (P/N 3102306-EN)*.

Signature data loop wire length

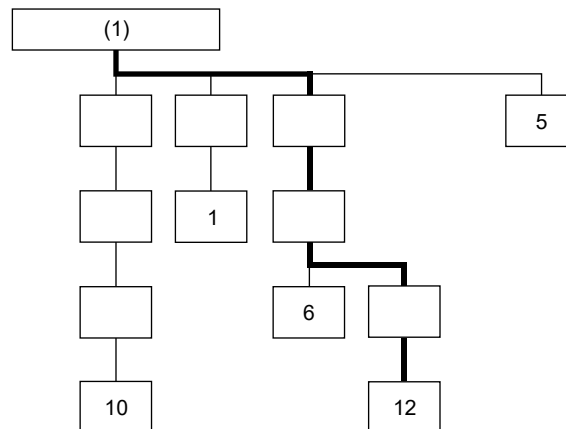
Signaling line circuit resistance and capacitance determines the maximum length of a Signature data loop. Circuit resistance affects the wire length of the longest loop branch. Circuit capacitance affects the total amount of wire that can be used on the loop.

Notes

- The design of the Signature data loop must not exceed either of the two measurements.
- There are no restrictions placed on the wiring used for the Signature data loop. Longer wire runs may be obtained using standard (non-twisted, non-shielded) wire pairs.

Determine the maximum allowable branch length

The maximum branch length is the wire distance measured from the Signature controller module to the last device on the longest loop path as shown below.



(1) Signature loop controller

Several factors influence the maximum allowable branch length:

- Wire gauge and type
- Number of Signature detectors and modules installed on the branch
- Number of SIGA-UMs or SIGA-MABs configured for two-wire smoke detectors installed on the branch

Table 74 through Table 77 provide the maximum allowable branch length for any detector, module, SIGA-UM, and SIGA-MAB and the wire gauge combination. Using the wire distances specified in the tables ensures that the signaling line circuit does not exceed the maximum circuit resistance of the Signature loop.

Note: To calculate the wire distance with respect to circuit resistance, the tables assume that the loop is end-loaded (all devices are clustered more towards the end of the loop) and the loop uses standard non-shielded wire.

To determine the maximum allowable length of a Signature data loop branch:

1. Identify the device located farthest from the Signature controller.
2. Determine the number of Signature detectors, modules, and SIGA-UMs or SIGA-MABs configured for two-wire smokes that lie on the same conductive path between the device identified in step 1 and the Signature controller.
3. Calculate the number of detector and module addresses. Some Signature modules require two addresses.
4. Determine the size of the wire used to construct the loop.
5. Find the maximum allowable wire distance for the longest branch in the lookup tables as follows:

If no SIGA-UMs or SIGA-MABs are installed, use Table 74.

If 1 to 5 SIGA-UMs or SIGA-MABs are installed, use Table 75.

If 6 to 10 SIGA-UMs or SIGA-MABs are installed, use Table 76.

If 11 to 15 SIGA-UMs or SIGA-MABs are installed, use Table 77.

Table 74: Maximum branch length with zero SIGA-UMs/SIGA-MABs configured for two-wire smokes

Signature detector addresses	Signature module addresses	Maximum allowable wire distance using non-twisted, non-shielded wire pairs					
		18 AWG		16 AWG		14 AWG	
		ft	m	ft	m	ft	m
1–25	0	7437	2267	11815	3601	18792	5728
26–50	0	7038	2145	11180	3408	17782	5420
51–75	0	6638	2023	10545	3214	16772	5112
76–100	0	6238	1901	9910	3021	15762	4804
101–125	0	5839	1780	9275	2827	14752	4497
0	1–25	7267	2215	11544	3519	18361	5597
1–25	1–25	6867	2093	10909	3325	17351	5289
26–50	1–25	6467	1971	10275	3132	16342	4981
51–75	1–25	6068	1849	9640	2938	15332	4673
76–100	1–25	5668	1728	9005	2745	14322	4365
101–125	1–25	5268	1606	8370	2551	13312	4057
0	26–50	6697	2041	10639	3243	16921	5157
1–25	26–50	6297	1919	10004	3049	15911	4850
26–50	26–50	5897	1798	9369	2856	14901	4542
51–75	26–50	5498	1676	8734	2662	13891	4234
76–100	26–50	5098	1554	8099	2469	12881	3926
101–125	26–50	4698	1432	7464	2275	11871	3618
0	51–75	5906	1800	9383	2860	14923	4549
1–25	51–75	5250	1600	8340	2542	13265	4043
26–50	51–75	4633	1412	7360	2243	11707	3568
51–75	51–75	4051	1235	6435	1961	10235	3120
76–100	51–75	3498	1066	5558	1694	8839	2694
101–125	51–75	2973	906	4723	1440	7512	2290
0	76–100	3931	1198	6245	1903	9932	3027
1–25	76–100	3404	1037	5407	1648	8601	2621
26–50	76–100	2899	883	4605	1404	7324	2232
51–75	76–100	2413	735	3833	1168	6096	1858
76–100	76–100	1945	593	3089	942	4913	1498
101–125	76–100	1493	455	2371	723	3771	1149
0	101–125	2631	802	4180	1274	6649	2027
1–25	101–125	2165	660	3439	1048	5470	1667
26–50	101–125	1713	522	2721	829	4328	1319
51–75	101–125	1274	388	2023	617	3218	981
76–100	101–125	847	258	1345	410	2140	652
101–125	101–125	431	131	685	209	1089	332

Table 75: Maximum branch length with 1 to 5 SIGA-UMs/SIGA-MABs configured for two-wire smokes

Signature detector addresses	Signature module addresses	Maximum allowable wire distance using non-twisted, non-shielded wire pairs					
		18 AWG		16 AWG		14 AWG	
		ft	m	ft	m	ft	m
1–25	0	6778	2066	10768	3282	17126	5220
26–50	0	6131	1869	9741	2969	15492	4722
51–75	0	5501	1677	8739	2664	13899	4236
76–100	0	4885	1489	7760	2365	12342	3762
101–125	0	4282	1305	6802	2073	10819	3298
0	1–25	5353	1632	8504	2592	13525	4122
1–25	1–25	4720	1439	7498	2286	11926	3635
26–50	1–25	4100	1250	6513	1985	10359	3157
51–75	1–25	3491	1064	5546	1691	8821	2689
76–100	1–25	2893	882	4597	1401	7311	2228
101–125	1–25	2306	703	3663	1116	5826	1776
0	26–50	3776	1151	5999	1829	9542	2908
1–25	26–50	3153	961	5009	1527	7966	2428
26–50	26–50	2539	774	4034	1230	6416	1956
51–75	26–50	1935	590	3075	937	4890	1491
76–100	26–50	1340	409	2130	649	3387	1032
101–125	26–50	754	230	1197	365	1905	581
0	51–75	2491	759	3957	1206	6293	1918
1–25	51–75	1868	569	2967	904	4720	1439
26–50	51–75	1254	382	1992	607	3168	966
51–75	51–75	648	198	1030	314	1638	499
76–100	51–75	50	15	80	24	126	39
101–125	51–75						
0	76–100	1386	422	2201	671	3501	1067
1–25	76–100	760	232	1208	368	1921	586
26–50	76–100	143	44	227	69	361	110
51–75	76–100						
76–100	76–100						
101–125	76–100						
0	101–125						
1–25	101–125						
26–50	101–125						
51–75	101–125						
76–100	101–125						
101–125	101–125						

Table 76: Maximum branch length with 6 to 10 SIGA-UMs/SIGA-MABs configured for two-wire smokes

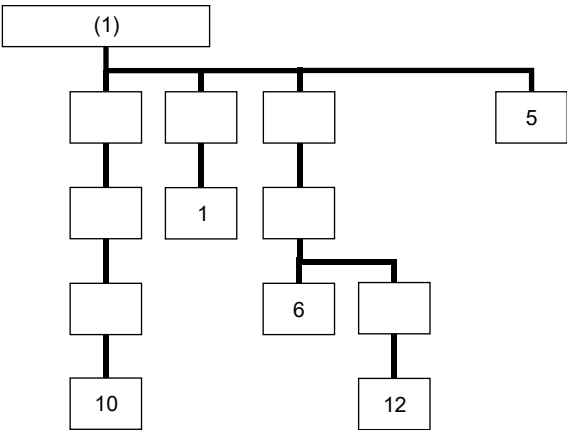
Signature detector addresses	Signature module addresses	Maximum allowable wire distance using non-twisted, non-shielded wire pairs					
		18 AWG		16 AWG		14 AWG	
		ft	m	ft	m	ft	m
1–25	0	5045	1538	8015	2443	12748	3886
26–50	0	4494	1370	7139	2176	11355	3461
51–75	0	3950	1204	6275	1913	9981	3042
76–100	0	3414	1040	5423	1653	8625	2629
101–125	0	2884	879	4581	1396	7286	2221
0	1–25	4106	1252	6523	1988	10375	3162
1–25	1–25	3542	1080	5627	1715	8950	2728
26–50	1–25	2985	910	4742	1445	7542	2299
51–75	1–25	2435	742	3868	1179	6152	1875
76–100	1–25	1891	576	3004	916	4778	1456
101–125	1–25	1353	412	2150	655	3419	1042
0	26–50	2869	874	4557	1389	7248	2209
1–25	26–50	2296	700	3648	1112	5802	1768
26–50	26–50	1730	527	2749	838	4372	1332
51–75	26–50	1170	357	1859	567	2957	901
76–100	26–50	617	188	979	299	1558	475
101–125	26–50	68	21	108	33	172	53
0	51–75	1796	547	2853	869	4537	1383
1–25	51–75	1214	370	1929	588	3067	935
26–50	51–75	638	195	1014	309	1613	492
51–75	51–75	69	21	109	33	173	53
76–100	51–75						
101–125	51–75						
0	76–100	833	254	1323	403	2105	642
1–25	76–100	242	74	385	117	613	187
26–50	76–100						
51–75	76–100						
76–100	76–100						
101–125	76–100						
0	101–125						
1–25	101–125						
26–50	101–125						
51–75	101–125						
76–100	101–125						
101–125	101–125						

Table 77: Maximum branch length with 11 to 15 SIGA-UMs/SIGA-MABs configured for two-wire smokes

Signature detector addresses	Signature module addresses	Maximum allowable wire distance using non-twisted, non-shielded wire pairs					
		18 AWG		16 AWG		14 AWG	
		ft	m	pi	ft	pi	m
1–25	0	3931	1198	6245	1904	9932	3028
26–50	0	3427	1045	5444	1660	8659	2640
51–75	0	2928	893	4651	1418	7397	2255
76–100	0	2432	741	3864	1178	6145	1873
101–125	0	1941	592	3083	940	4903	1495
0	1–25	3247	990	5158	1572	8204	2501
1–25	1–25	2722	830	4324	1318	6878	2097
26–50	1–25	2202	671	3498	1066	5563	1696
51–75	1–25	1686	514	2678	816	4259	1298
76–100	1–25	1174	358	1865	569	2966	904
101–125	1–25	666	203	1058	323	1683	513
0	26–50	2204	672	3502	1067	5570	1698
1–25	26–50	1664	507	2644	806	4205	1282
26–50	26–50	1129	344	1793	547	2852	870
51–75	26–50	598	182	950	290	1511	461
76–100	26–50	71	22	113	34	179	55
101–125	26–50						
0	51–75	1263	385	2007	612	3192	973
1–25	51–75	710	216	1128	344	1794	547
26–50	51–75	161	49	256	78	407	124
51–75	51–75						
76–100	51–75						
101–125	51–75						
0	76–100						
1–25	76–100						
26–50	76–100						
51–75	76–100						
76–100	76–100						
101–125	76–100						
0	101–125						
1–25	101–125						
26–50	101–125						
51–75	101–125						
76–100	101–125						
101–125	101–125						

Determining the total loop length

The total loop length is the sum of the lengths of all the wire segments installed in the data loop.



(1) Signature loop controller

The total length of all the cable installed in the Signature loop cannot exceed the values listed below:

Wire type	Wire Size		
	14 AWG	16 AWG	18 AWG
Twisted pair	13,157 ft (4,010 m)	13,888 ft (4,233 m)	20,000 ft (6,096 m)
Twisted-shielded pair	5,952 ft (1,814 m)	6,098 ft (1,859 m)	8,621 ft (2,628 m)
Non-twisted, non-shielded pair	20,000 ft (6,096 m)	20,000 ft (6,096 m)	20,000 ft (6,096 m)

If the cable manufacturer’s data indicates the capacitance per foot of the cable, the following method may be used to determine the maximum total loop length.

Note: In no case may the total loop length of a Signature data loop exceed 20,000 feet (6,098 meters).

$$L_{\text{max}} = 500,000 / C_{\text{pf}}$$

Where,

- L_{max} = maximum total cable length in feet
- C_{pf} = Cable capacitance in picofarads per foot

Note: A short circuit on a Signature data loop can disable the entire loop. In order to limit the effect of a single short circuit, SIGA-IB Isolator Bases or SIGA-IM Isolator modules can be installed at strategic points on the loop.

Notification appliance circuit calculations

Introduction

This topic shows you how to determine the maximum cable length of a notification appliance circuit for a given number of appliances.

Two methods are presented: worksheet and equation. The worksheet method is simpler, but your installation must meet the criteria listed on the worksheet. If your installation does not meet these criteria, you need to use the equation method.

The methods given here determine cable lengths that work under all operating conditions. The calculations ensure that the required operating voltage and current will be supplied to all notification appliances. To do this, we assume these two worst-case conditions:

- The voltage at the NAC terminals is the minimum provided by the power supply
- The notification appliances are clustered at the end of the NAC cable

Other, more detailed methods that distribute the appliance load along the NAC cable may indicate that longer cable runs are possible.

What you will need

Appliance and cable values

Whether you use the worksheet method or the equation method, you will need to know the following:

- The minimum operating voltage required for the appliances
- The maximum operating current drawn by each appliance
- The resistance per unit length of the wire used (Ω/ft)

This information can be found on the appliance installation sheets and on the cable specification sheet.

Power supply values

For either method, you'll need some fixed or calculated operating values for your specific power supply. The fixed values are:

- Maximum voltage = 27.4 V
- Rated voltage = 20.4 V
- Load factor = 0.37 V/A
- Power type = DC

The *maximum voltage* is the highest voltage measured at the NAC terminals. This value is not used in the calculations, but is given so you can ensure appliance compatibility.

The *rated voltage* is the theoretical operating minimum for the power supply, and is calculated as 85% of 24 volts [minus the diode drop].

The *load factor* is a measure of how the power supply voltage reacts when a load is applied. The load factor measures the voltage drop per ampere of current drawn by the load.

The *power type* reflects the type of power supplied to the NAC terminals at minimum voltage. The current draw of notification appliances can vary substantially with the type of power supplied: full-wave rectified (VFWR) or direct current (VDC). It is important to know the power type at minimum terminal voltage.

You will need to calculate the following values relating to your power supply and to the NAC circuit current. These are:

- Minimum voltage
- Voltage drop

The *minimum voltage* is the lowest voltage measured at the NAC terminals when the power supply is under the maximum load for that loop (i.e. for the appliances that constitute the NAC).

The *voltage drop* is the difference between the minimum voltage and 16 V. This value is for use with the worksheet only.

Worksheet method

Use this worksheet to determine the maximum cable length of a notification appliance circuit for a given number of appliances.

Use this worksheet only if all the appliances are regulated. That is, they must have a minimum operating voltage of 16 V. For other appliances, use the “Equation method.”

Worksheet 1: NAC cable length

		NAC1	NAC2	NAC3	NAC4	
Total operating current [1]						A
Load factor	×	0.37	0.37	0.37	0.37	V/A
Load voltage drop	=					V
Rated voltage		20.4	20.4	20.4	20.4	V
Load voltage drop	–					V
Minimum voltage	=					V
Regulated appliance voltage	–	16.0	16.0	16.0	16.0	V
Voltage drop [2]	=					V
Total operating current	÷					A
Maximum resistance	=					Ω
Wire resistance (Ω/ft) [3]	÷					
Maximum wire length	=					ft
	÷	2	2	2	2	
Maximum cable length	=					ft

[1] Total of the maximum operating currents for all appliances as specified for DC power. Refer to the appliance installation sheets for operating currents.

[2] This voltage drop is valid for regulated notification appliances only. For special application appliances, see “Equation method,” later in this topic.

[3] Use the manufacturer’s published wire resistance expressed in ohms per foot. For typical values, see Table 78 on page 178.

Equation method

Appliance operating voltage and current

Regulated notification appliances have an operating range from 16 V to 33 V. Use 16 V as the minimum appliance voltage when using regulated notification appliances.

When using special application appliances, refer to the installation sheets to determine the minimum appliance voltage required.

What if there are different types of appliances in the NAC, and each type has a different minimum operating voltage? In this case, use the *highest* minimum voltage required by any appliance.

The total current requirement for the appliances will be the sum of the individual maximum currents drawn by each appliance when using DC power. Use the maximum current for the appliance over the 16 V to 33 V range.

If all appliances draw the same maximum current, the total current is the maximum current multiplied by the number of appliances. If different appliance types have different maximum currents, the total current is the sum of the maximum current for each appliance type multiplied by the number of appliances of that type.

Wire resistance

Typical wire resistances are shown in the following table.

Table 78: Typical wire resistances

Wire gauge (AWG)	Resistance 1-strand uncoated copper		Resistance 7-strand uncoated copper	
	Ω per foot	Ω per meter	Ω per foot	Ω per meter
12	0.00193	0.00633	0.00198	0.00649
14	0.00307	0.01007	0.00314	0.01030
16	0.00489	0.01604	0.00499	0.01637
18	0.00777	0.02549	0.00795	0.02608

When performing these calculations, always refer to the actual cable supplier documentation and use the actual Ω/ft (or Ω/m) for the cable being used.

Calculating the maximum CAB cable length

1. Calculate the total current (I_{tot}) as the sum of the maximum operating currents for all the appliances.

$$I_{tot} = \sum I_a$$

Where:

I_a = appliance maximum current

Refer to the appliance installation sheets for I_a . Remember to use the maximum operating current specified for DC power.

2. Calculate the minimum voltage (V_m).

$$V_m = V_r - (I_{tot} \times K)$$

Where:

V_r = rated voltage

I_{tot} = total current (from above)

K = load factor

For the power supply, V_r is 20.4 V and K is 0.37 V/A.

3. Calculate the allowable voltage drop (V_d) between the NAC circuit source and the appliances.

$$V_d = V_m - V_a$$

Where:

V_m = minimum voltage (from above)

V_a = appliance minimum voltage

For regulated notification appliances, V_a is 16 V. For special application appliances, V_a is the lowest operating voltage specified on the appliance installation sheet.

4. Calculate the maximum resistance (R_{max}) the wire can have.

$$R_{max} = V_d / I_{tot}$$

Where:

V_d = voltage drop

I_{tot} = total current

5. Calculate the maximum length of the cable (L_c), based on the maximum resistance allowed, the resistance of the wire, and the number of wires in the cable (two).

$$L_c = (R_{max} / R_w) / 2$$

Where:

R_{max} = maximum resistance

R_w = wire resistance factor

Example: The NAC source is the 3-PPS/M. You are using regulated notification appliances. Assume that the maximum operating current for each appliance is 100 mA for DC power, and that 20 appliances will be placed on the NAC. The cable is 12 AWG wire, and the manufacturer specifies a wire resistance factor of 0.002 Ω /ft.

$$\begin{aligned} I_{tot} &= \Sigma I_a \\ &= 20 \times 0.1 \text{ A} \\ &= 2 \text{ A} \end{aligned}$$

$$\begin{aligned} V_m &= V_r - (I_{tot} \times K) \\ &= 22.4 \text{ V} - (2 \text{ A} \times 0.0 \text{ V/A}) \\ &= 22.4 \text{ V} - 0.0 \text{ V} \\ &= 22.4 \text{ V} \end{aligned}$$

$$\begin{aligned} V_d &= V_m - V_a \\ &= 22.4 \text{ V} - 16.0 \text{ V} \\ &= 6.4 \text{ V} \end{aligned}$$

$$\begin{aligned} R_{max} &= V_d / I_{tot} \\ &= 6.4 \text{ V} / 2.0 \text{ A} \\ &= 3.2 \Omega \end{aligned}$$

$$\begin{aligned} L_c &= (R_{max} / R_w) / 2 \\ &= (3.2 \Omega / 0.002 \Omega/\text{ft}) / 2 \\ &= 1600.0 \text{ ft} / 2 \\ &= 800.0 \text{ ft} \end{aligned}$$

So the maximum wire run for this NAC would be 800 ft (rounding down for safety).

25 or 70 VRMS NAC wire length

The maximum allowable wire length is the farthest distance that a pair of wires can extend from the amplifier to the last speaker on the notification appliance circuit without losing more than 0.5 dB of signal. Calculating the maximum allowable wire length using this method ensures that each speaker operates at its full potential.

Several factors influence the maximum allowable wire length:

- Wire size
- Output signal level of the amplifier driving the circuit
- Number of speakers installed on the circuit

To calculate the maximum allowable wire length for a 0.5 dB loss, use the following formula:

$$\text{Maximum length} = (59.25 \times \text{Amplifier output}^2) / (\text{Wire resistance} \times \text{circuit load})$$

Where,

- Amplifier output is the signal level in VRMS supplied by the amplifier driving the circuit
- Circuit load is the total watts required by the audio circuit
- Wire resistance is the resistance rating of the wire per 1000 ft pair, see Table 79.

For example, the maximum allowable wire length for an audio circuit consisting of a 30 W, 25 VRMS amplifier driving thirty 1 watt speakers, using 18-gauge wire would be 95 ft.

$$94.95 = (59.25 \times 25^2) / (13 \times 30)$$

Table 79: Wire resistance ratings

Wire Size	Resistance per 1,000 ft pair (ohms)
18 AWG (0.75 mm ²)	13.0
16 AWG (1.0 mm ²)	8.0
14 AWG (1.50 mm ²)	5.2
12 AWG (2.5 mm ²)	3.2

Table 80 and Table 81 on page 181 give the maximum allowable wire lengths for various wire sizes and loads. Use Table 80 when designing circuits for amplifiers set for 25 VRMS output. Use Table 81 when designing circuits for amplifiers set for a 70 VRMS output.

Table 80: Maximum allowable length at 25 VRMS, 0.5 dB loss

Wire size	Circuit load requirement									
	15 W		20 W		30 W		40 W		95 W	
	ft	m	ft	m	ft	m	ft	m	ft	m
18 AWG (0.75 sq mm)	190	58	142	43	95	29	71	22	Over max current limit	Over max current limit
16 AWG (1.0 sq mm)	309	94	231	70	154	47	116	35	48.7	15
14 AWG (1.5 sq mm)	475	145	356	109	237	72	178	54	75	23
12 AWG (2.5 sq mm)	772	235	579	176	386	118	289	88	121.8	37

Table 81: Maximum allowable length at 70 VRMS, 0.5 dB loss

Wire size	Circuit load requirement									
	15 W		20 W		30 W		40 W		95 W	
	ft	m	ft	m	ft	m	ft	m	ft	m
18 AWG (0.75 sq mm)	1489	454	1117	340	744	227	558	170	235	72
16 AWG (1.0 sq mm)	2420	738	1815	553	1210	369	907	276	382	116
14 AWG (1.5 sq mm)	3722	1134	2792	851	1861	567	1396	426	588.7	180
12 AWG (2.5 sq mm)	6049	1844	4537	1383	3024	922	2268	691	955	291

Cabinet battery

Use the following method to calculate the minimum ampere-hour capacity of a battery required in order to operate a control unit in the absence of AC power. Battery calculations must be performed separately for each cabinet in the system.

Determine the total amount of current in milliamps required by all of the components that derive power from the battery while the control unit is in standby mode. Multiply the total amount of standby current by the number of hours that the control unit is required to operate in standby mode while on battery power.

Determine the total amount of current in milliamps required by all of the components that derive power from the battery while the control unit is in alarm mode. Multiply the total amount of alarm current by the number of minutes that the control unit is required to operate in alarm mode while on battery power. Divide the result by 60 to convert minutes to hours.

Add the total amount of standby current and the total amount of alarm current then divide the result by 1000 to convert to ampere-hours. Multiply this number by 1.2 to add a 20% safety factor to the calculations.

Worksheet A: Control unit current

Instructions: Enter the number of option cards installed in the FACU under Quantity, then calculate the standby and alarm currents using the values below, and then calculate the total at the bottom.

Devices	Qty	Standby current (mA)	Alarm current (mA)	Qty x Standby current (mA)	Qty x Alarm current (mA)
4-CPU (includes 3-PPS/M(230))		211.0	211.0		
4-AUDTELS		85.0	101.0		
4-ANNCPU		125.0 at 24 VDC	125 at 24 VDC		
4-ANNAUDTEL		98.0	98.0		
4-CPUGRPH		123.0	123.0		
3-BPS/M(230)		50.0	50.0		
3-BBC/M(230)		70.0	70.0		
Display modules, EST4		3.0	5.6 (0.23 per LED)		
Display modules, EST3 retrofit only [1]		2.0 plus 1.5 for each active LED	2.0 plus 1.5 for each active LED		
3-LDSM		5.0	5.0		
4-LCD		40.0	93.0		
4-LCDAUDTEL		98.0	98.0		
4-3LCD		40.0	93.0		
4-NET-TP(-HC)		32.0	32.0		
4-NET-CAT / 4-FWAL-CAT		27.0	27.0		
4-NET-SM / 4-FWAL-SM		35.0	35.0		
4-NET-MM / 4-FWAL-MM		35.0	35.0		
4-NET-SMH / 4-FWAL-SMH		35.0	35.0		
4-NET-SMU / 4-FWAL-SMU		35.0	35.0		
4-NET-SMD / 4-FWAL-SMD		35.0	35.0		
4-FWAL1 / 4-FWAL2 / 4-FWAL3 / 4-FWAL4		92.0 at 24 VDC	92.0 at 24 VDC		
4-USBHUB – USB ports		44.0 (no nonisolated ports used)	44.0 plus 130.0 per nonisolated ports used		
4-NET-AD		92.0 at 24 VDC	92.0 at 24 VDC		
4-NET-XT		110.00	110.0		

Devices	Qty	Standby current (mA)	Alarm current (mA)	Qty x Standby current (mA)	Qty x Alarm current (mA)
4-MIC		8.0	38.0		
4-FT		9.0	168.0		
3-SSDC1 [2][3]		144.0	204.0		
3-SDDC1 [2][3]		264.0	336.0		
3-IDC8/4		48.0	408.0		
3-OPS		53.0	147.0		
3-MODCOM / 3-MODCOMP		60.0	95.0		
3-ZA20(A/B)		62.0	1,220.0		
3-ZA40(A/B)		62.0	2,480.0		
3-ZA95		85.0	5,540.0		
Other					
Totals (mA)					

[1] Add 5.0 mA if the display module is connected to a 3-LDSM.

[2] Standby and alarm current values are for a fully loaded loop.

[3] A 3-SSDC1 card with two 3-SDC1 or 3-SDC1-HC cards is the same as one 3-SDDC1 card.

Worksheet B: NAC/AUX power current

Instructions: For each NAC/AUX circuit used to provide power, enter the total amount of standby and alarm currents required by the devices powered by the circuit. Use the standby and alarm currents on the device installation sheet for your calculations. Use the DC RMS current values listed on the device installation sheet for your calculations.

Devices	Standby current (mA)	Alarm current (mA)
AUX 1 [1]		
AUX 2 [1]		
AUX 3 [1]		
AUX 4 [1]		
AUX 5 [1]		
AUX 6 [1]		
AUX 7 [1]		
AUX 8 [1]		
3-IDC8/4 - IDC/NAC 1 [2]		
3-IDC8/4 - IDC/NAC 2 [2]		
3-IDC8/4 - IDC/NAC 5 [2]		
3-IDC8/4 - IDC/NAC 6 [2]		
SIGA-REL module [1][3]		
CDR-3 module [1]		
3-ZA20(A/B) NAC [1]		
3-ZA40(A/B) NAC [1]		
Total (mA)		

[1] Enter values only if the power source is the 3-PPS/M, 3-BPS/M(-230), or 3-BBC/M(-230) power supply.

[2] Enter values only if the power source is the 3-PPS/M, 3-BPS/M(-230), or 3-BBC/M(-230) power supply (based on the 3-IDC 8/4 jumper settings).

[3] A maximum of ten SIGA-REL modules per signaling line circuit can be installed.

Worksheet C: CO device current load

Instructions: For each AUX circuit used to provide power to sounder bases for CO devices, enter the total amount of alarm current required by the devices. Use the alarm currents on the device installation sheet for your calculations.

CO devices	Alarm current (mA)
AUX 1 [1]	
AUX 2 [1]	
AUX 3 [1]	
AUX 4 [1]	
AUX 5 [1]	
AUX 6 [1]	
AUX 7 [1]	
AUX 8 [1]	
Total (mA)	

[1] Enter values only if the power source is the 3-PPS/M, 3-BPS/M(-230), or 3-BBC/M(-230) power supply.

Fiber optic cable worksheet

The fiber optic cable worksheet should be used to verify that the light attenuation factors do not exceed the fiber optic budget for any fiber optic cable segment.

Notes

- The contractor installing the fiber optic cable provides items A, B, and D.
- Fiber optic budget must be greater than the total link loss (F).

Fiber optic cable worksheet

Link Name	A Cable loss per unit distance [] dB/Ft [] dB/Km [] dB/Mi	B Distance [] Feet [] Km [] Miles	C Cable Loss A × B	D Number of Splices	E Contingency Splices	F Total Link Loss (dB) C+2[D+E]

Appendix B

System addressing

Summary

This appendix provides a list of logical addresses for system modules and devices.

Content

Address format 188
Module addresses 188
LRM device addresses 191
Control-display device addresses 191

Address format

EST4 addresses are in NNNCCCDDDD format, where:

- NNN is the node number. The node number is assigned when the installer downloads the CU database into the control unit.
- CCC is the local rail module slot address. The node number and LRM slot address make up the LRM's logical address.
- DDDD is the device's point address. The node number, LRM logical address, and LRM device point address make up the device or circuit's logical address.

Module addresses

Modules have a physical address and a logical address. The physical address identifies the card's location in the panel. The logical address identifies the card in the CPU database.

For a control unit with one node, see Figure 59 and Table 82 on page 189.

For a control unit with multiple nodes, see Figure 60 and Table 83 on page 190.

Figure 59: Logical addressing for control unit with one node (4-CAB21DL inner door/4-CAB21B backbox shown)

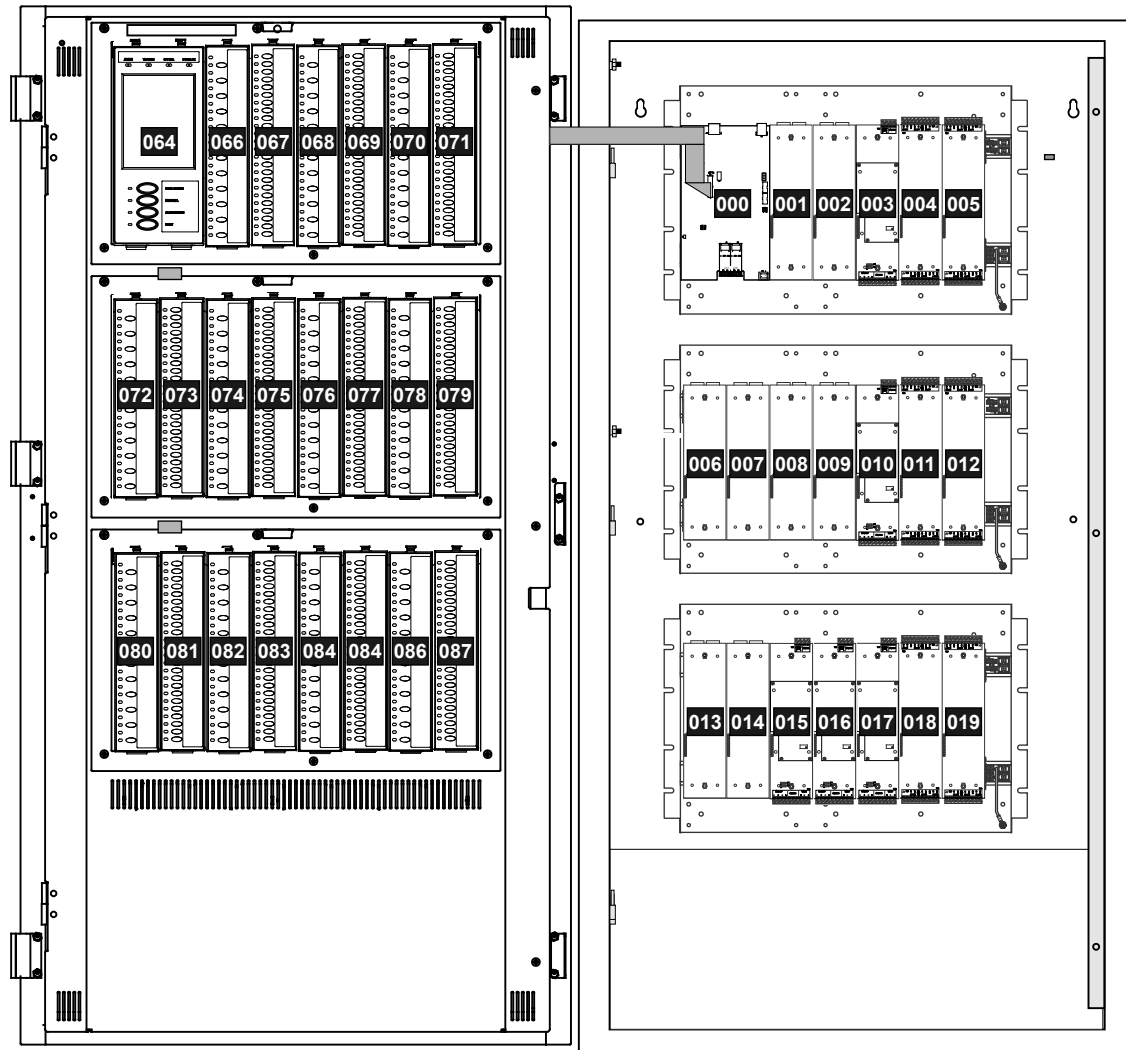
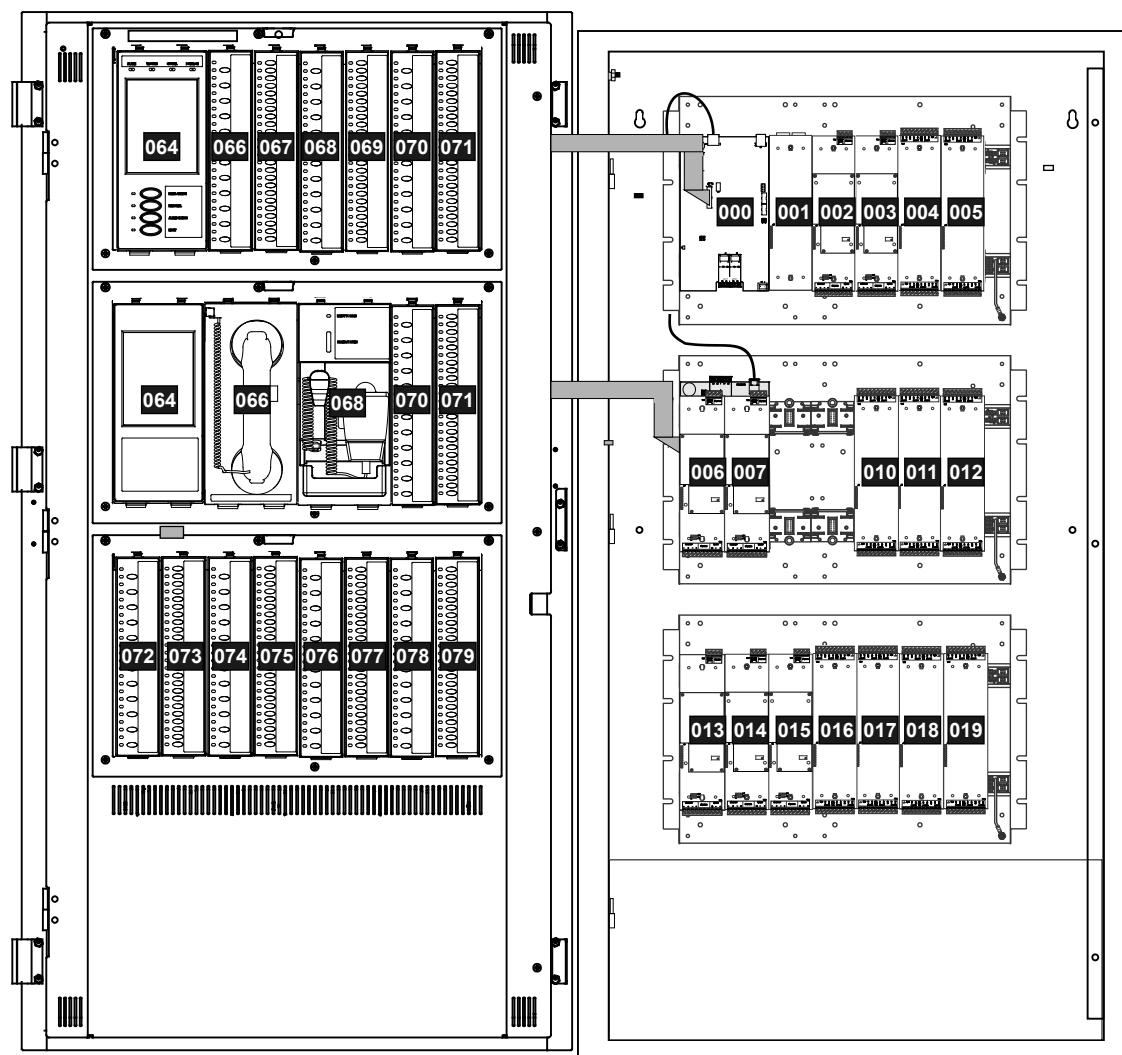


Table 82: Logical addressing for a control unit with one node

Card	Logical address
Operator layer modules on the inner door frame assembly	
4-LCD	NNN064(065)
Control-display modules	NNN066 to NNN087
Hardware LRMs on the 3-CHAS7 chassis	
4-CPU	NNN000
Local rail modules	NNN001 to NNN019

Figure 60: Logical addressing for control unit with multiple nodes (4-CAB21DL inner door/4-CAB21B backbox shown)**Table 83: Logical addressing for a control unit with multiple nodes**

Card	Logical address
Operator layer modules on the inner door frame assembly	
4-LCD	NNN064(065)
Control-display modules	NNN066 to NNN071
4-LCDANN	NNN064(065) [1]
4-FT	NNN066(067) [2][4]
4-MIC	NNN068(069) [3]
Control-display modules	NNN070 to NNN079
Hardware LRMs on the 3-CHAS7 chassis	
4-CPU	NNN000
Local rail modules	NNN001 to NNN007 [4]
Local rail modules	NNN010 to NNN019

[1] Logical addressing restarts with NNN064 when the 4-LCDx is connected to the CPU node module.

[2] The logical address for a 4-FT firefighter telephone is always NNN066.

[3] The logical address for a 4-MIC paging microphone is always NNN068.

[4] Local rail modules cannot be installed on the 3-CHAS rail behind a 4-FT firefighter telephone.

LRM device addresses

Table 84 below lists the device addresses for points for LRMs installed on the chassis rail.

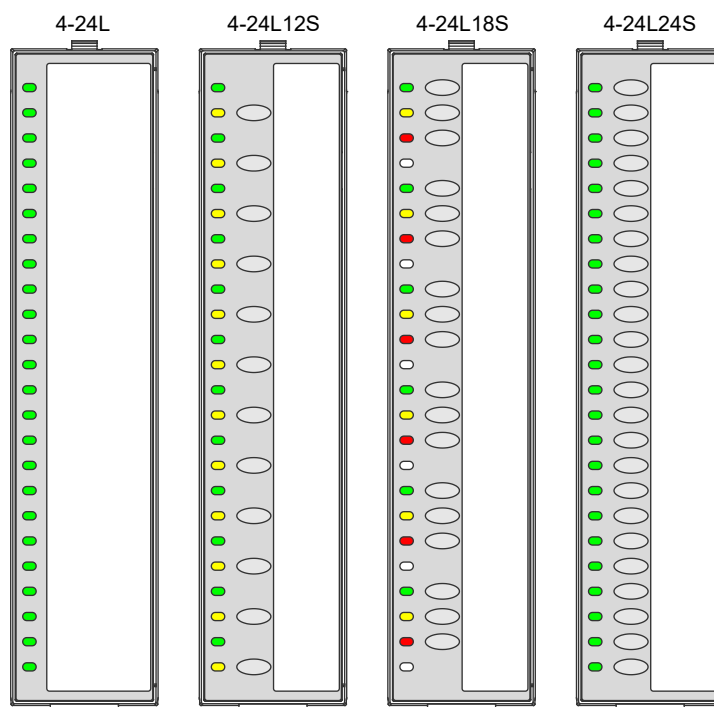
Table 84: LRM device point addresses

Card	Device or circuit	Address
3-IDC8/4	IDC/NAC 1	NNNCCC0001
	IDC/NAC 2	NNNCCC0002
	IDC 3	NNNCCC0003
	IDC 4	NNNCCC0004
	IDC/NAC 5	NNNCCC0005
	IDC/NAC 6	NNNCCC0006
	IDC 7	NNNCCC0007
	IDC 8	NNNCCC0008
3-SSDC1	Detectors Modules	NNNCCC0001 to NNNCCC0125 NNNCCC0126 to NNNCCC0250
3-SDDC1 Loop 1	Detectors	NNNCCC0001 to NNNCCC0125
	Modules	NNNCCC0126 to NNNCCC0250
	Detectors	NNNCCC0251 to NNNCCC0375
	Modules	NNNCCC0376 to NNNCCC0500
3-ZA20(A/B)	Amplifier output	NNNCCC0000
	24V DC output	NNNCCC0001
3-ZA40(A/B)	Amplifier output	NNNCCC0000
	24V DC output	NNNCCC0001
3-ZA95	Amplifier output	NNNCCC0000
	24V DC output	NNNCCC0001

Control-display device addresses

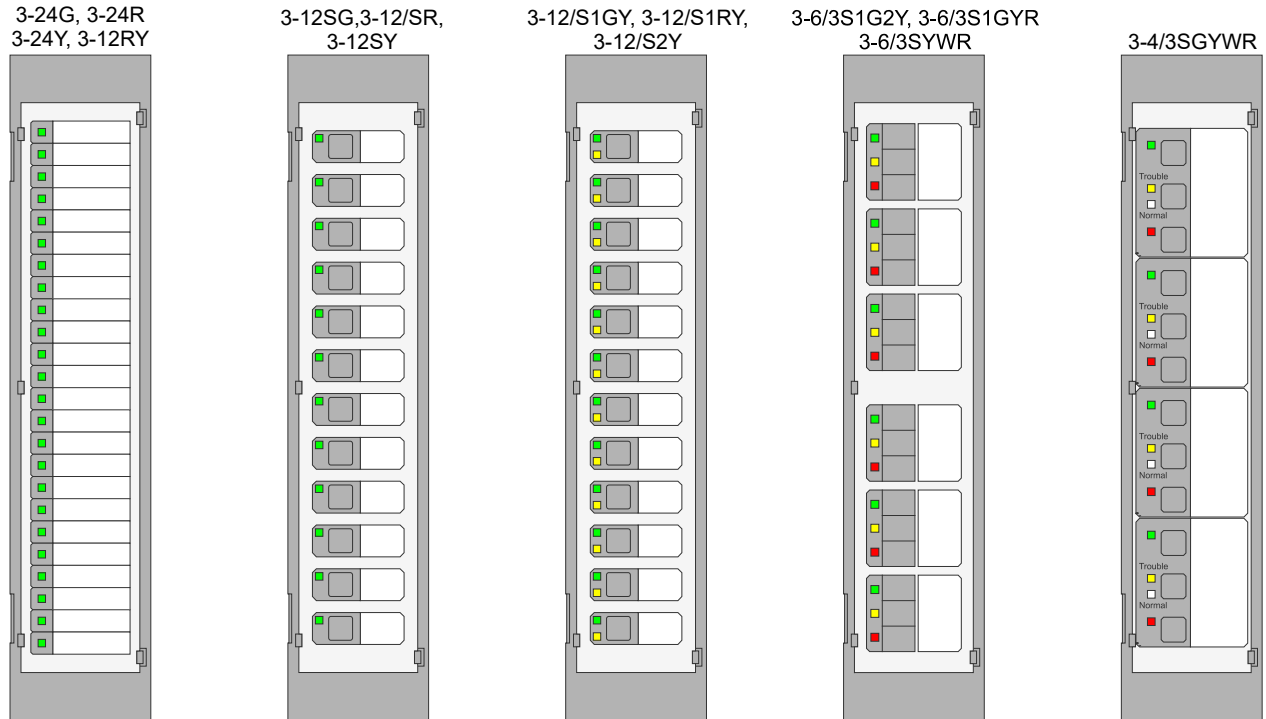
The tables in this section list switch and indicator addresses that the system assigns to operator layer control-display modules.

- EST4 system control-display modules LED and switch numbering see Table 85 on page 192
- EST3 migrated system control-display modules LED and switch numbering see Table 86 on page 194

Figure 61: EST4 operator layer control-display modules**Table 85: EST4 operator layer control-display addresses**

Module type	Switch group	Switch address	Indicator address
4-24L	N/A	N/A	0129 to 0152
4-24L12S	1	0002	0129 to 0130
	2	0004	0131 to 0132
	3	0006	0133 to 0134
	4	0008	0135 to 0136
	5	0010	0137 to 0138
	6	0012	0139 to 0140
	7	0014	0141 to 0142
	8	0016	0143 to 0144
	9	0018	0145 to 0146
	10	0020	0147 to 0148
	11	0022	0149 to 0150
	12	0024	0151 to 0152
4-24L18S	1	0001	0129
	2	0002	0130
	3	0003	0131
	–	–	0132
	4	0005	0133
	5	0006	0134
	6	0007	0135
	–	–	0136

Module type	Switch group	Switch address	Indicator address
	7	0009	0137
	8	0010	0138
	9	0011	0139
	–	–	0140
	10	0013	0141
	11	0014	0142
	12	0015	0143
			0144
	13	0017	0145
	14	0018	0146
	15	0019	0147
	–	–	0148
	16	0021	0149
	17	0022	0150
	18	0023	0151
	–	–	0152
4-24L24S	1 to 24	0001 to 0024	0129 to 0152

Figure 62: EST3 control-display LED and switch numbering**Table 86: EST3 operator layer control-display addresses**

Module type	Switch group	Switch address	Indicator address
3-24G, 3-24R, 3-24Y, 3-12RY	N/A	N/A	0129 to 0152
3-12SG, 3-12SR, 3-12SY	1	0001	0129
	2	0002	0130
	3	0003	0131
	4	0004	0132
	5	0005	0133
	6	0006	0134
	7	0007	0135
	8	0008	0136
	9	0009	0137
	10	0010	0138
	11	0011	0139
	12	0012	0140
3-12/S1GY, 3-12/S1RY, 3-12/S2Y	1	0001	0129 0130
	2	0002	0131 0132
	3	0003	0133 0134
	4	0004	0135 0136
	5	0005	0137 0138

Module type	Switch group	Switch address	Indicator address
	6	0006	0139 0140
	7	0007	0141 0142
	8	0008	0143 0144
	9	0009	0145 0146
	10	0010	0147 0148
	11	0011	0149 0150
	12	0012	0151 0152
3-6/3S1G2Y, 3-6/3S1GYR, 3-6/3SGYWR	1	0001	0129
		0002	0130
		0003	0131
	2	0004	0132
		0005	0133
		0006	0134
	3	0007	0135
		0008	0136
		0009	0137
	4	0010	0138
		0011	0139
		0012	0140
	5	0013	0141
		0014	0142
		0015	0143
	6	0016	0144
		0017	0145
		0018	0146
3-4/3SGYWR	1	0001	0129
		0002	0130 0131
		0003	0132
	2	0004	0133
		0005	0134 0135
		0006	0136
	3	0007	0137
		0008	0138 0139
		0009	0140
	4	0010	0141
		0011	0142 0143
		0012	0144

Appendix C

Pseudo points table

Summary

This appendix provides pseudo point addresses, sources, and descriptions used to troubleshoot pseudo point events.

Content

Pseudo points 198

Pseudo points

The table in this appendix provides pseudo point addresses, event types, sources, and descriptions that can be used to troubleshoot pseudo point events.

View the event's message details on the LCD screen to identify the pseudo point address. Cross-reference the pseudo point address to the Address column in Table 87 below to find details about the pseudo point.

Note: For instructions on how to view event details, see “Viewing event details” on page 48.

Table 87: EST4 pseudo points

Address	Label	Event type	Source	Description
0001	\$Startup	Startup	EST4	Control unit is energized or operator initiated a restart
0001	\$MicrophonePushToTalk	Local monitor	4-MIC	Operator pressed the push-to-talk switch on paging microphone
0001	\$Switch1	Common control switch	4-LCD/4-3LCD	User interface control button 1 pressed
0002	\$First Alarm	First alarm	EST4	Control unit or any control unit in the same network routing group changed to the alarm state
0002	\$AmplifierBackup	Local relay	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Amplifier's input relay selected backup amplifier input as its signal source
0002	\$Switch2	Common control switch	4-LCD/4-3LCD	User interface control button 2 pressed
0003	\$First Supervisory	First supervisory	EST4	Control unit or any control unit in the same network routing group changed to the supervisory state
0003	\$Switch3	Common control switch	4-LCD/4-3LCD	User interface control button 3 pressed
0003 - 0010	\$AmplifierChannel1 - \$AmplifierChannel8	Local relay	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Amplifier's input relay selected channel x
0004	\$First Trouble	First trouble	EST4	Control unit or any control unit in the same network routing group changed to the trouble state
0004	\$Switch4	Common control switch	4-LCD/4-3LCD	User interface control button 4 pressed
0005	\$First Monitor	First monitor	EST4	Control unit or any control unit in the same network routing group changed to the monitor state
0006	\$Evacuated	Evacuation	EST4	Operator pressed a switch that executed the EvacuationOn command
0007	\$Drilling	Drill	EST4	Operator pressed a switch that executed the DrillOn command
0008	\$AllCall	All call	EST4	Operator pressed a switch that executed an All Call function
0009	\$AlarmSilenced	Alarm silence	EST4	Operator pressed a switch that executed the AlarmSilenceOn command

Address	Label	Event type	Source	Description
0010	\$TwoStageTimer Expiration	Two stage timer expiration	EST4	Control unit's second stage activation timer expired
0011	\$Resetting	Reset	EST4	Operator pressed a switch that executed the ResetOn command
0011	\$AmplifierPage	Local relay	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Amplifier's input relay selected Page channel
0012	\$R1	R1	EST4	First phase of the 3-phase reset cycle starts
0013	\$R2	R2	EST4	Second phase of the 3-phase reset cycle starts
0014	\$R3	R3	EST4	Third phase of the 3-phase reset cycle starts
0015	\$FirstDisable	First disable	EST4	First point on a control unit or any control unit in the same network routing group changed to disable state
0016	\$ServiceGroupActive	Service group active	EST4	Operator enabled a Service Group from the control unit
0017	\$TwoStageTimer Active	Two stage timer active	EST4	Control unit's second stage activation timer started
0018	\$AllCallMinus	All call minus	EST4	Operator pressed a switch that executed the All Call Minus function
0019	\$CMSFirstTrouble	CMS first trouble	EST4	First point on a control unit or any control unit in the same network routing group changed to CMS trouble state
0020	\$AlarmSilenceInhibit	Signal silence inhibit	EST4	Control unit's alarm silence inhibit timer expired
0021	\$AlternateSensing Mode	Alternate sensing mode	EST4	Control unit's alternate sensing (sensitivity) mode activated
0022	\$SystemFunction1	System monitor	EST4	Operator pressed an LCD user interface button or control-display switch that executed a programmed Function 1 command
0023	\$SystemFunction2	System monitor	EST4	Operator pressed an LCD user interface button or control-display switch that executed a programmed Function 2 command
0024	\$SystemFunction3	System monitor	EST4	Operator pressed an LCD user interface button or control-display switch that executed a programmed Function 3 command
0025	\$SystemFunction4	System monitor	EST4	Operator pressed an LCD user interface button or control-display switch that executed a programmed Function 4 command
0026	\$RemoteReadLocked	Local monitor	EST4	IP read communications between the 4-CU and the control unit is blocked
0027	\$RemoteWrite Unlocked	Local monitor	EST4	IP write communications between the 4-CU and the control unit is unblocked
0028	\$PanelSilence	Local monitor	EST4	N/A (future use)

Address	Label	Event type	Source	Description
0029	\$AudioMessageSync	Local trouble	EST4	N/A (future use)
0030	\$NetworkReconciling	Local monitor	EST4	Network communication failure or packet loss caused the control unit to start reconcile process
0031	\$NetworkOutOfRangeNode	Local trouble	EST4	Control unit added to the network but not configured in the 4-CU
0032	\$UnconfiguredAlarm	General alarm	EST4	Alarm on a control unit is incompatible with system programming
0051-0300	\$NetworkCommunicationNode x	Communication failure	EST4	CPU is unable to communicate with the networked CPU module in control unit x
0129	\$IndicatorAlarm	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0129	\$ReadyToPage	Common control indicator	4-MIC	For internal use only
0130	\$IndicatorTrouble	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0131	\$IndicatorCPUFail	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0132	\$IndicatorFollowsAC	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0133	\$Indicator1	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0134	\$Indicator2	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0135	\$Indicator3	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0136	\$Indicator4	Common control indicator	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	For internal use only
0301-0550	\$NetworkIncompatibilityNode x	External database incompatibility	EST4	Database or firmware mismatch caused communication failure with control unit x
0600	\$Fail-safe	Fail safe	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Device asserted the rail alarm-not line and CPU module did not register alarm event

Address	Label	Event type	Source	Description
0600	\$Annunciator Supervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P)	Module is faulty, missing, or not properly configured
0601	\$LoopControllerReset Extension	Loop reset extension	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Loop controller stayed in reset mode longer than expected
0601	\$LRMCommunication	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P)	Communication failure on the chassis rail
0601	\$CANCommunication	Local trouble	4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDTEL 4-FT 4-MIC	Communication failure on the UI frame assembly rail
0602	\$ServiceDevice Supervision	Service device supervision	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator canceled Service Group test while a circuit under test remained active

Address	Label	Event type	Source	Description
0602	\$FirmwareMismatch	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDTEL 4-FT 4-MIC	Mismatch between module firmware and 4-CU database
0603	\$UserTriggered	User trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator forced a trouble into the system
0603	\$DownloadFail	Local trouble	3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDTEL	Firmware download to the module failed
0604	\$ExternalDatabase Incompatibility	External database incompatibility	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Different database is in one or more networked control units
0604	\$UnhandledCondition	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDTEL 4-FT 4-MIC	CPU module failure Note: Reset required [1]

Address	Label	Event type	Source	Description
0605	\$RebootFault	Reboot fault	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	CPU module is interrupted unexpectedly Note: Reset required [1]
0605	\$DataSupervision	Local trouble	3-PPS/M(230, 230) 3-BPS/M(230, 230) 3-BBC/M(230, 230) 3- LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL 4-FT 4-MIC	Corrupt database
0606	\$NetworkTransmitOver flow	Local trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Node unable to transmit queued messages within time limit set in the 4-CU Note: Reset required [1]
0606	\$CodeSupervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL 4-FT 4-MIC	Corrupt executable program
0607	\$NetworkReceive Congestion	Local trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Receiving node unable to process incoming packets within time limit set in the 4-CU Note: Reset required [1]
0607	\$SDC1Supervision	Local trouble	3-SSDC1 3-SDDC1	Missing or loose LIM card on loop controller module for SLC1

Address	Label	Event type	Source	Description
0608	\$WatchdogViolation	Task failure	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	EST4 task failed to execute properly
0608	\$SDC2Supervision	Local trouble	3-SSDC1 3-SDDC1	Missing or loose LIM card on loop controller module for SLC2
0609	\$Configuration Supervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230) 3-LDSM 3-EVDVR 3-EVDVRA 3-OPS 3-IDC8/4 3-ZA20B(A) 3-ZA40B(A) 3-ZA95 3-SSDC1 3-SDDC1 3-MODCOM(P) 4-24L series 4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL 4-FT 4-MIC	Module is in the wrong rail slot
0609 - 0627	\$Configuration MismatchCard1 - \$Configuration MismatchCard19	Local trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Card in slot xx cannot perform as programmed
0610	\$RailVoltage Supervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Rail voltage is >30 VDC or <21 VDC - Excessive rail current load - Faulty or misadjusted rail
0610	\$SupervisionLine1	Local trouble	3-MODCOM(P)	Line-cut fault detected on phone line 1
0610	\$LCDSupervision	Local trouble	4-LCD/4-3LCD 4-LCDANN 4-LCDAUDEL	Internal wiring fault detected on the LCD module
0610	\$AudioConnectionTo Rail	Local trouble	4-FT	Phone page activated for a period that exceeded time limit set in the 4-CU
0610	\$MicrophoneTrouble	Local trouble	4-MIC	Defective microphone or connections
0611	\$RailVoltageBelow Battery	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	Excessive rail current load
0611	\$SupervisionLine2	Local trouble	3-MODCOM(P)	Line-cut fault detected on phone line 2
0611	\$PhoneSupervision	Local trouble	4-FT	Internal wiring fault on the master firefighter telephone handset
0611	\$MicrophoneKey	Local trouble	4-MIC	Internal wiring fault on the paging microphone
0612	\$HeatSinkTemperature	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Clogged enclosure vents - Heat sink not fastened properly

Address	Label	Event type	Source	Description
0612	\$Receiver1Supervision	Local trouble	3-MODCOM(P)	Line 1 test transmission to CMS failed
0612	\$OffHookTrouble	Local trouble	4-FT	Defective hook switch
0613	\$LowBatteryCutoff	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	Battery voltage is below 22.5 VDC when on standby battery
0613	\$Receiver2Supervision	Local trouble	3-MODCOM(P)	Line 2 test transmission to CMS failed
0614	\$RS232Supervision	Local trouble	3-MODCOM(P)	Communication failure
0614	\$ACSupervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	AC line voltage is below 93 VAC at 50/60 Hz
0615	\$BatterySupervision	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Open detected on wiring - Battery voltage is below 22.5 VDC - Battery internal resistance too high (load test failure)
0615	\$IncomingRing	Local monitor	3-MODCOM(P)	Incoming call received
0616	\$AuxiliarySupervision2	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Excessive load - Short detected on wiring
0617	\$PowerSupplyFail	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Loose or missing cables between the power supply and CPU - Defective power supply
0617	\$DSPSupervision	Local trouble	3-MODCOM(P)	Failed DSP chip on module
0618	\$AuxiliarySupervision1	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Excessive load - Short detected on wiring
0619	\$DriverPowerSupply Fail	Local trouble	3-PPS/M(230,230) 3-BPS/M(230,230) 3-BBC/M(230,230)	- Loose or missing cables between power supply card and monitor card - Defective power supply card or monitor card
0620	\$WaitingForDownload		3-MODCOM(P)	4-CU database download in progress or incomplete
0620	\$IncomingAudio Supervision	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Missing digitized audio data
0621	\$AmplifierOvercurrent	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	- Short detected on wiring - Speaker wattage tap setting exceeds amplifier output rating 70 VRMS jumper setting used with 25 VRMS speakers
0621	\$ManualAnswer	Non-supervised output	3-MODCOM(P)	Module answered incoming calls
0622	\$PrimaryAudioOutput DC	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	- Open detected on DC NAC wiring - Missing or wrong value EOL resistor - Short detected on DC NAC wiring
0622	\$OutgoingCallIn Progress	Local monitor	3-MODCOM(P)	Dialer active

Address	Label	Event type	Source	Description
0623	\$PrimaryAudioOutput Analog	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	- Open detected on audio NAC wiring - Missing or wrong value EOL resistor - Short detected on audio NAC wiring - Output voltage jumper incorrectly set
0624	\$BackupAudioOutput Analog	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	- Open detected on audio NAC wiring - Missing or wrong value EOL resistor - Short detected on audio NAC wiring - Output voltage jumper incorrectly set
0625	\$DaughterBoard Supervision	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Defective board
0626	\$FuseSupervision	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Open fuse detected on amplifier
0627	\$ProgrammableLogic Supervision	Local trouble	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	Bad PAL chip on amplifier
0628 - 0646	\$DatabaseSyncCard1 - \$DatabaseSyncCard19	Card database incompatibility	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	CPU reporting mismatch between actual data and expected data
0629	\$RequestBackup Amplifier	Local monitor	3-ZA20B(A) 3-ZA40B(A) 3-ZA95	N/A
0640	\$JumperPosition Supervision	Local trouble	3-OPS	Jumpers incorrectly set
0641	\$AToDSupervision	Local trouble	3-OPS	Internal module failure
0642	\$CityTieSupervision	Local trouble	3-OPS	N/A
0647	\$Annunciator Supervision	Local trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Internal wiring fault on node
0648	\$GroundFault	Ground fault	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	- Wiring fault - Conductor connected to data card has continuity to ground
0649	\$AudioSupervision	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Short or open detected on a riser
0650	\$InternalFault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Module failure Note: Restart required [1]

Address	Label	Event type	Source	Description
0651	\$DatabaseSupervision	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Corrupt database
0652	\$CodeSupervision	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Corrupt executable program
0653	\$DownloadInProgress	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	4-CU database download in progress or incomplete
0654	\$UnexpectedModule	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Undefined module detected
0655	\$InvalidInstruction	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	- Invalid instruction for programmed response - Corrupt database Note: Restart required [1]
0656	\$MainBoardInternal Fault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	CPU detected internal fault on the main board Note: Restart required [1]
0657	\$CommunicationFault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Communication failed between CPU daughter card and CPU
0658	\$eth0GroundFault	Ground fault	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Ground fault on eth0
0659	\$eth1GroundFault	Ground fault	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Ground fault on eth1
0660	\$eth0	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Configured SFP not connected to eth0 port on the node card (network link failure)
0661	\$eth1	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Configured SFP not connected to eth1 port on the node card (network link failure)

Address	Label	Event type	Source	Description
0662	\$usb0	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	USB cable not connected to USB port on the node card
0663	\$usbNetwork	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Failed network connection with 4-USBHUB
0664	\$usbPowerCurrent Overload	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Connected 4-ANNCPU consuming too much current from the USB
0665	\$PowerSupplyFault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	- Loose or missing cable between power supply and CPU - Defective power supply
0666	\$eth0Card	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	- Configured SFP not connected to eth0 port on the node card - Connected SFP in eth0 port on the node card is not configured
0667	\$eth1Card	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	- Configured SFP not connected to eth1 port on the node card - Connected SFP in eth1 port on the node card is not configured
0668	\$WIFILink	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	N/A (future use)
0669	\$MainDatabase SupervisionFault	Local trouble	4-CPU	Corrupt database on the 4-CPU
0669	\$MainboardDataBase Fault	Local trouble	4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	CPU detected database fault on the main board
0670	\$MainBoardCode SupervisionFault	Local trouble	4-CPU	Corrupt firmware on the 4-CPU
0670	\$MainBoardCodeFault	Local trouble	4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Corrupt code on the 4-CPU
0670	\$ExecutingBootloader	Local trouble	3-SSDC1 3-SDDC1	4-CU attempting download
0671	\$MainBoard Configuration SupervisionFault	Local trouble	4-CPU	Bad microcode configuration on the CPU

Address	Label	Event type	Source	Description
0671	\$Mainboard ConfigurationFault	Local trouble	4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Bad microcode configuration on the CPU/node
0671	\$LineSupervisionSDC1	Local trouble	3-SSDC1 3-SDDC1	Wiring fault on SLC1
0672	\$NetworkNeighbor AuthenticationFault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	N/A (future use)
0672	\$MapFlawSDC1	Local trouble	3-SSDC1 3-SDDC1	- Mismatch detected between actual data and expected data on SLC1 - Defective wiring on SLC1 - Defective device on SLC1
0673	\$MainBoardFirmware Mismatch	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Mismatch between CPU firmware and 4-CU database
0673	\$MappingInProgress SDC1	Local monitor	3-SSDC1 3-SDDC1	SLC1 currently mapping field devices
0674	\$NetworkMessageLost	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Failed attempts to resend an out of sequence message
0674	\$MappingOffSDC1	Local monitor	3-SSDC1 3-SDDC1	SLC1 mapping manually disabled
0675	\$PerformanceFault	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	N/A (future use)
0675	\$MaintenanceAlert SDC1	Local monitor	3-SSDC1 3-SDDC1	Dirty detector on SLC1
0676	\$Unprogrammed DeviceSDC1	General alarm	3-SSDC1 3-SDDC1	Device on SLC1 circuit not defined in 4-CU is in alarm or trouble state
0676	\$eth0LinkFail	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Configured SFP connected to eth0 port on the node card is unable to communicate with the customer network
0677	\$GroundFaultSDC1	Ground fault	3-SSDC1 3-SDDC1	- Wiring fault on SLC1 - Conductor connected to SLC1 data card has continuity to ground
0677	\$eth1LinkFail	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Configured SFP connected to eth1 port on the node card is unable to communicate with the customer network

Address	Label	Event type	Source	Description
0678	\$RestoringLineSDC1	Local monitor	3-SSDC1 3-SDDC1	N/A
0678	\$eth0IsolateFault	Non-supervised output	4-CPU 4-ANNCPU 4-FWAL1,2,3,4	Isolates eth0 port to determine which Ethernet link is generating a ground fault
0679	\$SmokePowerSDC1	Local trouble	3-SSDC1 3-SDDC1	Fault detected on SLC1 smoke power circuit
0679	\$eth1IsolateFault	Non-supervised output	4-CPU 4-ANNCPU 4-FWAL1,2,3,4	Isolates eth1 port to determine which Ethernet link is generating a ground fault
0680	\$FailSafeDatabase	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Control unit in fail safe mode due to corrupted database or unexpected unit behavior (CPU is able to communicate with LRMS and recover)
0680	\$UnhandledCondition	Local trouble	3-LDSM 3-EVDVR 3-EVDVRA	CPU module failure Note: Restart required [1]
0680	\$UnusedSDC1	Local trouble	3-SSDC1 3-SDDC1	N/A
0681	\$LineSupervisionSDC2	Local trouble	3-SSDC1 3-SDDC1	Wiring fault on SLC2
0681	\$TestLogging	Non-supervised output	4-CPU 4-ANNCPU 4-FWAL1,2,3,4	N/A (future use)
0682	\$InternalTrouble	Local trouble	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Firmware failure on the main CPU Note: Reset required [1]
0683	\$MapFlawSDC2	Local trouble	3-SSDC1 3-SDDC1	- Mismatch between actual data and expected data on SLC2 - Defective wiring on SLC2 - Defective device on SLC2
0683	\$InternalMonitor	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	For internal use only
0683	\$MappingInProgressSDC2	Local monitor	3-SSDC1 3-SDDC1	SLC2 currently mapping field devices
0684	\$CANDownloadInProgress	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	4-CU database download in progress to devices on UI frame assembly rail
0684	\$MappingOffSDC2	Local monitor	3-SSDC1 3-SDDC1	SLC2 mapping manually disabled

Address	Label	Event type	Source	Description
0685	\$CPUCard	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Main CPU board mismatch
0685	\$MaintenanceAlertSDC2	Local monitor	3-SSDC1 3-SDDC1	Dirty detector on SLC2
0686	\$CULoggedIn	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4	4-CU log on to control unit executed
0686	\$UnprogrammedDeviceSDC2	General alarm	3-SSDC1 3-SDDC1	Device on SLC2 not defined in 4-CU is in alarm or trouble state
0687	\$GroundFaultSDC2	Ground fault	3-SSDC1 3-SDDC1	- Wiring fault on SLC2 - Conductor connected to SLC2 data card has continuity to ground
0688	\$RestoringLineSDC2	Local monitor	3-SSDC1 3-SDDC1	N/A
0689	\$SmokePowerSDC2	Local trouble	3-SSDC1 3-SDDC1	Fault detected on SLC2 smoke power circuit
0690	\$UnusedSDC2	Local trouble	3-SSDC1 3-SDDC1	N/A
0691	\$MapMismatchSDC1	Local trouble	3-SSDC1 3-SDDC1	Mismatch between actual data and expected data on SLC1
0692	\$MapMismatchSDC2	Local trouble	3-SSDC1 3-SDDC1	Mismatch between actual data and expected data on SLC2
0693	\$ExtraDevicesSDC1	Local trouble	3-SSDC1 3-SDDC1	Number of devices on SLC1 exceeds maximum allowed
0694	\$ExtraDevicesSDC2	Local trouble	3-SSDC1 3-SDDC1	Number of devices on SLC2 exceeds maximum allowed
0695	\$INFModeBypassedSDC1	Local trouble	3-SSDC1 3-SDDC1	Inhibition of device LED flashing was bypassed (LEDs will flash according to state) Note: Applies to DH and DS nonmapping devices
0696	\$INFModeBypassedSDC2	Local trouble	3-SSDC1 3-SDDC1	Inhibition of device LED flashing was bypassed (LEDs will flash according to state) Note: Applies to DH and DS nonmapping devices
0701 - 0710	\$Service1 - \$Service10	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Primary configuration communication failure with remote server for IP Receiver, Email, ECP, Printer, and Coder services
0711 - 0720	\$ServiceMonitor1 - \$ServiceMonitor20	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Communication failure with remote server for Web service

Address	Label	Event type	Source	Description
0726 - 0735	\$ServiceAlternate1 - \$ServiceAlternate10	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Communication failure with backup remote CMS or SMTP server for IP Receiver or Email services
0736 - 0745	\$ServiceAlternate Monitor1 - \$ServiceAlternate Monitor10	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	N/A (not used)
0751 - 0775	\$ServiceDisabled1 - \$ServiceDisabled25	Local trouble	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	Communication service disabled
0777	\$TelephoneCallsIn	Local monitor	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	N/A (future use)
0778	\$PageByTelephone	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center initiated a EVAC announcement from the master firephone
0779	\$AllCallActive	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the All Call function at the NCA
0780	\$AllCallMinusActive	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the All Call function at the NCA
0781	\$PageToEvacuated	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the EVAC function at the NCA
0782	\$PageToAlerted	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the Alert function at the NCA
0783	\$PageToEmergency	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the Emergency function at the NCA
0784	\$PageToOther	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Operator at a command center activated the Other function at the NCA

Address	Label	Event type	Source	Description
0785	\$ExternalToPage	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated a page from an external (third-party) firephone
0786	\$ExternalToAllCall	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the All Call function from external (third-party) microphone
0787	\$ExternalToAllCall Minus	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the All Call Minus function from external (third-party) microphone
0788	\$ExternalToEvacuated	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the EVAC function from external (third-party) microphone
0789	\$ExternalToAlerted	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the Alert function from external (third-party) microphone
0790	\$ExternalToEmergency	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the Emergency function from external (third-party) microphone
0791	\$ExternalToAuxiliary	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the Auxiliary function from external (third-party) microphone
0792	\$ExternalToOther	Audio source	4-CPU 4-ANNCPU 4-FWAL1, 2, 3, 4 4-NET-AD 4-CPUGRPH	Customer activated the Other function from external (third-party) microphone
0800	\$AudioTelFirmware Mismatch	Audio source	4-CPU 4-ANNCPU	Firmware mismatch between the CPU and LIM card
0801	\$AUDTEL Communication	Audio source	4-CPU 4-ANNCPU	AUDTELS card communication fault
0802	\$AUDTELFault	Audio source	4-CPU 4-ANNCPU	AUDTELS card detected an error
0803	\$AUDTELData Supervision	Audio source	4-CPU	Corrupted record in AUDTELS
0804	\$AUDTELCode Supervision	Audio source	4-CPU 4-ANNCPU	Corrupt firmware on the AUDTELS
0805	\$AUDTEL Configuration Supervision	Audio source	4-CPU 4-ANNCPU	Corrupt database record in the AUDTELS

Address	Label	Event type	Source	Description
0806	\$AUDTELMIC Supervision	Audio source	4-CPU 4-ANNCPU	Input fault on AUDTELS or ANNAUDTEL module's microphone input circuit
0807	\$AUDTELRiser Supervision	Local trouble	4-CPU	Open or short detected on the firephone riser circuit
0808	\$TelephoneRiserOn	Non-supervised output	4-CPU	Output used to turn off riser supervision when firephone is active
0809 - 0812	\$AudioInput1 - \$AudioInput4	Audio source	4-CPU	Short or open detected on external (third-party) audio input circuit x on the AUDTELS
0813	\$Preamplifier1	Audio output	4-CPU	Preamplifier 1 output signal detected
0814 - 0821	\$Preamplifier1Channel 1 - \$Preamplifier1Channel 8	Local relay	4-CPU	Preamplifier 1 selected audio source channel x
0822	\$Preamplifier1Channel Page	Local relay	4-CPU	Preamplifier 1 selected the page channel
0823	\$Preamplifier2	Audio output	4-CPU	Preamplifier 2 output signal detected
0824 - 0831	\$Preamplifier2Channel 1 - \$Preamplifier2Channel 8	Local relay	4-CPU	Preamplifier 2 selected audio source channel x
0832	\$Preamplifier2Channel Page	Local relay	4-CPU	Preamplifier 2 selected the page channel
0833	\$Preamplifier3	Audio output	4-CPU	Preamplifier 3 output signal detected
0834 - 0841	\$Preamplifier3Channel 1 - \$Preamplifier3Channel 8	Local relay	4-CPU	Preamplifier 3 selected audio source channel x
0842	\$Preamplifier3Channel Page	Local relay	4-CPU	Preamplifier 3 selected the page channel
0843	\$Preamplifier4	Audio output	4-CPU	Preamplifier 4 output signal detected
0844 - 0851	\$Preamplifier4Channel 1 - \$Preamplifier4Channel 8	Local relay	4-CPU	Preamplifier 4 selected audio source channel x
0852	\$Preamplifier4Channel Page	Local relay	4-CPU	Preamplifier 4 selected the page channel
1001 - 1999	\$IPReceiverAccount1 - \$IPReceiverAccount 999	IP account	4-FWAL1, 2, 3, 4	IP receiver CMS account x
2001 - 2999	\$EmailAccount1 - \$EmailAccount999	Email account	4-FWAL1, 2, 3, 4	Email account x
3001 - 3020	\$User1 - \$User20	Local monitor	4-CPU 4-ANNCPU 4-FWAL1,2,3,4 4-NET-AD 4-CPUGRPH	User x log on input

[1] If the event persists after a panel reset, call Edwards Technical Support at +1 800 655 4497.

Appendix D

UL 864 programming requirements

Summary

This appendix summarizes control unit programming options.

Content

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UL 864 programming requirements

For programming instructions, refer to the *4-CU Help*.

NOTICE TO USERS, INSTALLERS, AUTHORITIES HAVING JURISDICTION, AND OTHER INVOLVED PARTIES:

This product incorporates field-programmable software. In order for the product to comply with the requirements in the *Standard for Control Units and Accessories for Fire Alarm Systems*, UL 864, certain programming features or options must be limited to specific values or not used at all as indicated below.

Programmable feature or option	Permitted in UL 864? (Y/N)	Possible settings	Settings permitted in UL 864
Panel Silence Cancel	Y	Off 00:00:10:00 to 07:00:00:00 (dd:hh:mm:ss)	00:00:10:00 to 01:00:00:00 (dd:hh:mm:ss) [1]
AC Fail Delay	Y	Off 00:01:00 to 02:00:00 (dd:hh:mm)	00:01:00 to 00:03:00 (dd:hh:mm)
Event message routing	Y	All None User defined (1 to 15)	All None [2] User defined routes (1 to 15) [3]
Event message display filtering: Alarm, Supervisory, and Trouble options	Y	Enabled Disabled	Enabled Disabled [4]
Delays (programmed in rules)	Y	0 to 65,535 seconds	0 to 65,535 seconds [5]
Alarm verification	Y	None 4 to 56 seconds	4 to 44 seconds
Alternate alarm verification	Y	None 4 to 56 seconds	4 to 44 seconds
Automatic alarm silence	Y	Off 00:01 to 01:00 (hh:mm)	Off 00:03 to 01:00 (hh:ss)
Panel Silence Cancel	Y	Off 00:00:10:00 to 07:00:00:00 (dd:hh:mm:ss)	00:00:10:00 to 01:00:00:00
CMS event reporting priority (programmed in rules)	Y	1 to 255	1 to 255 [6]
CMS activate and restore messages (programmed in rules)	Y	Send on activation Send on restoration	Activation and restoration triggers must match the message type
Zone group member device types, Activation event: Alarm	Y	GENALARM SMOKE SMOKEVFY HEAT PULL STAGEONE STAGETWO WATERFLOW COALARM COSUPERVISORY	GENALARM SMOKE SMOKEVFY [7] HEAT PULL STAGEONE [7] STAGETWO [7] WATERFLOW COALARM COSUPERVISORY

Programmable feature or option	Permitted in UL 864? (Y/N)	Possible settings	Settings permitted in UL 864
AND group member device types, Activation event Alarm	Y	GENALARM SMOKE SMOKEVFY HEAT PULL STAGEONE STAGETWO WATERFLOW COALARM COSUPERVISORY	GENALARM SMOKE SMOKEVFY [8] HEAT PULL STAGEONE [8] STAGETWO [8] WATERFLOW COALARM COSUPERVISORY
AND group device activation count	Y	1 to 255	1 to 255 [9]
Matrix group member device types	Y	GENALARM SMOKE SMOKEVFY HEAT PULL STAGEONE STAGETWO WATERFLOW COALARM COSUPERVISORY	GENALARM SMOKE SMOKEVFY [7] HEAT PULL STAGEONE [7] STAGETWO [7] WATERFLOW COALARM COSUPERVISORY
Matrix groups: Device activation count	Y	3 to 10	3 to 10 [9]
Signature input modules: Security device types	N	Guard Security Security24Hour SecurityDay SecurityInterior SecurityInteriorMonitor SecurityPerimeter SecurityPerimeterMonitor	N/A
Signature input modules: Personality code 18	N	Alarm – soft short latching, European Style Class C (Class B)	N/A
Signature input modules: Maximum Delta Count	N	0 to 9	N/A
Signature input modules: Delay	N	None Delayed Follower	N/A
SIGA-IO(-MIO) modules: GenAlarm Personality codes 35 and 36	N	(35) Alarm – NO delayed latching input/output NO (36) Alarm – NO delayed latching input/output NC	N/A
CO Supervisory event	N	Latching Nonlatching	N/A
CO Monitor event	N	Latching Nonlatching	N/A
4-FWALx, 4-NET-AD modules: Display Partition Filtering	N	All None All user-defined partition routing groups	N/A

Programmable feature or option	Permitted in UL 864? (Y/N)	Possible settings	Settings permitted in UL 864
LCD screen configuration: Queues to Display	Y	Call In Request Alarm Emergency Supervisory Security Building Disable Trouble Ground Other	Call In Request Alarm Emergency Supervisory Building Disable Trouble Ground Other
LCD screen configuration (4-CPU, 4-ANNCPU, 4-CPUGPH, 4-FWALx, 4-NET-AD: Queues to Alert Signal Only)	Y	Call In Request Alarm Emergency Supervisory Security Building Disable Trouble Ground Other	Call In Request Alarm Emergency Supervisory Building Disable Trouble Ground Other
User Timeout	Y	Off 00:00:15 to 08:00:00 (hh:mm:ss)	Off 00:00:15 to 08:00:00 (hh:mm:ss) [10]

[1] Allowed only on control units that transmit trouble event signals off premises

[2] Allowed only with monitor device types and switches

[3] Allowed only if the user route includes the control unit

[4] Allowed only on nonrequired remote annunciators

[5] Allowed only when setting does not prevent the activation or transmission of alarm or supervisory signals within 10 seconds or trouble signals within 200 seconds

[6] When priorities are used, alarm events must have a higher priority than supervisory and trouble events

[7] Not allowed in Zone groups that are used to initiate the release of extinguishing agent or water

[8] Not allowed in AND groups that are used to initiate the release of extinguishing agent or water

[9] A minimum device activation count of 2 is required if the AND group or Matrix group is used to initiate the release of extinguishing agents or water

[10] Set User Time Out for 30 minutes or less when you are using access level passwords as the only means of preventing unauthorized access to the system.

Glossary

Term	Definition
active	Points that are in an alarm state.
activate	To turn on or energize. Pertains to outputs (including logical outputs).
alarm	The state of a fire alarm initiating device that has detected a smoke or fire condition.
alarm silence timer	A panel option that automatically silences the notification appliance circuits (NACs) after a preprogrammed time limit after the last alarm.
AND statement	A system input defined in the 4-CU that activates when <i>all</i> the input conditions as indicated in its AND statement list, are active.
card	See module.
CC	Command Center
CCS	Central Control Station
Central Control Station	Terminology used to identify a command center that may or may not reside in a notification control area, but it can control audio over its own or multiple NCAs.
Class A IDC	A circuit, connected directly to initiating devices that signals a trouble condition upon an open condition on the circuit. All devices wired on the circuit to continue to operate in the event of a single open.
Class A NAC	A circuit, connected directly to notification appliances that signals a trouble condition upon an open or shorted condition on the circuit. All appliances wired on the circuit to continue to operate in the event of a single open.
Class B IDC	A circuit, connected directly to initiating devices that signals a trouble condition upon an open condition on the circuit. All devices wired on the circuit to continue to operate up to the location of a break.
Class B NAC	A circuit connected directly to notification appliances that signal a trouble condition upon an open or shorted condition on the circuit. All appliances wired on the circuit to continue to operate up to the location of a break.
CMS	Central monitoring station
Command Center	Command centers are used to control audio within a Notification Control Area (NCA). The command center resides in a notification control area and it is the control unit used to request, grant, or deny control of the audio operation of an NCA, whether the one in which it resides or another NCA when the command center is a CCS.
command list	A 4-CU logic device used to create a predefined list of 4-CU commands. You can activate a command list from a rule, from another command list, or from an external command and control system.
CU	Configuration Utility. 4-CU software that lets programmers configure and program an EST4 control unit.

Term	Definition
DACT	Digital alarm communicator transmitter. A system component which transmits digital alarm, supervisory, and trouble signals to a central monitoring station over dial-up telephone lines. The 3-MODCOM(P) is a DACT.
database	User-defined, permanently stored, system parameters containing system definitions, device types, responses, messages, etc.
device	Circuits, buttons, or LEDs that exist on the chassis rail or UI frame assembly and all addressable devices connected by field wiring. Any Signature Series detector or module.
device address	A number that uniquely identifies a detector or module on a Signature data loop.
dialer	See DACT.
disable	Prevents an input, output, or system feature from functioning.
download	To send a project database configured in the 4-CU on your PC to the system control unit.
ECPxml	External Communications Protocol XML. A protocol that requires customer licensing to allow communication with the control unit.
enable	Permits an input, output, or system feature to function.
EVAC	Emergency Voice/Alarm Communications
fiber optic	Communication format that uses light signals carried on glass fibers to transmit and receive data.
group	A collection of Signature devices defined in the 4-CU that is treated as a single entity for programming purposes.
IDC	Initiating device circuit. An input circuit connected directly to any manual or automatic initiating device, whose normal operation results in an alarm or supervisory signal indication at the control unit. The electrical integrity of the circuit is monitored by the fire alarm system.
input	A signal generated by a field device and sent to the control unit for evaluation and responses as determined by the system database. Inputs to the system are detectors, modules, and switches.
label	A unique identifier for a device.
local system	A system that operates according to the provisions of NFPA 72.
loop	The wiring that connects devices to the fire alarm control unit.
matrix	A 4-CU logic device to create a correlation sheet that indicates the relationship between the activation of an input and the effect it will have upon all system outputs.
modem	Short for modulator/demodulator. A communications device that enables a computer to transmit information over a standard telephone line. Sophisticated modems are also capable of such functions as automatic dialing, answering, and redialing in addition to transmitting and receiving. The 3-MODCOM(P) includes a modem.
module	Modules (cards) installed on the backbox chassis rail and control-display modules installed on the inner door UI frame assembly.
NAC	Notification appliance circuit. A circuit connected directly to notification appliances. The electrical integrity of the circuit is monitored by the fire alarm system.
NCA	Notification Control Area
node	A control unit used to monitor system inputs for status changes and activate system outputs. The following are node modules: 4-FWALx, 4-CPU, 4-ANNCPU, 4-CPUGRPH, and 4-NET-AD.

Term	Definition
notification control areas	Grouped nodes installed throughout the premises that provide a structure for the distributed audio system and are used to define the audio subsystem's coverage area. A basic NCA has one or more command centers for controlling the operations of the members of the NCA. In some installations multiple NCAs can be managed by one or more CCSs.
output	A signal generated by the system based upon responses defined in the system database and sent to external field devices. Outputs are LEDs and modules.
output priority	A system of hierarchy that allows or prevents setting or resetting outputs. Output priorities range from low to high.
personality code	A number code used to set the configuration and operation of a SIGA module. The personality code is either factory installed or must be downloaded into SIGA modules for proper operation.
point	See device.
POTS	Plain Old Telephone Service. Standard telephone service employing analog signal transmission over copper loops.
power-limited	Wiring and equipment that conforms with and is installed to the <i>National Electrical Code</i> , Article 760, power-limited provisions.
proprietary system	A system that operates according to the provisions of NFPA 72.
pseudo point	An input or output point that is not a physical device. For example, ground fault and communication fault notifications.
Request Grant Deny	A 4-CU logic device that works in association with NCAs and command centers to arbitrate demands for control of audio communications in the life safety system.
reset	An active condition or command used to force an output to its OFF condition. An output's OFF state may be in the restored condition (normal condition, not under the influence of a response) or the reset condition. An output reset state contains a priority level.
response	A list of outputs or functions that occur as a result of the change of state of an input.
restore	A condition of an input, where the input is not active. It also refers to the condition of an output where the output is not in its set or reset condition and does not have a priority value associated with it.
RGD	See Request Grant Deny.
riser	An electrical path that contains power or a signal that is used by multiple outputs, zones, or circuits.
RS-232	A serial communications format normally used for serial peripheral devices.
sequence	A series of actions separated by time delays.
service group	A 4-CU logic device that defines a collection of devices that are configured for testing as a group using the system test function.
signaling line circuit	The wiring that connects devices to the fire alarm control unit.
Signature data loop	The wiring that connects Signature Series devices to the fire alarm control unit.
SMTP	Simple Mail Transfer Protocol. The Internet standard for email transmission.
supervisory circuit	An IDC input circuit used to monitor the status of critical fire protection equipment.
supervisory open (trouble)	A condition generated when a supervisory zone is open or in ground fault, or when a Signature Series device is not responding to a poll.
time control	An input activated by the time of day or day of the month.

Term	Definition
verification alarm	Upon receipt of an alarm by a smoke detector, verified detectors attempt to automatically reset. Receipt of a second alarm within the 60-second confirmation period after the automatic detector reset period is indicative of a verified alarm.
zone	A 4-CU logic device that defines a group of Signature Series detectors and modules that has a unique zone number and acts as a single entity for programming purposes whenever any component of the zone is activated

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